1
TERPENOIDS

1.1. GENERAL INTRODUCTION

Definition. Originally, the term "**terpene**" was employed to describe a mixture of isomeric hydrocarbons of the molecular formula $C_{10}H_{16}$ occurring in the terpentine[1] and many essential oils which are obtained from the sap and tissues of certain plants and trees. The oxygenated derivatives like alcohols, aldehydes, ketones, etc. at that time were called **camphors.**

As more compounds relating to terpenes and camphors were discovered with the pace of time, both the terms "**terpenes**" and "**camphors**" were amalgamated into a single term called '**terpenoids**". The modern definition of this term is as follows :

"*It includes hydrocarbons of plant origin of the general formula* $(C_5H_8)_n$ *as well as their oxygenated, hydrogenated and dehydrogenated derivatives*".

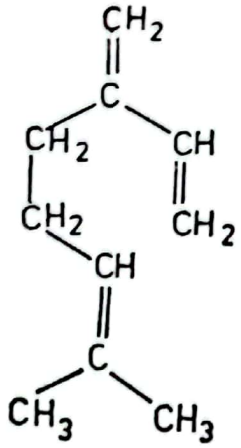
Another reason for rejecting the term "**terpene**" is that since the suffix "ene" signifies unsaturated hydrocarbons, the name "**terpene**" is inappropriate to include compounds such as alcohols, aldehydes, ketones, etc.

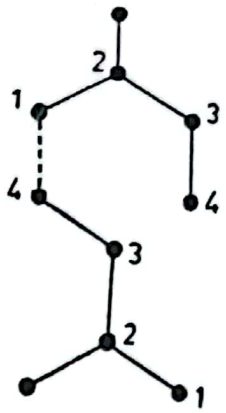
The term "**terpene**" is being used these days by some authors to represent terpenoids. However, this tendency is generally restricted to the hydrocarbons, $C_{10}H_{16}$.

As terpenoids are composed of isoprene units, these are sometimes called **isoprenoids**. Not only the carbon skeletons of terpenoids are divisible into isoprene units but the terpene hydrocarbons are usually exact multiplies of C_5H_8. An example is myrcene ($C_{10}H_{16}$) which has a carbon skeleton divisible into two isoprene units (Fig. 1).

Nomenclature. Many terpenoids are known by their trivial names because these were isolated and described in the literature long before their structures were elucidated. Generally these trivial

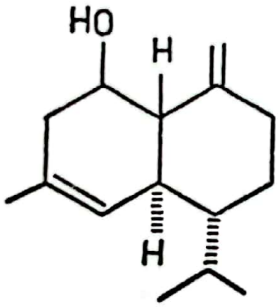
1, Terpentine is a German word which means turpentine.



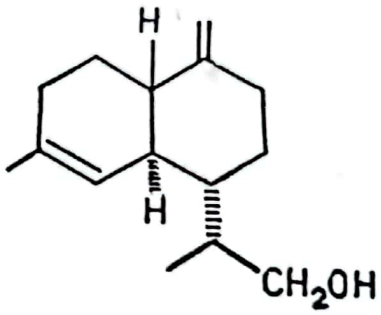


myrceen

names are derived from their botanical origins. For example, the crystalline sesquiterpene secondary alcohol belonging to the cadinane group was isolated from vetiver oil (vetiveria zizaniodides) and named '*khus*' oil in India. Therefore, its alcohol was called khusinol and another crystalline alcohol which also belonged to the same group and isolated from the same source was called khusol.

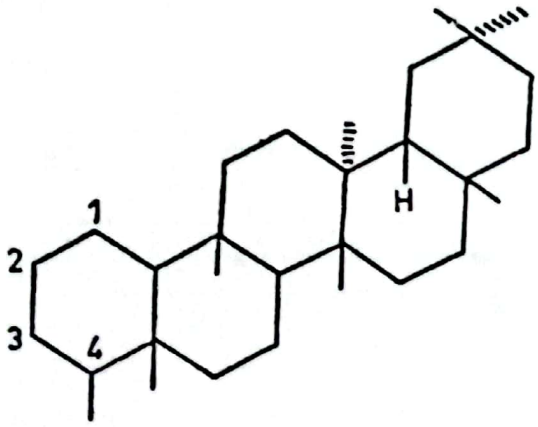


Khusinol



Khusol

In 1899, Istrati and Ostrogovich isolated two crystalline compounds from cork and named the more soluble of the two compounds as friedelin in honour of Friedel. Terpenoids of friedelin group have the following carbon framework and cerin, which is a keto alcohol, is belonging to this group.



Friedelin $>C=O$ at C_3
Cerin $>C=O$ at C_3
and $2\beta-OH$

Even these days the usual practice is to refer the terpenoids by modification of a trivial name.

Importance. In the recent past, biological activity of various substances has been related with terpenoids. Many sesquiterpenes and diterpenes are used as antibiotics. Several sesquiterpenes have been found to be active against experimental tumours and the plant growth hormones like gibberellins and diterpenoids.

As some terpenoids exhibit biological activity, *viz.*, insecticidal anthelmintic or antiseptic, those are used in pharmacy

Occurrence. Terpenoids are most widespread, chemically interesting and provide structures of great diversity. Although the majority of terpenoids occur in the plant kingdom, a few of them have also been obtained from other sources.

Most of the fragment components of plants are volatile with steam and may be isolated by steam distillation, solvent extraction, or other treatment of the plant. These components are called the **essential oils**. These have been used in perfumery from the earliest times. The simpler mono-and sesquiterpenoids are the chief constituents of the essential oils. However, the di-and tri-terpenoids, which are not steam volatile, are obtained from plant and tree gums and resins. Unlike mono-and sesqui-terpenoids, these compounds do not possess any perfumery value.

The **tetraterpenoids** constitute a group of compounds called **carotenoids**. This is studied under a separate group. This group includes carotenes xanthophylls and carotenoidic acids. Rubber is the most important polyterpenoid.

Some of the important essential oils with their terpenoid constituents are given in Table 1.1.

1.2. GENERAL PROPERTIES OF TERPENOIDS

As far as physical properties of terpenoids are concerned, most of them are colourless liquids which are lighter than water and boil between 150 and 180°C. A few are solids which are lighter than water, volatile in steam, usually highly refractive and insoluble in water but soluble in organic solvents. Many of these are optically active.

The various general chemical properties of terpenoids are as follows :

1. They are unsaturated compounds (open-chain or cyclic with one or more carbon atom rings) having one or more double bonds. Consequently, terpenoids undergo addition reactions with hydrogen, halogens, halogen acids, etc. Some of them form hydrates. They also form characteristic addition products with NO_x, NOCl and NOBr. These addition products are found to be useful

TABLE 1.1

Important Essential Oils with their Terpenoid Constituents

Essential oil	*Constituent terpenoid*
1. Bay	Eugenol
2. Bergamot	Linalool and its acetate
3. Caraway	Limonene, carvone
4. Camphor	Camphor
5. Cardamon	Terpineol
Cajeput,	
6. Citronella	Farnesol, citronellal and geraniol
7 Clove	Eugenol
8. Coriander	Linalool, α-pinene
9. Eucalyptus	Cineole
10. Geranium	Geraniol, citronellol
11. Ginger	Zingiberene
12. Jasmine	Linalool
13. Lavender	Linalool
14. Lemon	dl-Limonene, citral
15. Neroli	Nerolidol
16 Peppermint	Menthols and its esters
17. Rose	Geraniol, citronellol and farnesol
18 Sandal wood	Santalol
19. Sweet orange	*d*-Limonene

in the identification of terpenoids. A number of addition products have antiseptic properties

2. They undergo polymerisation, also dehydrogenation in the ring.

3. As they have olefinic bonds, they are very easily oxidised nearly by all the oxidising agents.

4. A number of terpenoids are labile and hence readily isomerised in the presence of acids into more stable forms.

5. On thermal decomposition, most of the terpenoids yield isoprene as one of the products.

$$\text{Terpenoid} \xrightarrow{\text{Heat}} \underset{\text{Isoprene}}{CH_2=\overset{CH_3}{\overset{|}{C}}-CH=CH_2}$$

1.3. ISOLATION

Due to their wide occurrence in nature, all the terpenoids could not be isolated and separated by a general method. However, mono-and sesqui-terpenoids have a common source, *i.e.*, essential oils and, therefore, their isolation has been generalised. This is carried out in two steps as follows :

1. Isolation of essential oils.
2. Separation of terpenoids from essential oils.

Let us discuss these steps one by one.

1. Isolation of essential oils. As plants having essential oils generally possess maximum concentration at some particular

time, e.g., Jasmine at sun-set, it becomes desirable to take the plant parts having essential oil at this particular time. In general, our methods based on different principles have been developed for the extraction of oils. These methods are described as follows :

(*a*) **Expression method**. This method is not used these days. However, it has the historical importance only.

In this method, the plant material is cut into small pieces. These are then crushed to get the juice which is screened to remove the larger particles. After screening, the juice is centrifuged in a high speed centrifugal machine when one-half of the oil is extracted and the rest half of the essential oil remains with the residue. From this residue, inferior quality of the oil is obtained by distillation.

Expression method is generally used to extract citrus, lemon and grass oils.

(*b*) **Steam distillation method**. This method is the one most widely used. In this method, the plant materials are macerated and then steam distilled to get the essential oils into the distillate from which these are extracted by using pure organic volatile solvents like light petroleum and the solvent is then removed by distillation under reduced pressure. One should employ steam distillation method carefully because it suffers from the following demerits :

(*i*) Some essential oils undergo decomposition during steam distillation.

(*ii*) Some constituents of the essential oils, *e.g.*, esters, which are resposible for the odour and fragrance of the oil may undergo decomposition, resulting in a perfume of inferior quality.

(*c*) **Extraction by means of volatile solvents**. This method is widely used in perfume industry. This method is generally used for such plants which yield an oil or give low quantities of oil on steam distillation due to decomposition of essential oils. In such cases, the plant material is directly treated with light petrol at 50°C. Under these conditions the oil is taken up by the solvent along with the soluble colouring materials. The essential oils from this extract are separated by removing the solvent by distillation under reduced pressure.

(*d*) **Adsorption in purified fats**. This method is also known as *enfleurage method* and is widely employed in France. By this method, the yield of the essential oil is generally higher. This method is used to extract a large number of essential oils like rose and jasmine.

In this method, the fat is warmed to 50°C in glass plates. Then, the surface of the fat is covered with flower petals and it is allowed to be kept as such for several days until it becomes saturated with essential oils. Then, the old

petals are replaced by fresh petals and this process is repeated. After removing the petals, the fat is digested with ethyl alcohol when all the oils present in fat are dissolved in alcohol. Some quantity of fat is also dissolved in alcohol. This can be removed by cooling the alcohol extract to 20°C, when the fat separates out. The alcoholic distillate is then finally fractionally distilled under reduced pressure to remove the solvent.

Recently, the fat has been replaced by coconut charcoal due to its greater stability and higher adsorptive capacity. After keeping the coconut charcoal in contact with petals for a number of days, the charcoal is submitted to steam to get essential oils. This method is superior to the enfleurage method.

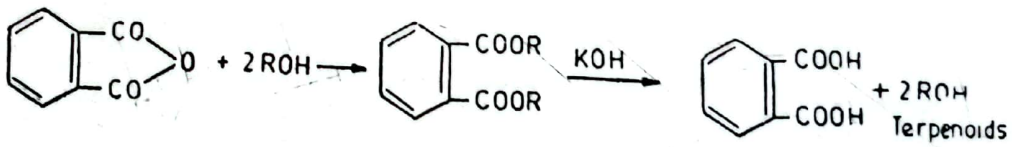
2. Separation of Terpenoids from Essential Oils. The essential oils obtained from the step 1 generally contain a number of terpenoids and these are separated by various physical and chemical methods.

(*a*) *Chemical methods.* These methods are not used these days to separate various terpenoids from essential oils. However, the various chemical methods are as follows :

(*i*) When essential oils containing terpenoid hydrocarbons are treated with nitrosyl chloride in chloroform, crystalline adducts of hydrocarbons having sharp melting points are obtained. These are separated and decomposed into their corresponding hydrocarbons.

(*ii*) When essential oils containing alcohols are treated with phthalic anhydride to form diesters, the primary alcohols react with phthalic anhydride readily, secondary alcohols less readily and tertiary alcohol do not react at all.

After extracting with sodium bicarbanate, diesters are decomposed by alkali to the parent terpenoid alcohols.



(*iii*) Terpenoid aldehydes and ketones are separated from essential oils by forming their adducts with the common carbonyl reagents like $NaHSO_3$, 2, 4-dinitrophenylhydrazine, phenylhydrazine, semicarbazide, etc. After separation, these are decomposed to regenerate terpenoid aldehydes and ketones.

(*b*) *Physical methods.* The various physical methods are as follows :

(*i*) *Fractional distillation methods.* The various terpenoids present in essential oils are separated by fractional distillation method. The terpenoid-hydrocarbons distil over first, followed by

the oxygenated derivatives. Distillation of the residue under reduced pressure yields the sesquiterpenoids and these are separated by fractional distillation.

On an industrial scale, specially designed stills are employed and an efficient condensing system is necessary to minimise loss of more volatile hydrocarbons.

Many times, the fractional distillation has to be carried out under reduced pressure and in the presence of an inert gas. These conditions are essential because many terpenes are sensitive to heat and atmospheric oxygen.

(*ii*) *Chromatography.* More recently, chromatography in its various forms has been widely used both for isolation and separation of terpenoids.

In adsorption chromatograpy, the essential oil is made to flow through a particular adsorbent when the different types of terpenoids are adsorbed at different places on the adsorbent to form different chromatograms. Then, the various chromatograms are eluted by different solvent systems to get different eluates (each eluate is having terpenoids of a single group). Each eluate is then subjected separately to adsorption chromatography when different bands due to the various terpenoids present in eluate are obtained which are then eluted to yield different terepenoids.

In adsorption chromatographic method, alumina and silica gel are generally used as adsorbents for separating the terpenoids, particularly triterpenoids.

Other chromatographic techniques such as vapour phase chromatography, partition chromatography and counter-current separation method have been used for the separation of terpenoids.

Gas chromatography has been particularly useful for isolating pure configurational forms of a given terpenoid from mixtures produced by synthesis.

1.4. ISOPRENE RULE

In 1887, Wallach enunciated the famous **isoprene rule** which may be stated as follows : "*The skeleton structures of all naturally occurring terpenoids are built up of isoprene units.*"

From the above rule it follows that the divisibility into isoprene units is regarded as a necessary condition to be satisfied by every naturally occurring terpenoid.

The isoprone rule has been deduced from the following facts :

(*a*) The empirical formula of almost all the naturally occurring terpenoids is C_5H_8.

(*b*) The thermal decomposition of almost all terpenoids gives isoprene as one of the products. For example, rubber (polyter-

penoid) on destructive distillation yields isoprene as one of the decomposition products.

$$(C_5H_8)_n \xrightarrow[\text{distillation}]{\text{Destructive}} \underset{\text{Isoprene}}{nC_5H_8}$$

Isoprene rule has been confirmed by the fact that under special experimental conditions, isoprene undergoes polymerisation to yield various terpenoids. For example,

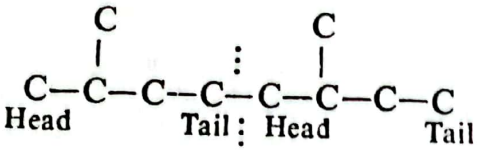
(*i*) Isoprene, when heated to 280°C, gets dimerised to yield a widely distributed terpenoid called dipentene.

$$\underset{\text{Isoprene}}{2\,C_5H_8} \xrightarrow[280C^\circ]{\text{Heat}} \underset{\text{Dipentene}}{C_{10}H_{16}}$$

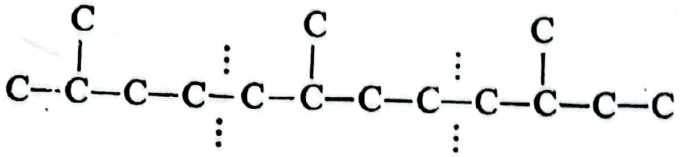
(*ii*) Isoprene may be polymerised to yield a rubber-like product.

$$n\,C_5H_8 \xrightarrow{\text{Polymerisation}} \underset{\text{Rubber (polyterpenoid)}}{(C_5H_8)_n}$$

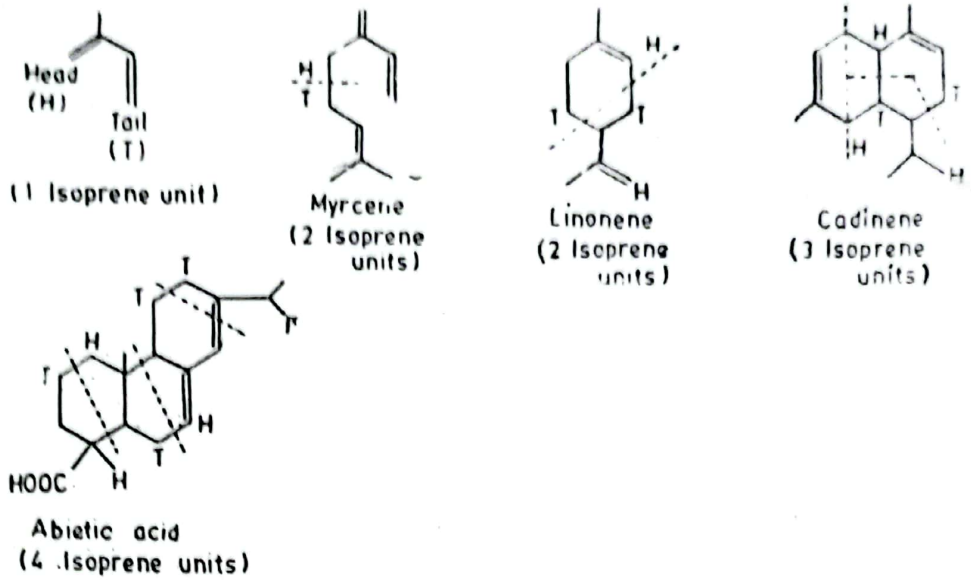
Special Isoprene Rule. This rule, proposed by Ingold (1925), shows how the isoprene units in the terpenoid molecule are linked together. For this purpose, the branched end of the isoprene unit is considered as the **head** and the other end as the **tail**. According to the special isoprene rule, **the isoprene units in terpenoids are linked in a head to tail fashion**. Thus, the basic carbon skeleton of a monoterpene according to the special isoprene rule will be as follows :



Similarly, a sesquiterpenoid with three isoprene units will have the following skeleton :



Special isoprene rule is a very suitable tool to limit the number of carbon skeletons of the structures of unknown terpenoids. Generally, the skeletons which do not follow isoprene rule are generally rejected.

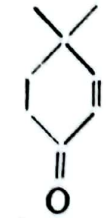


It should be noted, however, that this rule, which has proved very useful, can only be used as a guiding principle and not as a fixed rule. Therefore, there are departures and violations of this rule in many compounds. For example,

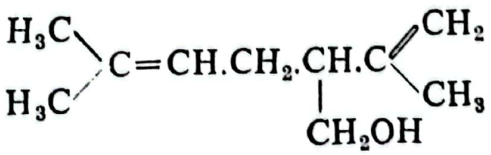
(*i*) Certain terpenoids are known whose carbon content is not a multiple of five. For example, cryptone, a naturally occurring ketonic terpenoid contains nine carbon atoms and, therefore, it cannot be divided into two isoprene units.

(*ii*) Certain terpenoids are known whose carbon content is a multiple of five but cannot be divided into two isoprene units.

(*iii*) Certain terpenoids are also known in which the special isoprene rule is violated. For example, lavandulol is composed of two isoprene units of isoprene with the unusual linkage of tail to tail but not head to tail. Also, the carotenoids are joined fail to tail at their centre.



Cryptone



Lavandulol

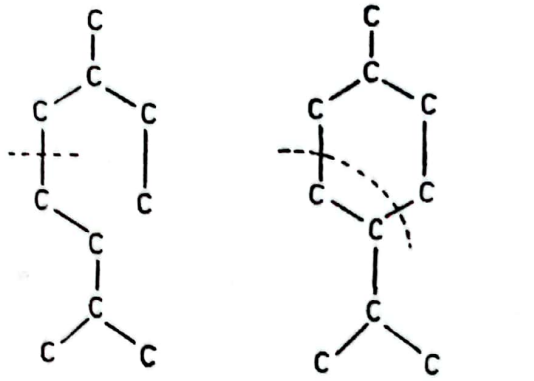
Gem-dialkyl Rule.

When two alkyl groups are attached to the same carbon atom, it is called **gem.dialkyl group**. The presence of gem-dialkyl group affects the stability of terpenoids. This was observed by Ingold who summarised his results in the form of empirical rule called **gem-dialkyl rule** which may be stated in words as follows :

"The gem-dialkyl group tends to render the cyclohexane ring unstable whereas it stabilises the three, four and five membered rings."

The gem-dialkyl group is helpful in two ways :

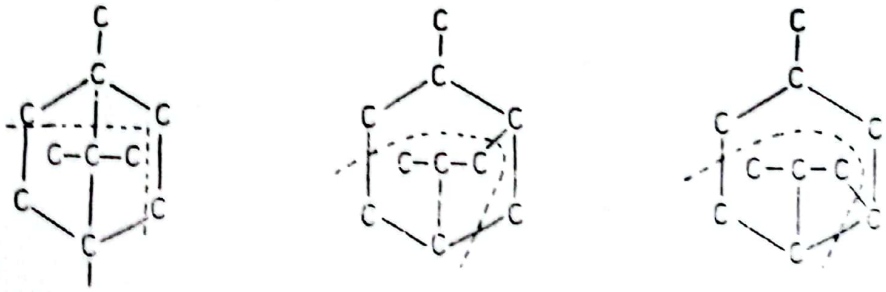
(*i*) This rule limits the number of possible structures of terpenoids. For example, the monoterpenoid open chain would give rise to only one possibility for a monocyclic monoterpenoid, *i.e.*, the *p*-cymene structure which is shown as follows :



Acyclic structure *p*-Cymene structure

Most natural monocylic monoterpenoids are derivatives of *p*-cymene.

(*ii*) This rule helps to find the carbon skeleton of cyclic terpenoids. Thus, the six membered terpenoids should contain the gem-dimethyl group outside the ring whereas the bicyclic terpenoids should contain gem-dimethyl group on the smaller rings, i.e., cyclopropane and cyclobutane rings, to render them sufficiently stable to be capable of occurrence in nature. For instance, closure of the 10C open chain would give rise to three possible structures ; actually all the three are known.



1.5. CLASSIFICATION OF TERPENOIDS

The terpenoids have the general formula $(C_5H_8)_n$ and the value of *n* is used as a basis of classification. This is given in Table 1.2.

TABLE 1.2

Classification of Terpenoids

S. No.	*Value of n*	*Number of carbon atoms*	*Class*	*Molecular formula of terpenoid*
(*i*)	2	10	Monoterpenoids	$C_{10}H_{16}$
(*ii*)	3	15	Sesquiterpenoids	$C_{15}H_{24}$
(*iii*)	4	20	Diterpenoids	$C_{20}H_{32}$
(*iv*)	5	25	Sesterterpenoids	$C_{25}H_{40}$
(*v*)	6	30	Triterpenoids	$C_{30}H_{48}$
(*vi*)	8	40	Tetraterpenoids (Carotenoids)	$C_{40}H_{64}$
(*vii*)	>8	>40	Polyterpenoids	$(C_5H_8)_n$

The sesterterpenoids have been discovered recently. The first example of this class called solanesol, an acyclic unsatuarated alcohol, was discovered in 1956. In addition to the hydrocarbons in each class given in Table 1.2, oxygenated compounds like alcohols, aldehydes, ketones are also known which occur in nature.

Each class of terpenoids has been further subdivided into subclasses according to the number of rings present in the molecule.

(*i*) **Acyclic Terpenoids.** These contain an open-chain structure.

(*ii*) **Monocyclic terpenoids.** These have *one* ring in the molecule.

(*iii*) **Bicyclic terpenoids.** These contain *two* rings in the molecule.

(*iv*) **Tricyclic terpenoids.** These contain *three* rings in the molecule.

(*v*) **Tetracyclic terpenoids.** These contain *four* rings in the molecule.

1.6. GENERAL METHODS FOR THE DETERMINATION OF STRUCTURE OF TERPENOIDS

The fundamental researches done by Wallach, Baeyer, Perkin, Semmler, Simonson, Ruzicka, etc. are of great importance in elucidating the complicated structures of terpenoids. All the methods used for these have been grouped into *four* classes :

1. Analytical methods.
2. Synthetical methods.
3. Physical methods.
4. Knowledge of a molecular rearrangement.
5. Synthesis.

We will discuss these one by one.

1. *Analytical methods* : These include the following methods :

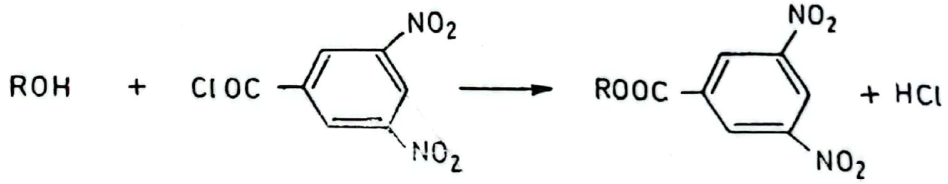
CamScanner

(*a*) **Molecular formula.** After obtainining the pure terpenoid from the natural source, its molecular formula is ascertained by the usual methods of qualitative and quantitative analysis, and also by means of mass spectrometry. If the terpenoid is optically active, its specific optical rotation is measured. Sometimes, optical rotation may be used to make distinction between the structures as in limonene and carvone. Then, the other physical constants like melting point, boiling point, specific rotation, etc., of terpenoids are also determined.

(*b*) **Nature of the oxygen atom.** If oxygen is present in a terpenoid, its functional nature is ascertained by the usual methods. The terpenoids generally contain hydroxy, aldehyde, keto or carboxylic groups. Other oxygenic functional groups, e.g.. $-OCH_3$ $-CONH_2$, $-NO_2$, etc. have not been reported in terpenoids. The various groups are detected in terpenoids by the following methods :

(*i*) The hydroxy groups can be detected by the formation of crystalline acetates with acetic anhydride and benzoates with 3, 5-dinitrobenzoyl chloride. These also yield crystalline substituted urethans with phenylisocyanate

$$ROH + (CH_3CO)_2O \longrightarrow \underset{\text{Acetate}}{ROCOCH_3} + CH_3COOH$$



3,5 – Dinitrobenzoate

$$ROH + O=C=N-C_6H_5 \longrightarrow \underset{\text{Substituted urethan}}{R-O-\overset{\overset{O}{\|}}{C}-NH-C_6H_5}$$

Further information about the nature of hydroxy group is revealed by the rate of esterification. For example, the primary alcohols undergo esterification more readily than secondary and tertiary alcohols. Thus, α-terpineol, a tertiary alcohol, is slowly esterified.

(*ii*) If a terpenoid forms crystalline addition products like bisulphite derivative, oxime and phenylhydrazone, this shows that terpenoid contains a **carbonyl group**.

$$>C=O + NaHSO_3 \longrightarrow \underset{\text{Sodium bisulphite derivative}}{>C\begin{matrix}OH\\SO_3Na\end{matrix}}$$

$$>C=O + H_2NOH \longrightarrow >C=NOH + H_2O$$

Oxime

$$>C=O + H_2N.NHC_6H_5 \longrightarrow >C=N.NHC_6H_5$$

Phenylhydrazone

The carbonyl group may be present either in the form of aldehyde or keto groups. This can be ascertained by oxidation. The aldehyde on oxidation yields monocarboxylic acid without loss of carbon atom whereas the ketone on oxidation yields a mixture of lesser number of carbon atoms.

$$RCHO + [O] \longrightarrow RCOOH$$

$$R-\overset{\overset{O}{\|}}{C}-CH_2R' + [O] \longrightarrow RCOOH + R'COOH$$

Terpenoids having $-CH_2CO-$ groups exhibit special properties. Such terpenoids form **oximes** with nitrous acid (liberated by action of hydrochloric acid on isoamyl nitrite) and benzylidene derivatives with benzaldehyde in the presence of alkali

$$\begin{array}{l}-CH_2\\ \quad|\\ -C=O\end{array} + O=NOH \xrightarrow[HCl]{C_5H_{11}OH} \begin{array}{l}-C=N-OH\\ \quad|\\ -C=O\end{array} + H_2O$$

Oxime

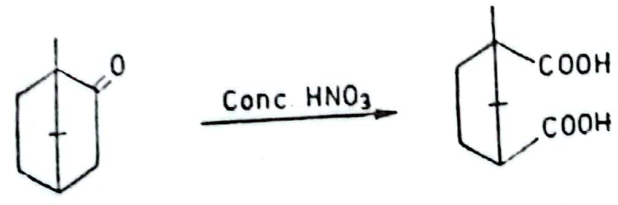
$$\begin{array}{l}-CH_2\\ \quad|\\ -C=O\end{array} + OHCC_6H_5 \xrightarrow{NaOH} \begin{array}{l}-C=CHC_6H_5\\ \quad|\\ -C=O\end{array} + H_2O$$

Benzylidene derivative

If $-CH_2-CO-$ group is present in a ring, the terpenoid on oxidation will yield a dicarboxylic acid without any loss of carbon atom/atoms. For example,

$$\begin{Bmatrix}-CH_2\\ \quad|\\ -C=O\end{Bmatrix} \xrightarrow{[O]} \begin{Bmatrix}-COOH\\ -COOH\end{Bmatrix}$$

Camphor on oxidation with concentrated nitric acid yields a dicarboxylic acid



Camphor

Camphoric acid

If a terpenoid contains $-COCH_3$ group, it can be detected by **Haloform reaction**, e.g.,

$$-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{Br_2+NaOH} -COONa+CHBr_3$$

On oxidation such a terpenoid will form acetic acid.

$$-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{[O]} -COOH+CH_3COOH$$

(*iii*) If the terpenoid is soluble in ammonia and gives effervescence with sodium bicarbonate, these facts reveal that the terpenoid contains carboxyl group. Number of carboxylic groups is estimated by titration against standard alkali. Sometimes it becomes essential to ascertain whether the carboxyl group is attached to a primary, secondary or tertiary carbon. This is done from the determination of rates of esterification of the acid which are in the following order :

$$\text{Tertiary} < \text{Sec.} < \text{Primary}$$

Tertiary —COOH groups are very difficult to be esterified and acids with such groups evolve carbon monoxide with concentrated sulphuric acid. Also, such acids do not form halogen substituted derivatives which are characteristic of the acids having α-hydrogen atoms.

(*c*) **Unsaturation.** The presence and number of olefinic double bonds in terpenoids have been ascertained by the formation of addition products with various regents like halogens, halogen acids, hydrogen, per acids and nitrosyl chloride (Tilden reagent).

(*i*) When a terpenoid possessing olefinic linkages is hydrogenated in the presence of a catalyst, an addition compound called **hydro compound** is formed. From the amount of hydrogen consumed, the extent of unsaturation in a terpenoid can be calculated. For example, the calalytic reduction of cadinene yields tetrahydro cadinene, indicating that cadinene contains two double bonds.

$$-\overset{|}{C}=\overset{|}{C}-+H_2 \xrightarrow[(Catalyst)]{Pd} -\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-\quad (H\ H)$$

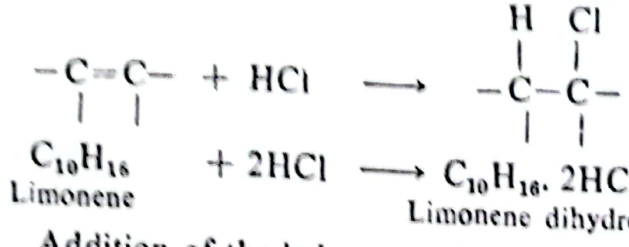
(*ii*) With halogens, terpenoids containing olefinic linkages form addition compounds. The measurement of the amount of halogen used gives the amount of unsaturation.

$$-\overset{|}{C}=\overset{|}{C}-+Br_2 \longrightarrow -\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-\quad (Br\ Br)$$

Baeyer and Villiger employed the method of **exhaustive methylation** to ascertain the nature of the carbon frame work present in a terpenoid molecule.

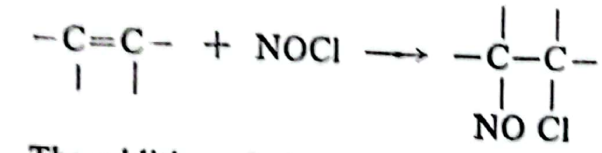
(*iii*) With halogen acids like HCl or HBr, terpenoids yield addition compounds which can be characterised very easily. These addition compounds also often serve for the identification of terpenoids.

In order to carry out halogenation, terpenoid is dissolved in a suitable solvent and then dry halogen acid gas is passed into it. For example, limonene with HCl yields a crystalline dihydrochloride, indicating that limonene contains two double bonds.

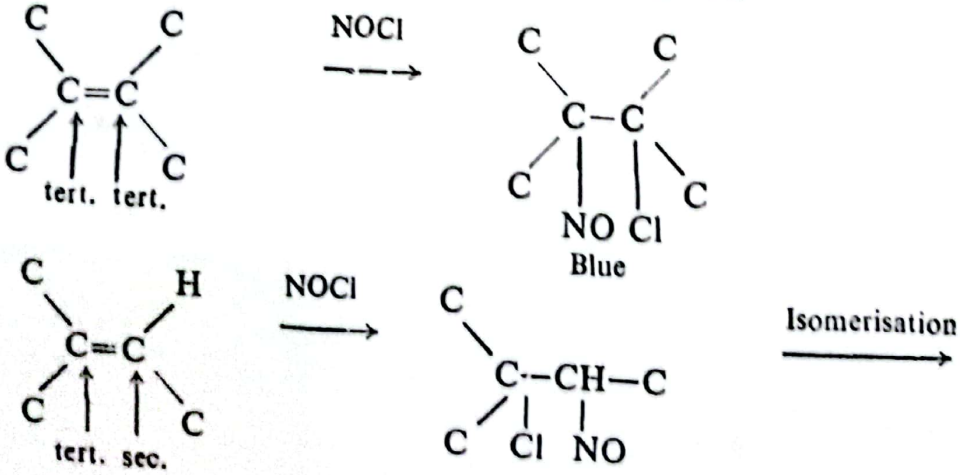


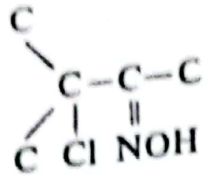
Addition of the halogen acids takes place according to Markownikoff's rule.

(*iv*) With nitrosyl chloride, terpenoids containing olefinic double bonds form addition compounds which are crystalline products having sharp melting points and may be used for identification and separation of terpenoids.



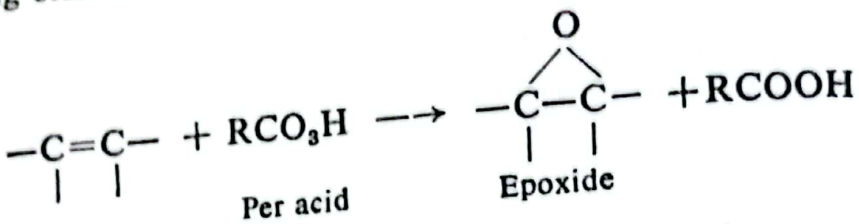
The addition of nitrosyl chloride (**Tilden's reagent**) also gives idea about the nature of the carbon atoms having double bonds in terpenoids. For example, if the two carbon atoms joined by an olefinic linkage are tertiary in nature, then a **blue coloured** nitroso chloride is obtained. On the other hand, if one of the carbon atoms joined to double bond is tertiary while the other secondary, a colourless nitrosyl shloride, is obtained.





The addition of nitrosyl chloride across a double bond takes place in accordance with the Markownikoff's rule.

(*v*) Peracids can be used to determine the amount of unsaturation in a terpenoid molecule. With peracids terpenoids containing olefinic double bonds form epoxides.



A known amount of terpenoid is treated with an excess of a peracid (perphthalic acid) and then the unused amount of acid is determined volumetrically. From the amount of peracid used, the number of olefinic double bonds can be determined.

In order to detect the presence of conjugated system of double bonds in a terpenoid, Diel's Alder reaction is used. Also, optical exaltation indicates the presence of double bonds in conjugation in terpenoids.

(*d*) **Number of rings.** From the above steps, the number of olefinic double bonds and nature of functional groups may be known. Then by considering the number of olefinic double bonds, nature of functional groups and the molecular formula of terpenoid, it is possible to find out the molecular formula of the **parent saturated hydrocarbon** of the terpenoid which on comparison with the formulae given in Table 1.3 for the various types of compounds reveals information about the number of rings present in the terpenoid.

TABLE 1.3

Relation Between General Formula of Compound and Type of Compound

General formula of the compound	*Type of compound*
C_nH_{2n+2}	Acyclic
C_nH_{2n}	Monocyclic
C_nH_{2n-2}	Bicyclic
C_nH_{2n-4}	Tricyclic
C_nH_{2n-6}	Tetracyclic

For example, the molecular formula of citral is $C_{10}H_{16}O$ which contains **two double bonds** and **one oxygen atom** as carbonyl group. Thus, the molecular formula of the parent hydrocarbon would be $C_{10}H_{16}O \equiv C_{10}H_{16} + 4H$ (for two double bonds) $+ 2H$ (for carbonyl oxygen)

$$\equiv C_{10}H_{22}$$

The molecular formula $C_{10}H_{22}$ corresponds to C_nH_{2n-2}, the general formula of acyclic compounds. Thus, citral must be acyclic.

Another example is cadinene which has the molecular formula $C_{15}H_{24}$ and contains two double bonds. Hence the molecular formula of parent saturated hydrocarbon is $C_{15}H_{24} + 4H$ (for two double bonds) or $C_{15}H_{28}$ which corresponds to the general formula C_nH_{2n-2} of bicyclic compounds and hence cadinene is **bicyclic**.

Measurement of the refractive index leads to a value for the molecular refraction which is calculated according to the following formula :

$$R_M = \frac{n^2-1}{n^2+2} \cdot \frac{M}{d}$$

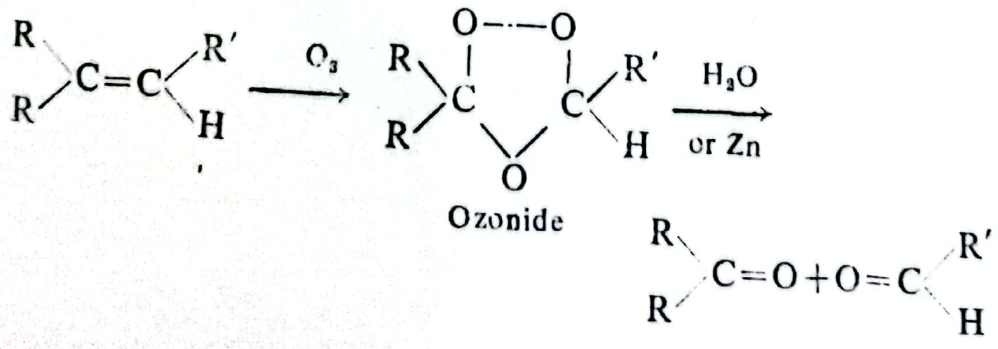
where n is the refractive index, d the density and M the molecular weight of the terpenoid.

From the value of molecular refraction, the nature of carbon skeleton particularly in sesquiterpenoids can be ascertained.

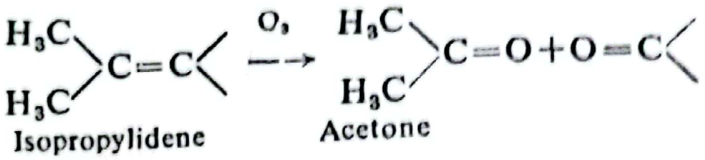
(*e*) **Oxidative degradation products.** After knowing the number of olefinic double bonds, various functional groups and the number of rings present in a terpenoid, the next step is to degrade the terpenoid to small fragments of known structure. Degradation may be carried out by various means but the most important method is that of oxidation.

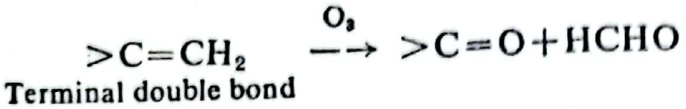
Most of the terpenoids, being olefinic in nature, a suitable oxidative method may be used to cleave the molecule into simpler fragments of known structures which give important clues to the structures of the terpenoids. The common oxidising agents employed in terpenoid chemistry are as follows :

(*i*) **Ozone.** This reagent attacks the olefinic linkage of the molecule to form ozonide which upon decomposition either by hydrolysis or catalytic reduction yields the corresponding carbonyl compounds.



The ozonolysis is generally used to locate the positions of double bonds in a molecule. For example, the formation of acetone and formaldehyde reveals the presence of isopropylidene group and the terminal double bond respectively.

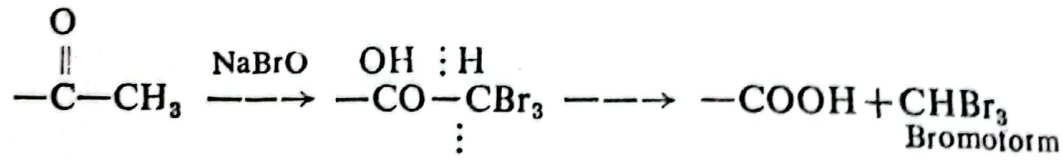


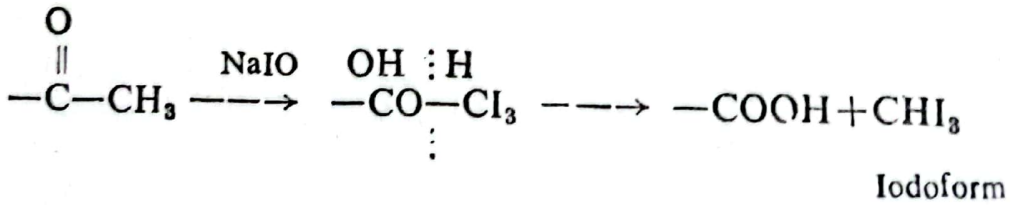


Sometimes, ozonolysis may be used in the determination of structure of terpenoids. For example, ozonolysis has been used in the determination of structure of citral.

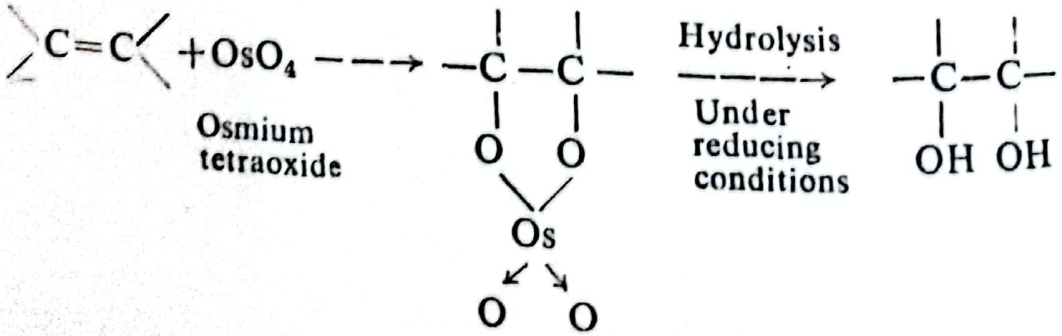
(*ii*) **Nitric Acid.** In a few cases, nitric acid is also used to affect oxidative degradation. As this is a very powerful oxidising agent, it degrades the terpenoid drastically to a mixture of aliphatic and aromatic acids. For example, camphor, when oxidised with nitric acid, is converted into camphoric and camphoronic acids.

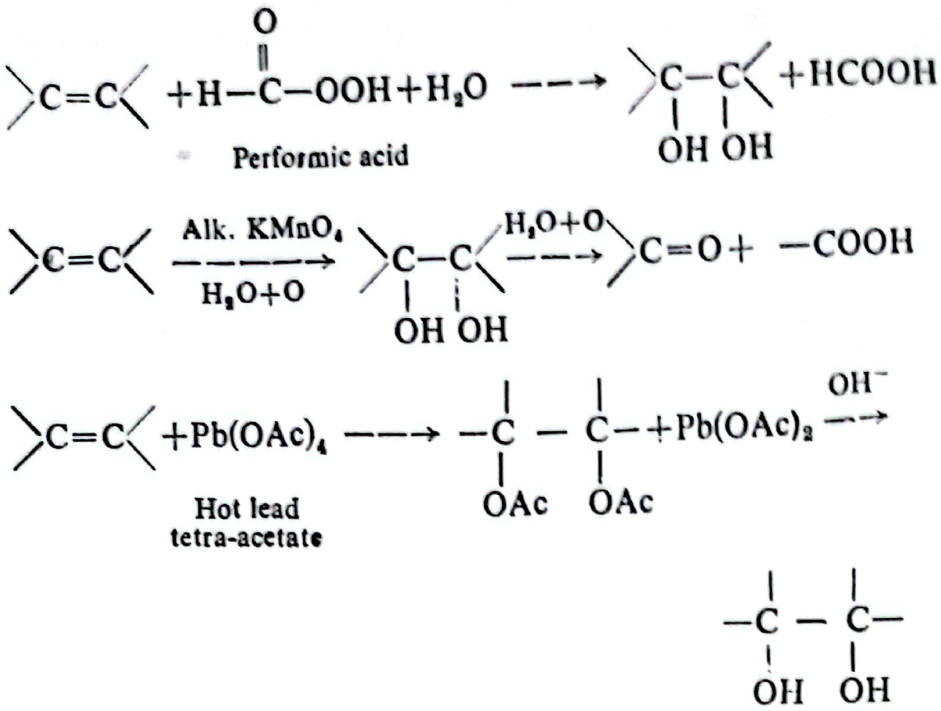
(*iii*) **Sodium hypohalite.** This reagent is used for the removal of one carbon atom as bromoform or iodoform from the methyl ketone (—$COCH_3$) group. This reaction is known as the haloform reaction and may be formulated as follows :



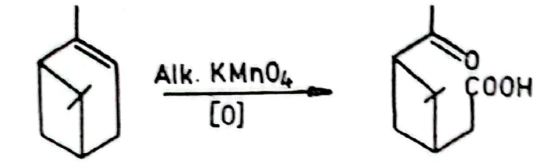


(*iv*) **Hydroxylating reagents.** Examples of these reagents are lead tetra-acetate in hot acetic acid, peracids, osmium tetraoxide, H_2O_2, $KMnO_4$, etc. When the hydroxylating reagents attack the olefinic double bonds of terpenoids, 1, 2-glycols will be obobtained. These glycols may be further cleaved to identifiable products.





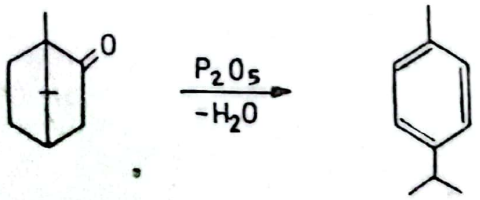
If the double bond is present in the ring, then on oxidation keto acids or dicarboxylic acids are obtained without the loss of carbon atoms. For example, α-pinene on oxidation yields keto acid without the loss of carbon atoms.



α- Pinene Pinonic acid

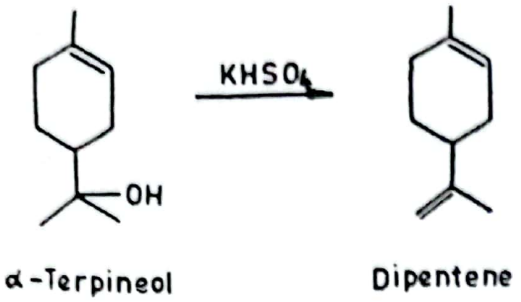
The chromic acid can be used to oxidise the acids, and ketones in different stages.

(*v*) **Dehydration.** When terpenoids containing the alcoholic and ketonic groups are heated with dehydrating agents, these lose a molecule of water and are converted into simple aromatic compounds. For example, comphor when heated with anhydrons zinc chloride or phosphorus pentoxide yields *p*-cymene.



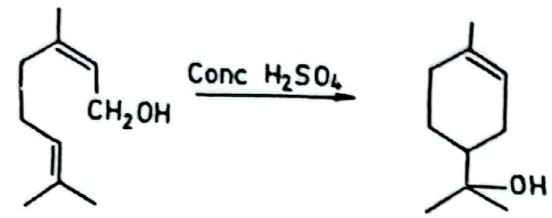
p- Cymene

Similarly, α-terpineol is dehydrated easily by potassium bisulphate to dipentene.



α-Terpineol Dipentene

When the open-chain terpenoids are dehydrated with dehydrating reagents, they yield cyclic compounds. For example, geraniol (an open chain terpenoid alcohol) on dehydration is converted into α-terpineol.



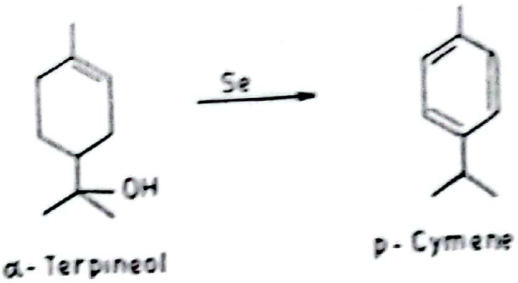
Geraniol α-Terpineol

(*g*) **Dehydrogenation.** When terpenoids are heated with zinc, iodine, bromine, sulphur, selenium, palladium, etc., they are converted into aromatic compounds of known structure. If the structures of aromatic compounds are not known, these can be determined by synthesis. By this method, the skeleton structure and position of the side chain in the original terpenoids can be determined.

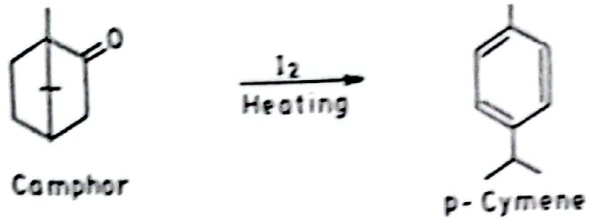
Selenium is a better dehydrogenating reagent as it is milder and results in a fewer side reactions. On the other hand, sulphur has been shown not to have this tendency.

In order to carry out dehydrogenation, the terpenoid is dissolved in a suitable high boiling solvent and heated in an inert atmosphere with dehydrogenating reagents at about 200--250°C. Six-membered alicyclic rings are dehydrogenated to more stable benzene rings. The oxygen as well as unsaturated functional groups present in terpenoids are reduced and the product is generally an aromatic compound. For example,

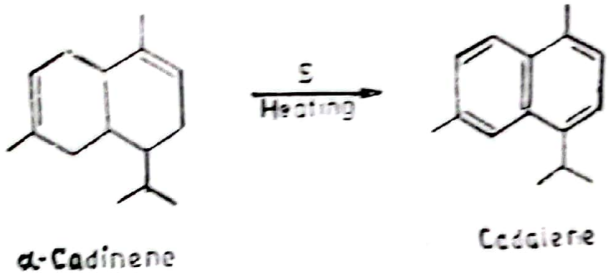
(*i*) α-Terpineol when heated with selenium yields *p*-cymene.



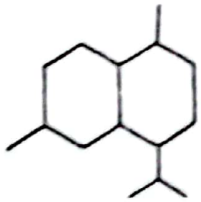
(*ii*) When camphor is heated with iodine, it is dehydrogenated to yield *p*-cymene. Thus,



(*iii*) The α-cadinene when dehydrogenated with sulphur forms cadalene which is 1, 6-dimethyl-4-isopropyl naphthalene.



From the above reaction, the carbon skeleton of cadinene is as follows :

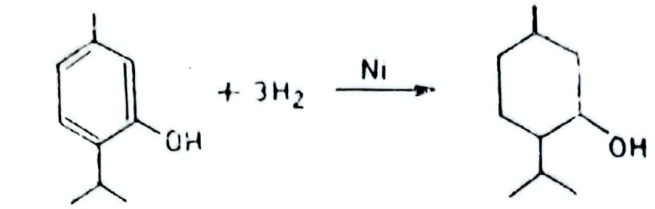


2. Synthetical Methods

The following synthetical reactions have been found to be of great value in elucidating the structures of terpenoids :

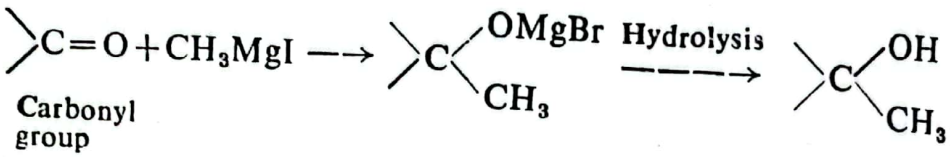
(*a*) **Catalytical hydrogenation.** When aromatic compounds are hydrogenated catalytically under suitable conditions, it is possible to obtain synthetic terpenoids. For example, a terpenoid

alcohol menthol may be prepared from thymol, an aromatic compound, by catalytic hydrogenation.



Thymol (An aromatic compound) — Menthol (A terpenoid alcohol)

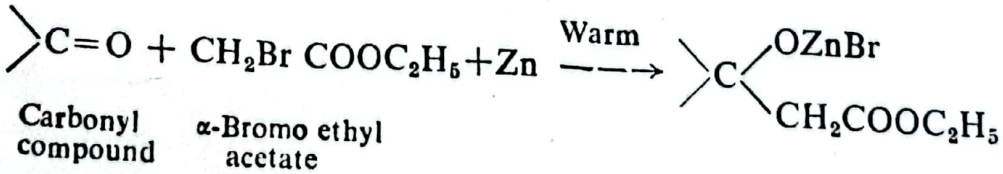
(*b*) **Grignard's reaction.** This reaction is of wide importance in the chemistry of terpenoids. This reaction was successfully employed by Perkin *et al.* to synthesise a large number of compounds which are related to terpenoids. By Grignard's reagent, methyl or isopropyl groups can be introduced into a compound having carbonyl group.



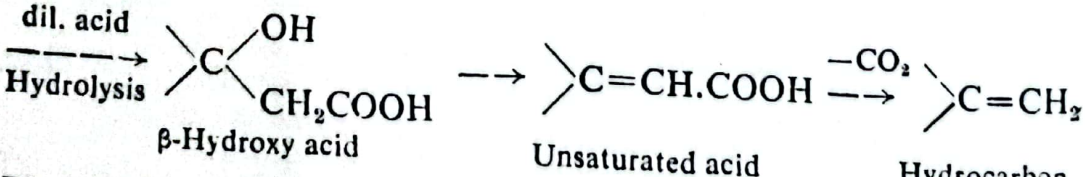
Carbonyl group

By the direct application of Grignard's reaction, α-terpineol, a naturally occurring terpenoid tertiary alcohol, can be readily prepared.

(*c*) **Reformatsky reaction.** Similar to Grignard's reaction, Reformatsky reaction is very useful in synthesising many terpenoids and related compounds like camphoronic acid, geranic acid, etc. In this reaction, α-halogen substituted ester is treated with a carbonyl compound (aldehyde, ketone or ester) in the presence of zinc to form a β-hydroxy ester. The latter compound when treated with dilute acid yields β-hydroxy acid which may be further converted into an unsaturated acid or a hydrocarbon.



Carbonyl compound — α-Bromo ethyl acetate



β-Hydroxy acid — Unsaturated acid — Hydrocarbon

Physical Methods :

A number of physical methods have been employed in elucidating the structure of natural terpenoids. All complicated structural

problems in terpenoids have been solved successively by physical methods. These methods also help to confirm the results of degradative studies.

The various methods for elucidating the structures of terpenoids are as follows :

(*a*) *Ultraviolet spectroscopy*. This technique is widely used in terpenoid chemistry for the detection of conjugation. The values of λ_{max} for various types of terpenoids have been calculated. Thus,

(*i*) For simple acyclic dienes, λ_{max} has been found to be 217—228 nm (ϵ 15,000—25,000). If the conjugated double bonds are not present in the same ring, *i.e.*, heteroannular diene, λ_{max} has been found to be 230—240 nm (ϵ 1300—20000). If the diene is homoannular in which both double bonds are present in the same ring, λ_{max} has been found to be 256—265 nm (ϵ 2500—10000).

(*ii*) For α, β-unsaturated carbonyl systems, λ_{max} has been found to be 220—250 nm (ϵ 10000—17500). Such systems also show a weak band at λ_{max} 315—330 nm (ϵ 15—100).

(*iii*) Substituents affect the absorption maximum of a diene system. Woodward (1942) studied the effect of substituents on λ_{max} of diene systems and concluded that the value of the absorption maximum is dependent on their number and type. He also developed a set of empirical rules to calculate λ_{max} from the molecular structure of the compound.

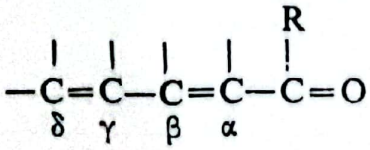
The values of λ_{max} for various polyenes and the increment values for substituents are given in Table 1.3.

TABLE 1.3

Absorption Values for Various Polyenes and the Increment Values for Substituents

S. No.	*Polyenes*	*Absorption value*
(*i*)	Basic value of homoannular dienes	253 nm
(*ii*)	Basic value of heteroannular (and acyclic) dienes	214 nm
(*iii*)	Increment for each C-substituent	5 nm
(*iv*)	Increment for each exocyclic double bond	5 nm
(*v*)	Increment for each double bond that extends conjugation	30 nm
	λ_{max} of compound	Total

The general formula of α, β-unsaturated ketones is



where R is an alkyl group or a ring residue and the parent system is C=C–C[R]=O. The various values of these systems are given in Table 1.4.

TABLE 1.4

Absorption Values of α, β-Unsaturated Ketones and Increment Absorption Values for Various Substituents

S. No.	Basic Chromophore	Absorption value
(*i*)	Basic value of parent system	215 nm
(*ii*)	Increment for each C-substituent at α-C	10 nm
(*iii*)	Increment for each C-substituent at β-C	12 nm
(*iv*)	Increment for each C-substituent at γ- or δ-C	18 nm
(*v*)	Increment for each exocyclic double bond	5 nm
(*vi*)	Increment for each double bond that extends conjugation	30 nm
	λ_{max} of compound	Total

The above mentioned rules have been successfully applied to various terpenoids. For example,

(*i*) The observed λ_{max} value for myrcene is 224 nm whereas the calculated λ_{max} value (an acyclic diene with one C-substituent) is 214+5=219 nm.

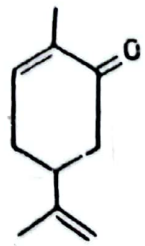
(*ii*) The observed λ_{max} value for β-phellandrene is 232 nm whereas the calculated λ_{max} value (a heteroannular diene with two substituents and one exocyclic double bond) is 214+2×5+5 or 229 nm.

(*iii*) The observed λ_{max} value for carvone is 235 nm whereas the calculated λ_{max} value is 237 nm which is obtained as follows :

Parent system	215 nm
C–substituent at α–C	10 nm
C–substituent at β–C	12 nm
	λ_{max}=237 nm

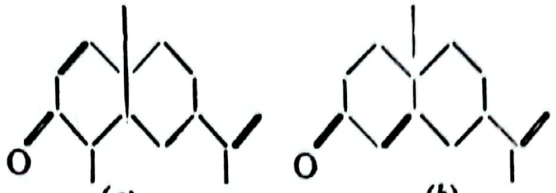
Myrcene

β-Phellandrene



Carvone

(*iv*) In α-cyperone the placement of the double bond in α, β-position to the keto group must be as shown in (*b*) rather than (*a*) because it exhibits λ_{max} at 252 nm which is matching with the predicted value of 254 nm. Further, (*a*) and (*b*) have been distinguished by NMR spectrum because (*b*) would show the presence of two olefinic protons as compared with four if the structure was (*a*).



(*a*) (*b*)

For (*a*)

Parent enone	215 nm
β-substituent	12 nm
	λ_{max}=237 nm

For (*b*)

Parent enone	215 nm
α-substituent	10 nm
2 β-substituents	24 nm
Exocyclic double bond	5 nm
	λ_{max}=254 nm

From the above examples, a good agreement is obtained between calculated and observed values. However, exceptions are five-membered rings in α, β-unsaturated ketones. The calculated λ_{max} for such compounds are found to be 10 nm longer than the observed value.

Allinger *et al.* (1965) established a quantitative theoretical basis for Woodward's rules in these compounds. These workers have also calculated λ_{max} values for a number of unsaturated compounds.

Infrared spectroscopy cannot be used with certainty to ascertain the presence of an isolated double bond in particularly tetra-substituted ethylenes. However, ultraviolet spectroscopy has been used successfully to detect the presence of this type of double bond (175—200 nm).

Ultraviolet spectroscopy has also been used to recognise α, β-unsaturated acids, esters and lactones. These compounds have λ_{max} in the region of 220 nm.

(*b*) **Infrared spectroscopy.** In terpenoid chemistry, this technique is used for detecting the presence of a hydroxyl group (3400 cm^{-1}) or an oxo group (saturated : 1750—1700 cm^{-1}; α, β-unsaturated ; 1700—1660 cm^{-1}).

Ultraviolet spretroscopy cannot be used for distinguishing heteroannular dienes and unsubstituted α, β-unsaturated ketones. However, infrared spectroscopy has been used to distinguish these compounds with certainty, Further, infrared spectroscopy has been used to distinguish between **cis** and **trans**-isomers. Also, this is used for detecting the presence of the isopropenyl group.

Furthermore, infrared spectroscopy is also used to recognise saturated and conjugated esters, aldehydes, ketones, acids, lactones and anhydrides.

Compounds	*Region of the frequency of the absorption maximum* (cm^{-1})
Acids (Saturated)	1725—1700
Esters	1950—1735
Anhydride-acyclic	1824—1748
Anhydride-succinic	1865—1782
Anhydride-glutaric	1802—1761
δ-Lactones	1750—1735
γ-Lactones	1780—1760
Cyclohexanones	1725—1700
Cyclopentanones	1750—1735
Cyclohexenones	1690—1665
Cyclopentenones	1710—1690
α, β-Unsaturated-γ-lactone	1760—1740
Aldehydes	1740—1720

Also, infrared spectroscopy is used for quantitative measurment, *e.g.*, in the determination of the number of methyl groups.

(*c*) **NMR spectroscopy.** This technique is of much use in identifying double bonds and in determining the nature of end-groups in terpenoid chemistry. Further, this is also used to determine the number of rings present and also to reveal the orientation of methyl groups in the terpenoid molecule. In some cases, NMR spectroscopy is successful in knowing their structures, *e.g.*, ψ-santonin.

Although infrared absorption reveals the presence of a hydroxyl group in a terpenoid, yet the number, position and the multiplicity of the CHOH protons in the NMR give the information on whether it is a primary, secondary or a tertiary alcohol.

(*d*) **Mass spectrometry.** This is quite a useful technique in terpenoid chemistry. This has been used successfully in elucidating

the structure of terpenoids by determining their molecular weights, molecular formulae, the nature of the various functional groups and the relative positions of double bonds. However, mass spectrometry should be employed with great care because even simple terpenoids yield complicated fragmentation patterns. It is to be remembered that a terpenoid can be identified by comparing its mass spectrum with an authentic compound of known structure.

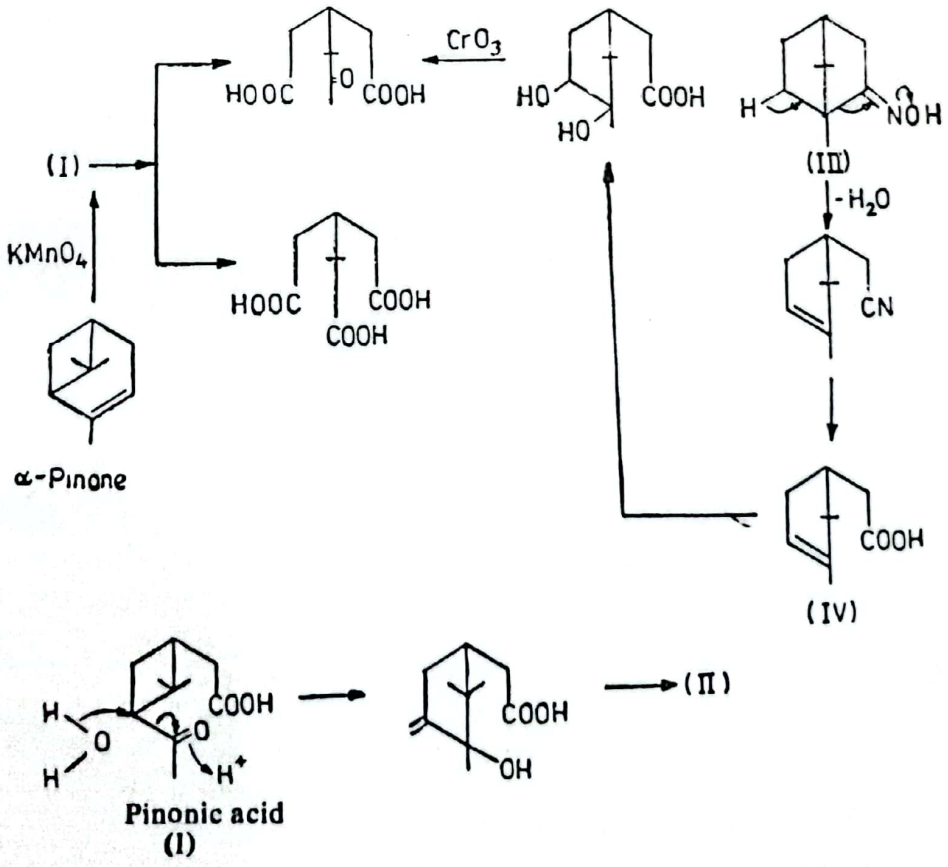
Although the number of methylene groups in the α-position(s) to the ketone is obtained by deuteration studies, mass spectroscopy being used to define the number of protons exchanged with deuterium, etc.

(*e*) **Optical rotation.** This technique has been applied to elucidate the structures of many terpenoids. Further, ORD studies are used to ascertain absolute configurations.

(*f*) **X-Ray analysis.** This is a very useful technique which has been used successfully for elucidating structure and stereochemistry of terpenoids.

4. *Knowledge of a molecular rearrangement.* During some degradative work, molecular rearrangement is very useful in structure elucidation An interesting example of this can be seen in the structure elucidation of α-pinene.

When α-pinene is oxidised with permanganate, it gives pinonic acid (I) as one of the products. Pinonic acid on oxidation with chromic acid is degraded to isoketocamphoric acid (II). Camphor



oxime (III) on acid-catalysed dehydration and hydrolysis yields α-campholenic acid (IV). Significantly, a rearrangement, subsequent to protonation of the carbonyl group has been involved in the oxidation of pinonic acid. Thus, this work clearly reveals the relationship of camphor with α-pinene and gives some idea about the structure of the latter.

5. *Synthesis.* Lastly, the tentative structure elucidated by the above analytical and physical methods may be confirmed by its unambiguous synthesis. However, in terpenoid chemistry, many of the syntheses are ambiguous. In such cases, analytical evidence will be considered in conjuction with the synthesis.

In modern times, the technique of stereoselective synthesis has been introduced. By using this technique, it becomes possible to synthesise particular configurational forms of many terpenoids.

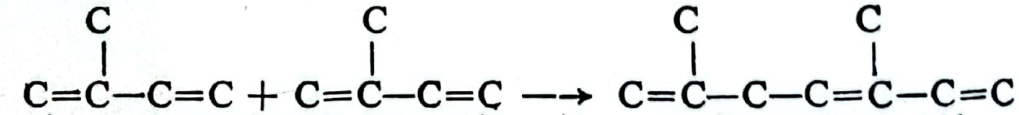
1.7. MONOTERPENOIDS

The monoterpenoids ($C_{10}H_{16}$ and its oxygenated derivatives) are composed of two isoprene units. These are the simplest of the naturally occurring terpenoids which are isolated from essential oils obtained from the leaves, roots and barks of various plants. These compounds have pleasant odours and are largely used in perfumery industry.

Monoterpenoids may be divided into three groups: (*a*) **acyclic,** (*b*) **monocyclic** and (*c*) **bicyclic.**

It is interesting to observe that the cyclic monoterpenoids undergo cyclisation readily whereas the bicyclic terpenoids undergo rerrangement very readily ; these two reactions are responsible for the delay in the interpretation of the results of earlier workers.

In monoterpenoids, two isoprene units are linked together. But each isoprene unit has two double bonds. Therefore, if the two isoprene units having four double bonds are linked together to form an open chain compound, one of the four double bonds is consumed and, thus, the acyclic monoterpenoid will have three double bonds. Thus, the formation of an acyclic monoterpenoid from two isoprene units may be depicted as follows :



Further, if acyclic monoterpenoid is converted into monocyclic terpenoids, one more double bond is consumed to form one more junction and thus monocyclic terpenoid would have two double bonds. Similarly, by following the same reasoning, bicyclic monoterpenoid and tricyclic monoterpenoid would have one and zero number of double bonds respectively. These facts have been summarised in Table 1.5.

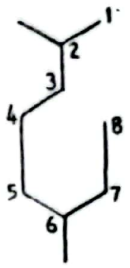
TABLE 1.5

Relation between Various Types of Monoterpenoids and Number of Double bonds

S. No.	*Monoterpenoid type*	*Number of double bonds*
(*i*)	Acyclic	3
(*ii*)	Monocyclic	2
(*iii*)	Bicyclic	1
(*iv*)	Tricyclic	0

1.8. ACYCLIC MONOTERPENOIDS

Only few examples of acyclic monoterpenoids are known. However, they are considered to be the most important isolates in perfumery. All the acyclic monoterpenoids contain the carbon-skeleton of 2, 6-dimethyloctane.



Some of the important hydrocarbons and their oxygenated derivatives are described in the following articles.

1.9. MYRCENE

Introduction. It is an acyclic monoterpenoid hydrocarbon which was first of all isolated from bay oil. Other sources are verbena, hops and turpentine oil. It is a liquid which boils at 166—168°C. It is a optically active compound.

Constitution of Myrcene. This has been established on the basis of the following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data and molecular weight determination, the molecular formula of myrcene has been found to be $C_{10}H_{16}$.

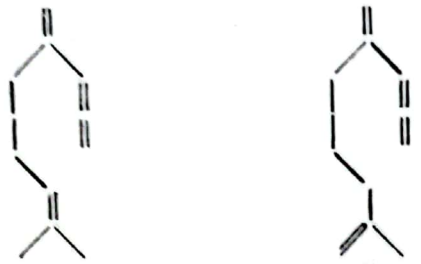
2. *Presence of three double bonds.* When catalytically reduced, myrcene adds on three moles of hydrogen to form decane, $C_{10}H_{22}$, indicating that myrcene contains three double bonds and it must be an acyclic compound.

$$\underset{\text{Myrcene}}{C_{10}H_{16}} + 3H_2 \xrightarrow{Pt} \underset{\text{Decane}}{C_{10}H_{22}}$$

3. *Presence of conjugated double bonds.* As myrcene is able to form an adduct with maleic anhydride, this shows that two of the three double bonds are conjugated. This has been further supported by the fact that myrcene shows optical exaltation, and exhibits a characteristic band (λ_{max} 224 nm) in the UV spectrum. Further, myrcene, when reduced with sodium and alcohol, yields dihydro-derivative, again, indicating that two of the double bonds are conjugated.

4. *Study of ozonolysis products.* On ozonolysis, myrcene yields acetone, formaldehyde and a ketodialdehyde, $C_5H_6O_3$. The latter compound on oxidation with chromic acid yields succinic acid and carbon dioxide.

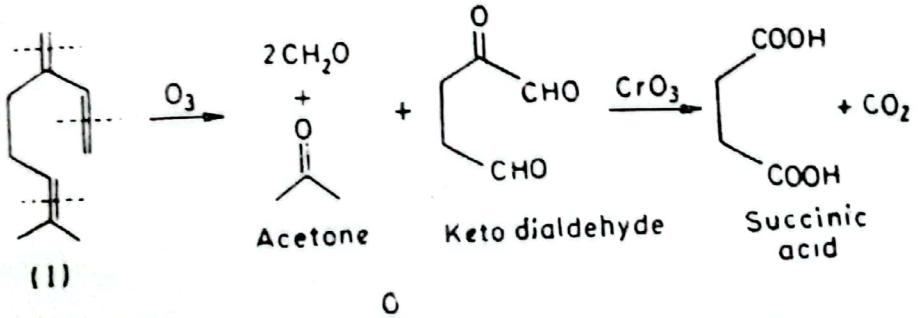
5. *Structure of myrcene.* On the basis of the above facts, myrcene may be assigned either structure (I) or structure (II).



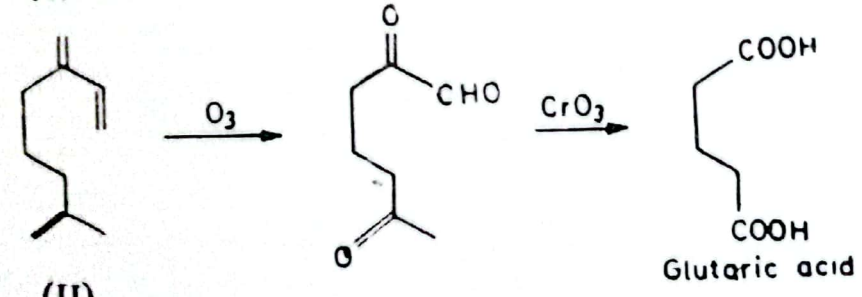
(I) (II)

Out of the above two structures, structure (I) is accepted on the basis of following facts :

(*a*) The results of ozonolysis can be explained only on the basis of structure (I). Had the structure (II) been correct, there would have yielded glutaric acid in place of succinic acid.

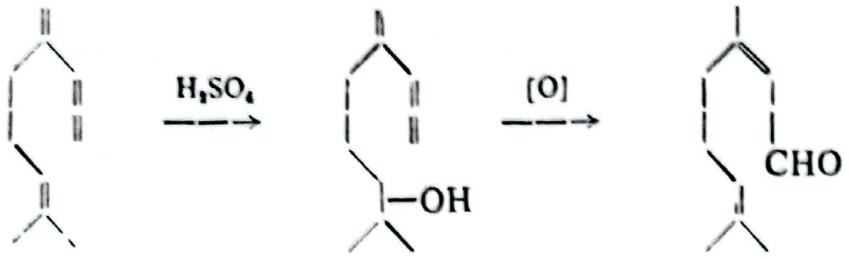


(I)



(II)

(*b*) On hydration with sulphuric acid, myrcene yields an alcohol which on oxidation yields citral. The structure of the latter compound (citral) is well known and its formation can only be explained on the basis of structure (I) for myrcene.

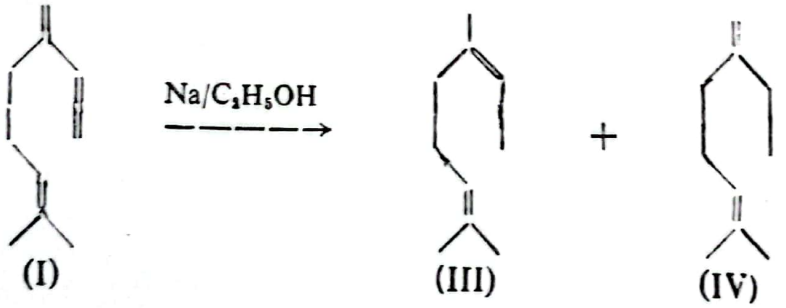


Myrcene (I) — Myrcenol (alcohol) — Citral

(*c*) The UV spectrum of myrcene exhibits absorption band having λ_{max} 224 nm (ϵ 14600). This can only be explained on the basis of structure (I) for which the calculated value of λ_{max} is 214+5 or 219 nm.

(*d*) If the structure (II) is correct, its infrared spectrum should exhibit a band at 890 cm^{-1} due to isopropenyl group. But this band is absent in the infrared spectrum of myrcene. Therefore, the structure (I) is correct.

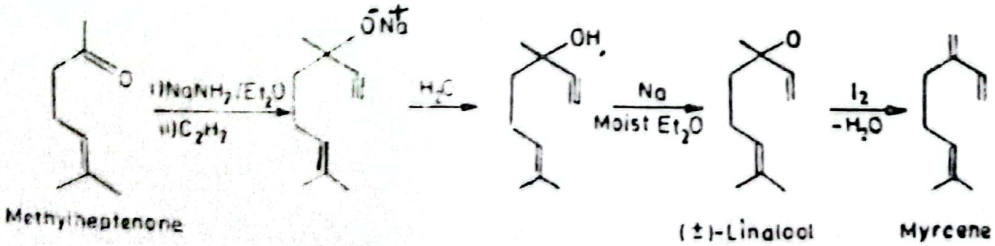
(*e*) Mainly on the evidence of Raman spectra, Dupont and Desreux (1936) have shown that, in the catalytic hydrogenation of myrcene, 1 : 4 addition of hydrogen is accompanied by 1:2 addition. The product is thus a mixture of dihydromyrcene (III) and 2-methyl-6-ethyl-$\Delta^{2,6}$-heptadiene (IV), in proportions which depend upon the nature of the catalyst. Formation of these products can only be explained if the structure (I) is correct.



(I) (III) (IV)

6. *Synthesis.* Finally, the structure (I) for myrcene has been confirmed on the basis of following syntheses :

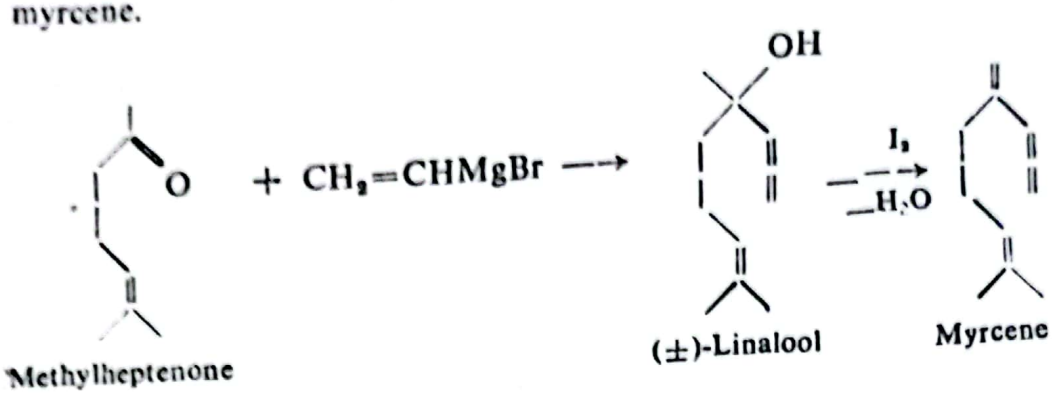
(*a*) **Ruzicka's synthesis (1919).** In this synthesis, the starting materials are methylheptenone and acetylene.



Methylheptenone — (±)-Linalool — Myrcene

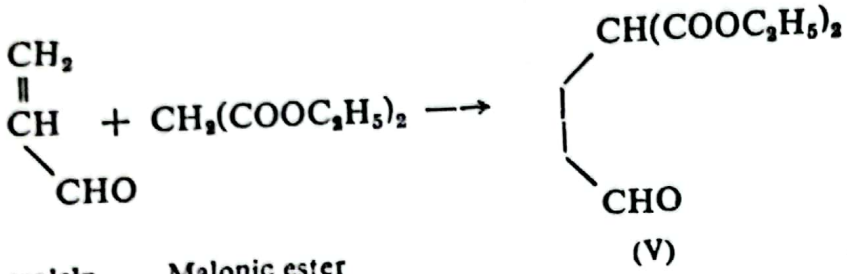
(*b*) **Normant's synthesis (1955).** In this synthesis, vinyl magnesium bromide, when treated with met hylheptenone, yields

linalool which on dehydration like Ruzicka's method yields myrcene.

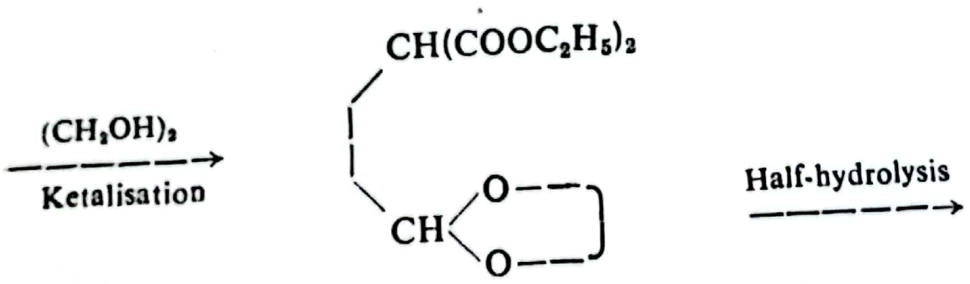


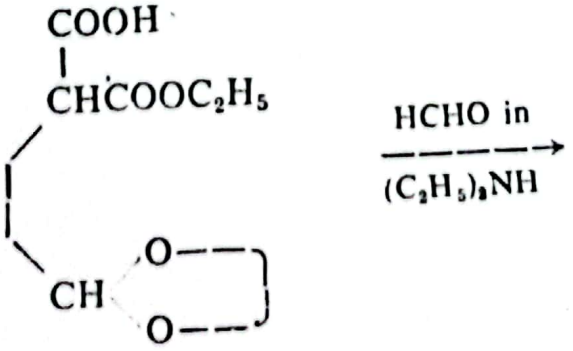
Methylheptenone (±)-Linalool Myrcene

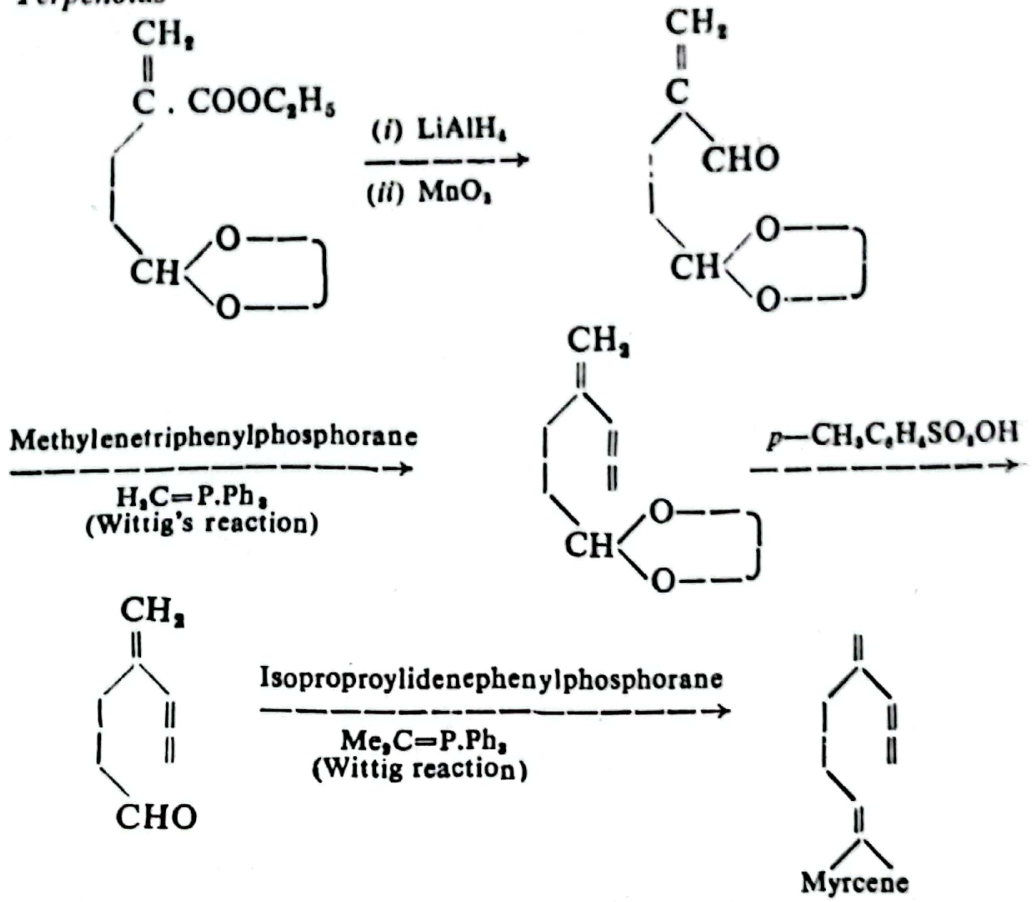
(*c*) **Vig's synthesis (1969).** In this synthesis, acrolein on condensation with malonic ester yields 2-carbethoxy-5-formyl-pentanoate (V). The latter compound then undergoes a series of reactions to yield myrcene.



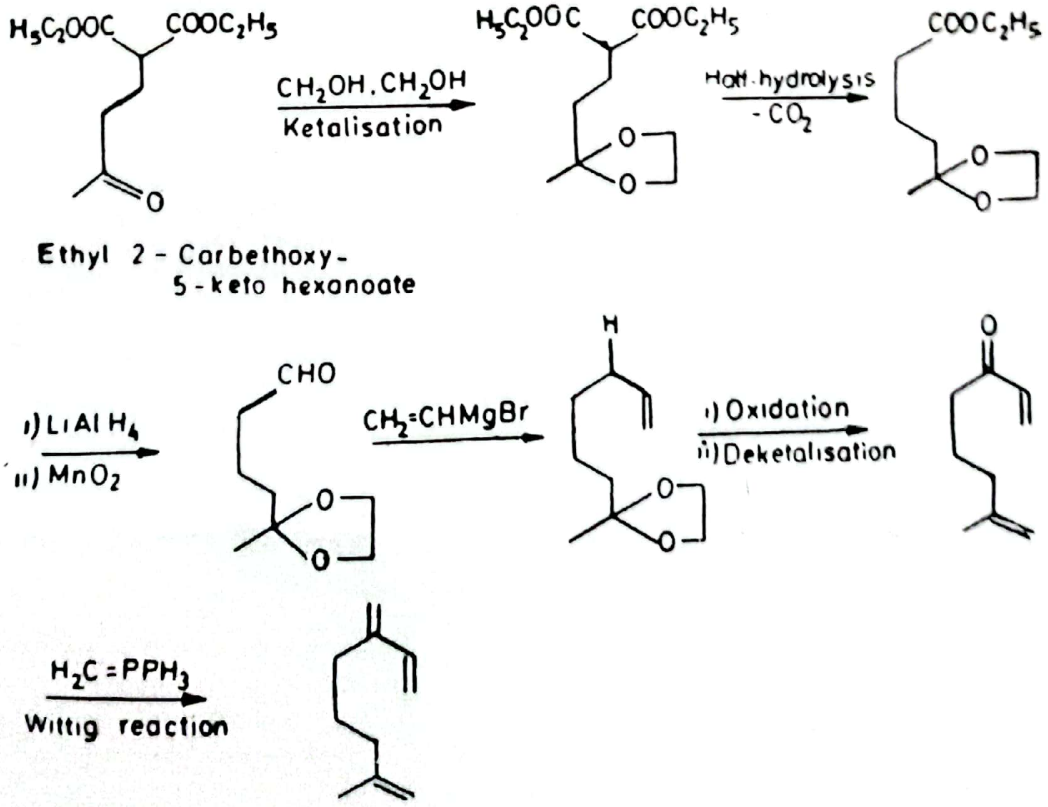
Acroleln Malonic ester (V)



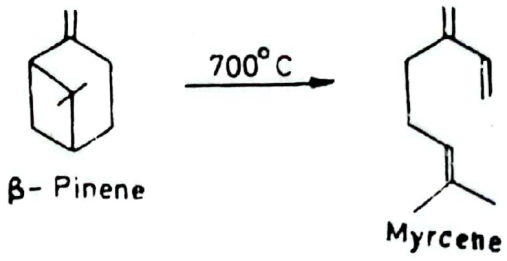




Vig *et al.* (1973) reported the synthesis of the α- or isopropenyl isomer of myrcene which does not occur in nature.



(*d*) **Industrial synthesis.** Myrcene is generally used for the synthesis of rubber-like polymers. For such a purpose, it is manufactured in 85 per cent yield by the pyrolysis of β-pinene at 700°C for 0.01 to 0.1 second.



1.10. OCIMENE

Introduction. It is an acyclic monoterpenoid which occurs in the leaves of *Ocimum basilicum* grown in Java. It is a liquid whose boiling point is 81°C at 30 mm pressure.

Ocimene is isomeric with myrcene. However, the two differ only in the position of their double bonds. As ocimene is less stable than myrcene, it is much difficult to get it in pure state. When exposed to air, it is converted into a yellow resin. However, it is quite stable when stored in the absence of oxygen/air.

Constitution of ocimene. Ocimene is an unstable compound, so much so that it has not yet been obtained in a pure form. Therefore, the elucidation of structure of this compound has offered considerable difficulty. However, its constitution has been deduced on the basis of following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data, the molecular formula of ocimene has been found to be $C_{10}H_{16}$.

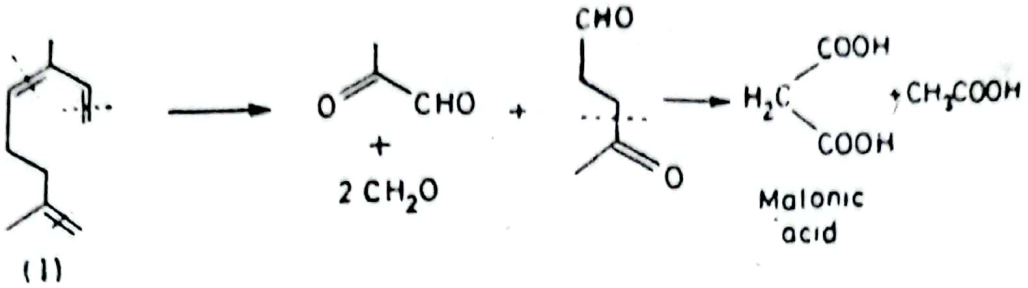
2. *Presence of three double bonds.* Similar to myrcene, ocimene when reduced catalytically adds on three moles of hydrogen to form decane ($C_{10}H_{22}$), indicating that it contains three double bonds.

$$C_{10}H_{16}+3H_2 \longrightarrow C_{10}H_{22}$$

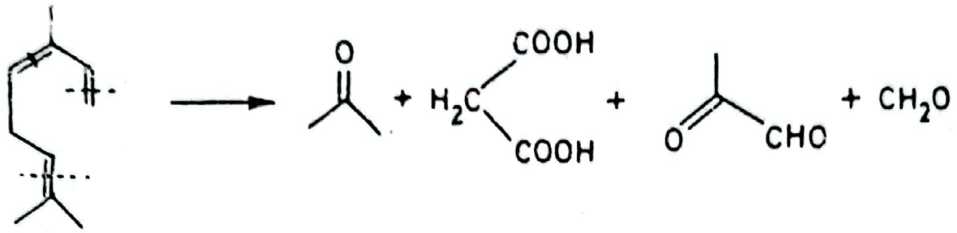
3. *Presence of two conjugated double bonds.* Similar to myrcene, ocimene forms an adduct with maleic anhydride, again, indicating that two of the double bonds must be conjugated. This conjugation has been further supported by its UV spectrum (λ_{max} 237 nm), optical exaltation and its reduction to dihydromyrcene with sodium/ethanol. The last reaction also reveals that ocimene and myrcene are positional isomers.

4. *Ozonolysis.* When ocimene is ozonolysed, it yields HCHO, methylglyoxal, laevulaldehyde, acetic acid, malonic acid and acetone. As the formation of these products could be accounted for on the basis of both the structures (I) (this has an isopropenyl end group) and (II) (this has the isopropylidene end group), ocimene is

therefore considered to be a mixture of these two isomers, (I) and (II).

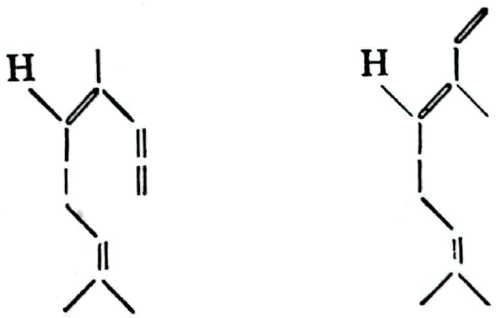


(I)



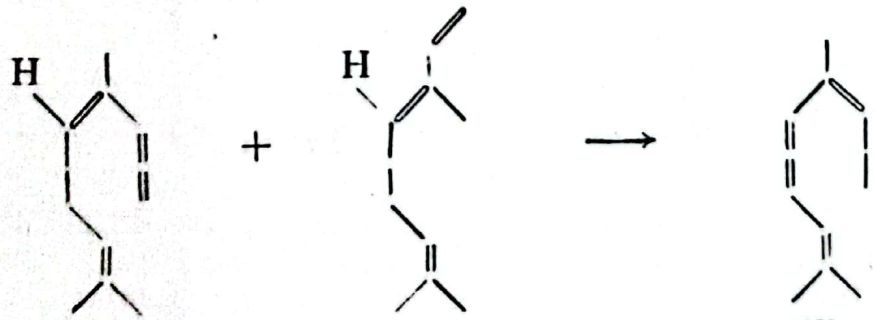
(II)

From the relative amounts of formaldehyde and acetone, it is believed that the structure (II) predominates in the mixture of structures (I) and (II) of ocimene. However, recent work could not confirm this. There is also sufficient evidence that ocimene is a mixture of geometrical isomers, α- and β- ocimene [structure (II)].



α-Ocimene β-Ocimene

5. *Action of heat.* When ocimene is heated, it undergoes isomerisation readily to yield allo-ocimene (III) in which the three double bonds are conjugated (λ_{max} 275 nm). This structure has been further supported by its synthesis. Again, the formation of compound (III) could only be explained if ocimene is considered to be a mixture of geometrical isomers, α- and β- ocimene.



α-Ocimene β-Ocimene (III)

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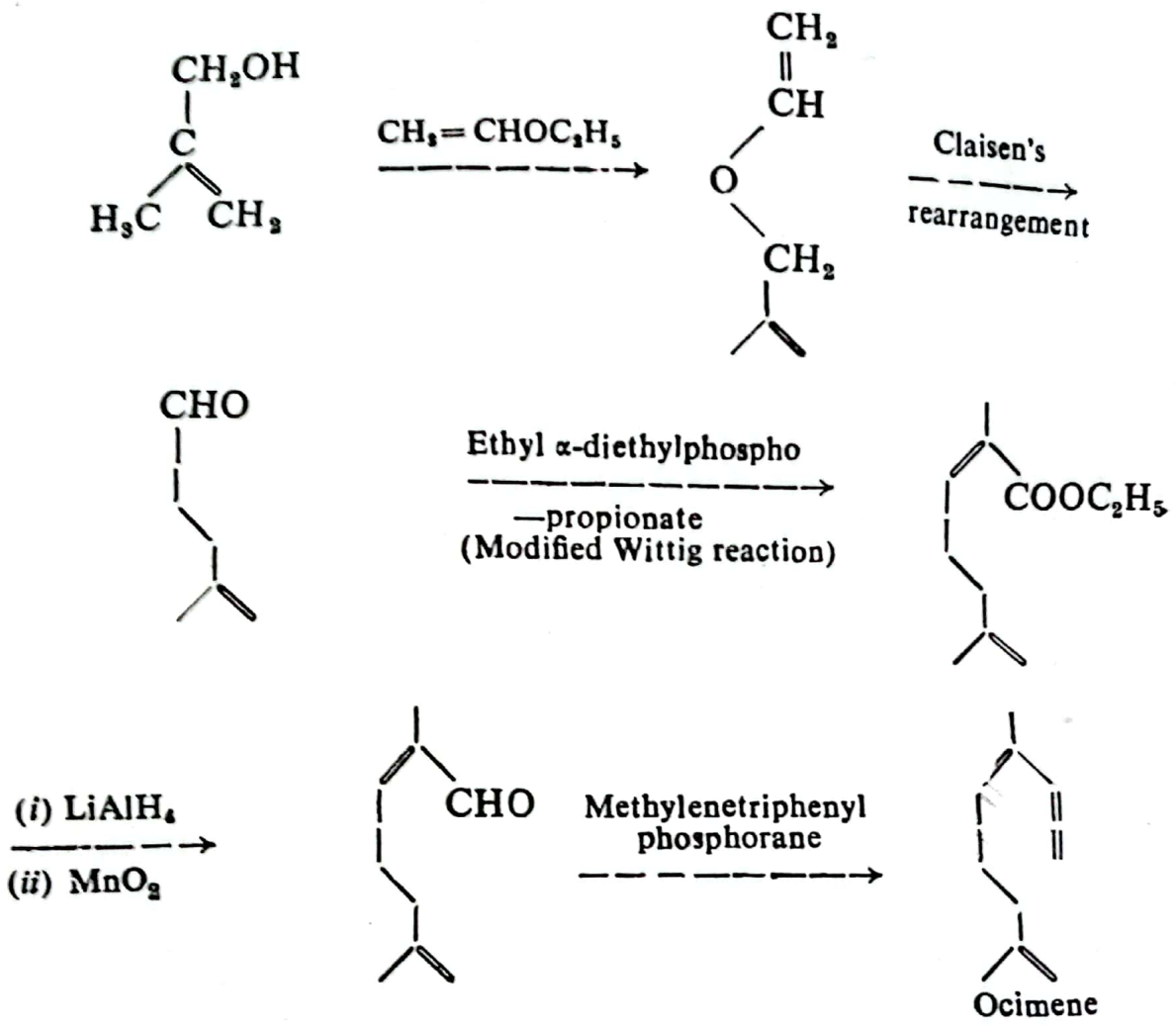
6. *UV spectrum.* The λ_{max} of ocimene is 237 nm at ϵ 40000 whereas the calculated value is 214+5+5 or 224 nm. As λ_{max} of ocimene (237 nm) is 13 nm longer than that for myrcene (224 nm) this indicates more substitution in the diene system.

7. *Synthesis.* In 1969, Vig *et al.* synthesised trans-α-ocimene from β-methyl allyl alcohol and ethyl vinyl ether. The ocimene synthesised by this method is the same as that isolated from the essential oil of lavender.



1.11. CITRAL

Introduction. It is considered to be the most important member of the acyclic monoterpenoids because the structures of remaining terpenoids in this group are based on the structure of citral.

It is the main constituent of lemon grass oil in which it occurs to an extent of 60—80 per cent. It is a pale yellow liquid (b.p. 228°C) having a strong lemon-like odour.

Isolation. From lemon grass oil, citral is obtained by its fractional distillation under reduced pressure. This is then purified by forming the bisulphite compound. The latter compound when decomposed with sodium bicarbonate yields pure citral.

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Constitution of Citral. This has been deduced on the basis of the work done by Semmler.

1. *Molecular formula.* From analytical data, the molecular formula of citral has been found to be $C_{10}H_{16}O$.

2. *Presence of two double bonds.* As citral adds on two moles of bromine, this shows that citral contains two double bonds.

$$C_{10}H_{16}O + 2\ Br_2 \longrightarrow C_{10}H_{16}O\ Br_4$$

Citral on ozonolysis yields acetone, laevulaldehyde and glyoxal. This reaction shows that citral is an acyclic compound containing two double bonds.

$$C_{10}H_{16}O \xrightarrow{O_3} \underset{CH_3}{\overset{CH_3}{\diagdown}}\!\!>C=O + OHC.CH_2.CH_2\overset{CH_3}{\underset{|}{C}}=O + OHC.CHO$$

Acetone — Laevulaldehyde — Glyoxal

3. *Presence of an aldehyde group.* With hydroxylamine, citral forms an oxime, indicating that it must contain one oxo group. With phenylhydrazine, it forms phenylhydrazone, further confirming the presence of oxo group.

$$>C=O + H_2NOH \longrightarrow >C=NOH + H_2O$$

$$>C=O + H_2NHNC_6H_5 \longrightarrow >C=N.NHC_6H_5 + H_2O$$

Citral on reduction with sodium amalgam yields an alcohol, geraniol, $C_{10}H_{18}O$. Citral is oxidised with silver oxide to yield geranic acid, $C_{10}H_{16}O_2$, containing the same number of carbon atoms as citral. Both these reactions reveal that oxo group in citral is therefore an aldehyde group.

$$C_{10}H_{16}O_2 \xleftarrow[Ag_2O]{[O]} C_{10}H_{16}O \xrightarrow[Na/Hg]{2\,[H]} C_{10}H_{18}O$$

Geranic acid — Citral — Geraniol

Citral reduces Fehling's solution, further confirming the presence of aldehydic group.

4. *Citral as an acyclic compound.* The presence of two double bonds and an aldehydic group in citral ($C_{10}H_{16}O$) leads to a saturated hydrocarbon of molecular formula $C_{10}H_{22}$ which corresponds to the general formula C_nH_{2n+2} for acyclic compounds, indicating that citral must be an acyclic compound.

5. *C-Skeleton of citral.* When citral is heated with potassium hydrogen sulphate, it cyclises to *p*-cymene, a cyclic compound of known structure.

$$C_{10}H_{16}O \xrightarrow[-H_2O]{KHSO_4} C_{10}H_{14}$$

Citral — *p*-Cymene

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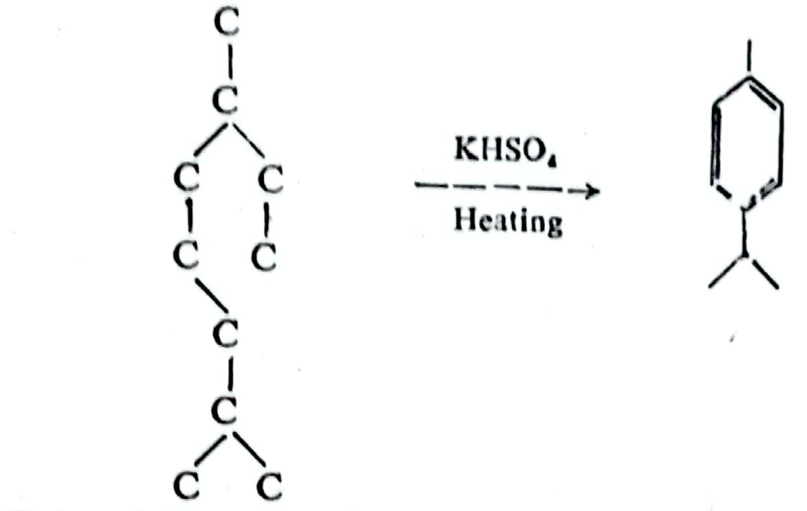
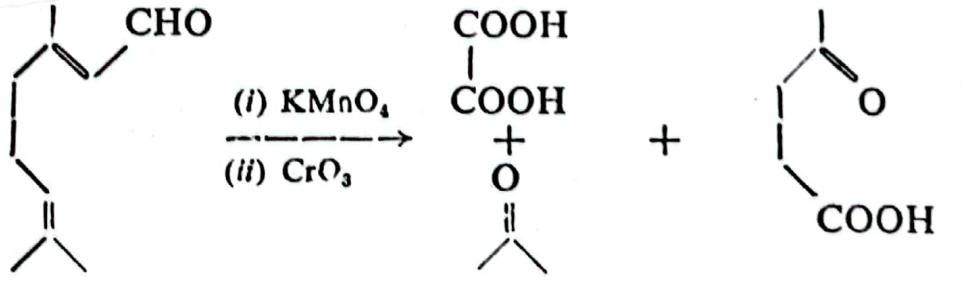
Formation of *p*-cymene and the products obtained from the ozonolysis reveal that C-skeleton (I) of citral contains two isoprene units which are joined by head to tail arrangement. The formation of *p*-cymene also reveals the relative positions of methyl and isopropyl groups in citral.



Carbon skeleton of citral (I) — *p*-Cymene

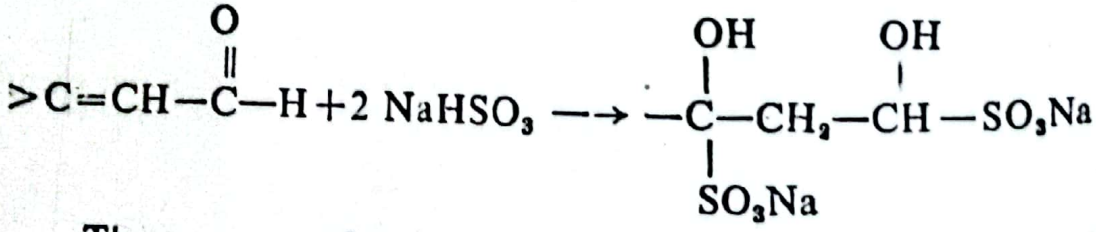
6. *Structure of citral.* Citral, on oxidation with alkaline permanganate followed by chromic acid, yields acetone, oxalic acid and laevulic acid. The formation of these oxidation products can only be explained if citral has structure (II).



Citral (II) — Acetone — Laevulic acid

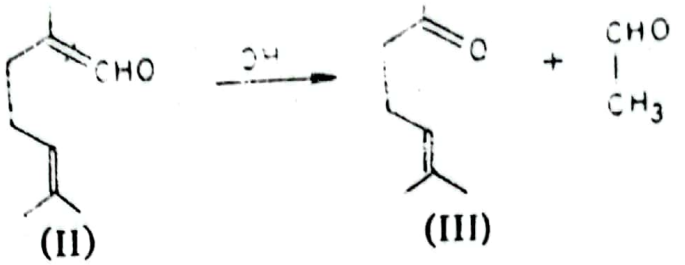
7. *Support for the structure (II).* Following evidences have been cited to confirm the structure (II) of citral :

(*a*) On treatment with sodium bisulphite, citral adds on two moles of sodium bisulphite successively, forming mono as well as di-bisulphite addition products respectively, indicating the presence of α, β-unsaturated aldehydic group.



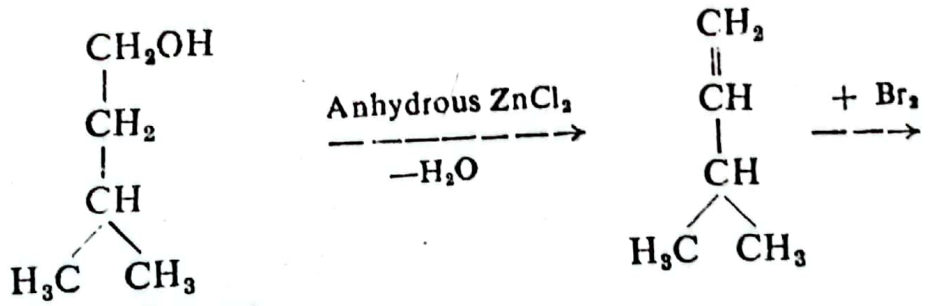
The presence of α, β-unsaturated aldehydic group in citral has been further confirmed by its UV absorption spectrum (λ_{max} 238 nm).

(*b*) Verley (1897) found that citral on boiling with aqueous potassium carbonate yields 6-methylhept-5-en-2-one (III) and acetaldehyde. The formation of these products could be accounted for if the structure (II) for citral is correct and it undergoes cleavage at α, β-double bond. This cleavage by alkaline reagents is a general reaction of α, β-unsaturated oxo compounds. Furthermore, methylheptenone, when it is oxidised itself, yields acetone and laevulic acid. The formation of these can only be explained on the basis of structure (II).

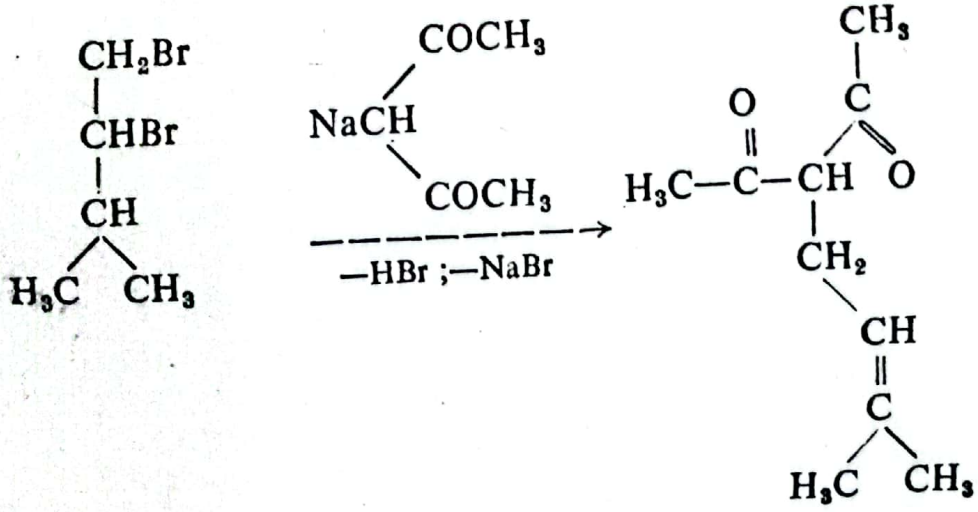


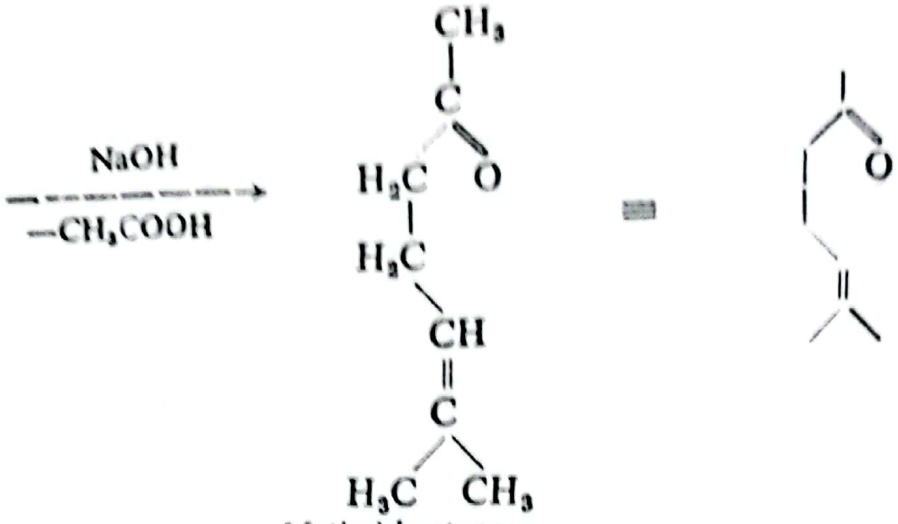
(II) (III)

(*c*) The structure (II) of citral is further confirmed by the synthesis of methylheptenone (III) from 3-methyl-1-butanol which is dehydrated followed by the addition of bromine to yield the product that on further condensation with sodium salt of acetyl acetone followed by hydrolysis yields methyl heptenone.



3-Methyl-1-butanol

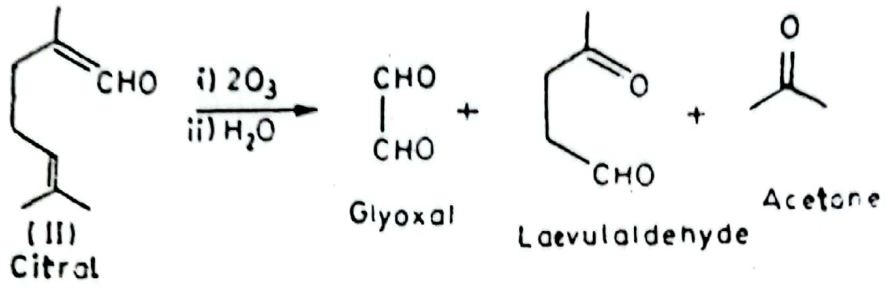




Methyl heptenone

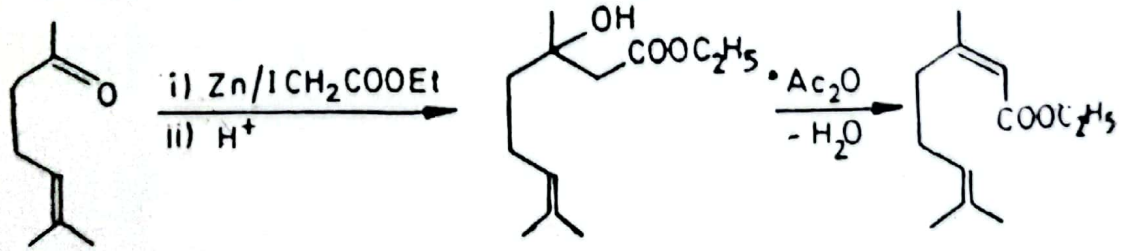
As citral on hydrolysis yields methylheptenone and acetaldehyde, it appears that citral is an aldol condensation of these two products, thereby, confirming the structure (II) for citral.

(*d*) The structure (II) for citral has been further confirmed by the fact that citral forms a diozonide $C_{10}H_{16}O.2O_3$ which on hydrolysis yields acetone, laevulaldehyde and glyoxal. The formation of these products is in agreement with the structure (II) of citral.

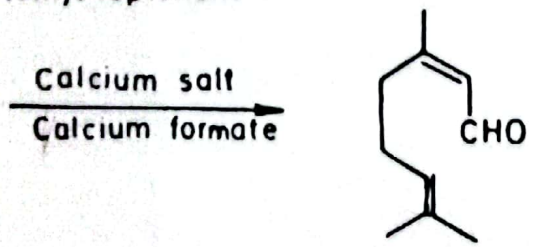


8. *Synthesis.* Finally, the structure (II) for citral has been confirmed on the basis of syntheses as follows :

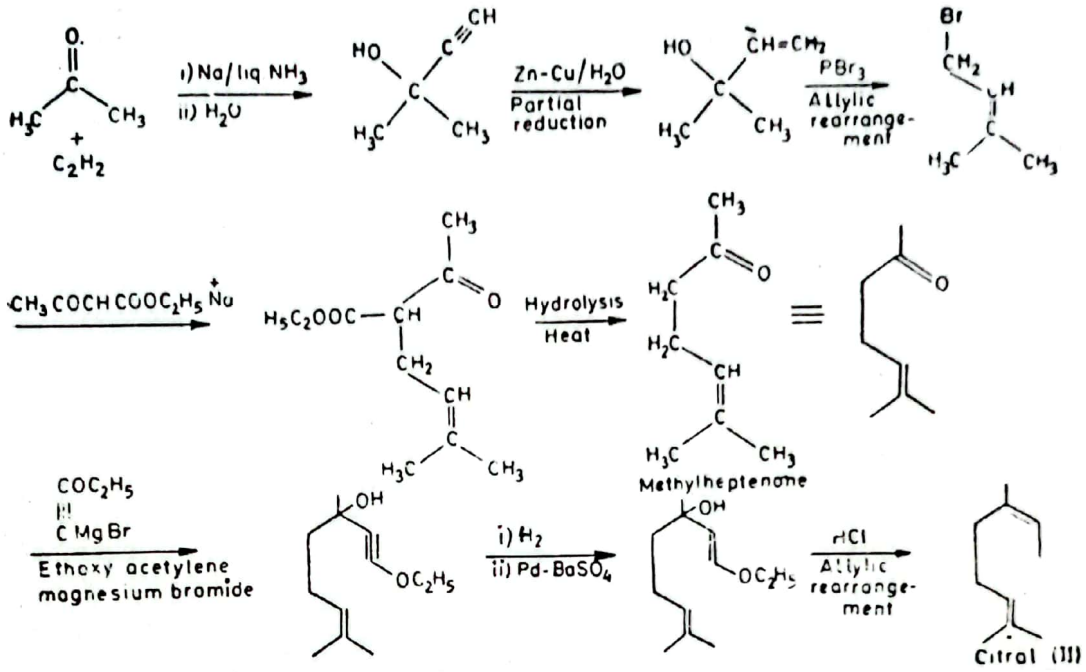
(*a*) *Barbier-Bouveault-Tiemann's Synthesis* (*1896, 1898*). According to this synthesis, methyl heptenone is converted into geranic ester by using Reformatsky's reaction (Barbier *et al.*, 1896). After this, Tiemann (1898) converted geranic ester into citral by distilling a mixture of calcium salts of geranic and formic acids.



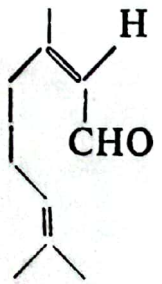
Methylheptenone



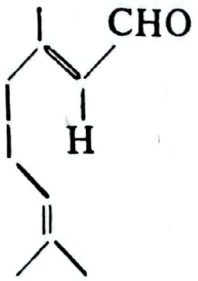
(*b*) *Arens—Van Dorp's synthesis* (*1948*). In this synthesis, acetone on condensation with acetylene in the presence of sodium and liquid ammonia yields the product which on reduction followed by treatment with PBr_3 undergoes allylic rearrangement. The product so obtained is treated with sodium salt of aceto-acetic ester and then hydrolysed to yield methyl heptenone. The latter compound on condensation with ethoxy acetylene magnesium bromide, followed by the partial reduction and acidification yields citral by allylic rearrangement.



Isomerism in Citral. Examination of formula of citral (II) reveals that citral may exist in two geometrical isomeric forms which differ in the arrangement of the aldehydic group about the double bond in the 2, 3-position. In one form, called the **cis-citral** or **citral-b** or **neral**, the aldehyde group is **cis** with respect to the methylene group of the main chain while in the other form called **trans-citral** or **citral-a** or **geranial**, the aldehyde group is ***trans*** with respect to the methylene group of the main chain.



Cis-citral ; citral-b ; neral ; Z-form



Trans-citral ; citral-a ; geranial ; E-form

Both **cis-** and **trans-**isomers occur in natural citral. **Cis-citral** has a b.p. 117—118°C at 20 mm pressure whereas **trans-citral** has a b.p. 118—119°C at 20 mm pressure. The existence of the two isomeric citrals in natural citral has been confirmed chemically due

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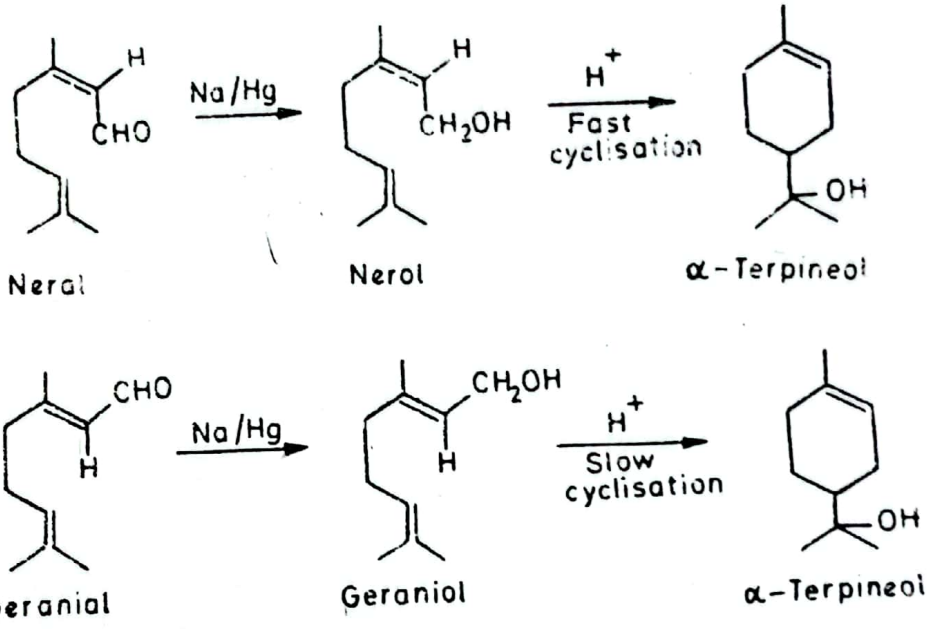
to the formation of two different semicarbazones and formation of geraniol and nerol on reduction. As the trans-citral yields geraniol on reduction, it is known as **geranial** or **citral-a**. Similarly as cis-citral on reduction yields nerol, it is called neral or citral-*b*.

The configurations of the two geometrical forms of citral have been determined from readiness with which the ring closure of the corresponding stereoisomeric primary alcohols, namely **geraniol** and **nerol**, takes place. The citrol, which cyclises easily to α-terpineol, is cis-isomer while the other which does not isomerise readily is called trans-isomer. As nerol (corresponding to neral) cyclises nine times as rapidly as geraniol (corresponding to geranial), nerol is assigned the cis-configuration and geraniol the trans-a ; hence neral is cis-citral whereas geranial is trans-citral.



It is believed that geraniol is cyclised to α-terpineol **via** nerol.

The assignments given above are confirmed on the basis of NMR spectra of citral-*a* and citral-*b*. For instance, the τ-values of CH_2 (*a*) and CH_3 (*b*) are different due to the different magnetic shielding effects of the carbonyl double bound in ·CHO group.

	$CH_2(a)$	$CH_3(b)$·
Citral-*a*	τ 7.76	τ 7.84
Citral-*b*	τ 7.42	τ 8.02

1.12. CITRONELLAL

Introduction. It is an optically active compound which occurs in the oil of citronella.

Constitution. This has been elucidated as follows :

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(*a*) Molecular formula of citronellal has been found to be $C_{10}H_{18}O$.

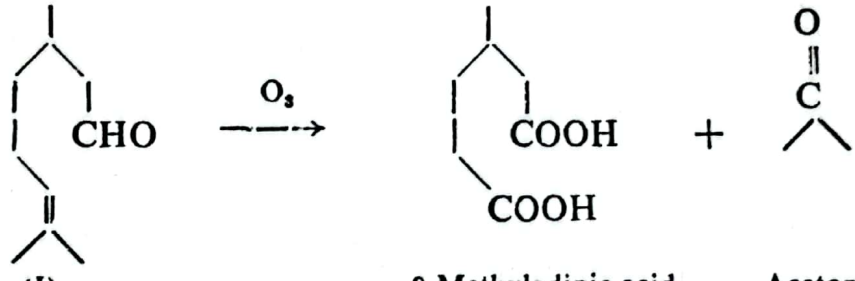
(*b*) When treated with hydroxylamine, it yields an oxime, indicating that it contains one $>C=O$ group. The oxime on dehydration with acetic anhydride yields nitrile, thus suggesting that keto group in citronellal is present in the form of **aldehyde group**.

$$\underset{\text{Citranellal}}{C_9H_{17}-CHO} \longrightarrow \underset{\text{Oxime}}{C_9H_{17}CH=NOH} \xrightarrow[-H_2O]{(CH_3CO)_2O} \underset{\text{Nitrile}}{C_9H_{17}C\equiv N}$$

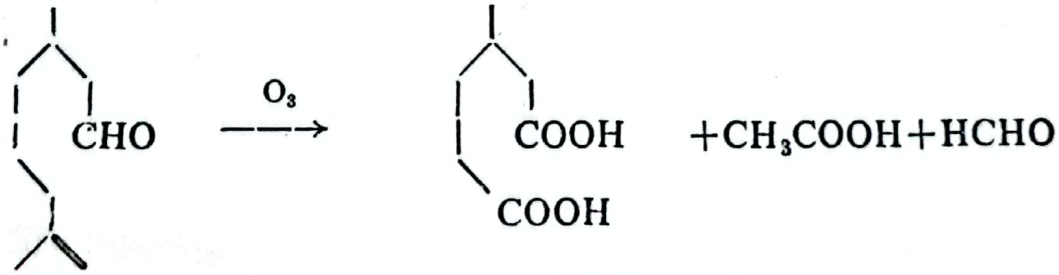
The nitrile on hydrolysis yields citronellic acid, $C_{10}H_{18}O_2$. The presence of aldehyde group in citronellal has been further confirmed by its reduction with sodium amalgam into the alcohol citronellol $C_{10}H_{12}O$ and its oxidation to citronellic acid, $C_{10}H_{18}O_2$.

$$\underset{\text{Citronellal}}{C_{10}H_{20}O} \xleftarrow{[O]} \underset{\text{Citronellal}}{C_{10}H_{18}O} \xrightarrow{[O]} \underset{\text{Citronellic acid}}{C_{10}H_{18}O_2}$$

(*c*) Ozonolysis of citronellal yields a mixture of acetone, β-methyladipic acid and some HCHO. The formation of these products reveals that citronellal is a mixture of two isomeric forms, (I) (containing isopropylidene end group) and (II) (containing isopropenyl end group).



(I) β-Methyladipic acid Acetone



(II) β-Methyladipic acid

The isopropenyl isomer (II) was named **rhodinal** but it is no longer believed to be present in natural citronellal. This has been further proved on the basis of following facts :

(*a*) IR spectral studies revealed that natural citronellal contains only isopropylidene structure (I).

(*b*) Again, oxidation of citronellal with chromic acid to 3-methyladipic acid and acetone can be explained only on the basis of structure (I).

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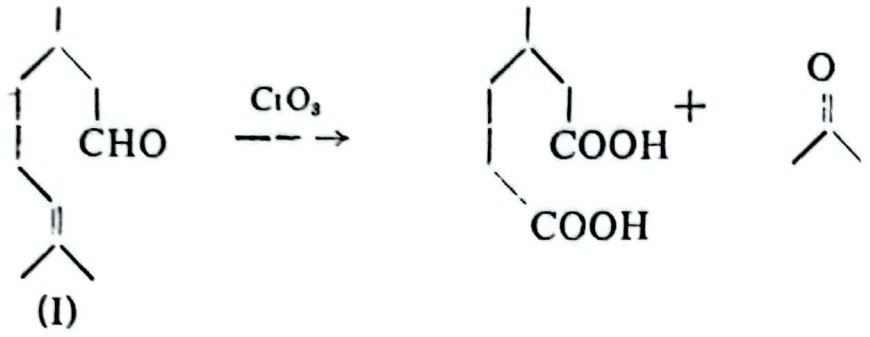
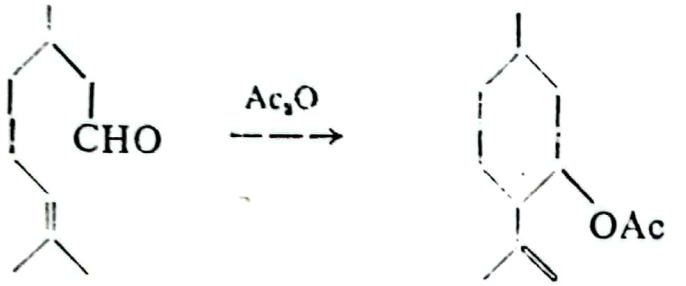


(I)

(*c*) The conversion of citronellal into isopulegol acetate under the action of acetic anhydride can be explained only on the basis of structure (I).



(I) Isopulegol acetate

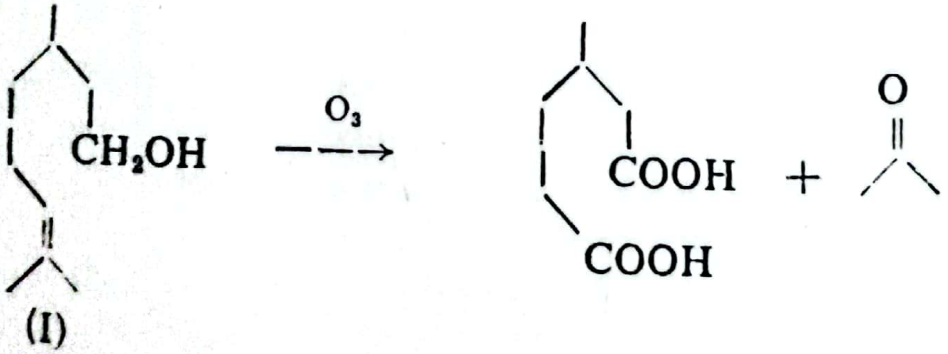
1.13. CITRONELLOL

Introduction. It is an optically active liquid. It occurs in the (−)-form in rose and geranium oils and in the (+)-form in citronella oil.

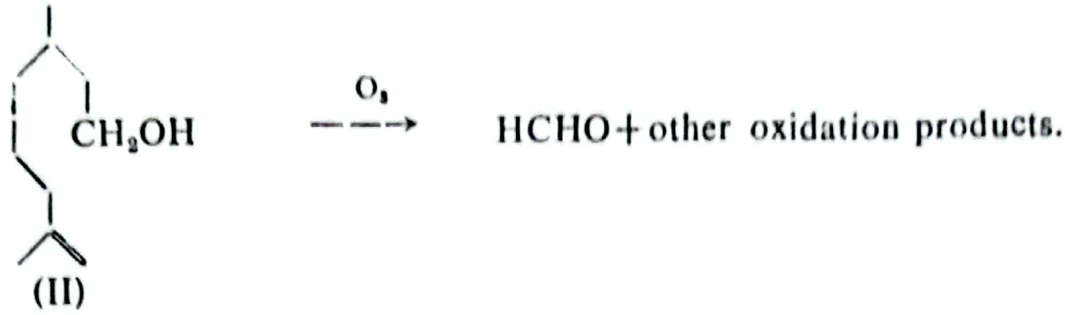
(+)-Citronellol is prepared on a large scale by carrying out reduction of citronellal with sodium-aluminium amalgam. Generally, citronellol occurs along with geraniol.

Constitution. 1. The molecular formula of citronellol has been found to be $C_{10}H_{20}O$.

2. On ozonolysis, citronellol yields acetone, and β-methyl adipic acid as the main products, and formaldehyde as minor product. The formation of acetone and β-methyl adipic acid could be explained only on the basis of structure (I) (containing isopropylidene group) whereas the formation of formaldehyde could be explained only on the basis of structure (II) (containing isopropenyl group). Thus, it is regarded that citronellol is a mixture of structures (I) (major isomer) and (II) (minor isomer).



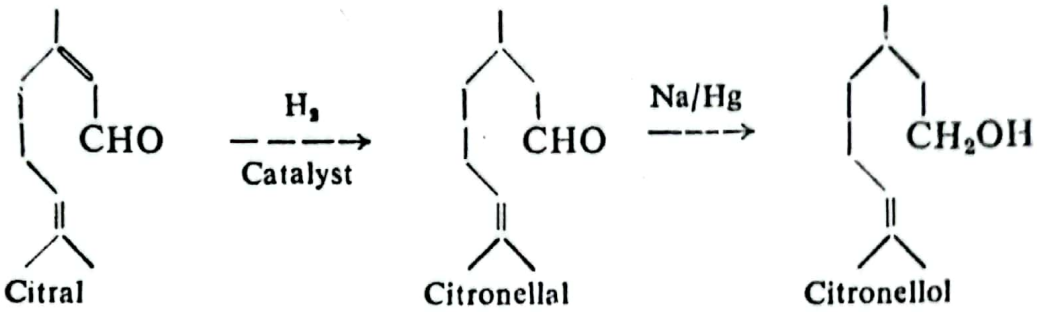
(I)



(II)

The isopropenyl isomer (II) was named **rhodinol** but its presence in natural citronellol, although claimed by I.R. spectral studies, is no longer accepted. However, it has been synthesised.

3. Finally, the structure (I) for citronellol has been confirmed by its synthesis from citral.



Citral Citronellal Citronellol

1.14. GERANIOL

Introduction. Geraniol occurs either free or in the form of its esters in many essential oils like rose, lemon-grass, geranium, lavender and citronella oils.

Geraniol is a pleasant smelling colourless liquid which boils at 229·–230°C at 757 mm pressure. After extracting from essential oils, it is purified by crystallising it as a complex compound with anhydrous calcium chloride.

Geraniol is also prepared commercially by the reduction of citral (corresponding aldehyde) with sodium amalgam.

Geraniol and its esters (e.g., acetate, formate, butyrate) are mainly used in perfumery, particularly as a component of artificial rose scents.

Constitution. 1. Its **molecular formula** has been found to be $C_{10}H_{18}O$.

2. As it adds on two moles of hydrogen, two moles of bromine, etc., to form addition products, this shows that it contains two double bonds.

$$C_{10}H_{22}O \xleftarrow[Ni]{H_2} \underset{\text{Geraniol}}{C_{10}H_{18}O} \xrightarrow{2\,Br_2} C_{10}H_{18}O.Br_4$$

3. On oxidation, it yields an aldehyde (citral-a) having the same number of carbon atoms, indicating that geraniol is a primary

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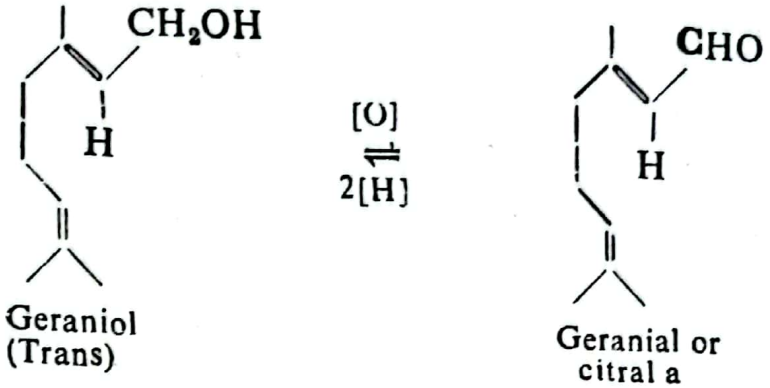
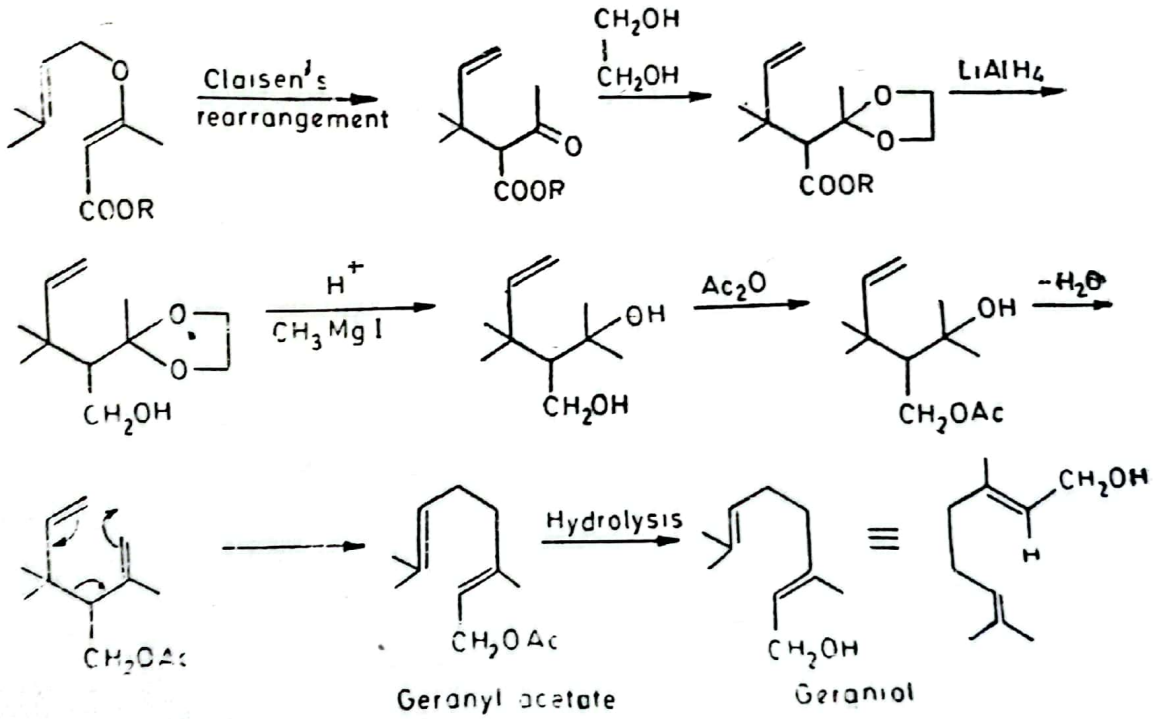
alcohol. Furthermore, the aldehyde $C_{10}H_{16}O$ has been identified as **citral-a** or **geranial** which could be reconverted into geraniol by reduction. Thus, geraniol must be the corresponding alcohol of citral-a or geranial. But the structure of the citral-a is well established. Therefore, the structure of geraniol may be represented as follows :



Geraniol (Trans) — Geranial or citral a

4. Finally, the structure of geraniol has been confirmed by its synthesis (Brack and Schinz, 1951) as follows :



1.15. NEROL

Intrduction. It is the geometrical isomer of geraniol. It occurs naturally in various essential oils, e.g., oil of neroli, bergamot, etc. Its boiling point is 225-226°C.

Isolation. In essential oils, nerol is generally accompanied by its geometrical isomer geraniol. In order to obtain pure nerol, the two must be separated. For this, oil is made to treat with anhydrous calcium chloride when crystalline complex with geraniol is obtained. This is filtered and the residue is decomposed with water to yield geraniol. The filtrate, when treated with N-diphenyl-amino carbonyl chloride, crystallises out nerol as its diphenyl

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carbamate. The latter on decomposition with water yields nerol.

$$\text{Oils} + CaCl_2$$

$$\downarrow \text{Filter}$$

Filtrate (Nerol)	Residue
$\downarrow [(C_6H_5)_2N.COCl]$	(Crystallised complex
Crystalline diphenyl	of geraniol)
$\downarrow H_2O$	$\downarrow H_2O$
Nerol	Geraniol

Nerol is used in perfumeries.

Constitution. 1. From analytical data, molecular formula of nerol has been found to be $C_{10}H_{18}O$.

2. Nerol is a geometrical isomer of geraniol on the basis of the following facts :

(*a*) Both nerol and geraniol add on two moles of hydrogen to yield the same saturated alcohol $C_{10}H_{22}O$, indicating that both nerol as well as geraniol contain **two double bonds.**

(*b*) Both add on four atoms of bromine to yield $C_{10}H_{18}O.Br_4$.

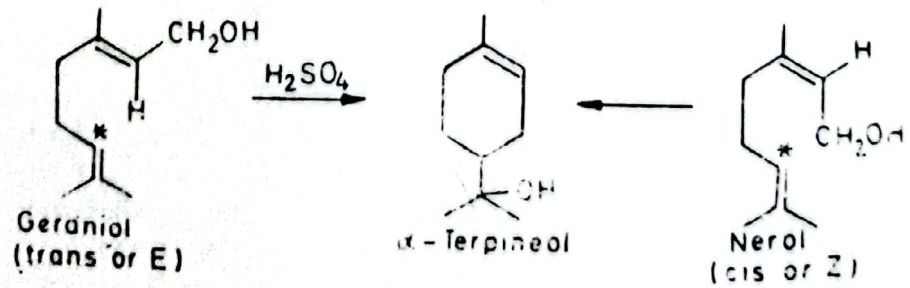
$$C_{10}H_{18}O + 2\ Br_2 \longrightarrow C_{10}H_{18}O.Br_4$$

(*c*) Both geraniol and nerol yield the same oxidation products with the same number of carbon atoms. Thus, the positions 2 and 7 of double bonds in nerol and geraniol remain the same. Also, both contain a primary alcoholic group.

(*d*) On reduction with sodium amalgam, citral yields some amount of nerol along with geraniol.

(*e*) Both geraniol and nerol on hydration with dil. H_2SO_4 undergo cyclisation to yield the same product, *i.e.*, α-terpineol.

On the basis of above facts, it may be concluded that geraniol and nerol are geometrical isomers. But the cyclisation of nerol with dilute sulphuric acid takes place about 9 times faster than that of geraniol. Therefore, nerol is given the **cis-configuration** because its faster rate of cyclisation is due to the proximity of the alcoholic group to the carbon (*) which is involved in the ring formation. On the other hand, geraniol is assigned the **trans-configuration.**



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3. The mechanism for the hydration of geraniol and nerol to α-terpineol involves the formation of an intermediate allyl carbonium ion.

1.16. LINALOOL

Introduction. It is an optically active compound. Its (−)-form occurs in rose oil whereas its (+)-form in orange oil. It is a liquid having b.p. 198-199°C.

It is isomeric with geraniol and nerol both. It is the most chemically reactive member of the acyclic oxygenated monoterpenoids.

As linalool does not form any crystalline derivative, it is usually isolated by fractional crystallisation.

Constitution of Linalool. This has been deduced on the basis of following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data, the molecular formula of linalool has been found to be $C_{10}H_{18}O$.

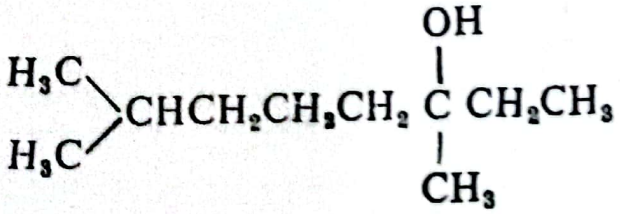
2. *Presence of two double bonds and one tertiary alcoholic group.* On bromination, it yields tetrabromide, indicating the presence of **two double bonds**. Further, PCl_5 reacts with linalool. This reaction reveals that linalool contains **one hydroxyl group**.

As linalool on catalytic dehydrogenation yields saturated alcohol, which on dehydration yields olefin, $C_{10}H_{20}$, these reactions reveal that linalool contains **two double bonds** and **one tertiary alcoholic group**.

$$\underset{\text{Linalool}}{C_{10}H_{17}OH} \xrightarrow[4[H]]{Pd} \underset{\text{Tetrahydro-linalool}}{C_{10}H_{21}OH} \xrightarrow{-H_2O} \underset{\text{Olefin}}{C_{10}H_{20}}$$

The presence of tertiary alcoholic group in linalool has been further confirmed by the fact that the oxidation of olefin, $C_{10}H_{20}$ (produced as above) yields isohexyl ketone which on treatment with Grignard's reagent yields a saturated alcohol, tetrahydro linalool having a tertiary alcoholic group.

$$\underset{\text{Olefin}}{C_{10}H_{20}} \xrightarrow[[O]]{KMnO_4} \underset{\text{Isohexylmethyl ketone}}{(H_3C)_2HC{-}CH.CH_2CH_2CH_2{-}\overset{O}{\overset{\|}{C}}{-}CH_3} \xrightarrow[(ii)\ H_2O]{(i)\ C_2H_5MgI}$$



Tetrahydrolinalool

3. *Relation with geraniol.* Linalool is isomeric with geraniol and contains a tertiary alcoholic group instead of primary alcoholic group. This has been proved on the basis of following facts :

(*a*) When linalool is heated with acetic anhydride, it yields geranyl acetate and the latter is converted into the former by heating with steam at 200°C under pressure.

$$\text{Linalool} \underset{\text{Steam ; 200°C}}{\overset{Ac_2O}{\rightleftharpoons}} \text{Geranyl acetate}$$

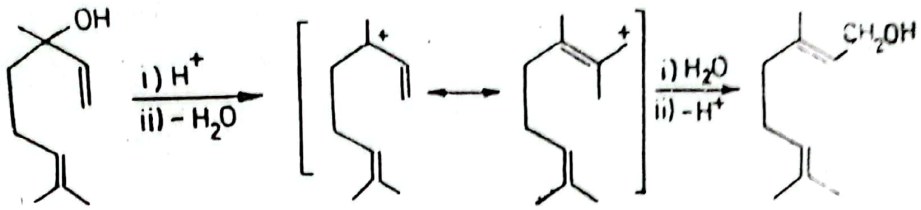
(*b*) Linalool when heated with hydrogen chloride in toluene solution at 100°C yields geranyl chloride. The latter compound is also obtained by the action of hydrochloric acid on geraniol. These reactions reveal that linalool is closely related to geraniol.

$$C_{10}H_{18}O + HCl \longrightarrow C_{10}H_{17}Cl + H_2O$$

Geranyl chloride when treated with moist silver oxide in benzene solution is again reconverted into linalool.

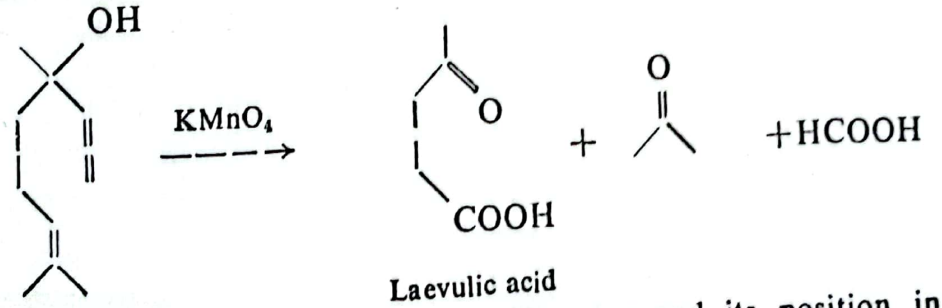
(*c*) Linalool on oxidation with chromic acid yields citral-a (geranial).

On the basis of the above facts, it may be concluded that linalool is isomeric with geraniol and contains a tertiary alcoholic group instead of primary alcoholic group. But the structure of geraniol is well known. Therefore, the possible structure of linalool may be obtained on the basis of an allylic rearrangement.



Linalool Geraniol

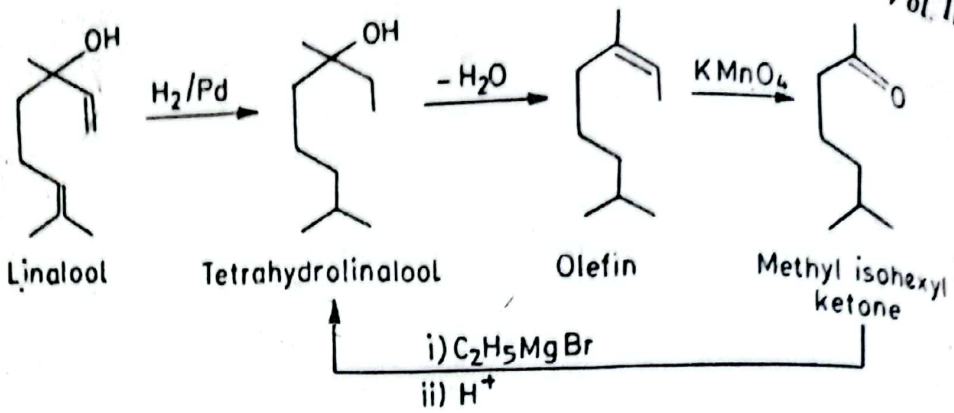
4. *Confirmation of structure.* The structure of linalool (given above) has been supported by its oxidation with permanganate to laevulic acid.



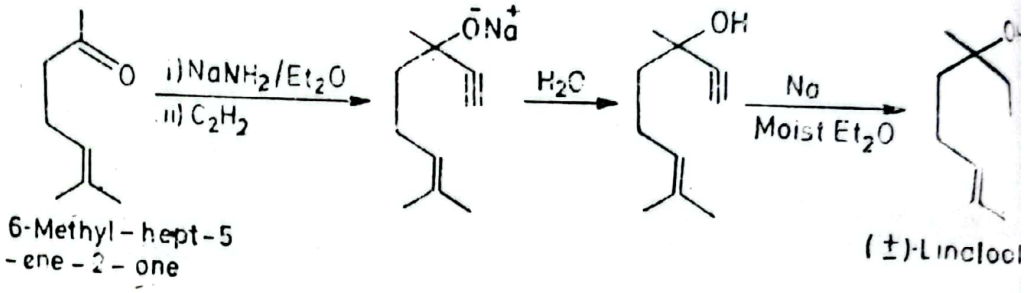
Laevulic acid

The presence of a tertiary alcoholic group and its position in the structure of linalool (given as above) have been confirmed by dehydrating tetrahydrolinalool (obtained by the catalytic reduction of linalool) and then oxidising the alkene produced to methyl isoheptyl ketone, a compound of known structure.

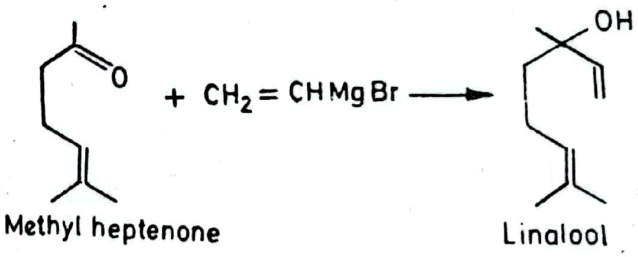
5. *Synthesis.* Finally the structure of linalool has been confirmed on the basis of various syntheses given as follows :



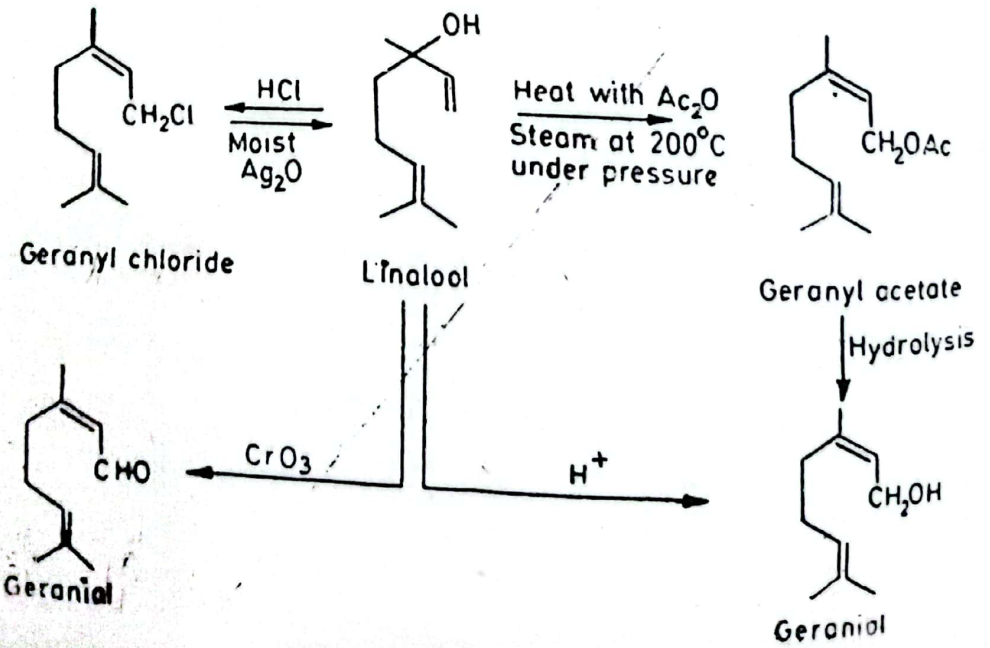
(*a*) *Ruzicka synthesis* (1919). In this synthesis, sodioderivative of the methylheptenone is condensed with acetylene, followed by partial reduction of the triple bond.



(*b*) *Normant's synthesis* (*1955*). In this synthesis, linalool is obtained in one step by the action of vinylmagnesium bromide on methylheptenone.



Reactions. The reactions given in step (3) can be represented as follows :

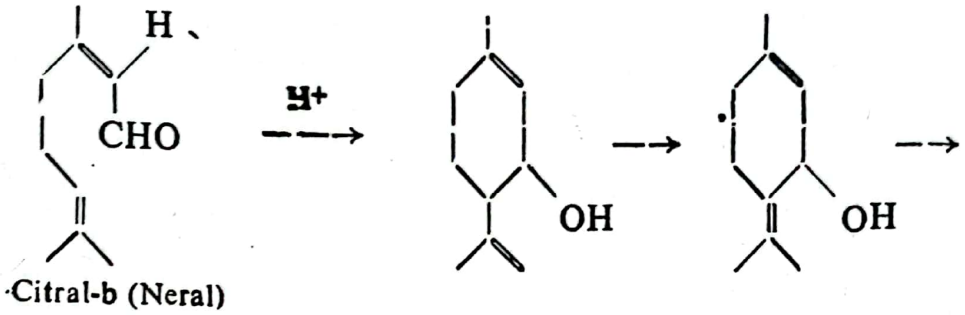


1.17. CYCLISATION REACTIONS OF ACYCLIC MONOTERPENOIDS

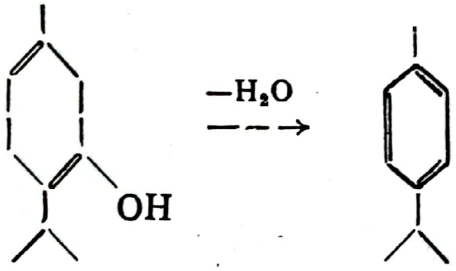
Acyclic monoterpenoids have a tendency to undergo cyclisation even under the mild conditions. Hence, these should be studied very carefully. The nature of the cyclisation product obtained depends upon the nature of the attacking reagent.

Some interesting examples of cyclisation reactions of acyclic monoterpenoids are as follows :

(*a*) **Citral to p-cymene.** When citral is treated with strong acids or heated with dehydrating agents like $KHSO_4$, *p*-cymene is obtained.

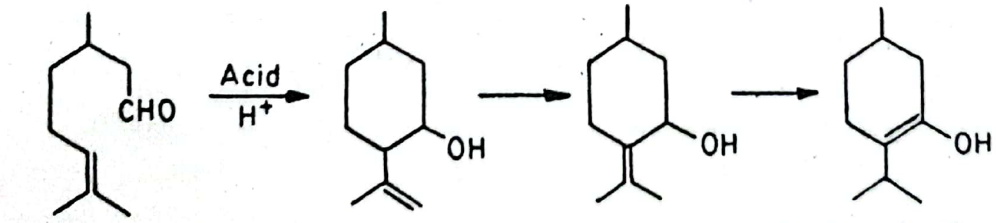


Citral-b (Neral)

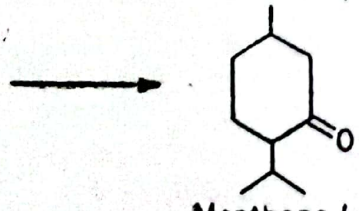


p-Cymene

(*b*) **Citronellal to isopulegol.** In the presence of acids, citronellal undergoes cyclisation to yield isopulegol which is then isomerised to pulegol and menthone. In the conversion of citronellal to isopulegol, H^+ ion acts a catalyst.

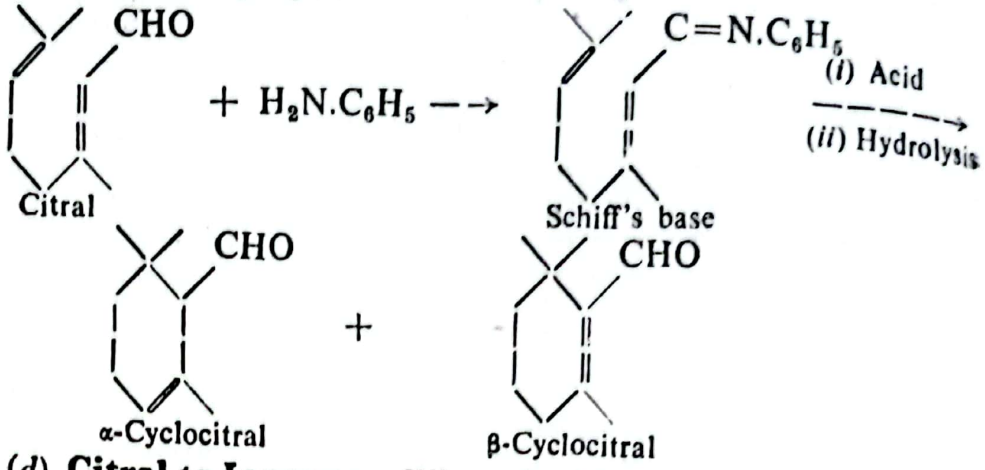


Citronellal Isopulegol Pulegol Menthone (enol form)

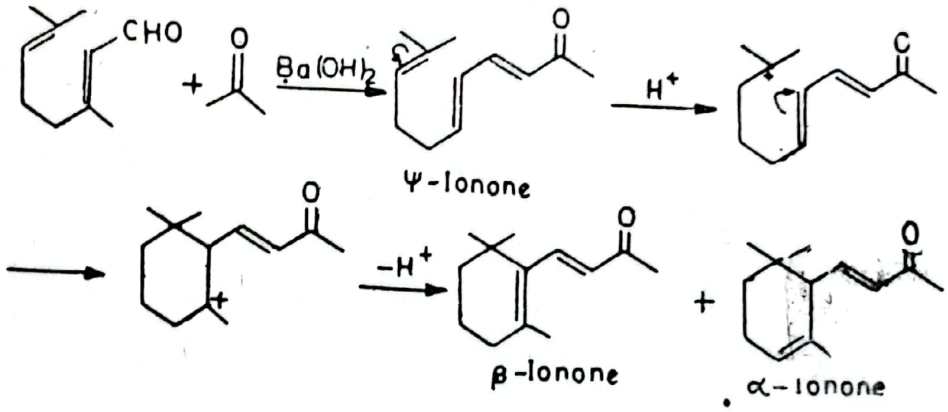


Menthone (ketol form)

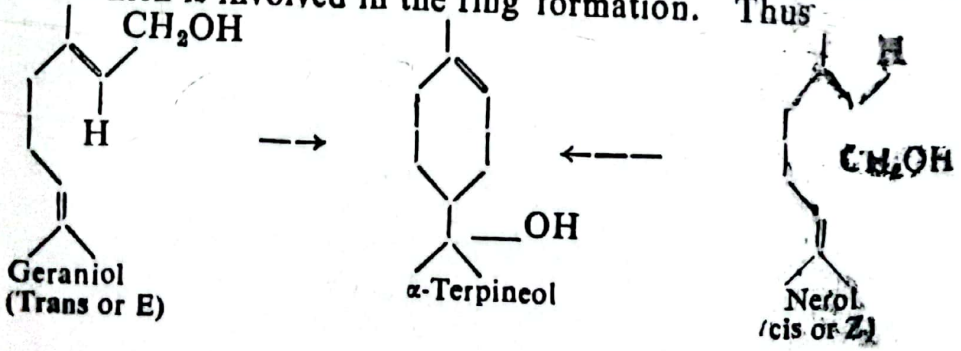
(*c*) **Citral to cyclocitrals.** When the aldehyde group in citral is protected by its treatment with aniline, it forms anil or Schiff's base. The latter product, when treated with acid, undergoes cyclisation in a quite different manner as citral undergoes to yield a mixture of α- and β- cyclocitrals on hydrolysis.



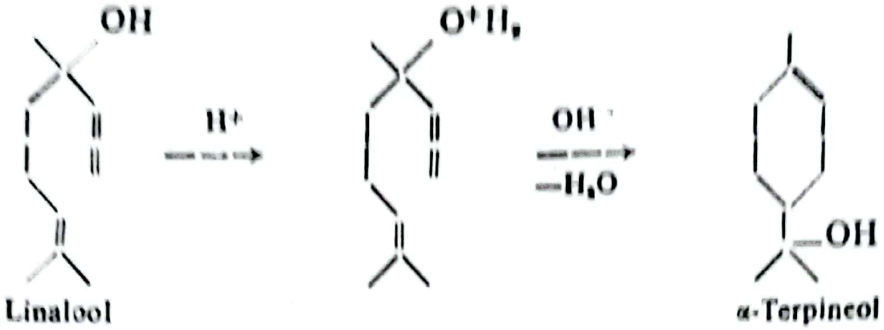
(*d*) **Citral to Ionones.** When citral is treated with acetone in the presence of barium hydroxide, ψ-ionone is formed. The latter product when heated with dilute sulphuric acid in the presence of glycerol yields a mixture of α- and β- ionones. The proportion of α to β depends upon the nature of the cyclising agent used, e.g., with sulphuric acid, β- ionone is the main product whereas with phosphoric acid α-ionone is main product.



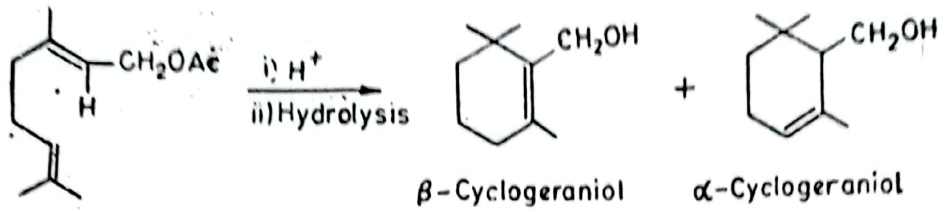
(*e*) **Geraniol and nerol to α- terpineol.** Both geraniol and nerol can be cyclised to α- terpineol in the presence of acids. The rate of cyclisation by means of dilute sulphuric acid takes place about 9 times as fast with nerol as it does with geraniol. The faster rate with nerol is due to the proximity of the alcoholic group to the carbon which is involved in the ring formation. Thus



(*f*) Linalool to α-terpineol. With acids, linalool undergoes cyclisation to yield α-terpineol.

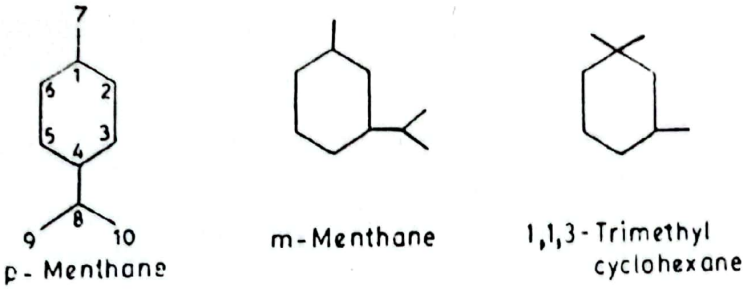


(g) Geranyl acetate to cyclogeraniols. When geranyl acetate is treated with acid, a mixture of cyclogeraniols is obtained. This conversion is similar to the conversion of citral into a mixture of cyclocitrals.

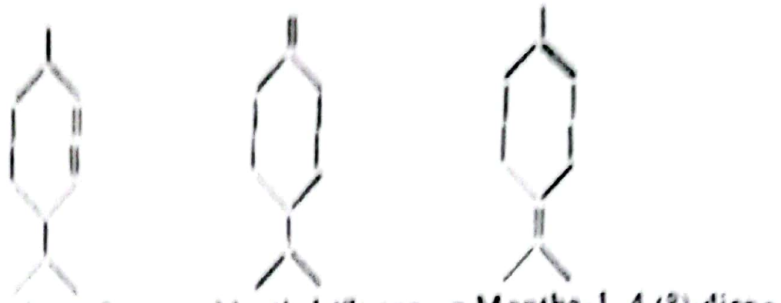


1.18. MONOCYCLIC MONOTERPENOIDS

Introduction. Depending upon the nature of the carbon skeleton, monocyclic monoterpenoids have been classified into three important classes : *viz.*, *p*-menthane (1-methyl-4-isopropyl cyclohexane), *m*-menthane (1-methyl-3-isopropyl cyclohexane) and 1, 1, 3-trimethyl cyclohexane derivatives.



Nomenclature. In the structure of *p* menthane, numbering being shown. However, ambiguity may arise if a compound derived from *p*-menthane contains one or more double bonds. In order to prevent ambiguity, the position of the **first carbon atom** joining to the double bond is indicated in the usual way whereas the **second** carbon atom joined to the double bonds is also shown but is placed in **parentheses**. The following examples illustrate this method of nomenclature.



p-Menthene-2 ; *p*-Menth-2-ene ; 2-*p*-Menthene

p-Menth-1 (7)-ene

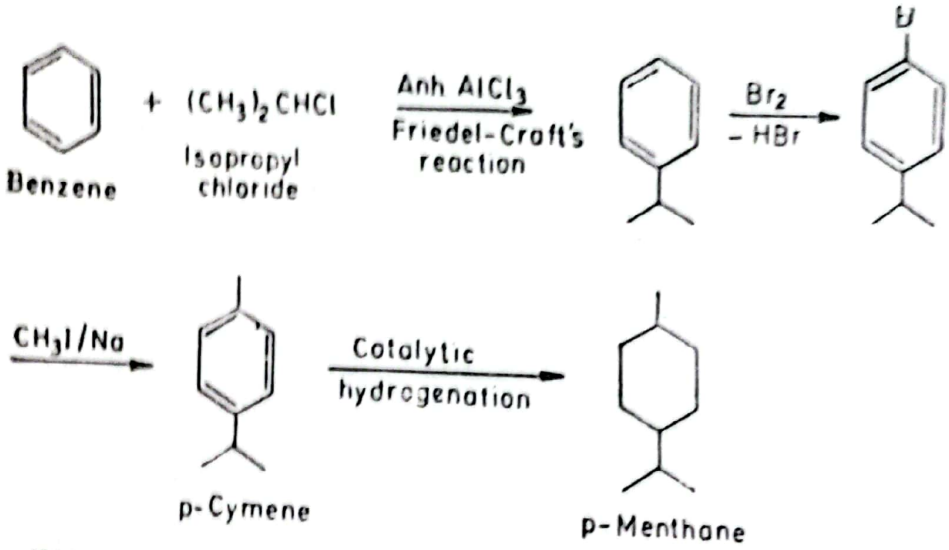
p-Mentha-1, 4 (8)-diene

1.19. *p*-MENTHANE

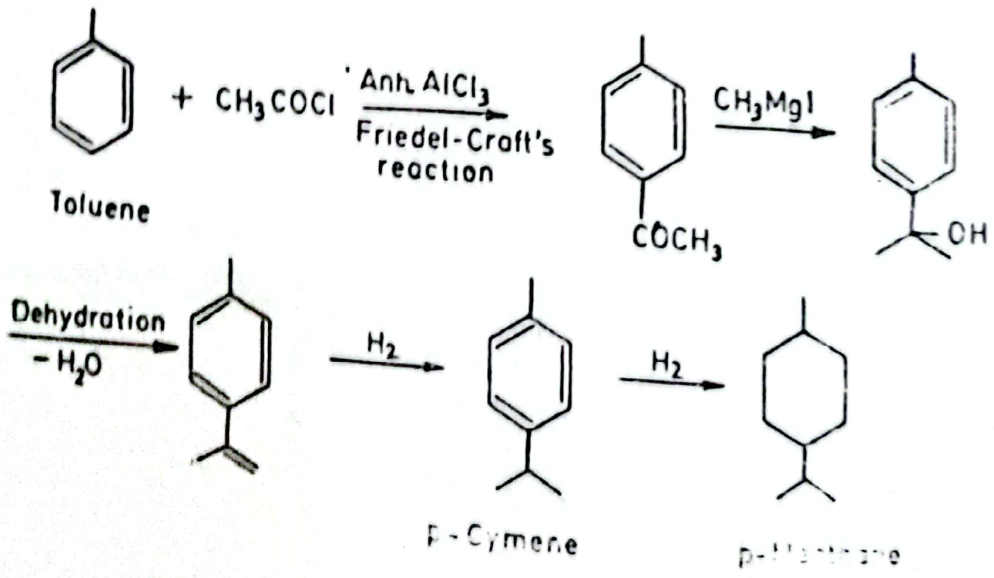
Introduction. It is a synthetic compound which does not occur in nature. It is the parent substance of monocyclic terpenoids, having molecular formula $C_{10}H_{20}$ which boils at 170°C.

Synthesis. *p*-menthane may be prepared synthetically by the following methods :

(*a*) *From benzene.* The various steps for its synthesis are as follows :



(*b*) *From toluene.* The various steps for its synthesis are as follows :



1.20. LIMONENE

Introduction. It is the most important monoterpenoid which is widely distributed in nature. Its +)-form occurs in lemon and orange oils the, (−)-form occurs in peppermint oil whereas the (±)-form occurs in turpentine oil. The racemic modification of limonene is known as **dipentene**. This name was given to the inactive form before its relation to the active form (limonene) was established.

Limonene is a liquid which boils as 175.5−176.5°C.

This is produced by the racemisation of the optically active forms at about 250°C.

Constitution of Limonene. This has been elucidated on the basis of following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data, the molecular formula of limonene has been found to $C_{10}H_{16}$.

2. *Presence of two olefinic bonds.* This has been revealed on the basis of following facts :

(*a*) It adds on four bromine atoms to form a tetrabromide.

(*b*) On catalytic reduction, it adds on four atoms of hydrogen to form tetrahydro derivative.

(*c*) With hydrochloric acid, a dihydrochloride is formed.

(*d*) With hydrobromic acid, it yields a dihydrobromide.

$$C_{10}H_{16}Br_4 \xleftarrow{2Br_2} C_{10}H_{16} \xrightarrow{H_2/Pt} C_{10}H_{20}$$

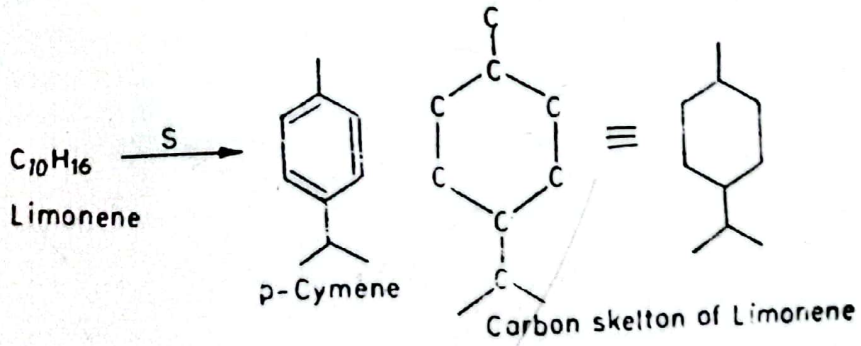
Limonene tetrabromide — Limonene — Tetrahydro limonene

$$C_{10}H_{16}\,2HCl \xleftarrow{2HCl} C_{10}H_{16} \xrightarrow{2HBr} C_{10}H_{16}.2HBr$$

Limonene dihydrochloride — Limonene — Limonene dihydro bromide

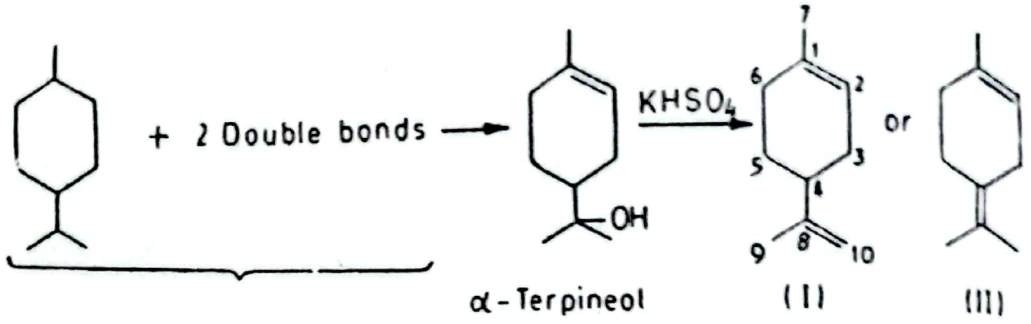
3. *As a monocyclic derivative.* The molecular formula of the saturated parent hydrocarbon corresponding to limonene is $C_{10}H_{20}$ which corresponds to the general formula C_nH_{2n} (n=10) for the monocyclic compound. Hence, the limonene contains monocyclic system.

4. *Nature of the carbon skeleton.* Limonene, when heated with sulphur, undergoes dehydrogenation to yield *p*-cymene, thus indicating that the carbon skeleton of *p*-cymene is present in limonene.



5. *Position of the double bonds.* Now the only problem is to ascertain the positions of the two double bonds in the fully saturated nucleus of *p*-menthane (*p*-cymene). This has been done by its relation to α-terpineol and carvone.

When limonene is hydrated by dilute sulphuric acid, it yields α-terpineol which on dehydration with potassium hydrogen sulphate regenerates limonene. But the structure of α-terpineol is known with certainty. Therefore, the possible hydration and dehydration products may be represented as follows :

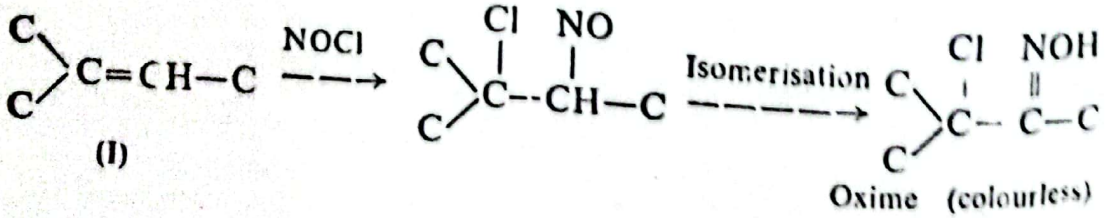


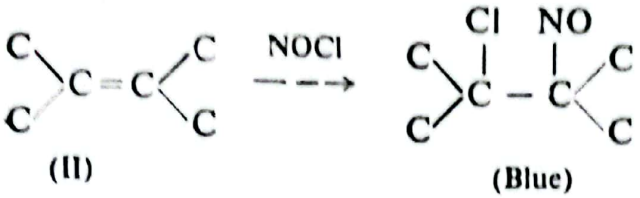
Out of the two structures, either (I) or (II) must be limonene. But both of these structures have one double bond between C_1 and C_2. Therefore, in limonene there should be a double bond between C_1 and C_2. This position of one of the double bonds in limonene has been further confirmed by the fact that limonene when oxidised with $KMnO_4$ yields terpenylic and terebic acids (also obtained by the oxidation of α- terpineol during which only the double bond between C_1 and C_2 is affected).

The position of one double bond is fixed between C_1 and C_2. Now the question arises : **what is the position of the second double bond** ? This has been found to be between C_8 and C_{10} on the basis of following facts :

(*a*) As limonene is optically active, it must possess at least one asymmetric carbon atom which is possible only in structure (I) (C_4 is asymmetric). The structure (II) has no such asymmetric carbon atom and hence must be optically inactive. Actually, it is found to be so and this compound (II) was identified as terpinolene. Hence, structure (I) must be limonene.

(*b*) Limonene, when treated with nitrosyl chloride, yields a colourless product which indicates that the double bond, at which nitrosyl chloride is added, is between a tertiary and a secondary carbon atom. Hence, a double bond cannot be situated between C_4 and C_8 as in (II) because this compound (II) when treated with nitrosyl chloride should yield the blue coloured product. But a colourless product is obtained. Therefore, limonene must be (I).

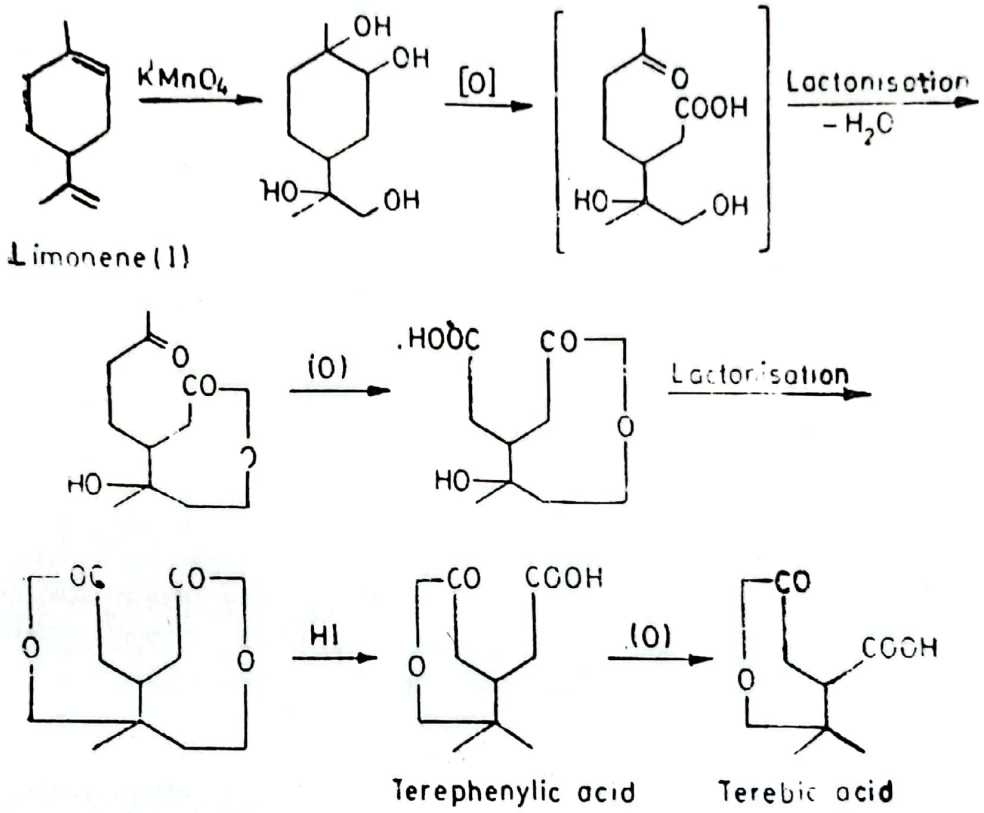




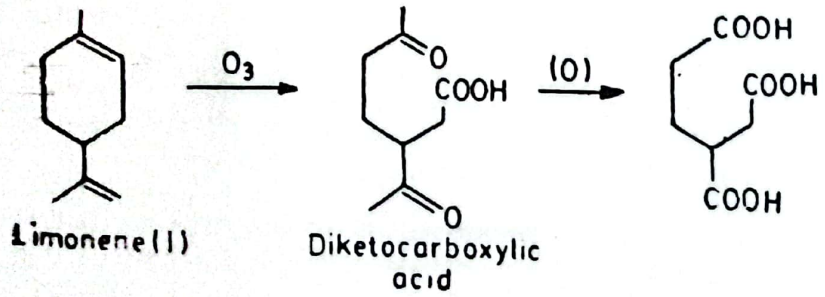
Further limonene nitrosochloride, when treated with ethanolic caustic potash, is converted into carvone through carvoxime. The structure of carvone is well known which does not contain any double bond between C_4 and C_8. Thus, the structure (I) is the correct structure of limonene.

6. *Explanation of reactions.* On the basis of structure (I), all the reactions of limonene can be explained.

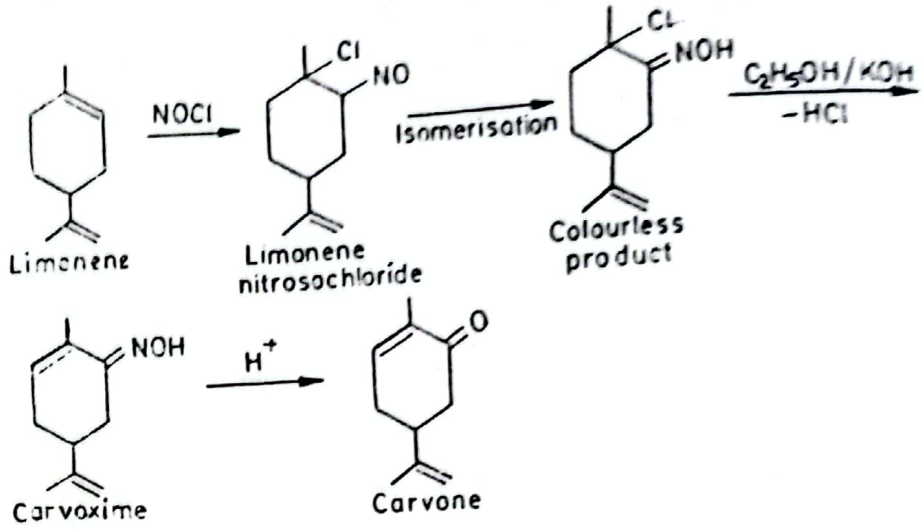
(*a*) Oxidation to terpenylic and terebic acids.



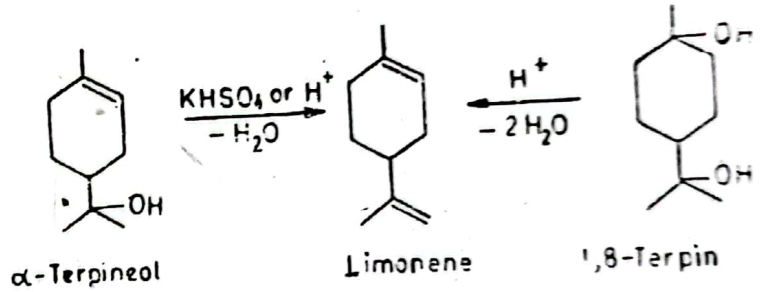
(*b*) Ozonolysis to a diketocarboxylic acid.



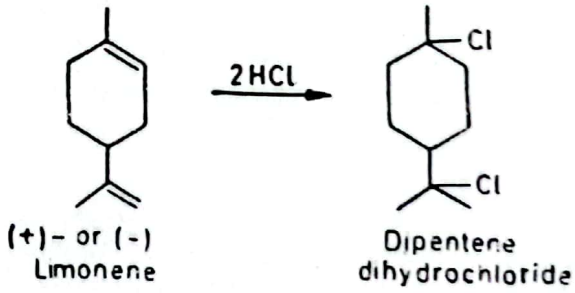
(c) Formation of carvone.



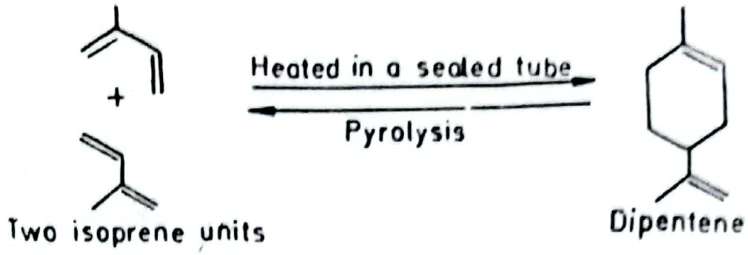
7. *Synthesis.* Finally, the structure (I) of limonene may be confirmed by its synthesis which involves the dehydration of α-terpineol, or 1, 8- terpin.



Connection between Limonene and Dipentene. This has been revealed by the fact that (+)- or (−)- limonene, when treated with hydrogen chloride in the presence of moisture, yields limonene dihydrochloride which is identical with dipentene dihydrochloride. This reaction shows that dipentene is (±)- limonene.



Dipentene [(±)- limonene] is prepared by dimerisation of isoprene in a sealed tube. However, the yield is very poor.



The above reaction is reversible and when dipentene is pyrolysed, it yields about 60 per cent of isoprene.

1.21. α- TERPINEOL

Introduction. It is an optically active monoterpenoid that is widely distributed in nature in the (+)-, (−)- and (±)-forms. It is a solid having m.p. 35°C of the racemic modification.

The (+)- form occurs in oils of petitgrain and neroli, (−)-form in camphor oils and (±)- form in cajuput oil.

α- Terpineol is widely used in the preparation of the lilac purfumes and lilac-scented soaps.

Constitution of α-Terpineol. This has been elucidated on the basis of following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data and molecular weight determination, the molecular formula of α- terpineol has been found to be $C_{10}H_{18}O$.

2. *Presence of a double bond.* As α- terpineol adds on one mole of bromine and of nitrosyl chloride to form their respective addition compounds, it follows that α- terpineol contains one double bond.

$$C_{10}H_{18}O + NOCl \longrightarrow C_{10}H_{18}O.NOCl$$
$$C_{10}H_{18}O + Br_2 \longrightarrow C_{10}H_{18}O.Br_2$$

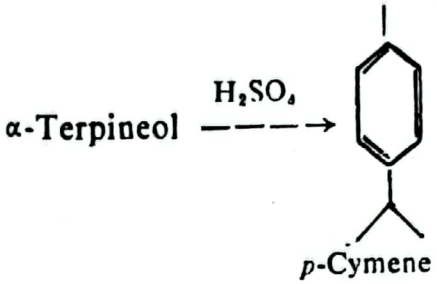
3. *Presence of tertiary —OH group.* α-Terpineol, when treated with phenylisothiocyanate, yields phenylurethane, indicating the presence of a hydroxyl group in terpineol. Further, α-terpineol is insoluble in sodium hydroxide solution. This shows that the hydroxyl group cannot be a phenolic one. Also, terpineol undergoes dehydration readily, when heated with sulphuric acid in glacial acetic acid, forming terpenoids like dipentene and terpinolene. Moreover, α- terpineol does not undergo oxidation readily. All these facts reveal that the hydroxyl group in α-terpineol is present as a **tertiary alcoholic group.**

$$\underset{\text{Terpineol}}{C_{10}H_{18}O} \xrightarrow{-H_2O} \underset{\text{Terpinolene}}{C_{10}H_{16}}$$

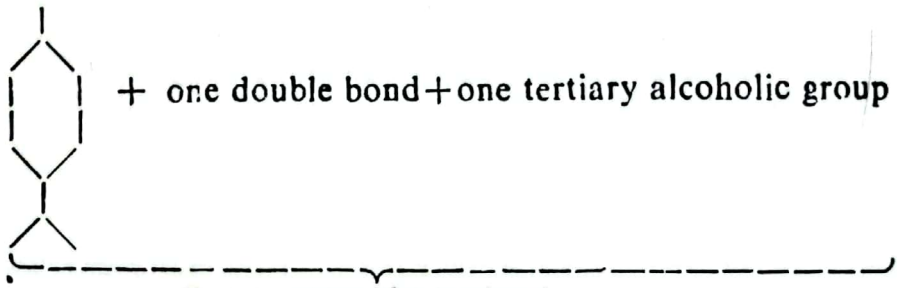
4. *α- Terpineol as a monocyclic compound.* As α- terpineol contains one double bond, the molecular formula of its saturated hydrocarbon is $C_{10}H_{20}$. This corresponds to C_nH_{2n}, the general

formula of the monocyclic cycloalkanes, and so it follows that α-terpineol is a monocyclic compound.

5. *Carbon skeleton of α-terpineol.* When α-terpineol is heated with sulphuric acid, it forms some *p*-cymene. Taking this in conjuction with the tentative proposal that α-terpineol is monocyclic, it may be inferred that α-terpineol contains the *p*-cymene skeleton. Thus, it may be concluded that **α-terpineol may be p-menthane** (fully saturated *p* cymene) **with one double bond and a tertiary alcoholic group.** Thus, the part structure of α-terpineol may be



written as follows :



Part structure of α-terpineol

6. *Position of a double bond and tertiary alcoholic group.* These have been ascertained by Wallach by means of graded oxidation. Wallach degraded α-terpineol having C_{10} to the well known C_7 acid called terebic acid The series of degradation reactions of α-terpineol are as follows :

α-Terpineol $\xrightarrow[KMnO_4]{1\%\ Alk}$ Trihydroxy compound $\xrightarrow{CrO_3}$ [Ketohydroxyacid]

C_{10} (I) C_{10} (II) C_{10} (III)

$\longrightarrow$ keto-lactone $\xrightarrow[Alk.\ KMnO_4]{Warm}$ Terpenylic acid $\xrightarrow{KMnO_4}$ Terebic acid

C_{10} (IV) C_8 (V) + CH_3COOH C_7 (VI)

From the above set of degradative products, following conclusions could be obtained :

(*a*) α-Terpineol (I), when oxidised with 1 per cent alkaline $KMnO_4$, undergoes hydroxylation at double bond to produce the

trihydroxy compound (II), $C_{10}H_{20}O_3$. The latter compound on further oxidation with chromic acid (chromium trioxide in acetic acid) yields a compound with the molecular formula, $C_{10}H_{16}O_3$ (IV). This compound contains a ketonic group, and does not give any reaction with sodium bicarbonate and is, therefore, neutral. However, when the compound (IV) is refluxed with excess of standard alkali followed by back titration, it is found that the amount of alkali consumed corresponds to the presence of one carboxyl group. Thus, the compound (IV) may be the lactone of a monocarboxylic acid which has been produced due to the lactonisation of γ- or δ-hydroxy acid (III). Thus, the lactone (IV) may be a γ- or δ-lactone.

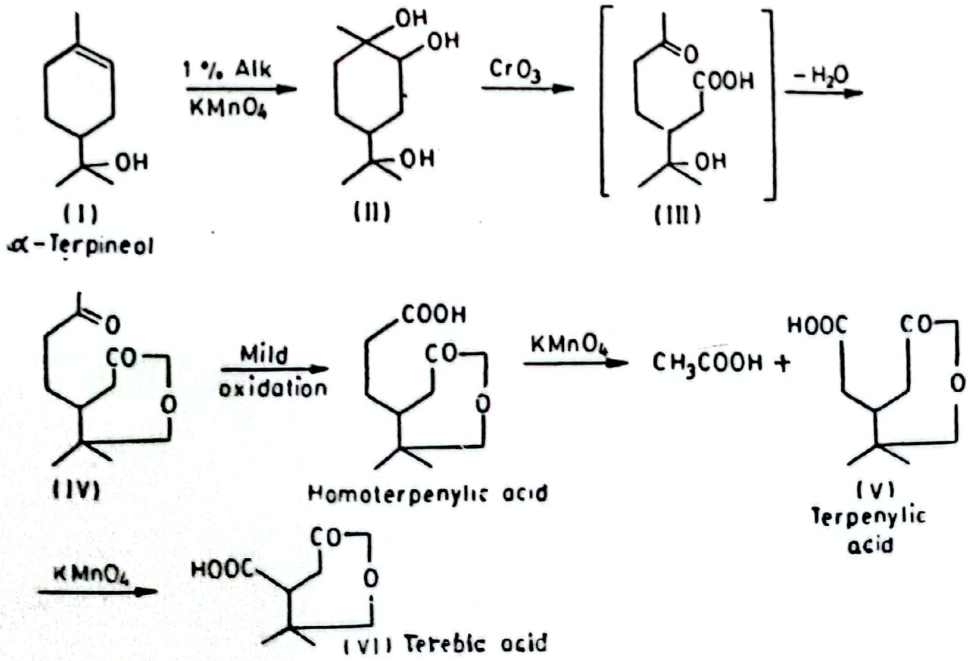
(*b*) As compound (IV) is formed due to the breaking of the glycol bond in (II) without any loss of carbon atoms in the process, it means that the double bond must be present in the ring of compound (I).

(*c*) When the compound (IV) is warmed with alkaline permanganate, it yields acetic acid and a compound $C_8H_{12}O_4$ (V).

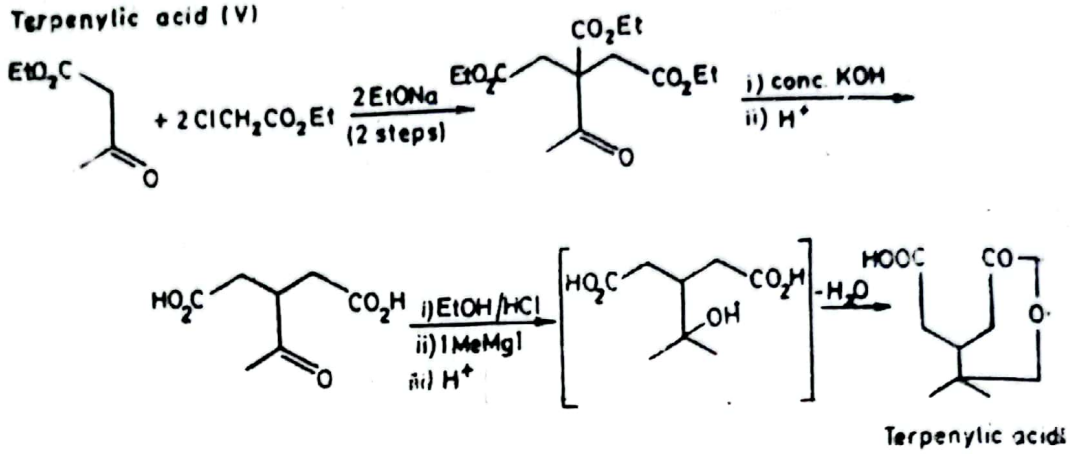
The formation of acetic acid reveals that the compound (IV) contains CH_3CO group, *i.e.*, the compound (IV) is a methyl ketone. But we have already proved that it is a lactone. Therefore, the compound must be **hometerpenyl methyl ketone.** The structure of the latter compound has been further confirmed by its synthesis, given by Simonsen *et al.*, (1932).

When the properties of compound (V) are studied, it is found that this compound (V) is the lactone of a monohydroxydicarboxylic acid. The latter compound when oxidised yields terebic acid $C_7H_{10}O_4$ (VI) which is also the lactone of a monohydroxydicarboxylic acid.

All the above reactions have been formulated on the basis of formula (I) (*p*-menth-1-en-8-ol) as the structure of α-terpineol. The structures of terpenylic (V) and terebic (VI) acids have been confirmed by their synthesis.

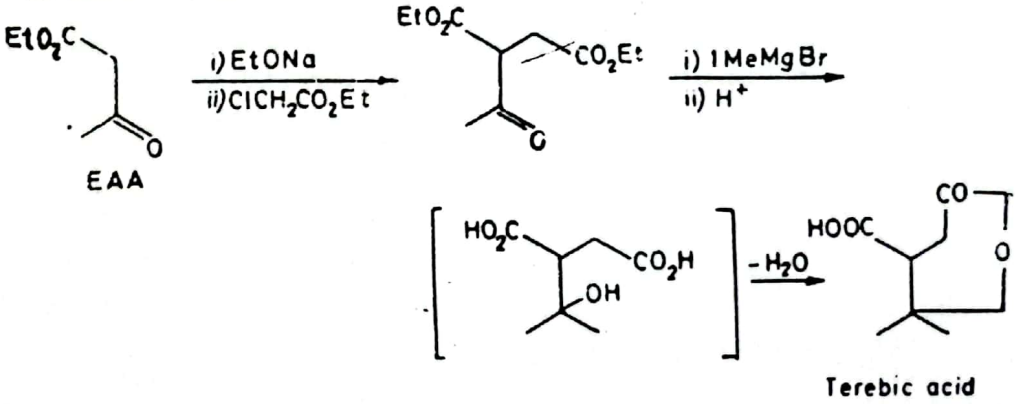


Terpenylic acid (V)



Terpenylic acid

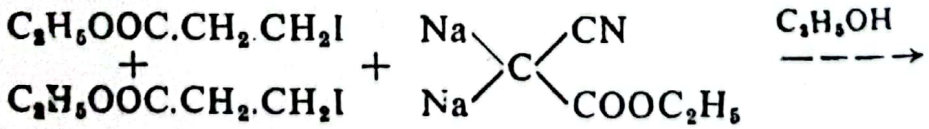
Terebic acid (VI)



Terebic acid

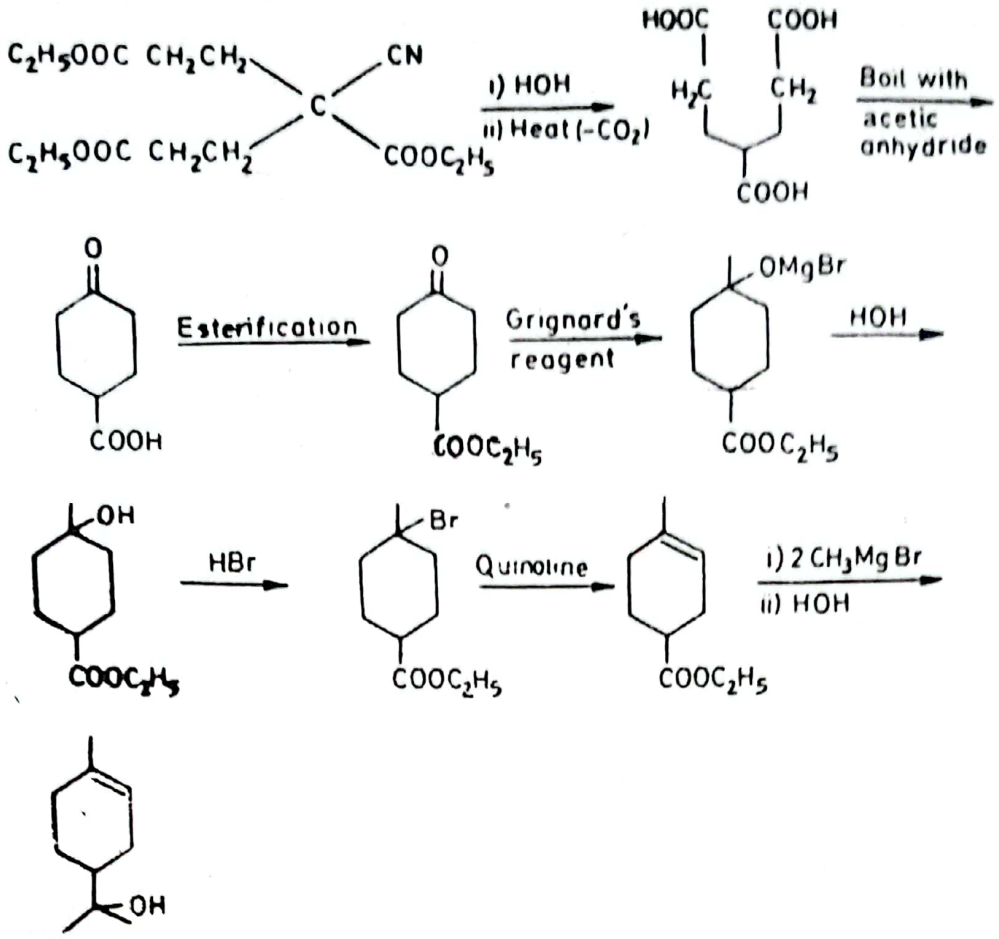
7. *Synthesis.* Finally, the structure of α-terpineol (I) has been confirmed by its various syntheses as follows :

(*a*) Perkin's synthesis (1904).



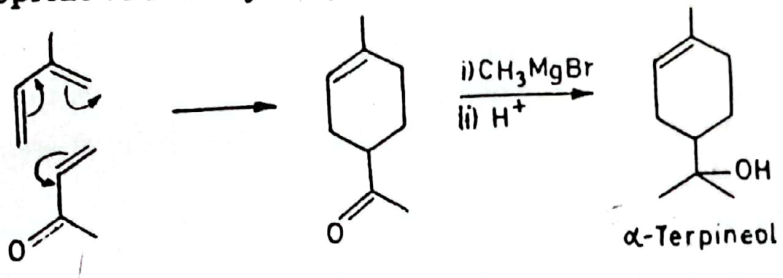
2 Moles of β-iodopropionic acid

Disodium derivative of cyano acetic ester



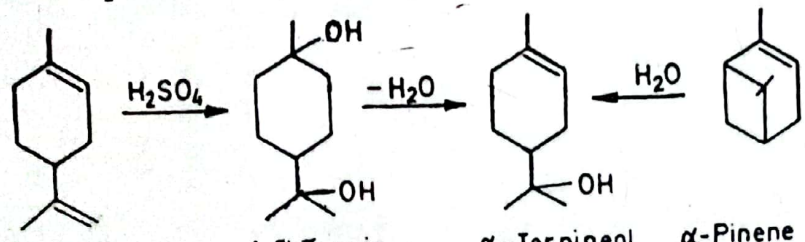
(±)-α-Terpineol (1)

(*b*) **Alder-Vogt's synthesis.** This synthesis makes use of the Diel's-Alder reaction. The starting materials for this synthesis are isoprene and methyl vinyl ketone.



α-Terpineol

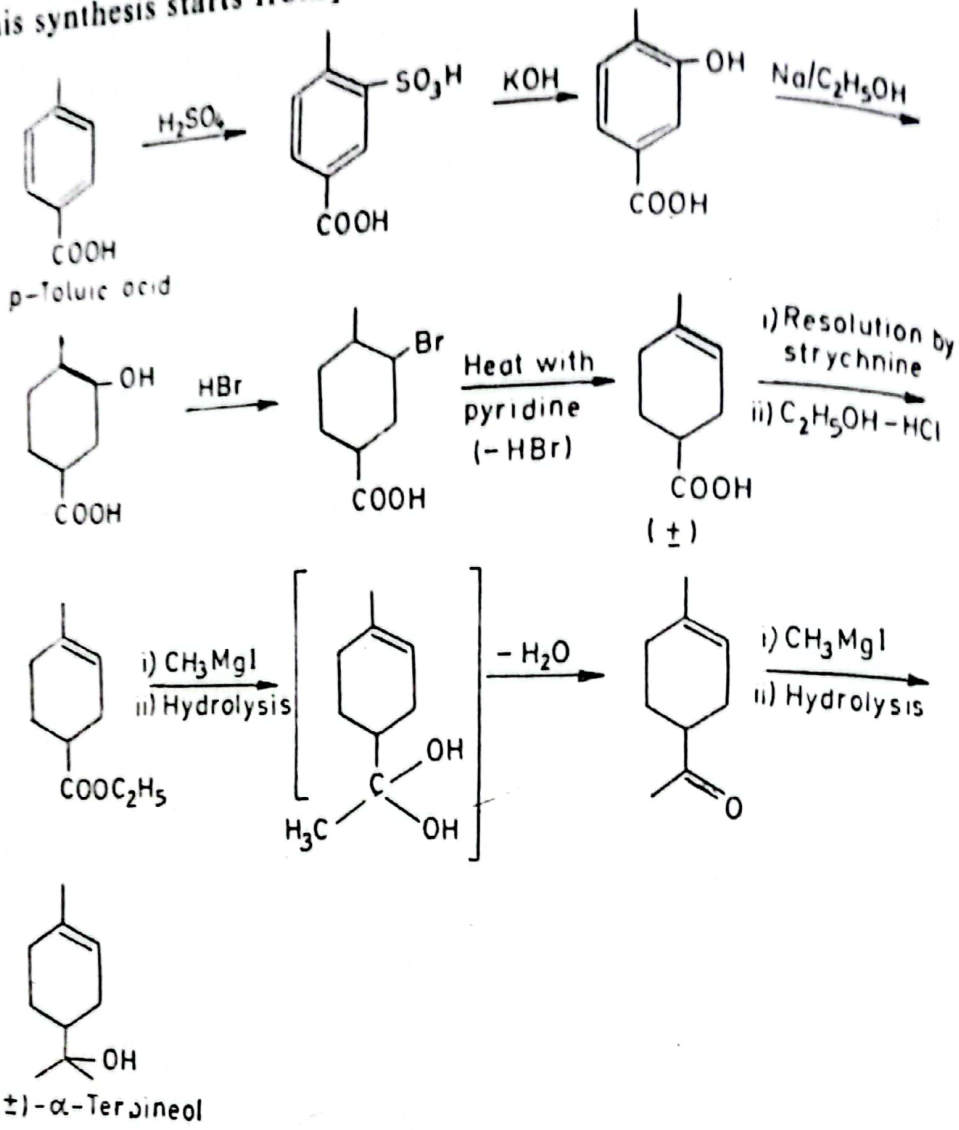
(*c*) **Commercial method.** α-Terpineol required in perfume and cosmetics is prepared by the dehydration of terpin or by the hydration of α-pinene. Terpin is prepared by the hydration of dipentene.



Dipentene 1,8-Terpin α-Terpineol α-Pinene

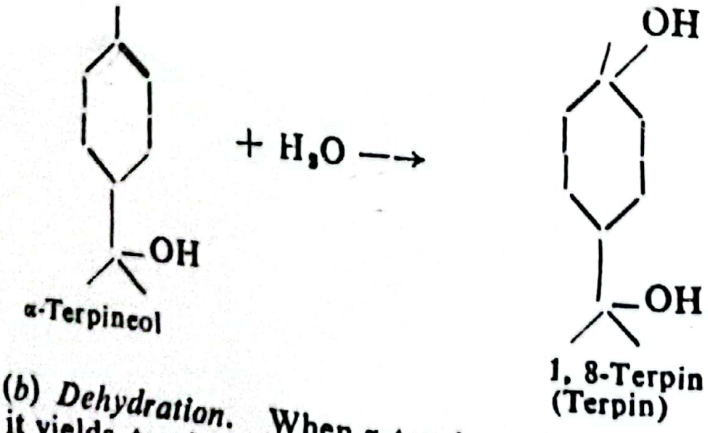
Terpin on dehydration, yields a mixture of α-, β-and γ-terpineols.

(*d*) **Perkin Junior-Meldrum-Fisher's synthesis** (1908).
This synthesis starts from *p*-toluic acid.

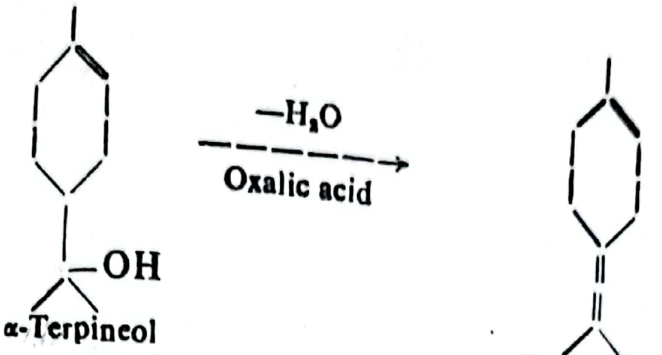


Reactions of α-Terpineol. Important reactions of α-terpineol are its hydration, dehydration and conversion into carvone.

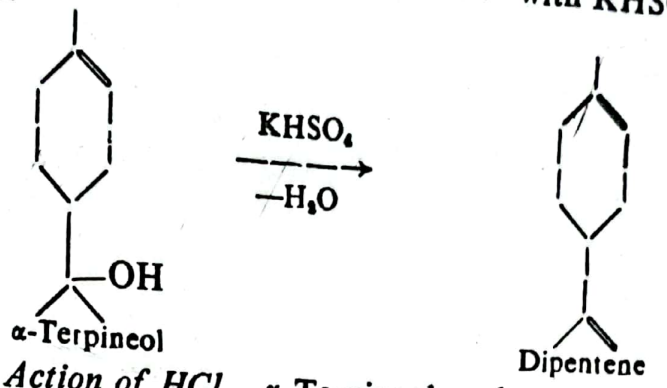
(*a*) *Hydration of α-terpineol.* When α-terpineol is hydrated, it yields 1, 8-terpin (terpin).



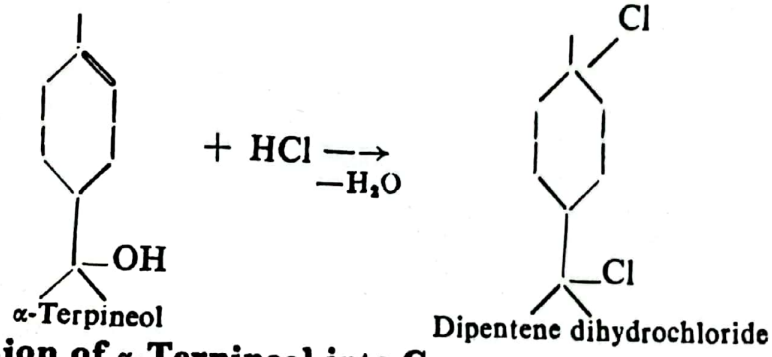
(*b*) *Dehydration.* When α-terpineol is dehydrated with oxalic acid, it yields terpinolene,



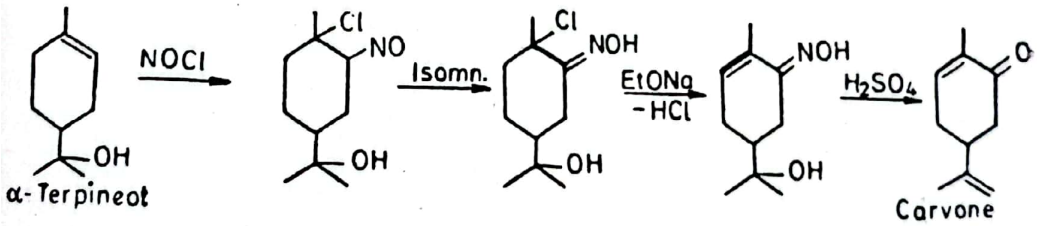
When α-terpineol is dehydrated with $KHSO_4$, dipentene is produced.



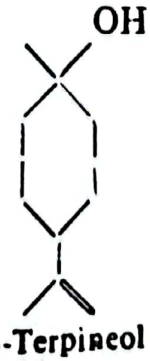
(c) *Action of HCl.* α-Terpineol, when treated with hydrochloride, yields dipentene dihydrochloride.



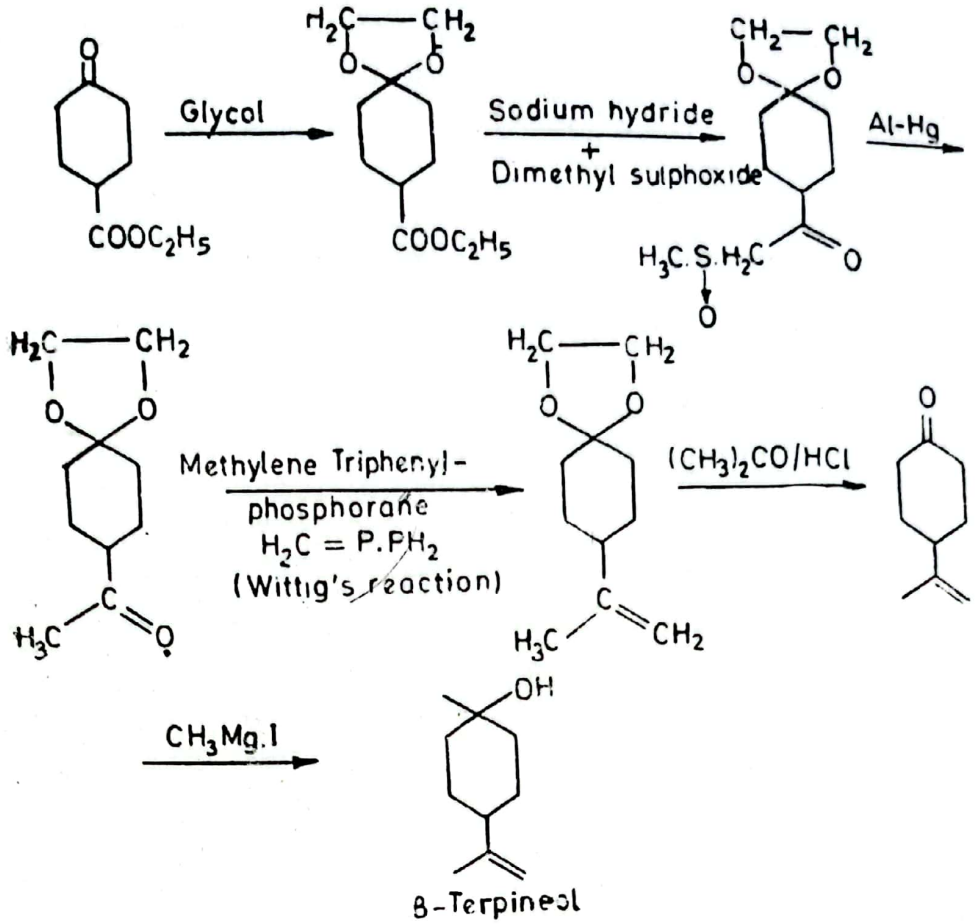
Conversion of α-Terpineol into Carvone.



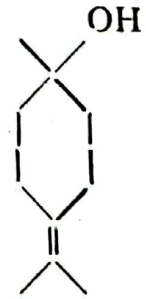
β-*Terpineol.* It is isomeric with α-terpineol and γ-terpineol. Its formula is as follows :



It occurs in nature in the nutmeg oil. It is a low melting solid having m.p. 32—33°C. It has been synthesised as follows :

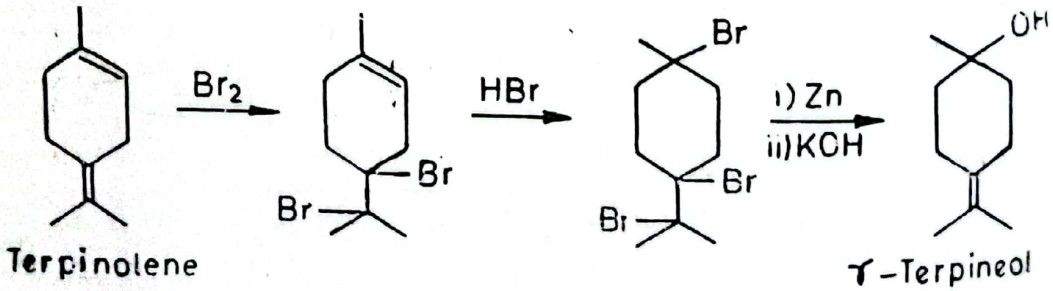


γ-Terpineol. It is isomeric with α- and β-terpineols. It occurs naturally in *Cupressus torulosa.* Its formula is as follows :

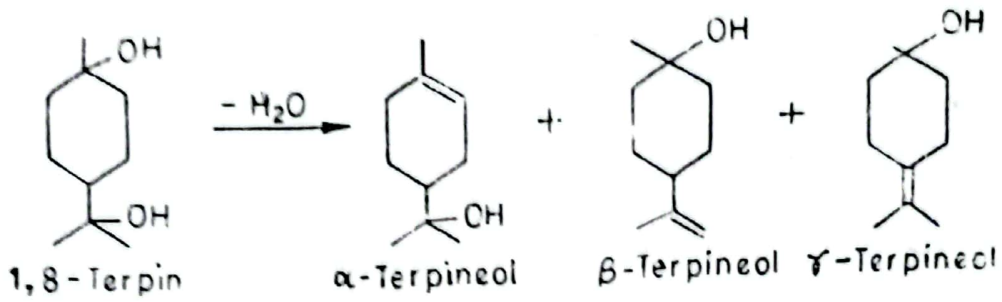


γ-Terpineol

According to Baeyer, γ-terpineol is prepared from terpinolene by the following method :



γ-Terpineol along with α-terpineol and β-terpineol may also be formed by the dehydration of 1, 8-terpin.



1.22. 1, 8-TERPIN OR TERPIN OR 1, 8-MENTHANEDIOL

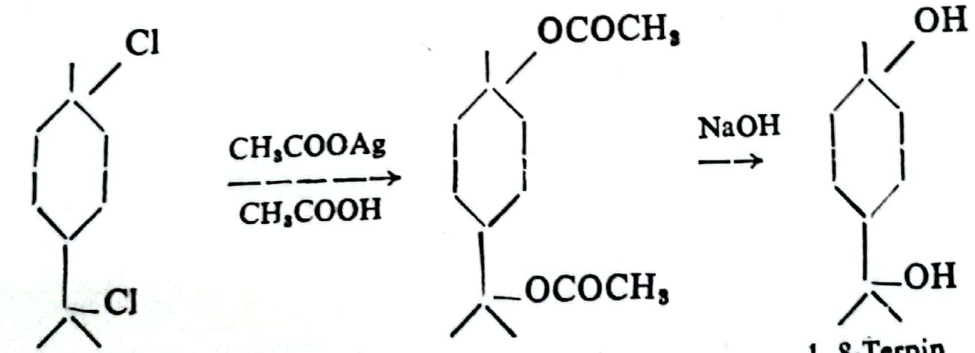
Introduction. Although the presence of 1, 8-terpin has been reported in some essential oils, yet it is accepted that it does not occur in nature but its presence in oils is due to secondary reactions.

Constitution of 1, 8-Terpin.

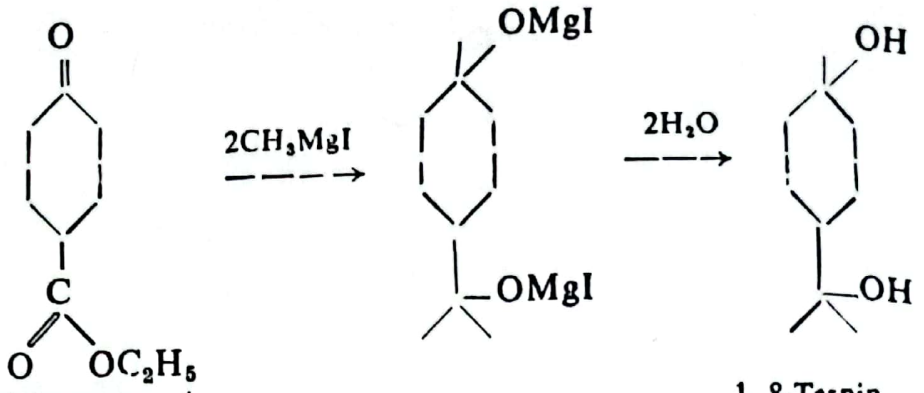
1. The molecular formula of terpin has been found to be $C_{10}H_{20}O_2$.

2. With acetyl chloride, it forms a diacetyl derivative, indicating that terpin molecule contains two hydroxyl groups.

3. Its structure has been established on the basis of its intimate relation with limonene dihydrochloride, a compound of known structure. Limonene dihydrochloride, when heated with silver acetate in acetic acid, yields the ester which upon hydrolysing yields 1, 8-terpin.



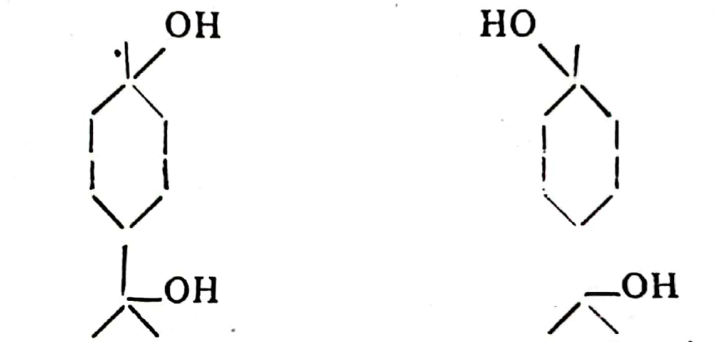
4. Finally, the structure of 1, 8-terpin is established on the basis of its synthesis from cyclohexanone 4-carboxylic ester (Perkin and Key).



Cyclohexanone -4-carboxylic ester

1, 8-Terpin

Isomerism in 1, 8-Terpin. This compound exists in two geometrical isomeric forms, corresponding to the *cis*- and *trans*-dipentene dihydrochlorides. The cis-1, 8- terpin melts at 105°C and it readily undergoes combination with one mole of water to yield terpin hydrate. The trans-1, 8- terpin melts at 158—159°C and does not form a hydrate.

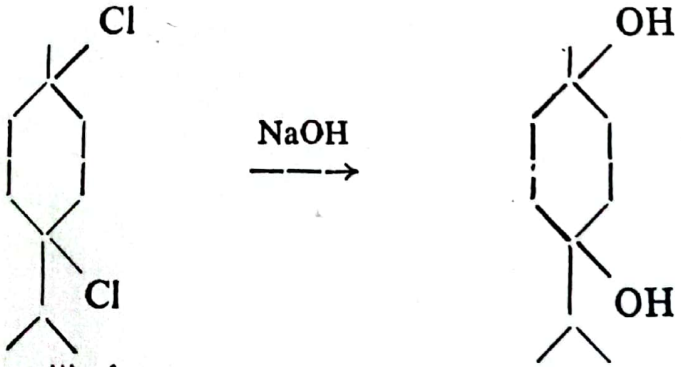


Cis-1, 8- terpin

Trans -1, 8-terpin

1.23. 1', 4-TERPIN

This is the positional isomer of 1, 8-terpin. Its structure has been elucidated on the basis of its preparation by the action of dilute alkali on terpinene dihydrochloride.



Terpinene dihydrochloride

1, 4-Terpin

1.24. 1, 8-CINEOLE

Introduction. It is isomeric with α-terpineol. But it does neither contain a hydroxyl group nor a double bond. It occurs in eucalyptus oils. It is a liquid which boils at 174.4°C.

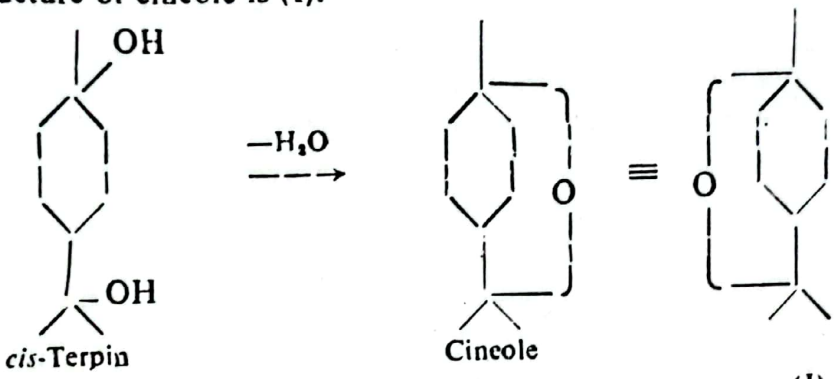
Constitution of 1, 8-Cineole. Its structure has been elucidated on the basis of following analytical and synthetic evidences :

1. From analytical data, the molecular formula of 1, 8-cineole has been found to be $C_{10}H_{18}O$.

2. It is isomeric with α-terpineol.

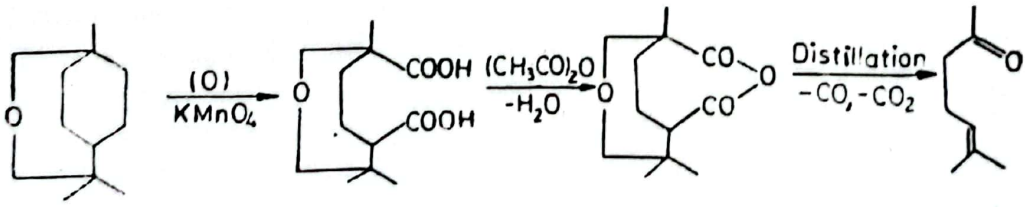
3. It contains neither a hydroxyl group nor a double bond.

4. As it is not attacked by sodium or by the usual reducing agents, this shows that the oxygen atom in cineole is inert. This inertness further suggests that this oxygen atom is of the ether type. This has been further confirmed from the fact that dehydration of *cis*-1, 8-terpin yields 1, 8-cineole ; this reaction also predicts that the structure of cineole is (I).



cis-Terpin Cineole (I) (I)

5. The above structure of cineole has been confirmed by its oxidation reactions. Cineole, when oxidised with potassium permanganate, yields cineolic acid (II) which on distillation with acetic anhydride, yields cineolic anhydride (III). The latter compound, when distilled at atmospheric pressure, yields 6-methylhept-5 en-2-one (IV), a compound of known structure. These reactions were interpreted by Wallach on the basis of structure (I) of 1, 8-cineole.

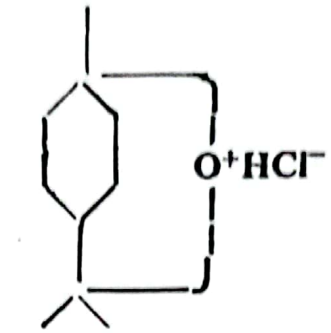


Cineole (I) Cineolic acid (II) Cineolic anhydride (III) Methyl heptenone (IV)

6. Finally, the structure (I) of cineolic acid has been further confirmed by Rupe *et al.* (1900—1908) by its degradation reactions.

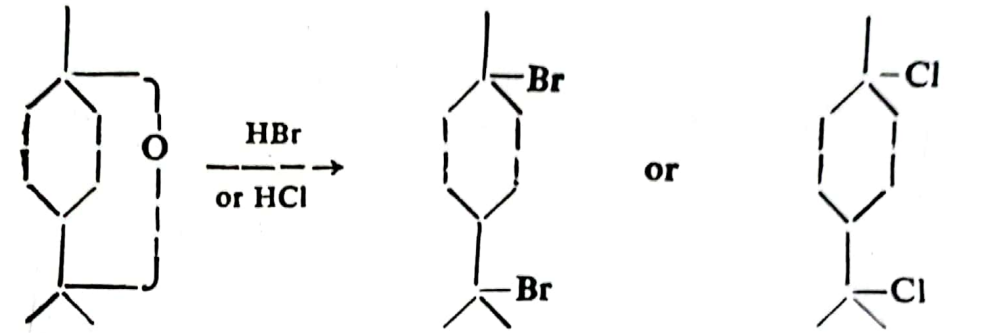
Reactions of 1, 8-Cineole. Due to its strong basic properties, cineole possesses remarkable property of forming stable oxonium salts (addition compounds) with hydrogen halides, bromine and

many substituted phenols. Thus, the oxonium salt which is prepared by the addition of hydrogen chloride on cineole may be represented as follows :



Oxonium salt

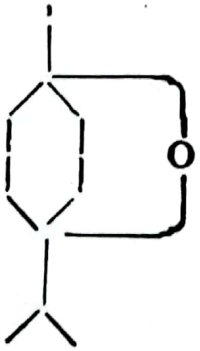
However, if hydrobromic acid or hydrochloric acid in acetic acid solution is added to cineole, no oxonium salt is formed but instead *cis*-dibromide or *cis*-dichloride of dipentene is obtained.



Cineole | *cis*-Dibromide of dipentene | *cis*-Dichloride of dipentene

As the addition compounds of 1, 8-cineole are very stable, they are used for identification and isolation.

1, 4-Cineole. Corresponding to 1, 4-terpin, there is also a 1, 4-cineole which occurs naturally.



1.25. ASCARIDOLE

Introduction. The cineoles are oxides whereas ascaridole is the only terpenoid peroxide which occurs naturally in chenopodium oil. It is an oily liquid having b.p. 96—97°C at 8 mm pressure. It is generally used in the treatment of hookworms.

Constitution

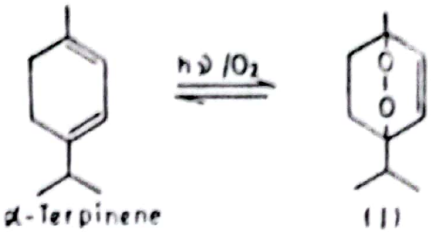
1. Its molecular formula is $C_{10}H_{16}O_2$.

2. When it is heated to 130—150°C, it decomposes with explosive violence.

3. When it is reduced catalytically, it yields 1, 4-terpin which on dehydration with oxalic acid yields 1, 4-cineole. These reactions support that structure (I) is ascaridole.

$$\text{Ascaridole} \xrightarrow[H_2\text{-Pd}]{\text{Reduction}} \text{1,4 Terpin} \xrightarrow[\text{acid}]{\text{oxalic}} \text{1,4-cineole}$$

4. Ziegler *et al.* (1944) synthesised ascaridole by the irradiation of α-terpinene in dilute solution in the presence of sensitisers like chlorophyll, dyes, etc.



α-Terpinene (I)

1.26. MENTHOL AND MENTHONE

Introduction. Menthol is an optically active compound. Only its (−)-form occurs naturally in peppermint oils. It is a saturated compound having m.p. 43°C. It is used as an antiseptic and anaesthetic.

Menthone, which is the oxidation product of menthol, also occurs naturally in the essential oils of peppermint (upto 35 per cent), geranium and pennyroyal mainly in the (—)-form. Menthone also finds use as an antiseptic and flavouring agent.

Constitution of Menthol and Menthone

1. The molecular formula of menthol has been found to be $C_{10}H_{20}O$ whereas that of menthone has been found to be $C_{10}H_{18}O$.

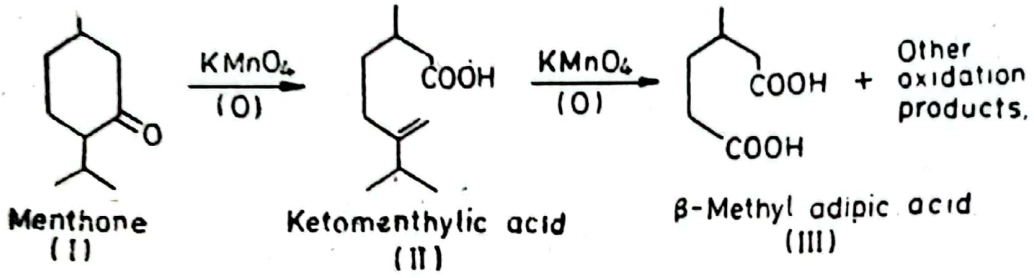
2. As menthol forms esters readily with acids, this means that it must possess an alcoholic group. Further, the oxidation of menthol yields a ketone, menthone ($C_{10}H_{18}O$), indicating that the hydroxyl group in menthol is secondary alcoholic in nature.

$$\underset{\text{Menthol}}{C_{10}H_{20}O} \xrightarrow{[O]} \underset{\substack{\text{Menthone}\\(\text{ketone})}}{C_{10}H_{18}O}$$

The above oxidation reaction reveals that menthol and menthone are related to each other as a ketone (menthone) and a secondary alcohol (menthol).

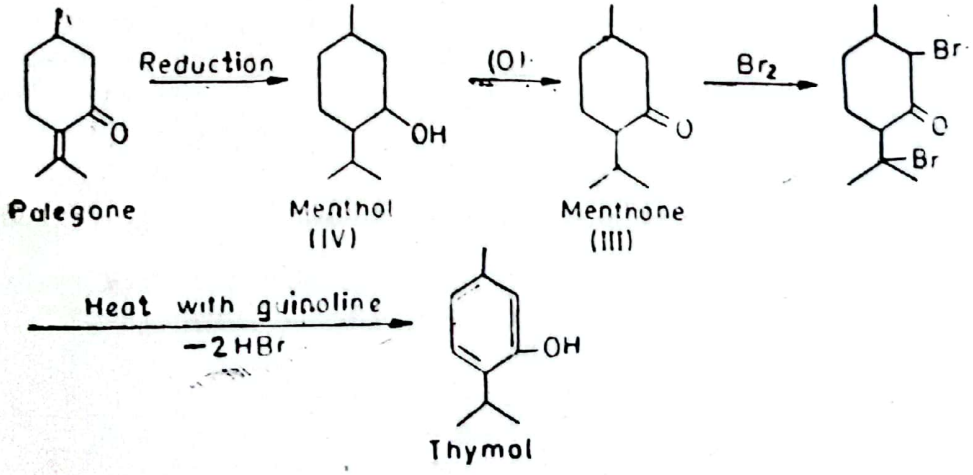
3. On dehydration followed by dehydrogenation, menthol as well as menthone yield *p*-cymene, indicating the presence of *p*-cymene skeleton, (*i.e.*, menthane skeleton) in the two compounds. This has been further supported by the fact that menthol as well as menthone on reduction with hydrogen iodide give *p*-menthane.

4. When menthone is oxidised with $KMnO_4$, it yields a keto acid, $C_{10}H_{18}O_3$; this compound has been shown to possess one keto group and one carboxyl group and is called ketomenthylic acid (II). The latter compound undergoes oxidation readily with permanganate to yield 3-methyladipic acid (III) and some other acids. These oxidation reactions can be explained if structure (I) represents menthone.



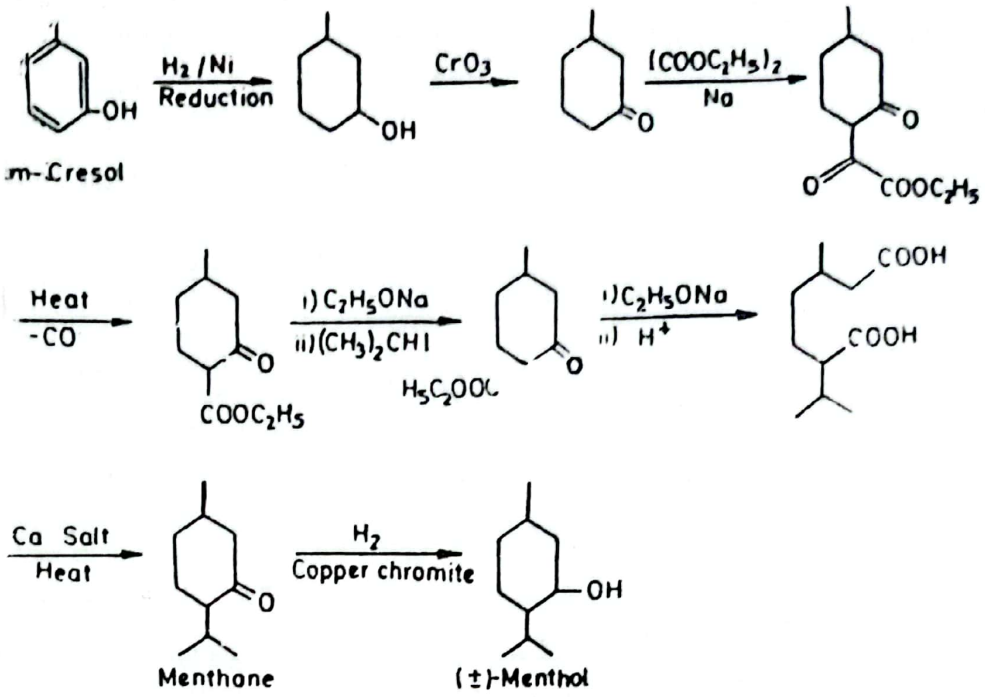
Menthone (I) — Ketomenthylic acid (II) — β-Methyl adipic acid (III)

5. As menthone is the oxidation product of menthol, the structure of menthol should be (IV). This is also supported by the fact that pulegone on reduction yields menthol and since the structure of pulegone is known, it should be same as (IV). Actually, this has been found to be so. Further, the structure (IV) explains all its reactions, e.g., its formation from pulegone and its aromatisation to thymol by using bromine and quinoline.



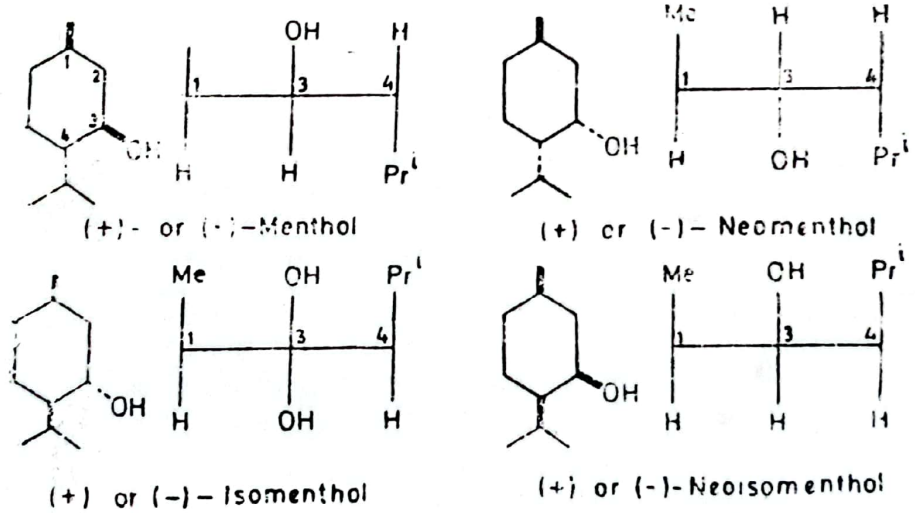
Pulegone — Menthol (IV) — Mentnone (III) — Thymol

6. Finally, the structures of menthone and menthol have been confirmed by their syntheses given by Kotz and Hese from *m*-cresol.



The (±)-menthol has been resolved by employing phthalic acid.

Stereochemistry of Menthol. If we carefully observe the structure of menthol (as given above), it is found to contain three chiral centres (1, 3 and 4) and thus its eight optically active forms (four racemic modifications) are possible theoretically. Actually, all eight enantiomers are known and their configurations are as follows :



In the above enantiomorphs, the horizontal lines are representing the plane of the cyclohexane ring.

In correlation with glyceraldehyde, it is found that (—)-menthol belongs to the L-series and has the absolute configuration given as above. Similarly, other enantiomorphs are also assigned absolute

configurations on the basis of a study of chemical and optical relationships and the Auwers-Skita rule. These configurations have also been confirmed by conformational analysis. Eliel applied the following postulates to the enantiomorphs of menthol :

(*i*) The esterification of an axial group takes place less readily than that with an equatorial group.

(*ii*) The esterification proceeds *via* the conformation of the molecule in which the reactive hydroxyl group occupies equatorial position.

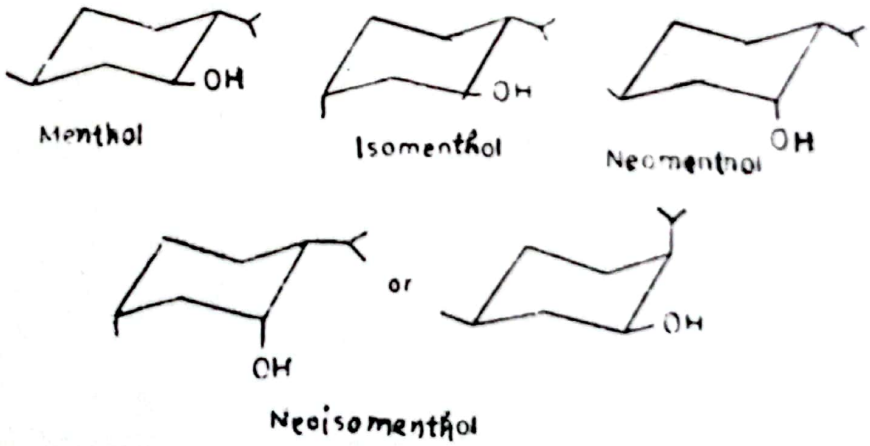
(*iii*) The rate differences may be due to the spending of energy which becomes necessary to place the other substituents, if necessary, into the axial conformation,

On the basis of the above postulates, the order of the rates of esterification would be as follows :

menthol > iso- > neoiso- > neo-.

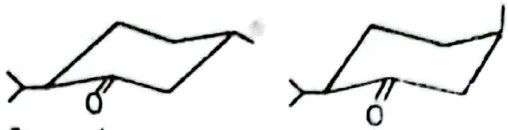
Read *et al.* carried out the esterification of the enantiomorphs of menthol with dinitrobenzoyl chloride and observed that the relative rates of esterification are same as given above, *viz.*, (—)-menthol, 16.5 : (+)-isomenthol, 12.3 ; (+)-neoisomenthol, 3.1 ; (+)-neomenthol, 1.0.

On the basis of the above studies, Eliel assigned the following configurations.



The above configurations have been confirmed by Cole *et al.* (1956) from their infrared spectra and conformation studies.

Stereochemistry of Menthone. In the structure of menthone, there are two dissimilar chiral centres (1, 4). Thus, it must exist in four optically active forms (and two racemic modifications). Actually all are known and they correspond to the menthones and isomenthones which are geometrical isomers : each one is existing as a pair of enantiomers. These configurations have been assigned on the basis of physical evidence that the **cis**-isomer has higher density and refractive index than the **trans**-isomer.



Trans-isomer menthone — Cis-isomer isomenthane

Theoretically it is postulated that menthone (e, e-form) should be more stable than isomenthone (e, a-form). Actually, this has been confirmed by Willhalm *et al.* from the mass spectra of these two compounds.

1.27. PULEGONE

Introduction. This is an optically active compound having b.p. 221 – 222°C. This occurs in pennyroyal oils.

Constitution of Pulegone.

1. From analytical data, the molecular formula of pulegone has been found to be $C_{10}H_{16}O$.

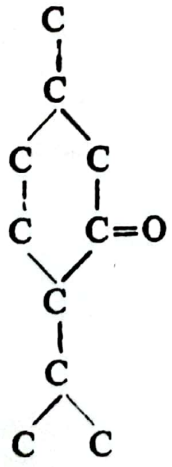
2. It adds on one molecule of bromine to form a dibromide, indicating that it contains a double bond.

3. With hydroxyl amine, it forms oxime, indicating the presence of a >C=O group in it.

4. When catalytically reduced, it is first changed into menthone (a ketone) and then into menthol (a secondary alcohol).

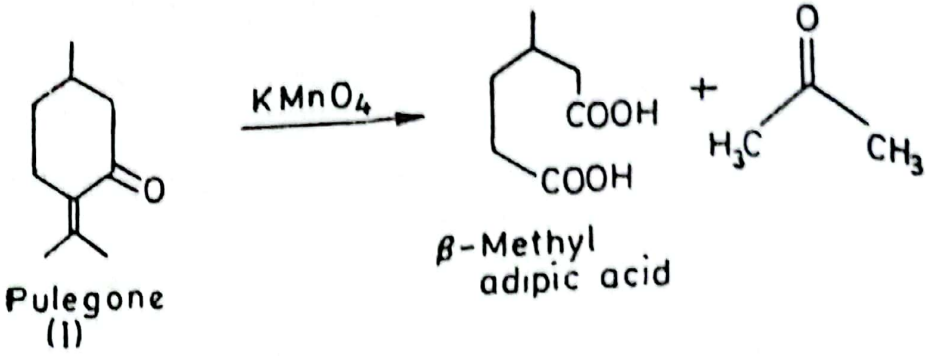
$$\underset{\text{Pulegone}}{C_{10}H_{16}O} \xrightarrow{H_2} \underset{\text{Menthone}}{C_{10}H_{18}O} \longrightarrow \underset{\text{Menthol}}{C_{10}H_{20}O}$$

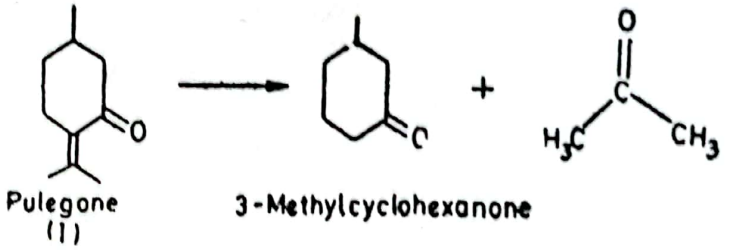
The formation of menthone and menthol reveals that pulegone has the *p*-menthane type carbon skeleton. Further, the position of keto group in pulegone is also same as in menthone,



5. When pulegone is oxidised with potassium permanganate, it yields acetone and β-methyl adipic acid. On the other hand, pulegone, when boiled with aqueous ethanolic potassium hydroxide,

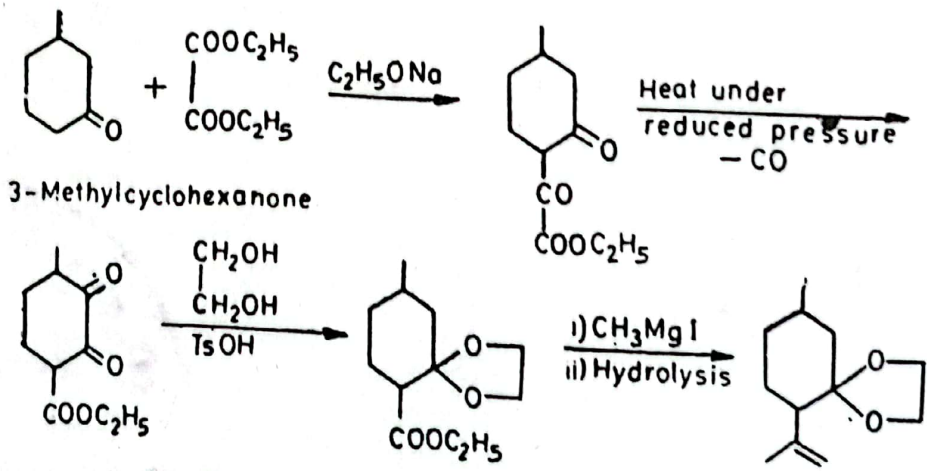
yields acetone and 3-methyl cyclohexanone. The formation of these products could be explained if the pulegone has structure (I).

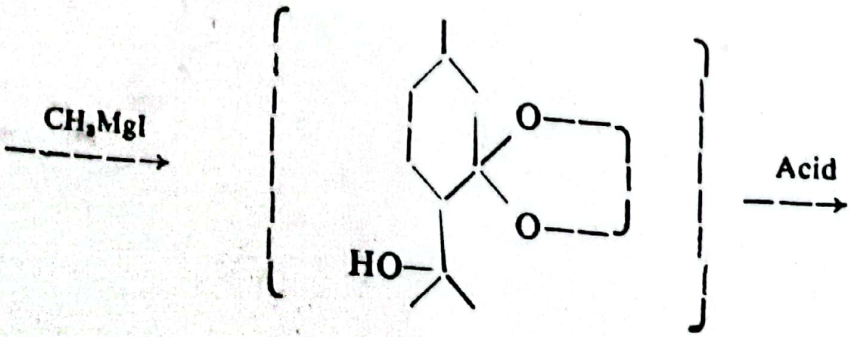




The presence of α, β-unsaturated ketonic group has been confirmed by its UV absorption spectum : **obs**, λ_{max} 252 (ϵ 5130) nm, **calc.**, λ_{max} 215+10+2×12=249 nm.

6. Finally, the structure (I) of pulegone has been confirmed by its synthesis given by Black *et al.* (1956) from 3-methylcyclohexanone.

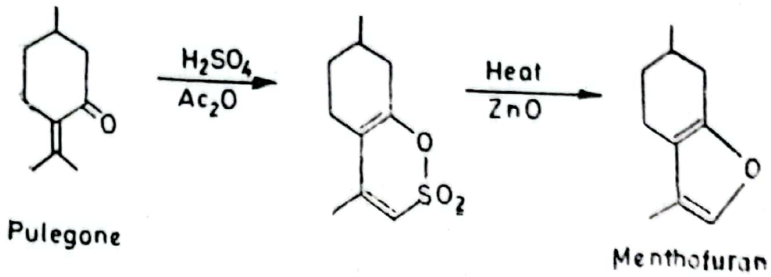




Pulegone + Isopulegone

The above mixture of pulegone and isopulegone, when treated with sodium ethoxide, yields pure pulegone only.

Reactions of Pulegone. The most important reaction of pulegone is its conversion into menthofuran, a constituent of American peppermint oil. Pulegone when treated with H_2SO_4 in acetic anhydride yields cyclopulegenol sulphonic ester. The latter compound on distillation with zinc oxide forms menthofuran.

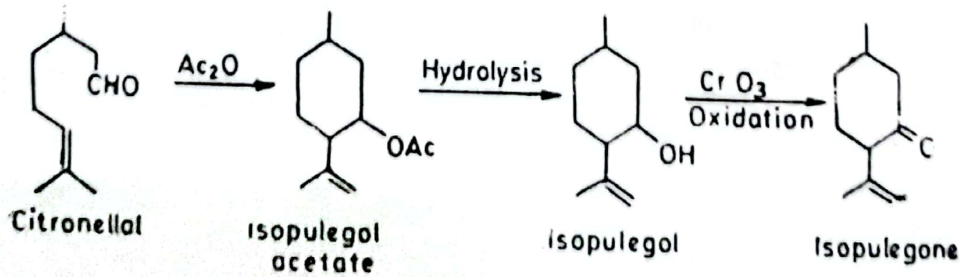


Pulegone Menthofuran

1.28. ISOPULEGONE

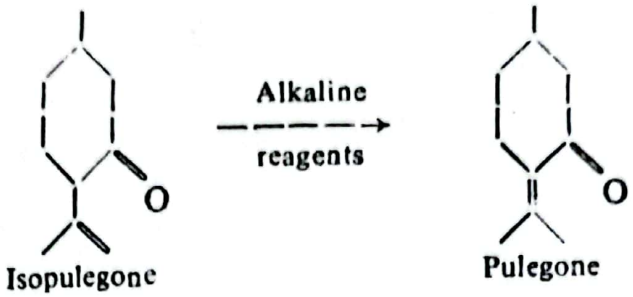
This is not isolated from natural sources. However, it is believed that it exists in equilibrium with the pulegone because the α, β- and β, γ-unsaturated substances always exist in the equilibrium, mainly on the side of the former.

Isopulegone may be synthesised from citronellal. The latter compound on boiling with acetic anhydride undergoes cyclisation to yield isopulegol acetate which upon hydrolysis yields isopulegol. The oxidation of the latter compound with CrO_3 yields isopulegone.



Citronellal Isopulegol acetate Isopulegol Isopulegone

Isopulegone can be isomerised to pulegone by alkaline reagents.



Isopulegone Pulegone

1.29. PIPERITONE

Introduction. It is an unsaturated ketone which occurs naturally in the eucalyptus oils, especially in the broad-leaved peppermint. The (+)-piperitone has been isolated from the essential oil of a Himalayan grass called *Andropogon Jwarancusa*.

Constitution of Piperitone.

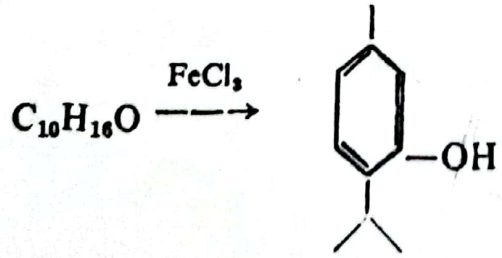
1. *Molecular formula.* From analytical data, the molecular formula of piperitone has been found to be $C_{10}H_{16}O$.

2. *Presence of one double bond.* When it is hydrogenated catalytically, it takes up one mole of hydrogen to yield menthone, indicating that piperitone contains one double bond.

$$C_{10}H_{16}O \xrightarrow{H_2—Ni} C_{10}H_{18}O$$

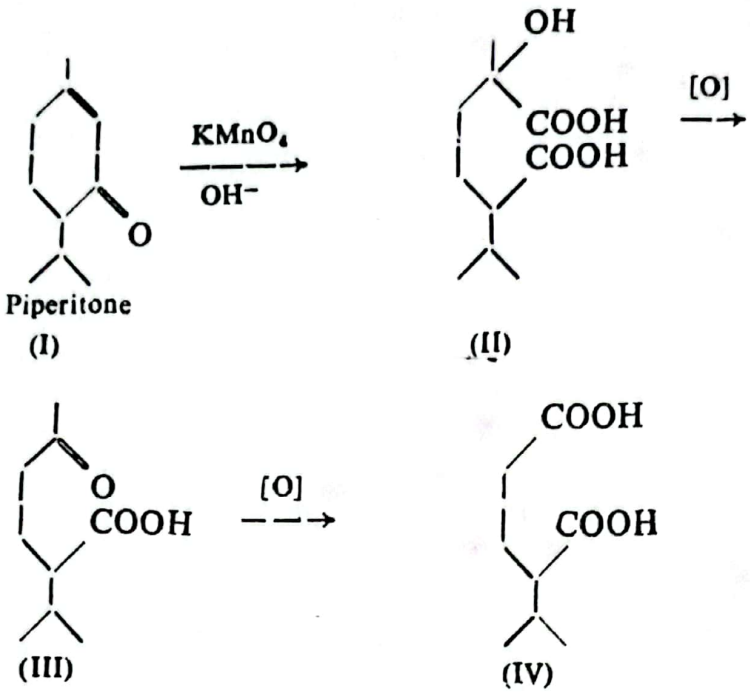
3. *Presence of a keto group.* As piperitone forms oxime with hydroxylamine and yields secondary alcohol on reduction, this shows that piperitone contains one **ketonic group.** Further, UV absorption spectrum of (λ_{max} 235 nm) piperitone reveals the presence of an α, β-unsaturated ketonic group.

4. *Nature of the carbon frame work.* When oxidised with ferric chloride, it gives thymol, a compound of known structure. This reaction shows that piperitone is *p*-menthene-3-one but does not show the position of the double bond.

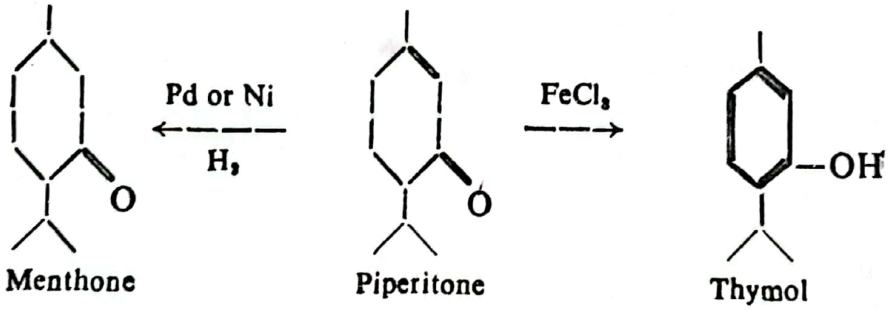


Piperitone Thymol

5. *Position of the double bond.* This has been ascertained by Schimmel (1910) who found that piperitone when oxidised with alkaline potassium permanganate yields 2-hydroxy-2-methyl-5-isopropyladipic acid (II), 4-acetyl-2-isopropylbutyric acid (III) and 2-isopropylglutaric acid (IV). These oxidation results could be explained only if piperitone is *p*-menth-1-en-3-one (I).

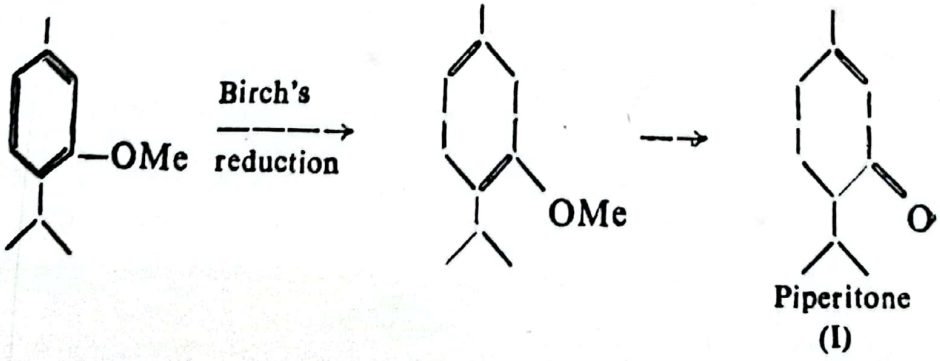


The structure (I) for piperitone also explains the formation of menthone and thymol.

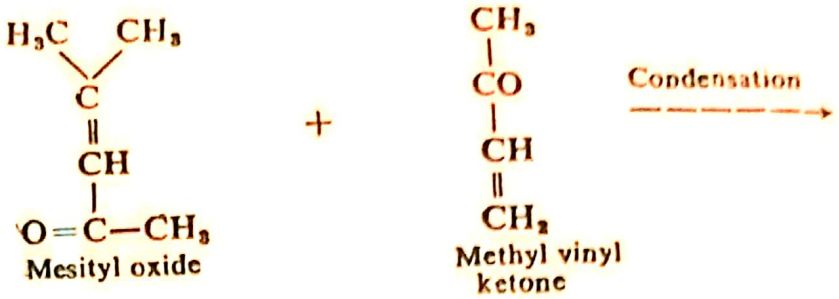


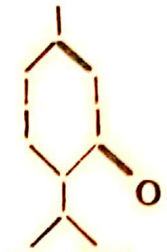
6. *Synthesis.* Finally, the structure (I) for piperitone has been confirmed by its syntheses as follows :

(*a*) **Birch's synthesis.** This synthesis involves the Birch's reduction.



(*b*) **Bergmann's synthesis.** According to this synthesis, piperitone is formed directly by the condensation of mesityl oxide and methyl vinyl ketone.





Piperitone
(I)

7. *UV spectrum.* The structure (I) for piperitone has been found to be in agreement with the observed UV absorption spectrum ; **observed** value : λ_{max}=235 nm at ε 15000 ; calculated value : λ_{max}=215+2×12=239 nm.

1.30. CARVONE

Introduction. This has been found to occur in the racemic and optically active forms in various essential oils, e.g., spearmint and caraway oil. The latter oil contains a large quantity of *d*-carvone.

Constitution of carvone. This has been elucidated on the basis of analytical data and synthetic evidences as follows :

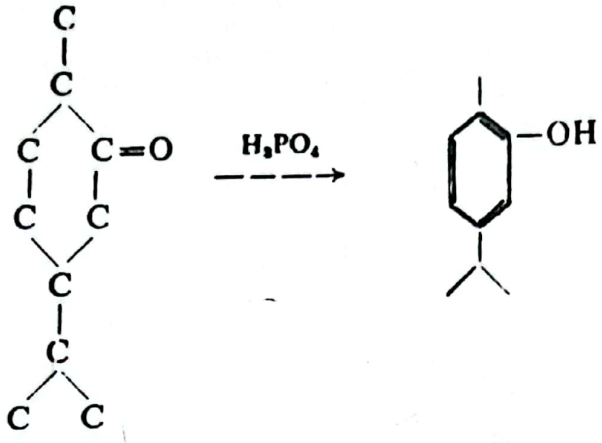
(1) The **molecular formula** of carvone has been found to be $C_{10}H_{14}O$.

(2) As carvone adds on four bromine atoms to yield a tetra-bromo derivative, this indicates that it contains **two double bonds.**

(3) Carvone gives usual reactions of a ketone, *i.e.*, it forms oxime, hydrazone, etc.

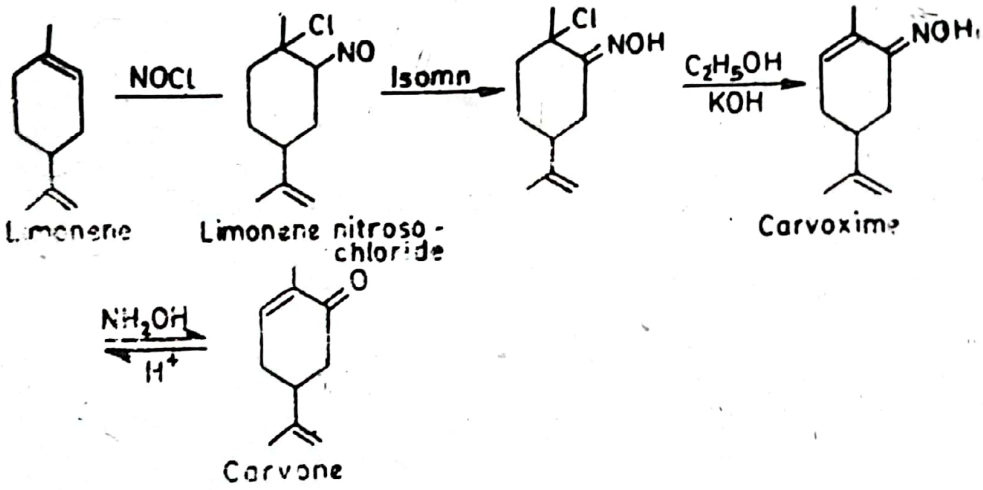
(4) The presence of two double bonds and a ketonic group in carvone implies that the molecular formula of its parent saturated hydrocarbon is $C_{10}H_{20}$. As the latter formula corresponds to the general formula C_nH_{2n} for monocyclic compounds, it means that carvone must be **monocyclic.**

(5) When carvone is heated with phosphoric acid, it yields *carvacrol.* The formation of this product reveals that carvone probably contains the **p-menthane**, *i.e.*, **p-cymene structure** in which the keto group is present in the ring in the *ortho*-position with respect to the methyl group.



Carvone skeleton Carvacrol

(6) Now the next problem is to ascertain the position of two double bonds in the *p*-menthane structure which is having a keto group at C-2. This is ascertained by its close relationship to limonene. For example, limonene nitrosochloride when treated with ethanolic alkali yields the oxime which is also obtained by the reaction of carvone with hydroxyl amine.



Limonene Limonene nitroso-chloride Carvoxime

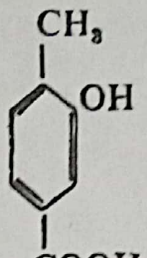
Carvone

As the same oxime is obtained from limonene and carvone, this reveals that, like limonene in carvone, one of the double bonds must be present between C_3 and C_4 whereas the other double bond between C_1 and C_6 because the ketonic group is already present at C_2.

(7) *Evidence for the presence of a double bond in the 8-position.* This has been obtained from degradative reactions of carvone carried out by Tiemann and Semmler.

(I) C_{10} (II) C_{10} (III) C_{10}

$$\xrightarrow[CH_3COOH]{CrO_3} \underset{(IV)\ C_9}{\text{Ketonic alcohol}} \xrightarrow{NaOBr} \underset{(V)\ C_8}{\text{Hydroxy acid}} \xrightarrow[190^\circ C]{Br_2/H_2O}$$



(VI)

The above set of reactions yields the following results :

(*a*) As the reduction of carvone (I) is done with sodium and ethanol, it yields **dihydrocarveol** $C_{10}H_{18}O$ (II) which is a secondary alcohol and contains one double bond. This means that **one** of the double bonds and the keto group of carvone have been reduced.

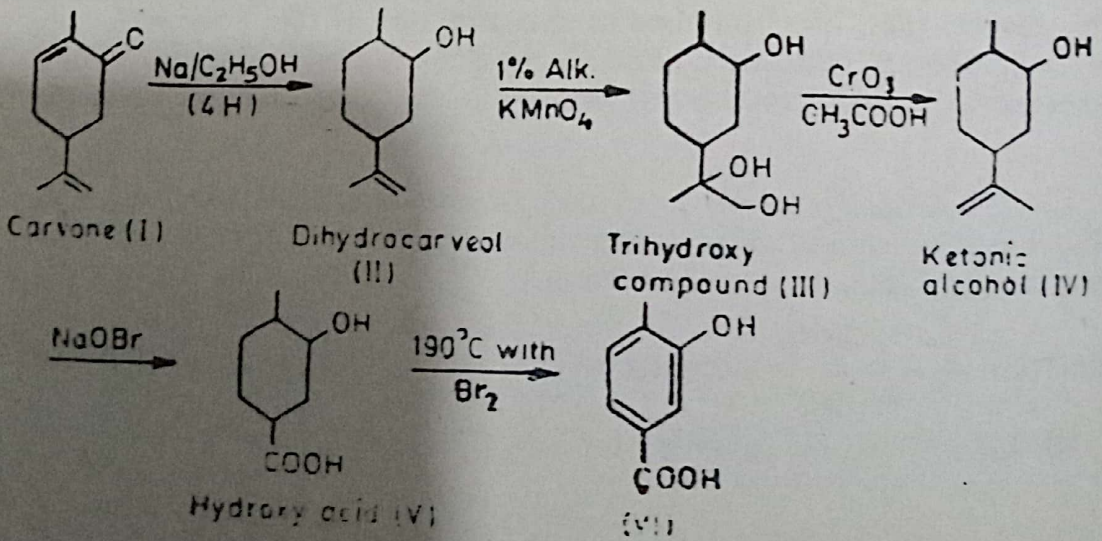
(*b*) The dihydrocarveol (II), when oxidised with 1 per cent alkaline $KMnO_4$, yields the trihydroxy compound, $C_{10}H_{20}O_3$ (III).

(*c*) In the trihydroxy compound (III), when oxidised with chromic acid, there occurs the scission of the glycol bond to yield a compound $C_9H_{16}O_2$ (IV). The latter compound has been shown to contain a keto group and a hydroxyl (alcoholic) group.

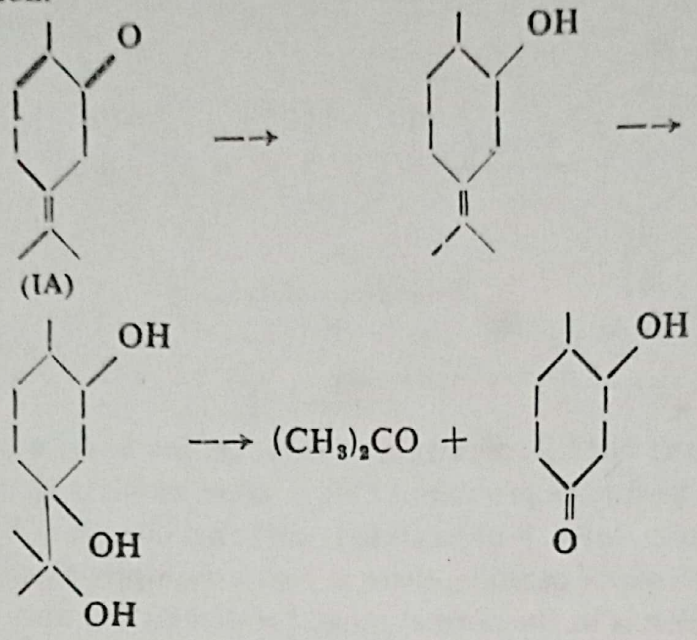
(*d*) The ketonic alcohol (IV), when oxidised with sodium hypobromite, loses one carbon atom to produce the compound $C_8H_{14}O_3$ (V). The latter compound has been shown to be a hydroxymonocarboxylic acid, and since one carbon is lost in its formation, its precursor (IV) must therefore be a ketone.

(*e*) Finally, dehydrogenation of (V), when heated with bromine water at 190°C under pressure, yields *m*-hydroxy-*p*-toluic acid (VI) (a known compound).

In order to explain the formation of the above products, Tiemann and Semmler assumed that one double bond in carvone is present in the 8-position. Thus, the structure (I) for carvone explains its degradation reactions as follows :



Had the double bond been present in the 4-position as (IA), then compound (IV) and hence (VI) would not have been obtained because there would have been loss of three carbon atoms during the oxidation.

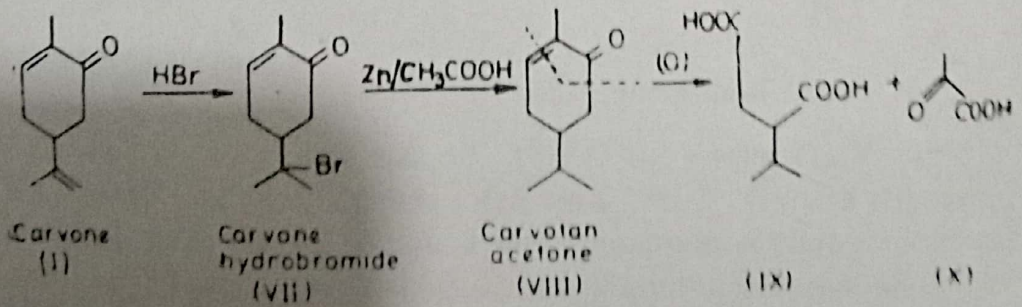


If we examine the structures (I) and (IA), we observe that structure (I) contains a chiral centre whereas structure (IA) does not have a chiral centre because it is symmetric and so cannot exhibit optical activity. But carvone is known in optically active forms. Therefore, structure (I) is correct whereas structure (IA) must be rejected on these grounds.

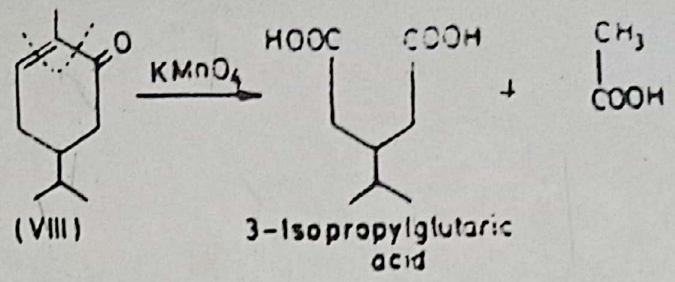
8. *Evidence for the double bond in 6-position.* The position of the other double bond in carvone at 6-position has been confirmed by the following reactions :

Carvone adds on one mole of hydrogen bromide to yield carvone hydrobromide $C_{10}H_{15}OBr$ (VII). The latter, when treated with zinc dust and methanol, yields carvotanacetone, $C_{10}H_{16}O$ (VIII) by replacement of the bromine atom (VII) by hydrogen (VIII). The compound carvotanacetone, when oxidised with permanganate, yields isopropylsuccinic acid (IX) and pyruvic acid (X).

All the above reactions could be explained if the ring contains the double bond in 6-position. Had the double bond present in the 7-position, formic acid and not pyruvic acid would have been obtained.



Further support for the 6-position has been obtained from the work of Simonsen *et al.* (1922) who got 3-isopropylglutaric acid and acetic acid when carvotanacetone was oxidised with permanganate.

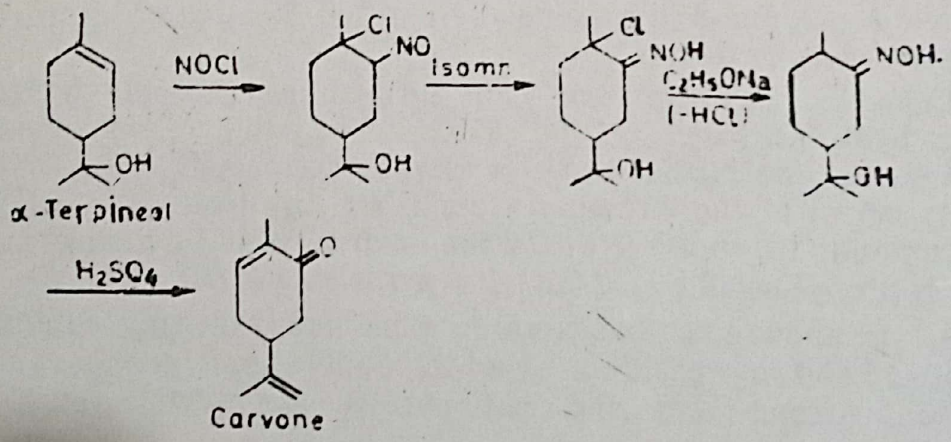


(VIII) 3-Isopropylglutaric acid

9. In the N.M.R. spectrum of carvone, there is a multiplet signal at τ 3.25 for the protion at C-6, a value which is characteristic of a β-proton in α, β-unsaturated carbonyl compounds. In the N.M.R. spectrum of carvone, there is also a multiplet signal for C-8 at τ 4.78, which is in the normal range for olefinic protons.

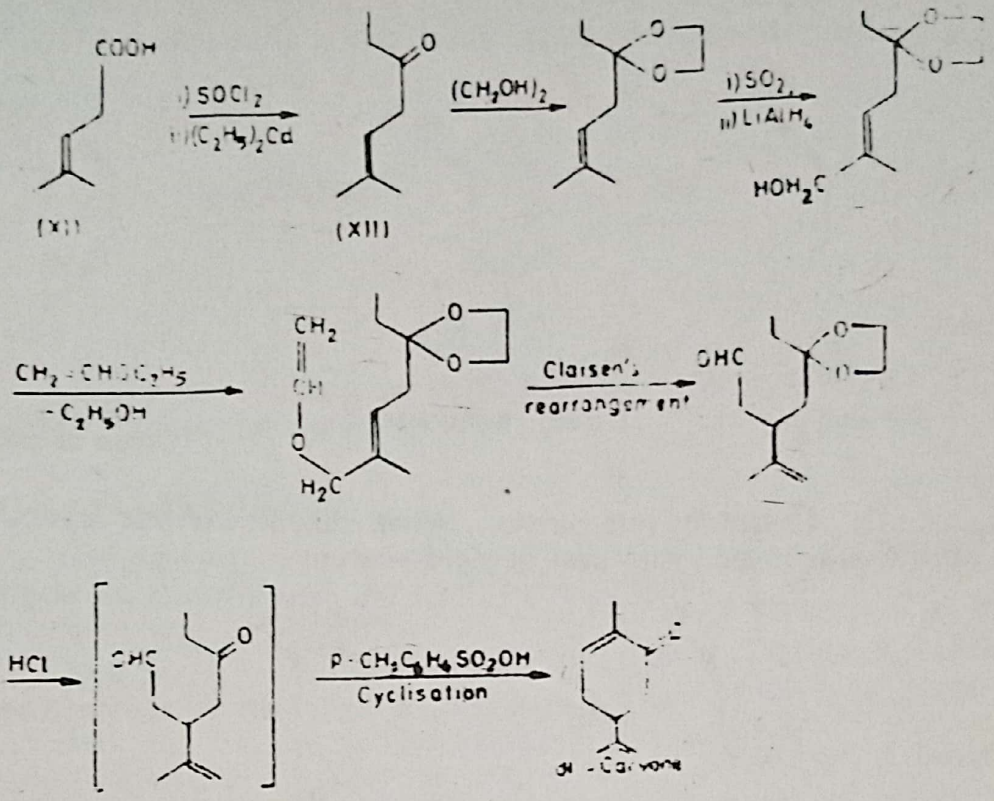
10. *Synthesis.* Finally, the structure of carvone has been confirmed by its various syntheses :

(*a*) From α-terpineol.



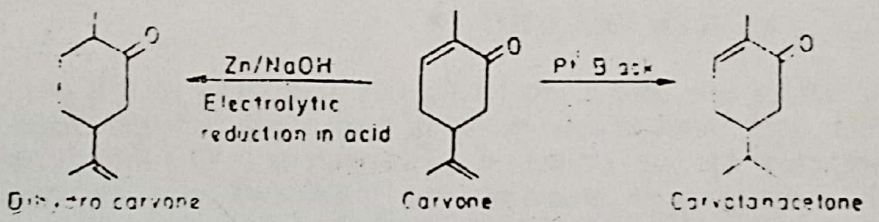
(*b*) From limonene. See step (6).

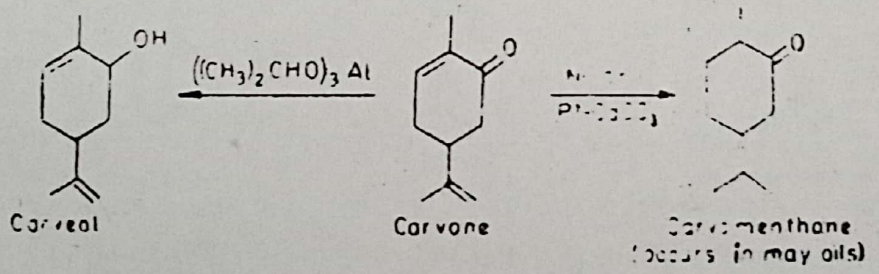
(*c*) Vig's synthesis (1966). In this synthesis, dl-carvone is prepared directly from 2-methyl-5-keto-Δ^2-heptene (XII). The latter compound is prepared by the action of thionyl chloride followed by diethyl cadmium on 2-methyl-Δ^2-pentenoic acid (XI).



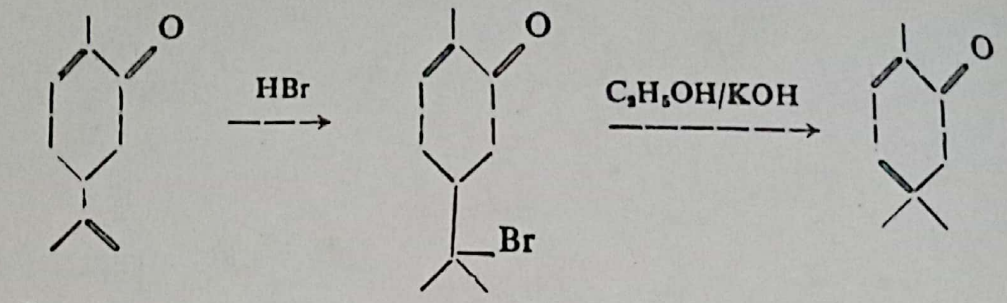
Important Reactions of Carvone.

1. *Reduction of carvone.* Under different conditions, carvone undergoes reduction to yield different products.



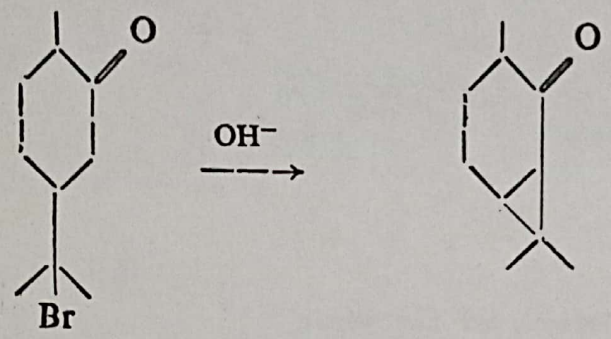


2. *Conversion into eucarvone.* When carvone hydrobromide is treated with ethanolic alkali, eucarvone is obtained.



Carvone — Carvone hydrobromide — Eucarvone (Occurs naturally)

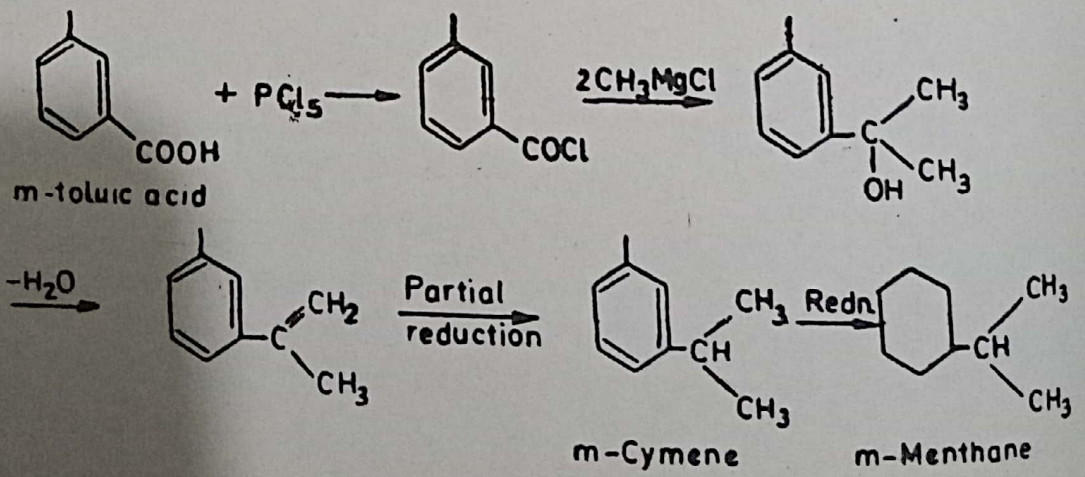
3. *Conversion into carone.* When dihydrocarvone hydrobromide is treated with base, it yields carone.



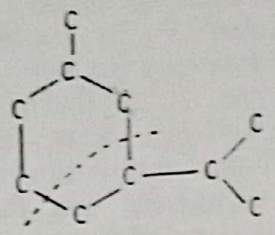
Dihydrocarvone hydrobromide — Carone

1.31. *m*-MENTHANE GROUP

There are only a few terpenoids belonging to this group. The parent compound of this group of terpenoids is *m-menthane* which is prepared by the reduction of *m*-cymene. The latter compound is the constituent of many essential oils and may be synthesised from m-toluic acid according to the following steps :



Although the structure of *m*-menthane is composed of two isoprene units, yet these are not joined head to tail. Same has been found in all the terpenoids belonging to this group.

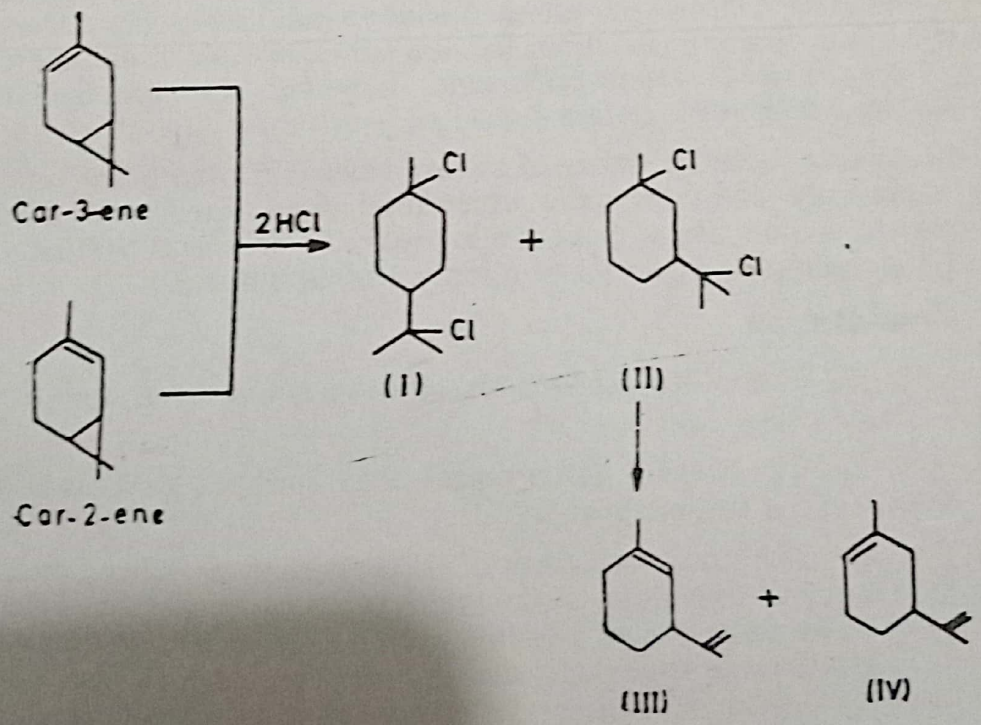


In the next article, we shall describe a typical example, sylvestrene of *m*-menthane group.

1.32. SYLVESTRENE

Sylvestrene appears to be the only monocyclic monoterpenoid which does not have the *p*-cymene structure but have the *m*-cymene carbon skeleton.

Sylvestrene exists in (+), (−) and (±)-forms ; the racemic modification is also called *carvestrene*. The (+)-form of sylvestrene was first obtained from Sweedish pine needle oil. The occurrence of sylvestrene in pine needle oil as such is doubtful. When sylvestrene is extracted by heating pine oil with hydrogen chloride, this yields a mixture of dipentene dihydrochloride (I) and sylvestrene dihydrochloride (II). Simonsen *et. al.* (1923, 1925) showed that these compounds, (I) and (II), were produced by the action of hydrogen chloride on car-3-ene. Thus, these workers came to the



conclusion that the terpene originally present in pine oil is car-3-ene but not sylvestrene. Sylvestrene may be prepared from its dihydrochloride by heating the latter with aniline ; the removal of hydrogen chloride from the ring would result two possible positions for the ring double bond. From analytical data, it was concluded that the isopropenyl constitutes the side chain and sylvestrene is a mixture of the two forms (III) and (IV). It was further shown that pine oil also contains car-2-ene. Thus, pine oil contains a mixture of car-3-ene and car-2-ene which are readily converted into sylvestrene and hence it is apparent that the precursor of sylvestrene (itself a mixture) is a mixture of the two carenes.

Perkin, Junior, *et.al.* (1913) have synthesised the enantiomers of sylvestrene. They have also revealed that an equimolecular mixture of the dihydrochloride of (+)-and (—)-sylvestrene is identical with carvestrene dihydrochloride.

1.33. METHYLATED CYCLOHEXANE

This group contains a large number of terpenoids. The parent compounds of these terpenoids are 1, 1, 3-trimethylcyclohexane and 1, 1, 3, 6, tetramethyl cyclohexane. Examples of this group of terpenoids are cyclogeraniol, ionones and irones. All these will be discussed in the following articles.

1.34. IONONES

Three isomeric ionones, *viz.*, α, β, and γ, are known. Out of these, the α- and β- ionones occur in the essential oils of *Boronia megastigma* and in costus oil whereas γ-ionone has been isolated from the mixture of ionones by vapour phase chromatography. In the nature, β-ionone predominates but commercially both β- and α-ionones have been synthesised. β-Ionone has wood-like odour and its dilute ethanolic solution has the odour of violets and therefore, it is extensively used in perfumery. However, β-ionone is also used in the synthesis of vitamin A.

Due to the presence of an asymmetric carbon atom, α-ionone is optically active. On the other hand, due to the lack of an asymmetric carbon atom, β-ionone is optically inactive. Actually, the (+)-, (−)-and (±)-forms of α-ionone occur naturally.

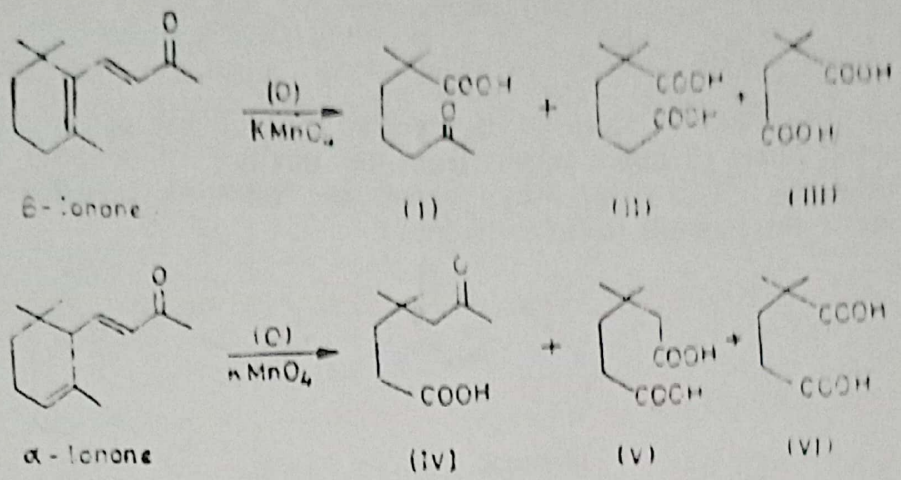
Constitution

(1) The molecular formula of ionones has been found to be $C_{13}H_{20}O$.

(2) Their usual tests reveal that ionones have two double bonds and a ketonic group.

(3) The exact structures of these ionones have been established by a study of oxidation products produced by potassium permanganate. The oxidation of β-ionone yields a mixture of geronic acid (I), 2, 2-dimethyladipic acid (II) and 2, 2-dimethylsuccinic acid (III). On the other hand, α-ionone yields a mixture of isogeronic acid (IV), 3, 3-dimethyladipic acid (V) and 2, 2-dimethylglutaric acid (VI), the

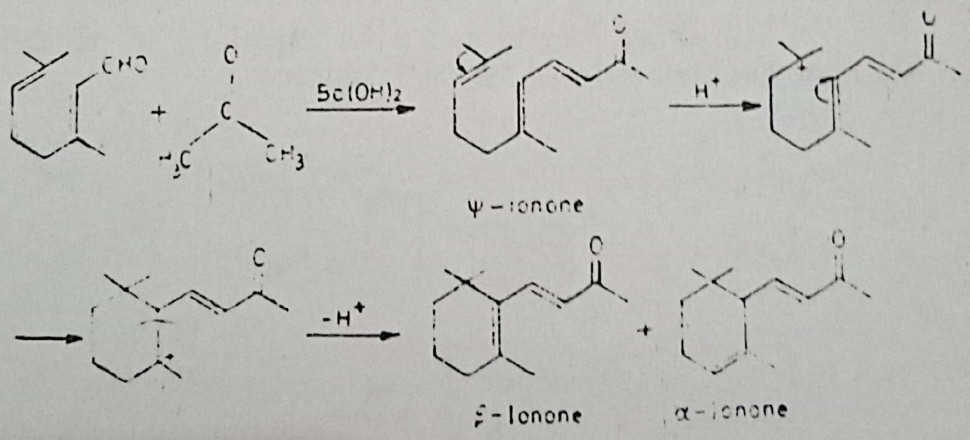
formation of these products could be explained if the following structures have been assigned to these ionones.



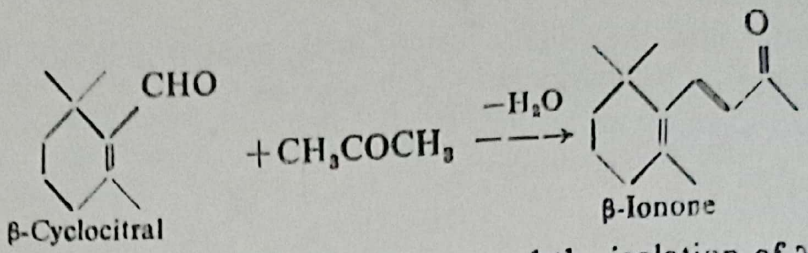
(4) The structures of these two ionones have been confirmed by the positions of the maxima of their ultraviolet spectra. These observed values are : α—, 228.5 (ε 14,000) *nm* and β—, 296 (ε 11000) *nm*. These values are in agreement with calculated values, *i.e.*, the calculated value for the α-isomer is 215+12=227 *nm* whereas that for the β-isomer (with extended conjugation) is

$$215+30+18+2\times18 \text{ or } 299 \text{ } nm.$$

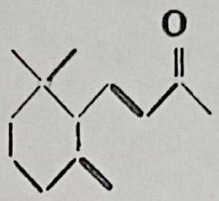
(5) Finally, the structures of α-and β-ionones have been confirmed by their synthesis from citral. The latter compound when condensed with acetone in the presence of barium hydroxide yields ψ-ionone which on heating with dilute sulphuric acid in the presence of glycerol yields a mixture of α-and β-ionones. The proportion of α to β depends upon the nature of the reagent used for cyclising the ψ-ionone. For example, with sulphuric acid, β-ionone is the main product whereas with phosphoric acid, α-ionone is the main product.



The α-and β-ionones may also be prepared by the condensation of respective α-and β-cyclocitral with acetone.



In 1962, Theimer *et al.* reported the isolation of γ-ionone by vapour phase chromatography from the mixture of ionones obtained above. This group also assigned the following structure (this ionone corresponds to the γ−irone) :



1.35. IRONES

Similar to ionones, there are three irones, *viz.*, α-, β-and γ-. Out of these, only γ-irone is the natural form and occurs in the oil obtained from the **orris root**. Like ionones, irones also possess odour of violet and therefore used in perfumery.

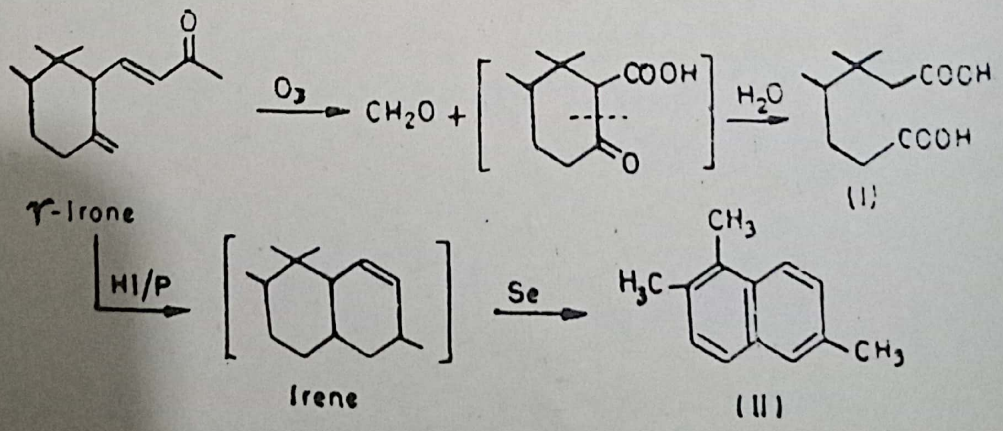
Constitution. As only the γ-isomer is the most important, it is alone described here. The constitution of this was established by Ruzika *et al.* (1947) on the basis of following facts :

(1) The molecular formula of γ-irone has been found to be $C_{14}H_{22}O$.

(2) The usual reactions reveal that γ-irone contains **two double bonds** and a **ketonic group**.

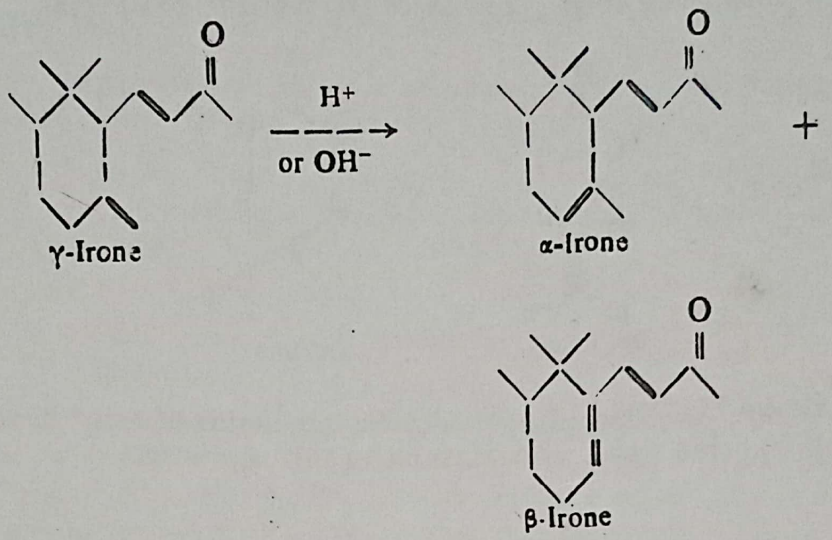
(3) When γ-irone is ozonolysed, it yields a mixture of formaldehyde and 3, 3, 4-trimethylpimelic acid (I). Also γ-irone, when reduced with hydriodic acid and red phosphorus, followed by dehydrogenation with selenium, yields 1, 2, 6−trimethylnaphthalene (II).

All the above reactions can be explained if the following structure has been assigned for the γ-isomer :

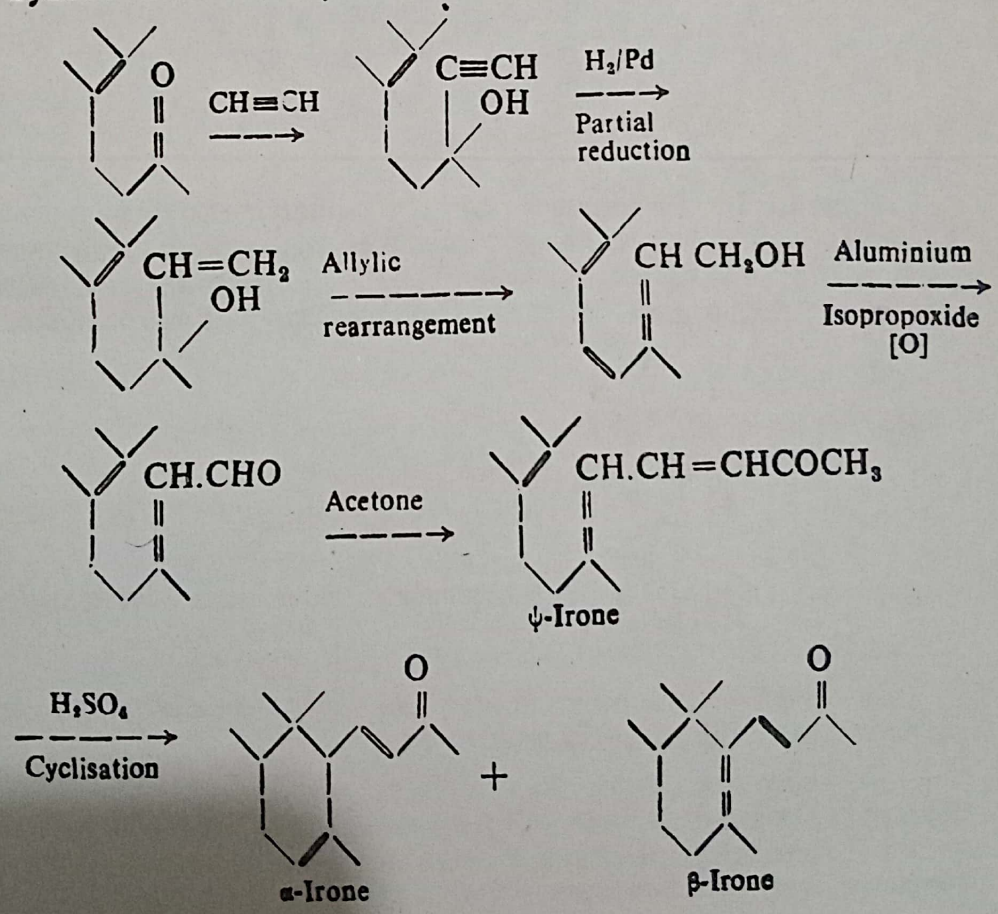


(4) The above strueture of γ-irone has been confirmed by its UV spectrum. The observed value of λ_{max} is 226.5 *nm* which is in close agreement with the calculated value of 227 *nm*.

(5) Finally the structure of γ-irone has been established by its synthesis. γ-Irone, when treated with acid or alkali, isomerises to a mixture of α-and β-irones which have been synthesised by Ruzicka (1947) and the structure of γ-irone has been established indirectly.



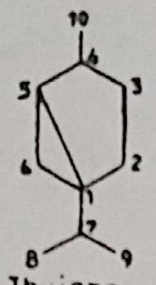
Synthesis of α- and β-irones.



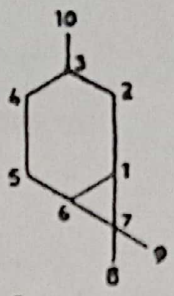
1.36. BICYCLIC MONOTERPENOIDS

In bicyclic monoterpenoids, the size of the **first ring** is six-membered and is same in all these terpenoids but the size of **second** ring varies among these terpenoids. On the basis of the size of the **second ring**, the bicyclic monoterpenoids may be divided into three classes.

Class I. This includes bicyclic terpenoids of the type having 6- + 3- membered rings. Examples are thujane, and carane.

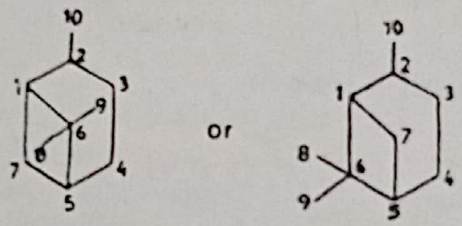


Thujane



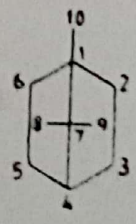
Carane

Class II. This includes bicyclic terpenoids of type having 6- + 4- membered rings. An example of this is pinane.

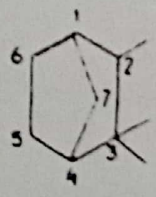


Pinane

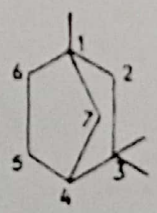
Class III. This includes bicyclic monoterpenoids of the type having 6- + 5- membered ring. Examples are bornane (camphane), norbornane derivative (isocamphane), norbornane derivative (fenchane), norbornane derivative (isobornylane) and norbornane.



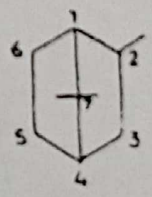
Bornane (Camphane)



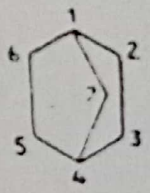
Norbornane derivative (Isocamphane)



Norbornane derivative (Fenchane)



Norbornane derivative (Isobornylane)



Norbornane

In bicyclic monoterpenoids, two rings are not lying in one plane but are lying perpendicular to each other.

No nenclature. According to IUPAC system of nomenclature, the names **thujane**, **carane** and **pinane** are retained whereas the names of remaining are changed and considered to be derivatives of **bornane** and **norbornane**. For example, camphane is called

bornane, isocamphane is 2, 3, 3- trimethylnorbornane, fenchane is 1, 3, 3- trimethylnorbornane and isobornylane is 2, 7, 7- trimethylnorbornane.

1.37. CAR-3-ENE

Introduction. This occurs in sweedish pine needle oil. This is a liquid having b.p. 170°C. This occurs in both *d*- and *l*-isomers.

Constitution. This has been elucidated on the basis of following analytical and synthetic evidences :

1. *Molecular formula.* From analytical data, the molecular formula of carene has been found to be $C_{10}H_{16}$.

2. *Presence of one double bond.* When carene is catalytically reduced, it takes up one mole of hydrogen, indicating that it (carene) contains one double bond.

$$\underset{\text{Carene}}{C_{10}H_{16}} + H_2 \xrightarrow{\text{Catalyst}} C_{10}H_{18}$$

3. *Presence of a six membered ring system.* When carene is treated with hydrogen chloride, it yields a mixture of sylvestrene dihydrochloride and dipentene dihydrochloride. The formation of these products reveals that one of the rings is six membered in the centre.

4. *As a bicyclic compound.* As the molecular formula $C_{10}H_{18}$ of the parent saturated hydrocarbon corresponding to carene belongs to the series $C_n H_{2n-2}$ of bicyclic compounds, this shows that carene must be a bicyclic compound.

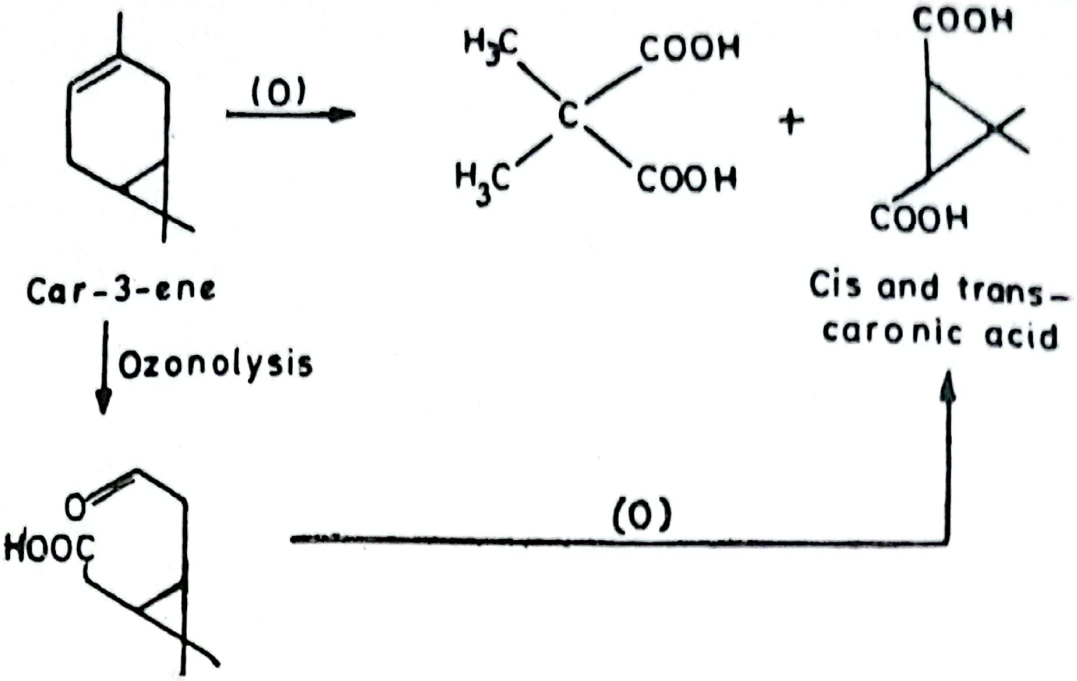
5. *Presence of cyclopropane ring.* In step (3), it has been proved that one of the rings is six-membered. In order to ascertain the nature of the other ring, oxidative degradation of carene is carried out.

When carene is oxidised with potassium permanganate, it yields dimethyl malonic acid, and *cis* and *trans*-forms of caronic acids.

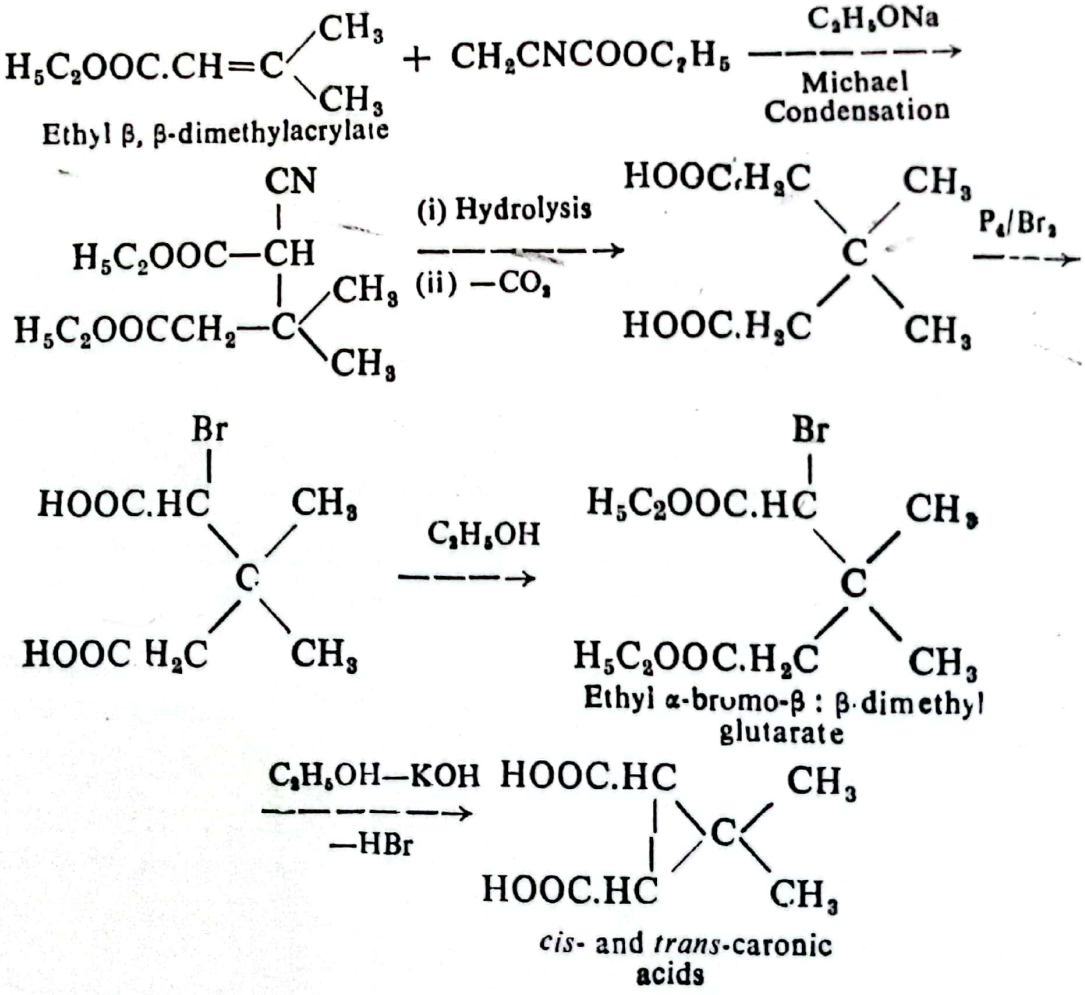
The ozonolysis of carene yields the keto acid which is finally oxidised to caronic acid. This reaction shows that the position of of double bond is C_3.

All the above reactions can be explained on the basis of structure (I) for car-3-ene.

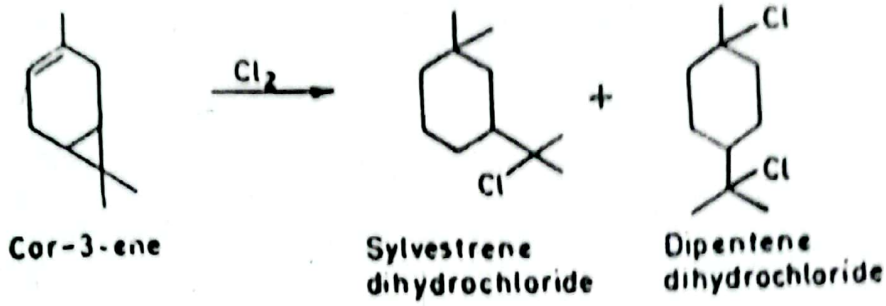
CamScanner



6. *Synthesis.* Finally the above structure for caronic acid, has been confirmed by its synthesis given by Perkin and Thorpe (1899) from ethyl β, β-dimethyl acrylate.



Reaction of car-3-ene. On the basis of the structure (I) for car-3-ene, the foregoing reaction may be represented as follows :



1.38. (+)-CAR-2-ENE

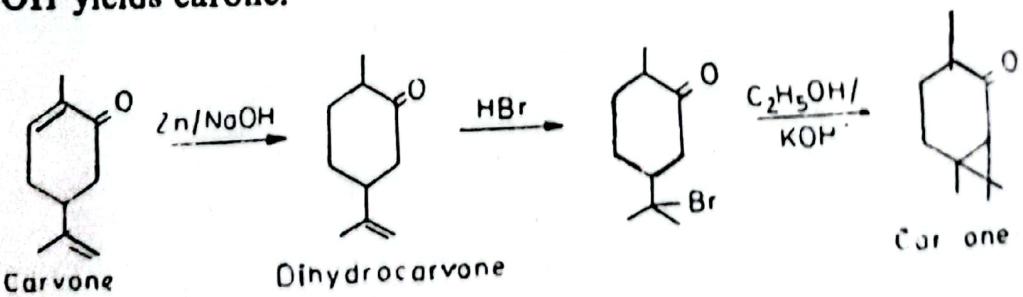
Introduction. It is an isomer of car-3-ene. Its boiling point is 165.5-167° at 707 mm pressure. Its occurs in various essential oils.

Constitution. It is established on the same lines as given in car-3-ene. With HCl, car-2-ene forms sylvestrene dihydrochloride. On the other hand, car-3-ene with HCl forms a mixture of sylvestrene dihydrochloride and dipentene dihydrochloride.

In the NMR spectrum of car-2-ene, different signals, *i.e.*, τ 9.23 and τ 8.97 are observed for the two gem-dimethyl groups of the bridge. The reason for the different signals is that one of the methyl groups (τ 9 23) is closer to the double bond and is therefore shielded with respect to the other methyl group.

1.39. CARONE

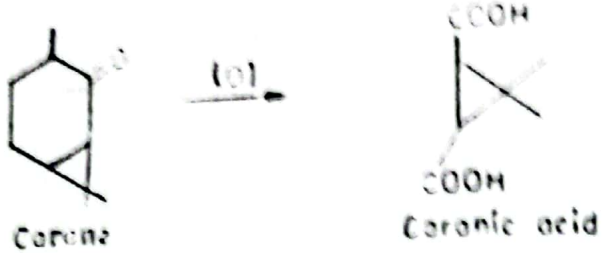
Introduction. Carone is a bicyclic compound which does not occur naturally. It is a synthetic compound. It is of importance because of its relationship with carone. Baeyer *et al.* (1894) synthesised carone for the first time from carvone, The latter compound on treatment with zinc and sodium hydroxide yields dihydrocarvone which on treatment with HBr followed by alcholic KOH yields carone.



Carone is a liquid having b.p. 99-100°C at 15 mm pressure.

Constitution. This has been established by Baeyer *et al.* (1896) similarly as that of carenes who got caronic acid on oxidation of carone with permanganate. Baeyer concluded that caronic acid was a cyclopropane derivative.

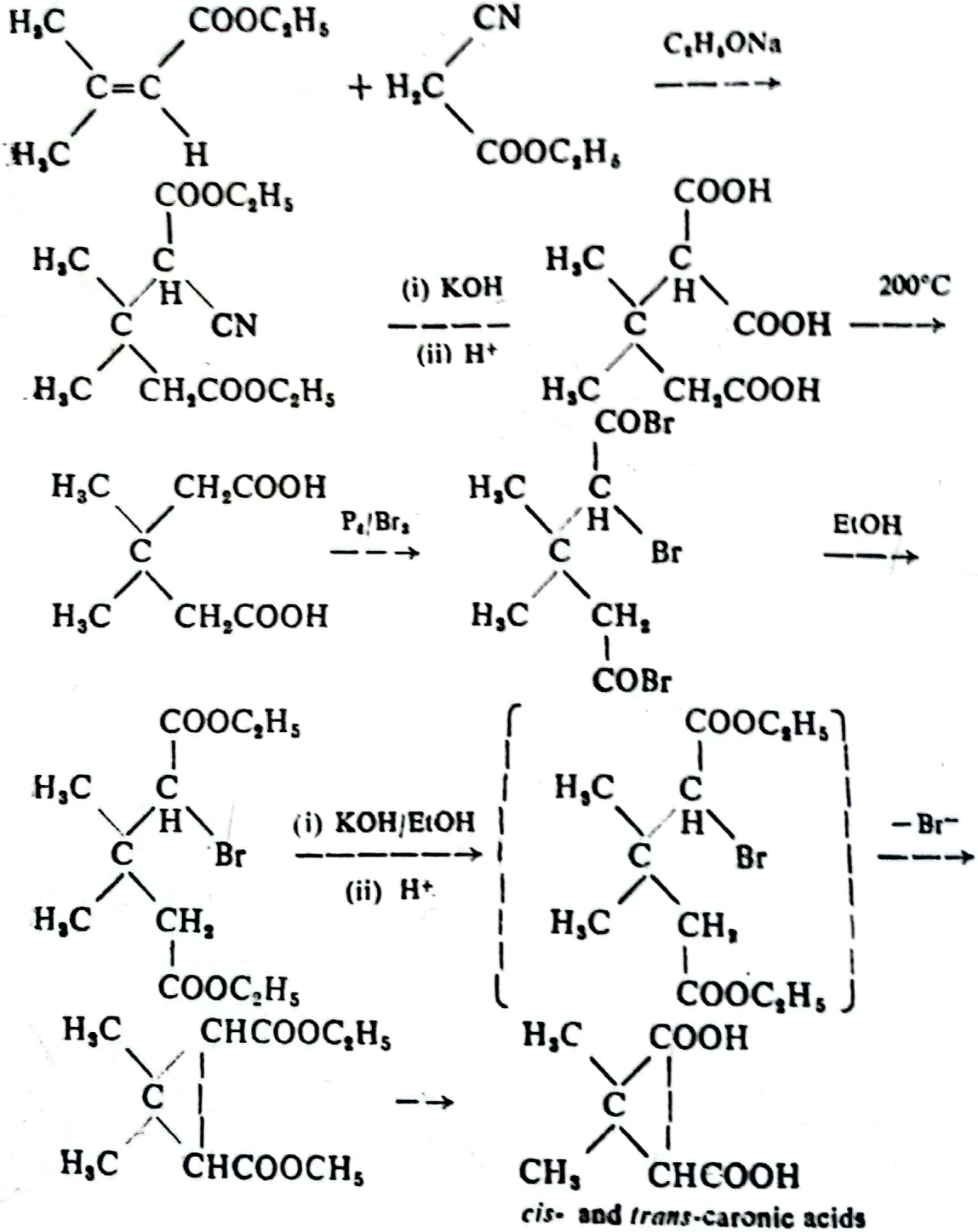
CamScanner



Carone Caronic acid

The above structure of carone has been further confirmed by the fact that its UV absorption spectrum is similar to that of α, β-unsaturated ketone, its λ_{max} is in the region 210—220 mm.

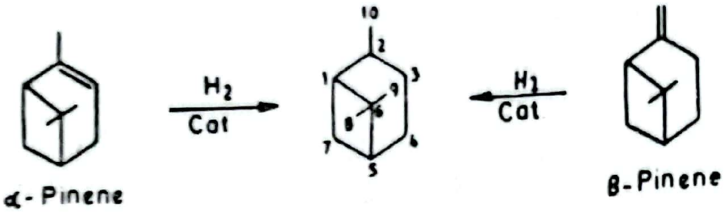
Finally the structure of carone given as above has been further confirmed by its synthesis given by Perkin Jr and Thorpe, (1899) from ethyl β, β-dimethylacrylate and ethyl cyanoacetate by using the Michael condensation.



cis- and *trans*-caronic acids

1.40. THE PINANE GROUP

The parent compound of this group is pinane which does not occur free naturally but is obtained by the catalytic hydrogenation (nickel or platinum) of either α- or β-pinene.



Pinane exists in two geometrical isomeric forms, *cis* and *trans*, and each of these exists as a pair of enantiomers.

Pinane is a bicyclic, saturated hydrocarbon of formula $C_{10}H_{18}$ having six and four membered rings.

1.41. α-PINENE

Introduction. It is the most important member of pinane class which is the chief constituent of the oil of turpentine. It occurs in both the (+)- and (−)- forms in all turpentine oils.

The inactive or (±)- isomer is obtained from turpentine oil which is the volatile fraction obtained by steam distillation of oleoresin produced by pine trees. The inactive α-pinene is separated by repeated fractional distillation of turpentine oil. Then, the purification of α-pinene so obtained is done by formation of its crystalline nitrosyl chloride derivative with nitrosyl chloride. This addition product is then decomposed by aniline to yield pure α-pinene.

α-Pinene is a liquid which boils at 156°C. When α-pinene is exposed to air, it undergoes autoxidation.

Constitution. For sake of convenience, the analytical evidence for the constitution of α-pinene has been divided into two sections. However, each section leads independently to the structure and if the two sections are taken together, this provides very powerful evidence for the structure assigned.

Method I

(1) **Molecular formula.** From analytical data and molecular weight determination, it follows that the molecular formula of α-pinene has been found to be $C_{10}H_{16}$.

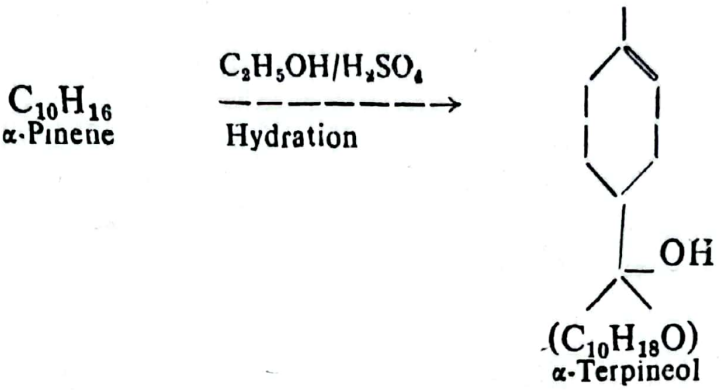
(2) **Presence of a double bond.** As α-pinene adds on two bromine atoms, this means that α-pinene contains one double bond. This is further revealed by the fact that it forms addition crystalline derivative with nitrosyl chloride.

$$C_{10}H_{16}NOCl \xleftarrow{NOCl} C_{10}H_{16} \xrightarrow{Br_2} C_{10}H_{16}Br_2$$

α-Pinene nitrosochloride — α-Pinene — α-Pinene dibromide

(3) **Presence of a bicyclic compound.** The molecular formula of α-pinene is $C_{10}H_{16}$. But it contains one double bond. Therefore, its parent hydrocarbon is $C_{10}H_{18}$ which corresponds to the general formula C_nH_{2n-2} of compounds containing two rings. Therefore, it follows that α-pinene is bicyclic.

(4) **Presence of six-membered ring.** When α-pinene is heated with alcoholic sulphuric acid, it yields α-terpineol, a compound of well known structure.

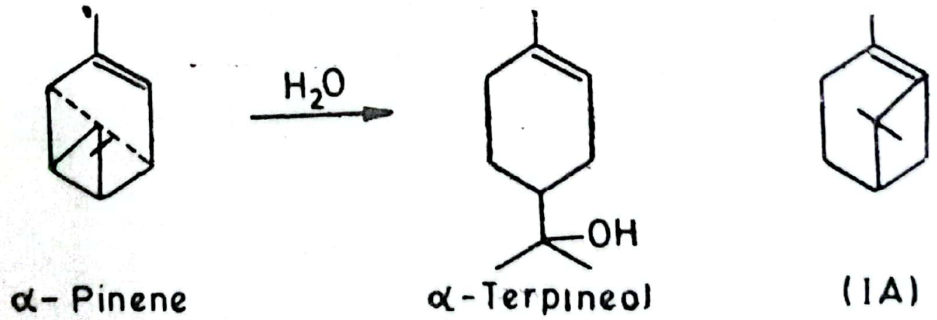


The formation of α-terpineol from α-pinene leads to the following facts :

(*a*) In α-pinene, a six-membered ring having the double bond of α-terpineol is present.

(*b*) In α-terpineol, the hydroxyl group is present at C_6. But this is not present in α-pinene.

Therefore, it means that the carbon atom *i.e.* C_6 must be involved in the formation of the second ring of α-pinene. There are three possible points of union for this C_6, giving rise to **two** three-membered and **one** four-membered ring (as represented by dotted lines in the following structure), and at the same time the position of the double bond in α-pinene has been shown by conversion into α-terpineol. The exact structure of the second ring, *i.e.*, whether 4- or 3-membered is established by the oxidative degradation of α-pinene.



(I)

Actually there are **four** possible points for union of C_6. Out of these, three are shown in structure (I) whereas the fourth being at the double bond to form a four-membered ring. However, the

fourth was rejected on the basis of Bredt's rule* (1924). According to this rule, the structure (IA) cannot exist because it has a large amount of strain.

(*c*) The **gem**-dimethyl group ($-CHMe_2$) of α-terpineol is not present in the six-membered ring of α-pinene and therefore it must be present in other ring.

(5) **Oxidative degradation of α-pinene.** (Nature of the second ring). Baeyer (1896) concluded that the second ring is four-membered ring in α-pinene on the basis of the following series of reactions :

$$\alpha\text{-Pinene} \xrightarrow[KMnO_4]{1\%\ Alk} \underset{(II)\ C_{10}}{\text{Pinene glycol}} \xrightarrow[Alk.\ KMnO_4]{Warm} \underset{(III)\ C_{10}}{\text{Pinonic acid}} \xrightarrow{NaOBr}$$

(I) C_{10}

$$\text{Pinic acid} + CHBr_3 \xrightarrow[(iii)\ PbO_2]{(i)\ Br_2;\ (ii)\ Ba(OH)_2} \underset{(V)\ C_8}{\textit{cis}\text{-Norpinic acid}}$$

(IV) C_9

The above set of oxidative degradation led to the following conclusions :

(*i*) When (I) is oxidised to (II), the double bond of the former is hydroxylated.

(*ii*) We have already proved that the double bond is present in the six-membered ring. Therefore, the oxidation of (II) to (III) occurs due to scission of the glycol bond. At the same time, a small amount of pinoylformic acid is also formed, *i.e.*, MeCO of (III) is now HOOC—CO.

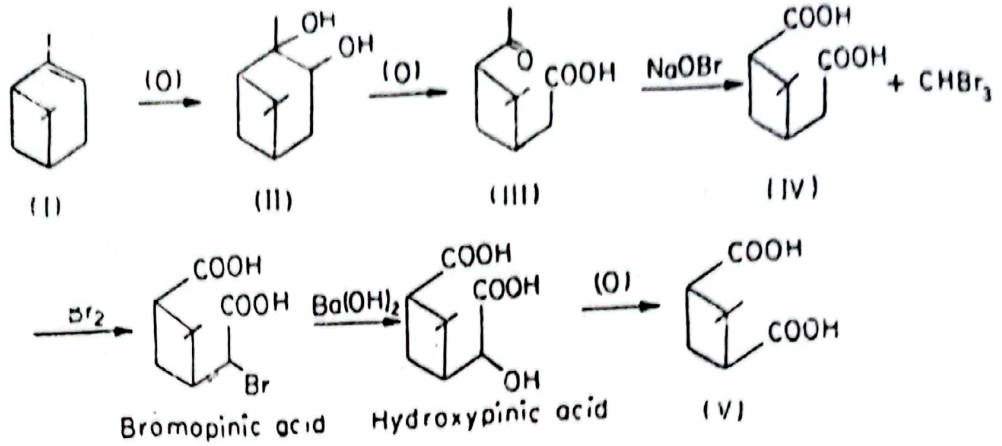
(*iii*) The formation of pinic acid (IV) and bromoform from the oxidation of pinonic acid (III) reveals that the acetyl group is present in pinonic acid.

(*iv*) Pinic acid has been shown to be a saturated dicarboxylic acid. This, when treated first with bromine, then with barium hydroxide and finally with the oxidising agent like PbO_2, yields *cis*-norpinic acid, $C_8H_{12}O_4$.

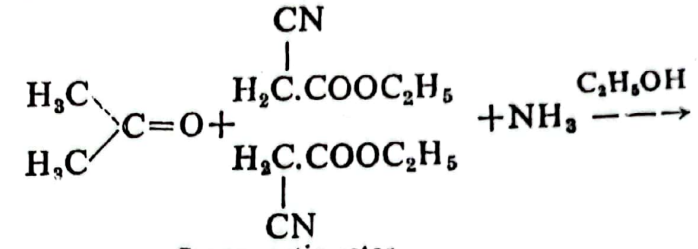
As *cis*—norpinic acid has been shown to be a saturated dicarboxylic acid, its formula may be written as $C_6H_{10}(COOH)_2$. But α-pinene contains two methyal groups attached to a carbon atom in the second ring (I) and its six-membered ring containing the double bond has been opened by the above oxidation. Therefore, the norpinic acid (with the second ring intact) would contain these two methyl groups Hence, the formula of norpinic acid may be written as $(CH_3)_2\ C_2H_4\ (COOH)_2$ If methyl and carboxyl groups are considered as substituents, the parent saturated hydrocarbon

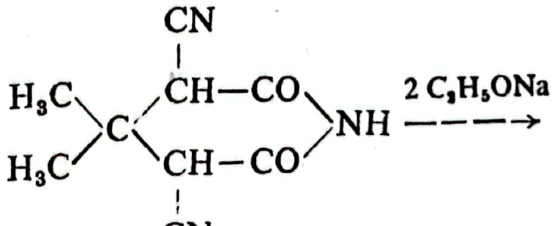
* According to Bredt's rule, a double bond cannot be formed by a carbon atom occupying the bridge-head of a bicyclic system.

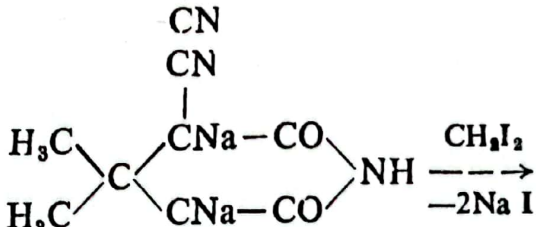
from which norpinic acid has been derived is C_4H_8, corresponding to cyclobutane. Therefore, norpinic acid may be a dimethyl cyclobutane dicarboxylic acid. On the basis of this fact, pinic acid would be a cyclobutane derivative in which one side-chain is $-CH_2COOH$. Therefore, Baeyer argued that pinic and norpinic acids possessed a cyclobutane ring. He further postulated structures (I) to (V) to explain the above reactions by accepting structure (I) for α-pinene.

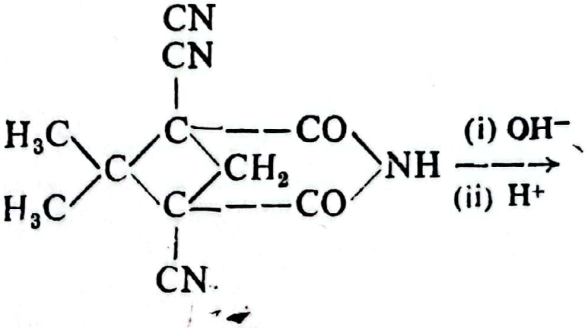


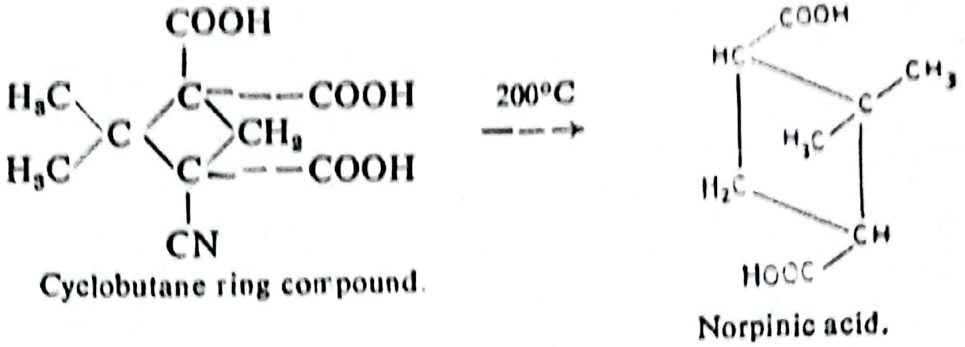
In order to confirm the above reactions, the synthesis of norpinic acid was carried out by Kerr in 1929.











Cyclobutane ring compound. Norpinic acid.

The norpinic acid obtained by this method was *trans-isomer*. The latter compound when heated with acetic anhydride yields this *cis* anhydride of norpinic acid which upon hydrolysis gives the *cis*-norpinic acid.

Method II (*Wagner's work*)

(1) **Molecular formula.** This has been found to be $C_{10}H_{16}$.

(2) **Presence of a double bond.** As α-pinene adds on one mole of bromine, this shows that α-pinene contains one double bond.

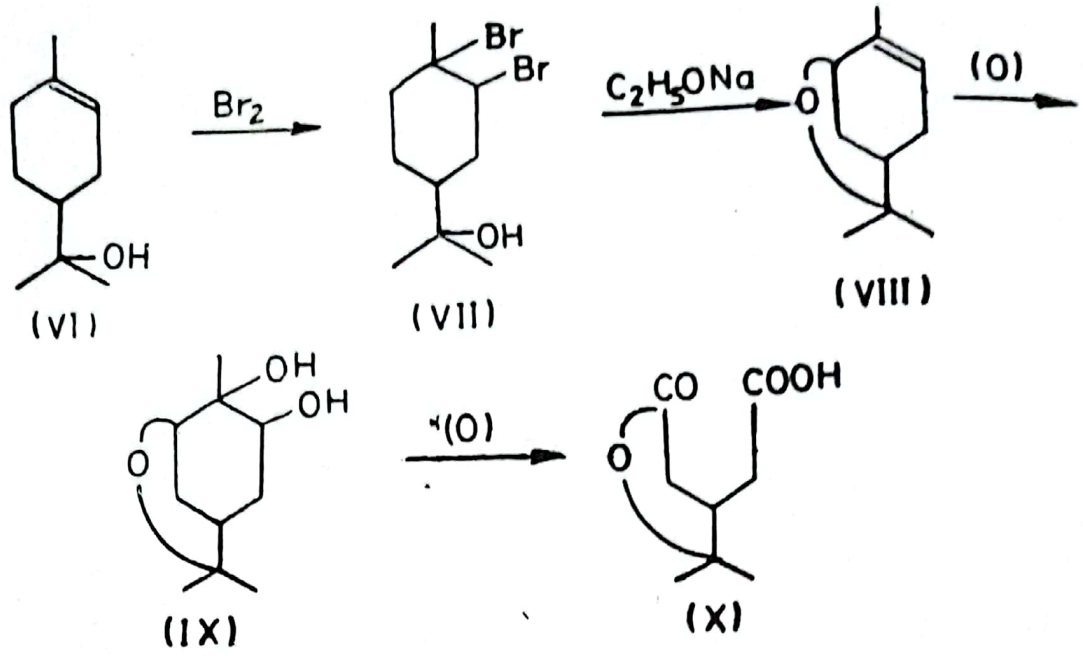
(3) **As a bicyclic compound.** The parent hydrocarbon of α-pinene is $C_{10}H_{18}$ which corresponds to the general formula C_nH_{2n-2} for bicyclic compounds and therefore α—pinene is **bicyclic.**

(4) **Conversion into pinol.** When nitrosyl chloride is added to α-pinene, α-pinene nitrosochloride is obtained along with the by-products. The latter when steam distilled yields **pinol**, $C_{10}H_{16}O$.

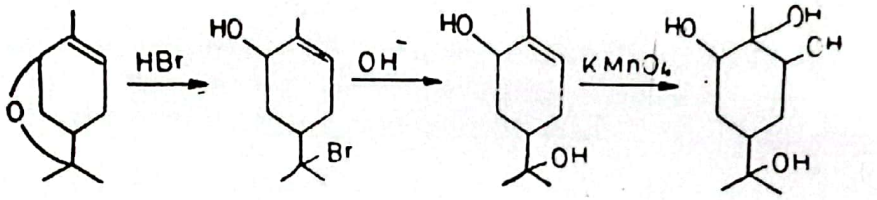
Pinol contains one double bond because it adds on one mole of bromine to form pinol dibromide. The latter compound, when treated with lead hydroxide, is converted into pinol glycol $C_{10}H_{16}O(OH)_2$ which upon oxidation yields terpenylic acid.

Pinol (VIII) is also obtained by the action of sodium ethoxide on α-terpineol dibromide (VII). Further, pinol (VIII) when oxidised with permanganate, yields pinol glycol (IX) which upon further oxidation yields terpenylic acid (X).

All the above facts can be explained if (VI) being the structure of α-terpineol.



Further support for pinol (III) is derived from the fact that sobrerol (pinol hydrate) gives a tetrahydric alcohol, sobrerythritol. Sobrerol is itself prepared by treating pinol with hydrogen bromide followed by sodium hydroxide. All these reactions may be formulated as follows :

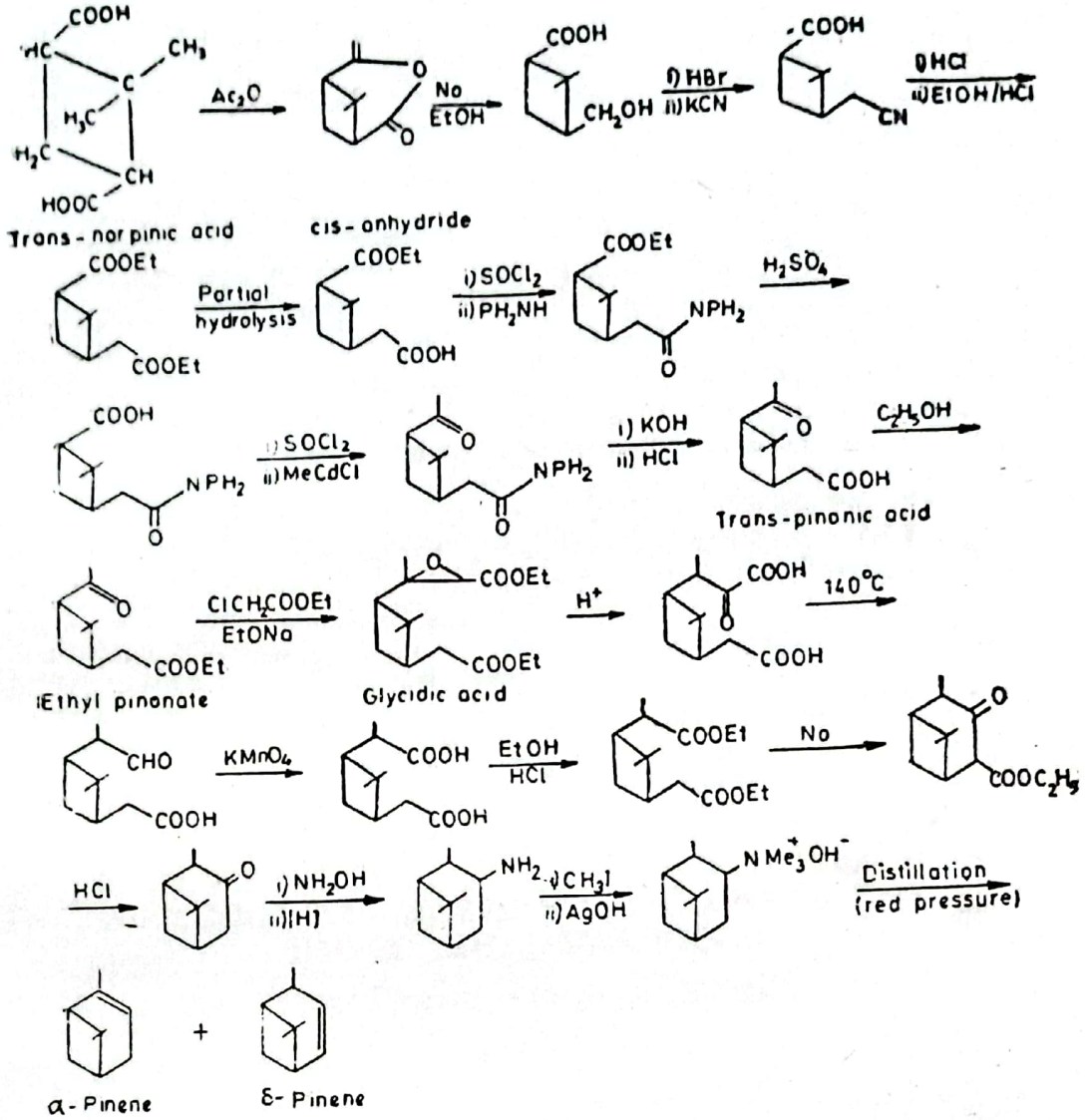


Pinol — Pinol hydrobromide — Sobrerol — Sobrerythritol

All the above reactions of α-pinene could be explained if the formula of α-pinene is considered to be (I) as given by Wagner. However, Wagner's work provides no direct evidence for the existence of the cyclobutane ring. Such evidence was provided by Baeyer given in **Method I.**

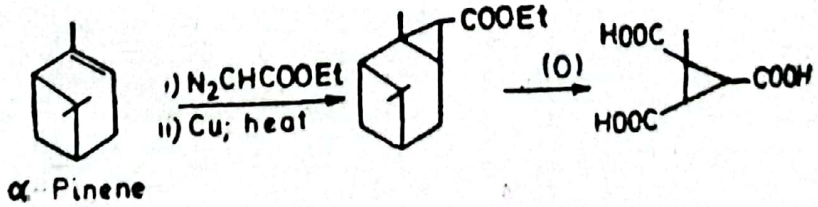
Total Synthesis of α-Pinene.

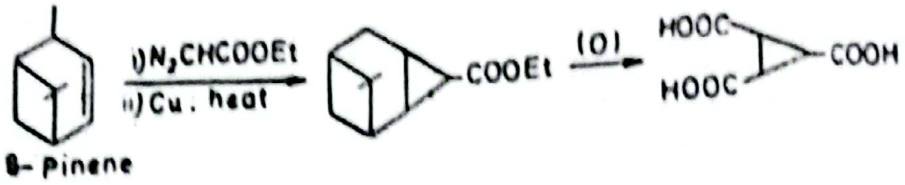
Finally the structure (I) for α-pinene has been confirmed by its synthesis. Guha *et al.* (1937) sythesised pinic acid from norpinic acid whereas Rao (1943) synthesised pinonic acid from synthetic pinic acid. Ruzicka *et al.* (1920—1924) synthesised α-pinene from pinonic acid. Thus, a total synthesis of α-pinene is achieved. The various steps for the synthesis of α-pinene are as follows :



In the above synthesis, a mixture of two compounds, α-and δ-pinene is obtained. Now question arises, which is α and which is δ. The e are distinguished on the basis of evidence that diazo-acetic ester undergoes combination with compounds having a double bond to form pyrazoline derivatives and these when alone or with copper powder undergo decomposition to yield cyclopropane derivatives.

When the two pinenes are subjected to the above treatment and the resulting compounds are oxidised, α-pinene yields 1-methyl-cyclopropane-1, 2, 3-tricarboxylic acid and δ-pinene cyclopropane-1, 2, 3-tricarboxylic acid. These products are in accord with the structures assigned to α-and δ-pinene.

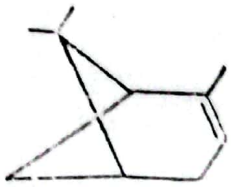




Stereochemistry

From the structure of α-pinene, it follows that it contains two dissimilar chiral centres and, therefore, two pairs of enantiomers are possible. However, only one pair is known. The reason for this is that four-membered ring can only be fused to the six-membered one in the cis-position, *i.e.*, trans fusion is impossible. Thus, only the enantiomers of the *cis*-isomer are known.

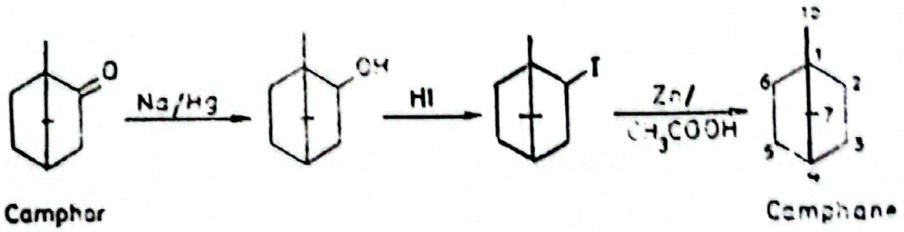
On the basis of electron diffraction studies, Arbusov and Naumov have recently found that α-pinene exists in a bridged half-chair-type conformation.



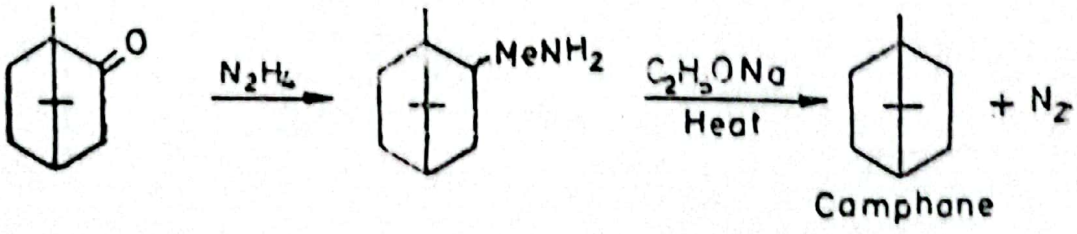
THE BORNANE (CAMPHANE)—NORBORNANE (ISOCAMPHANE) GROUP

1.42. CAMPHANE (BORNANE)

This does not occur naturally but is prepared by synthesis from camphor which constitutes the most important example of this group.



Camphane is also obtained from camphor by means of Wolff-Kishner's reduction.



Camphane is a saturated hydrocarbon. It is a solid having m.p. 156°C and it is optically inactive.

1.43. CAMPHOR

Introduction. This occurs in nature in the camphor tree of Formosa and Japan. It is present in all the parts of the tree. However, the highest proportion is present in the trunk.

Isolation. When the wood and leaves of camphor tree are boiled with water in a vessel covered with a dome, camphor sublimes and collects on the surface of the dome. Further purification of camphor is done by resublimation.

The quantity of camphor isolated from camphor tree is not sufficient to meet the world demand. Therefore, it is prepared from α-pinene (which is obtained from the turpentine oil) by the following series of reactions :

(*a*) α-Pinene $\xrightarrow[10°C]{\text{HCl gas}}$ Bornyl Chloride $\xrightarrow[-HCl]{\text{AcONa}}$ Camphene $\xrightarrow[H_2SO_4]{\text{AcOH}}$ Isobornyl acetate $\xrightarrow{\text{NaOH}}$ Isoborneol $\xrightarrow{PhNO_2}$ Camphor

(*b*) α-Pinene $\xrightarrow[10°C]{\text{HCl gas}}$ Bornyl Chloride $\xrightarrow[-Hcl]{\text{AcONa}}$ Camphene $\xrightarrow{\text{HCOOH}}$ Isobornyl formate $\xrightarrow{\text{NaOH}}$ Isoborneol $\xrightarrow[Ni\ ;\ 200°C]{O_2}$ Camphor

(*c*) α-Pinene $\xrightarrow{\text{Isomn.}}$ Camphene $\xrightarrow[rearr.]{\text{AcOH}}$ Isobornyl acetate $\xrightarrow{\text{NaOH}}$ Isoborneol $\xrightarrow[dehydrogen.]{\text{Catalytic}}$ Camphor

Properties. It is a solid having m.p. 180° and is optically active ; the (+) and (−) forms occur naturally whereas racemic camphor is the usual form of synthetic camphor.

Uses. Its various uses are as follows :

(*i*) As a plasticiser for the manufacture of celluloid and photographic films.

(*ii*) For the manufacture of smokeless powders and explosives.

(*iii*) As a mild disinfectant and stimulant for the heart muscles.

(*iv*) As an insect repellent.

CamScanner

(*v*) As a solvent in the Rast camphor method for the determination of molecular weights.

Constitution of Camphor. A tremendous amount of work was done before the structure of camphor was successfully elucidated by Bredt (over 30 structures have been proposed) on the basis of following analytical and synthetic evidences :

1. **Molecular formula.** From analytical data and molecular weight determination, it follows that the molecular formula of camphor is $C_{10}H_{16}O$.

2. **Saturated characteristics.** Camphor does not add reagents like bromine, nitrosyl chloride, etc. However, it forms mono-substitution products like mono-bromocamphor, mono-chloro-camphor, camphor-sulphonic acid. The formation of these products reveals that camphor is a saturated compound and does not contain any double bond. This is further supported by the facts that (*i*) it does not react with 1% alk. $KMnO_4$ and (*ii*) Molecular refraction also favours this.

3. **Presence of a keto group.** The nature of oxygen atom is found to be *cyclic ketonic* on the basis of following facts :

(*i*) It forms an oxime with hydroxylamine

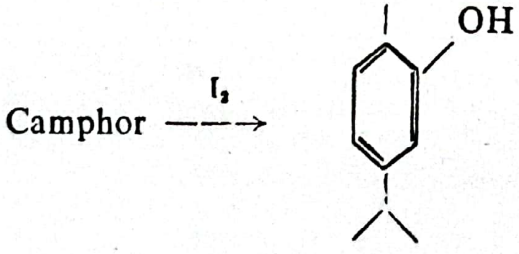
$$\underset{\text{Camphor}}{C_{10}H_{16}O} + H_2NOH \longrightarrow \underset{\text{Camphor oxime}}{C_{10}H_{16}=N-OH}$$

(*ii*) It forms semicarbazone with semicarbazide

(*iii*) It forms phenylhydrazone with phyelhydrazine.

(*iv*) Camphor, when oxidised with nitric acid, yields a dicarboxylic acid called camphoric acid without loss of carbon atoms. Further, camphor when reduced with sodium amalgam, yields a secondary alcohol called *borneol*. Hence, camphor must be a cyclic ketone.

(*v*) When camphor is distilled with iodine, it yields carvacrol.

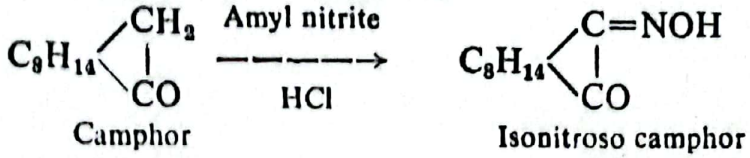


The presence of phenolic group in carvacrol reveals the presence of ketonic group in camphor.

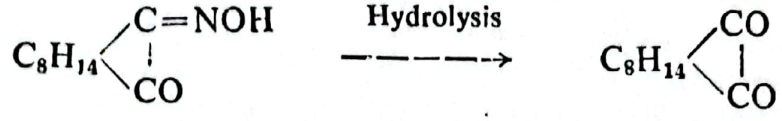
4. **Bicyclic system.** The molecular formula of saturated parent hydrocarbon of camphor is $C_{10}H_{18}$ which corresponds to the general formula C_nH_{2n-2} of bicyclic compounds and, therefore, camphor is a bicyclic compound.

5. **Presence of $-CH_2CO$ group.** When camphor is treated with amyl nitrite and hydrochloric acid, it yields an iso-nitroso

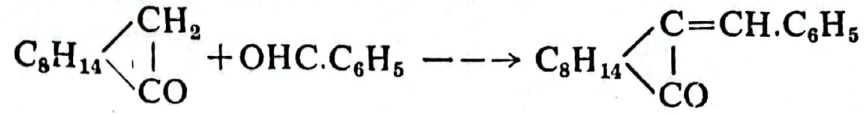
(oximino) camphor in which two hydrogen atoms have been replaced by=NOH group. This reaction reveals that the >C=O group is directly attached to a $-CH_2$ group.



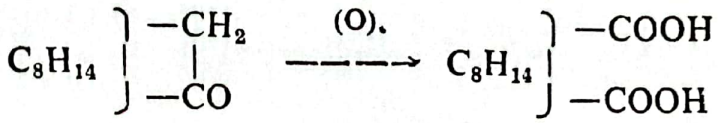
Isonitroso camphor when hydrolysed yields camphor-quinone, thus confirming the presence of a $-CO-CH_2$ group.



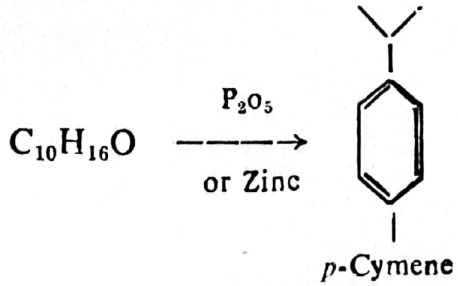
When camphor is treated with benzaldehyde, it yields a benzylidene derivative. Again, this reaction is characteristic of the reactive methylene group attached to a CO group. This shows that camphor must contain $-CH_2CO$ group.



The presence of $-CO-CH_2$ grouping in a ring of camphor has been further confirmed by the fact that camphor, when oxidised, yields camphoric acid, a dibasic acid having the same number of carbon atoms. The oxidation takes place by opening of the ring.



6. **Presence of a six-membered ring.** When camphor is distilled with zinc chloride or phosphorus pentoxide, it yields *p*-cymene. The formation of the latter product reveals the presence of six-membered ring, methyl and gem-dimethyl groups in camphor.



When camphor is heated with hydriodic acid, it yields tetra-and hexa-hydro *m*-xylene, confirming that camphor contains a six-membered ring.

7. **Nature of the carbon-frame in camphor.** When camphor is oxidised with nitric acid, it yields a crystalline dibasic acid, camphoric acid, $C_{10}H_{16}O_4$. As camphoric acid possesses the same number of carbon atoms as camphor, it means that the keto group must be present in one of the rings in camphor. Further,

camphoric acid is a dicarboxylic acid and its molecular refraction reveals that it is also saturated. Thus, during the conversion of camphor into camphoric acid there occurs the opening of ring containing the keto group and, therefore, camphoric acid must be a monocyclic compound.

When camphoric acid is further oxidised with nitric acid, camphoronic acid is obtained.

$$\underset{C_{10}H_{16}}{\text{Camphor}} \xrightarrow[[O]]{HNO_3} \underset{C_{10}H_{16}O_4}{\text{Camphoric acid}} \xrightarrow[[O]]{HNO_3} \underset{C_9H_{14}O_6}{\text{Camphoronic acid}}$$

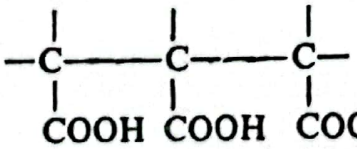
In order to elucidate the structure of camphor, the structures of camphoric acid and camphoronic acid should be known with certainty.

8. **Structure of camphoronic acid.** This can be elucidated as follows :

(*a*) The molecular formula of camphoronic acid is $C_9H_{14}O_6$.

(*b*) As camphoronic acid has been shown to be a saturated tricarboxylic acid, its molecular formula may be written as $C_6H_{11}(COOH)_3$ and its parent hydrocarbon will, therefore, be C_9H_{14} which corresponds to the general formula (C_nH_{2n+2}) for acyclic compound, indicating that camphoronic acid is an acyclic compound.

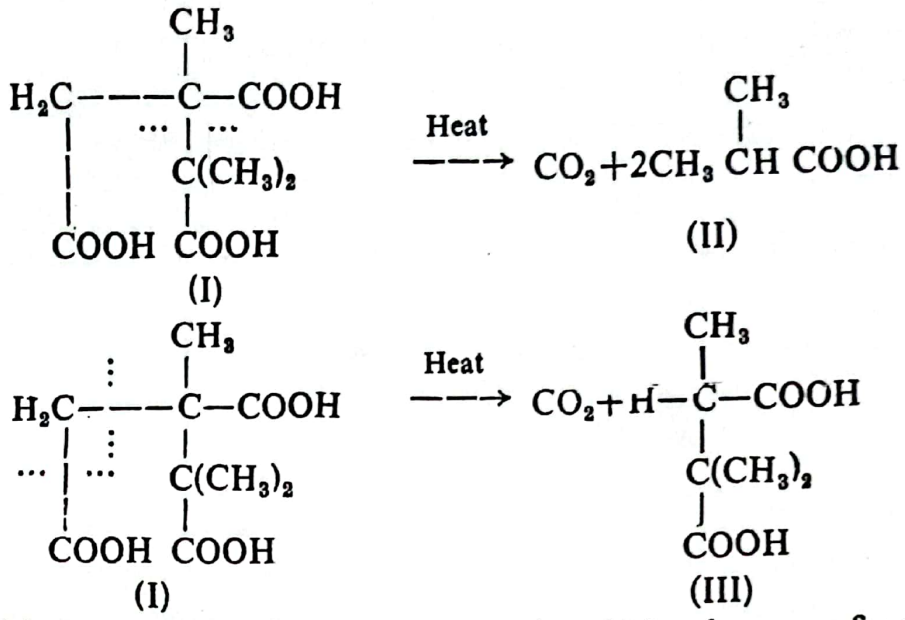
(*c*) As camphoronic acid does not undergo decarboxylation under ordinary conditions, it means that its three carboxyl groups should be attached to three different carbon atoms.



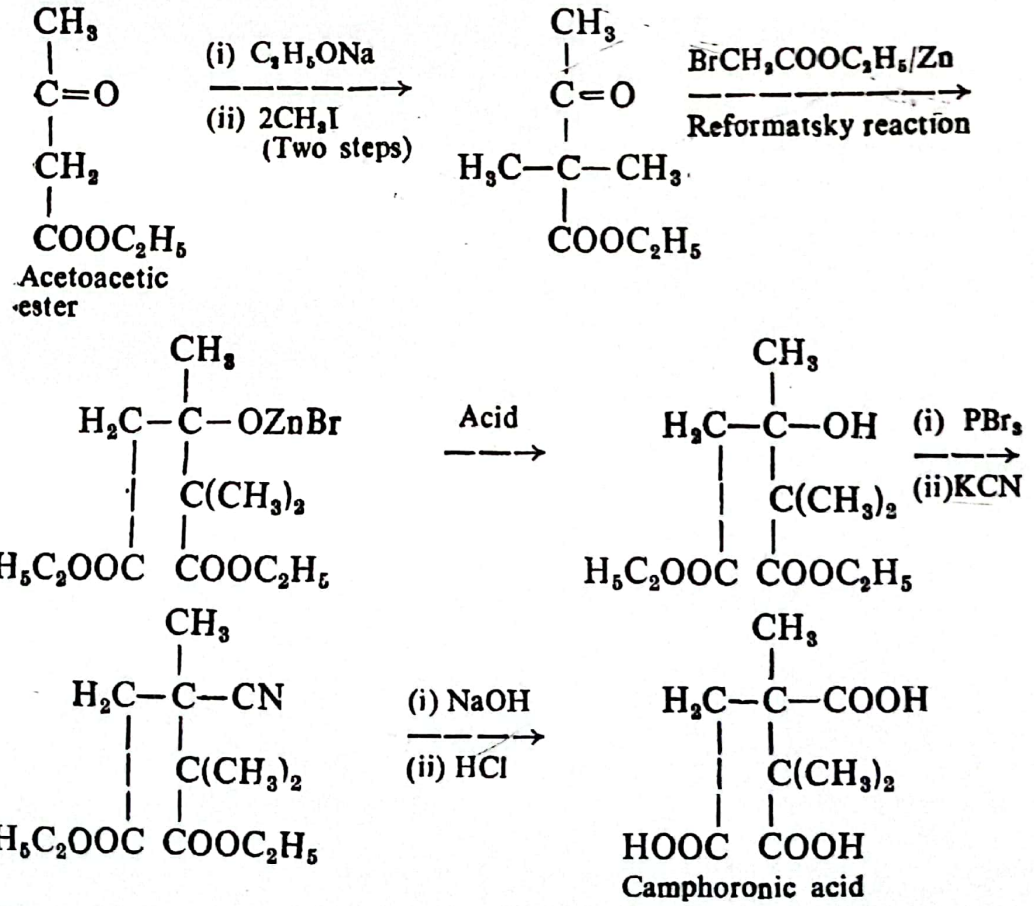
(*d*) When camphoronic acid is distilled at atmospheric pressure, it yields isobutyric acid (II), trimethylsuccinic acid (III), carbon dioxide and carbon.

$$\underset{\text{Camphoronic acid}}{2[C_9H_{14}O_6]} \xrightarrow[\text{Distil}]{\text{Heat}} \underset{\text{Isobutyric acid (II)}}{2\,[H_3C{-}\overset{\overset{CH_3}{|}}{CH}{-}COOH]} + \underset{\text{Trimethyl succinic acid (III)}}{\begin{array}{l} \quad\quad CH_3 \\ \quad\quad | \\ H{-}C{-}COOH \\ \quad\quad | \\ H_3C{-}C{-}CH_3 \\ \quad\quad | \\ \quad\quad COOH \end{array}} + 2\,CO_2 + C$$

In order to explain the formation of these products, Bredt (1893) suggested that camphoronic acid is α, α, β-tricarballylic acid (I).



(*e*) The structure (I) for camphoronic acid has been confirmed by its synthesis given by Perkin Junior and Thorpe (1897). In this synthesis, acetoacetic ester is converted into the dimethyl derivative which is then subjected to Reformatsky reaction with ethyl bromoacetate. The product so obtained is converted into halide and then into a cyanide. Finally, the latter compound when hydrolysed yields camphoronic acid.

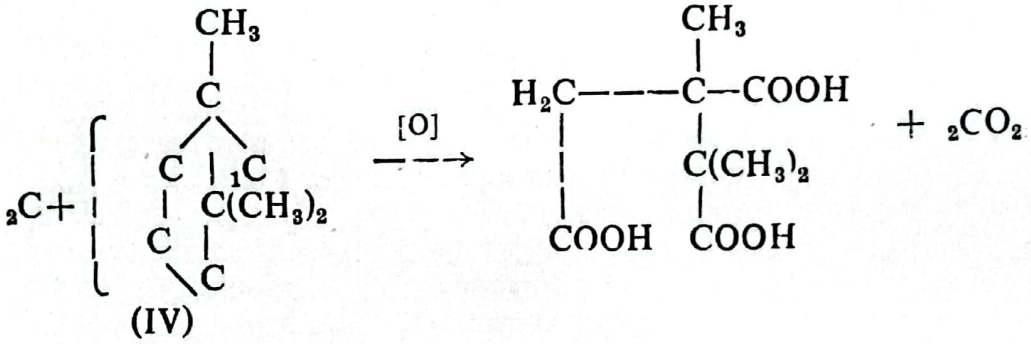


9. **Structure of camphoric acid.** It is elucidated as follows :

(*a*) The molecular formula of camphoric acid is $C_{10}H_{16}O_4$.

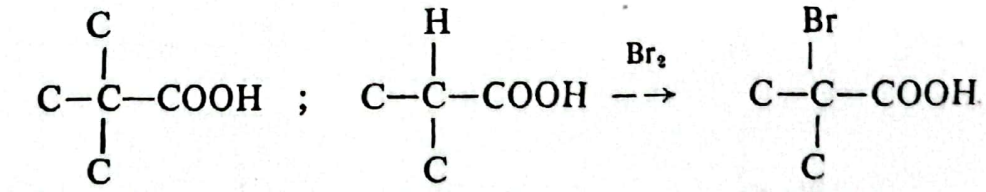
(*b*) By usual tests, camphoric acid has been shown to be a saturated dicarboxylic acid.

(*c*) If camphoronic acid has structure (I), then camphoric acid (and hence camphor) must possess three methyl groups. On this basis, the molecular formula of camphoric acid, $C_{10}H_{16}O_4$, may be written as $(CH_3)_3C_5H_5(COOH)_2$. The parent (saturated) hydrocarbon corresponding to camphoric acid is C_5H_{10} which corresponds to general formula of C_nH_{2n} for a monocyclic derivative and, therefore, camphoric acid is a monocyclic derivative. This also agrees with the previous evidence that camphoric acid is monocyclic. Thus, the oxidation of camphoric acid to camphoronic acid may be written as follows :



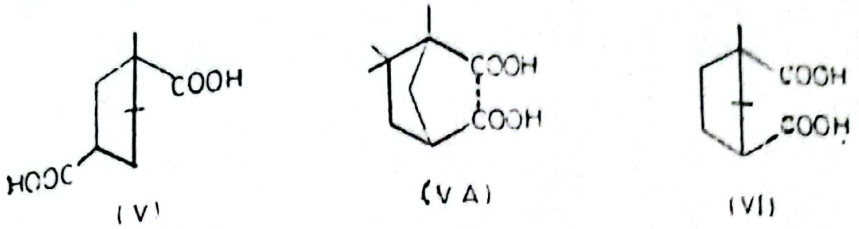
The skeleton (IV) plus one carbon, having arrangement with two carboxyl groups, will lead to the structure of camphoric acid.

(*d*) As camphoric acid is able to form a monoester readily but it forms diester with difficulty, this shows that the two carboxyl groups are not similar, *i.e.*, one is **primary** or **secondary** and the other is **tertiary.** This is confirmed by the fact that camphoric anhydride forms only one monobromo derivative with phosphorus/bromine. The formation of this monobromo derivative is only possible if one of the carboxyl groups is secondary, *i.e.*, the carbon atom of one carboxyl group must be $_1C$.



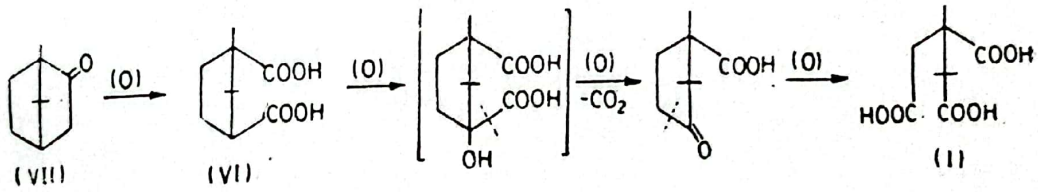
t-Carboxyl group (esterified with difficulty) ; *Sec*-Carboxyl group (easily esterified) → Monobromo derivative

(*e*) In the step (*d*), the position of one carboxyl group has been ascertained. Now the next problem is to ascertain the position of other carboxyl group in camphoric acid. Its position should be such that when there occurs the opening of cyclopentane ring of camphoric acid to yield camphoronic acid, one carbon atom is readily lost. If this fact is taken into account, then there are two reasonable structures for camphoric acid, (V) and (VI). The structure (V) may be rewritten as (VA).



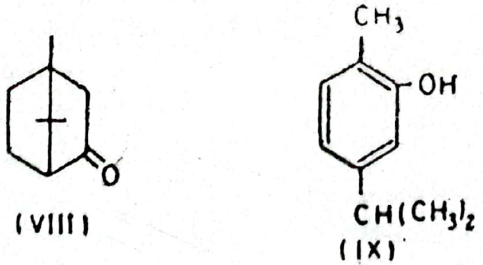
(V) (VA) (VI)

10. **Structure of camphor.** The camphoric acid is obtained by the oxidation of $-CH_2-CO-$ group of camphor. Therefore, the structure of camphor would be obtained by joining the carbon atoms of the carboxyl groups in camphoric acid. Hence, the structure of camphor obtained from (VA) would contain a six-membered ring with a gem dimethyl group. But this structure could not explain the conversion of camphor into *p*−cymene. Thus, the structure (VA) and hence (V) was rejected. On the other hand, structure (VI) explains all the foreging facts and the structure (VII) of camphor corresponding to structure (VI) would be the correct structure. This argument was given by Bredt who proposed the following reactions to reveal the relationships between camphor, camphoric acid and camphoronic acid.



(VIII) (VI) (I)

Bredt also proposed structure (VIII) for camphor. This structure also explains all the foregoing facts but he rejected this in favour of (VII). The simple reason for its rejection is that camphor when distilled with iodine yields carvacrol (IX). The formation of this compound can be explained only on the basis of structure (VII) but not on (VIII).

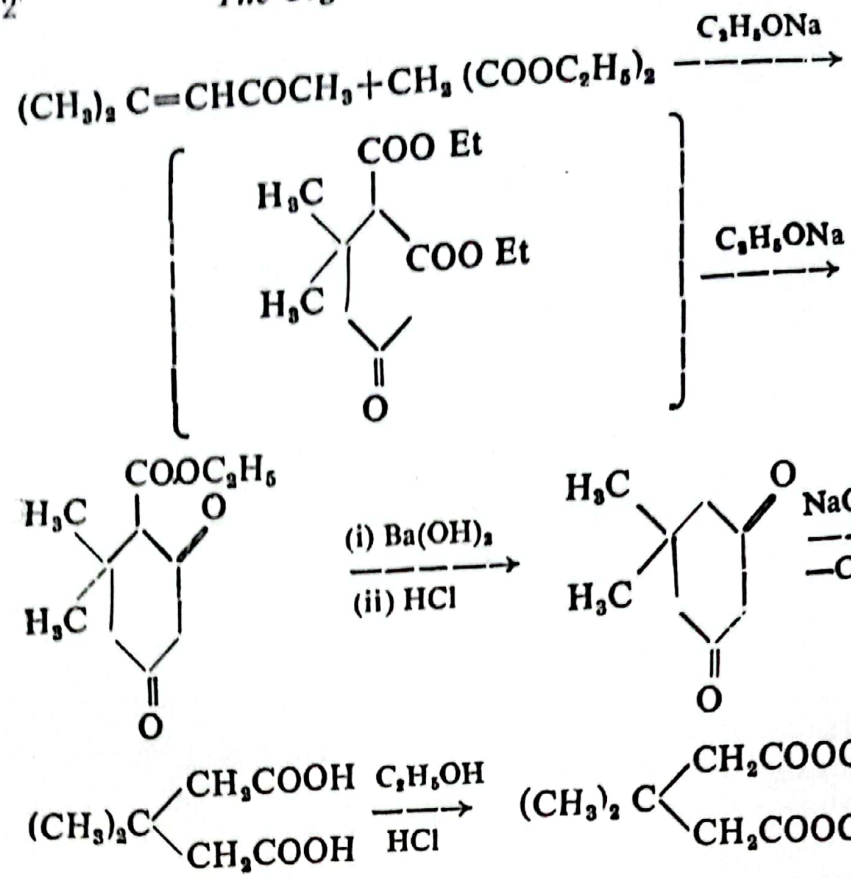


(VIII) (IX)

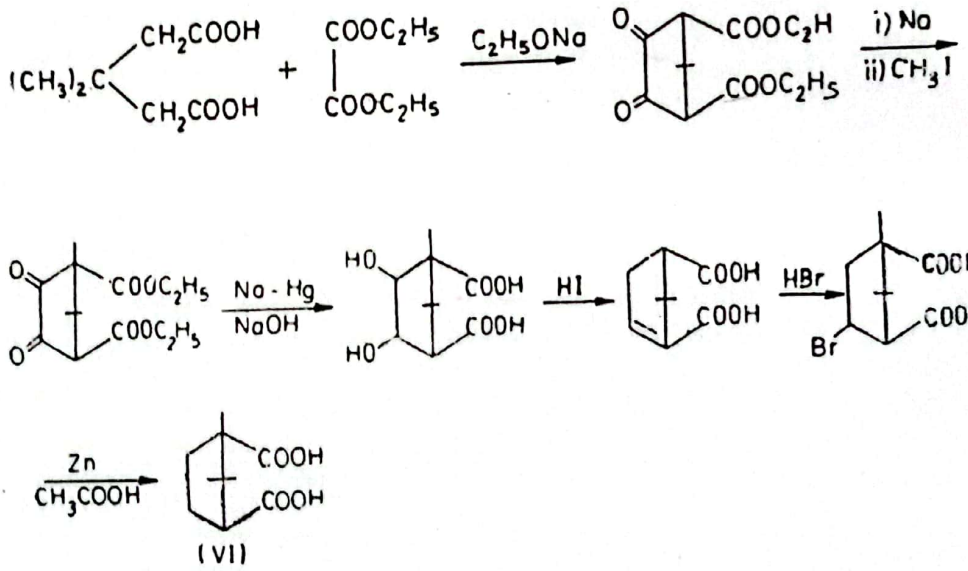
Finally, the structures of camphoric acid (VI) and camphor (VII) have been confirmed by their respective synthesis.

11. **Synthesis of (±)-camphoric acid.** This was given by Komppa. His synthesis involves the two steps :

(*a*) **First step** involves the preparation of 3, 3-dimethyl-glutaric acid from mesityl ketone and ethyl malonate.

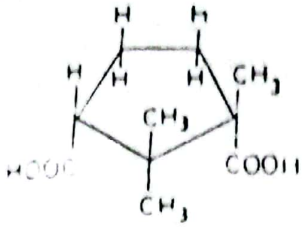


(*b*) **Second step** Involves the conversion of 3, 3-dimethyl-glutaric acid into camphoric acid as follows :

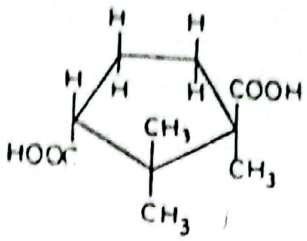


There are two geometrical isomeric forms of camphoric acid, *i.e.*, **cis** and **trans**. Neither of these has any elements of symmetry. Hence, there may be four optical isomers of camphoric acid. However, all are known that would correspond to the (+)- and (−)-forms of camphoric acid and isocamphoric acid. Further, camphoric acid on heating yields anhydride whereas isocamphoric acid does

not yield anhydride. This means that camphoric acid is *cis*-isomer whereas isocamphoric acid is the *trans*-isomer.

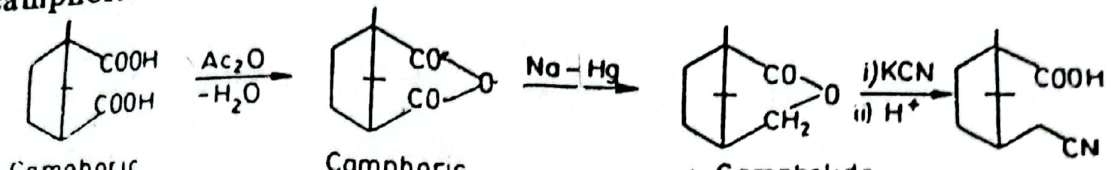


Camphoric acid
m.p. 187°C

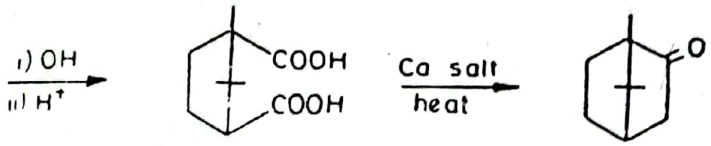


Isocamphoric acid
m.p. 171°-172°

12. **Synthesis of camphor.** (*a*) This was synthesised from camphoric acid by Haller (1896).

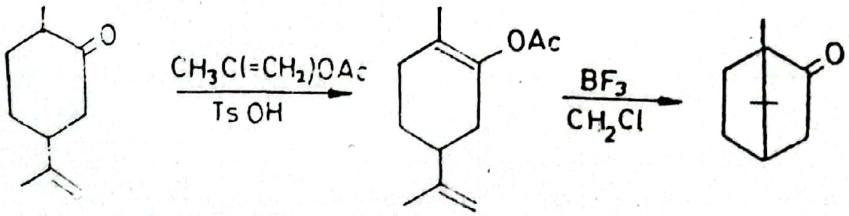


Camphoric acid — Camphoric anhydride — α-Campholide



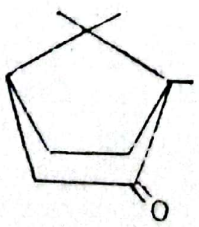
Homocamphoric acid — Camphor

(*b*) Another synthesis was given by Money *et al.* (1969). This involves a two-step synthesis of (±)-camphor from dihydrocarvone.



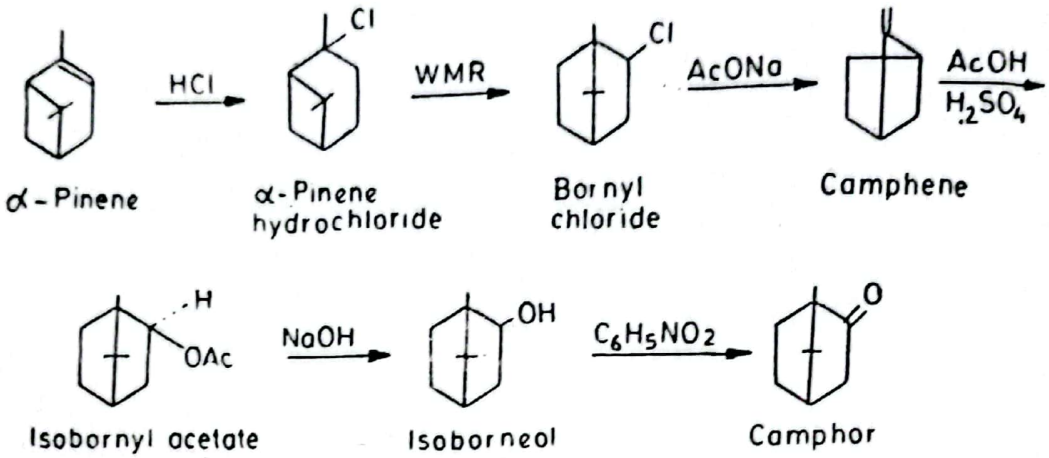
13. **Stereochemistry of camphor.** Similar to camphoric acid, camphor contains two dissimilar chiral centres. Therefore, its two pairs of enantiomers are known. However, only one pair of enantiomer due to cis-form is possible because other pair of enantiomer is not possible due to the **impossibility** of the trans fusion of the gem-dimethylmethylene bridge to the cyclohexane ring.

Camphor as well as its derivatives exist in the boat conformation. Since the gem-dimethyl bridge must be *cis*, the cyclohexane ring must exhibit the boat form.



Camphor

14. **Commercial preparation of camphor.** On a large scale, camphor is prepared from α-pinene which is commercially available from turpentine oil. The reaction involves number of Wagner-Meerwein rearrangements (WMR) which cause a complete change in carbon skeleton. The various steps for the synthesis of camphor are as follows :



For other methods, see isolation of camphor.

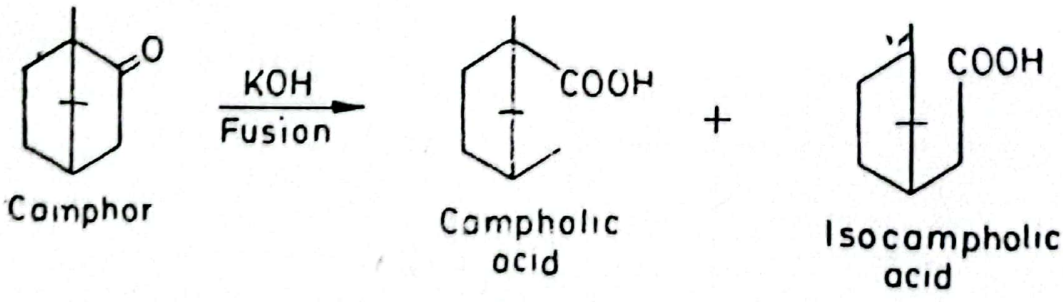
Reactions of Camphor. These are of two types :

(*a*) Fission reactions

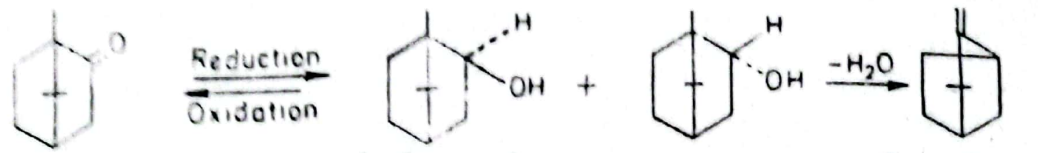
(*b*) Reduction reactions.

We shall discuss these one by one.

(*a*) When camphor is heated with KOH, fission will take place at that bond which involves the least substituted carbon atom adjacent to the carbonyl group. This fission occurs according to the general rule for acyclic ketones. Thus, camphor yields a mixture of campholic and isocampholic acids (Guerbet) ; the former being the major product



(*b*) When camphor is reduced, it yields a mixture of two isomeric alcohols, borneol and isoborneol (both these alcohols belong to the camphane group). The relative proportion of borneol to isoborneol is mainly dependent upon the nature of the reducing agent. Both borneol and isoborneol on dehydration yield camphene and on oxidation yield camphor.



camphor isoborneol Camphene

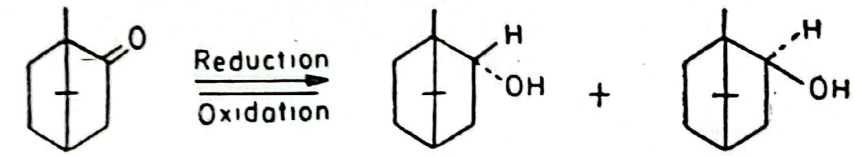
1.44. BORNEOLS AND ISOBORNEOLS

Introduction. Borneols and isoborneols are stereoisomeric compounds of the formula, $C_{10}H_{18}O$. Both these types of terpenoids are optically active and are known to exist in the (+)- and (−)- forms.

Borneols occur widely distributed in essential oils. For example, (+)-borneol occurs in a tree called *Dryobalanops camphore* which grows abundantly in a Sumatra and Borneo, also in rosemary and spike oils. The (−)- and racemic-borneols occur in valerian oil. The borneols are similar to camphor in odour and burning taste.

Isoborneols have been isolated from only one essential tree.

Constitution. Their constitutions have been elucidated on the basis of the fact that both when oxidised yield camphor. The latter compound on reduction gives back a mixture of borneol and isoborneol. It means that both borneol and isoborneol are the stereoisomeric secondary alcohol corresponding to the ketone, camphor.



Camphor Borneol m p 208.5°C Isoborneol m p 217°C

The relative amounts of borneol and isoborneol produced by the above reaction depends upon the nature of the reducing agent used. For example, electrolytic reduction yields mainly borneol whereas catalytic dehydrogenation with platinum yields mainly isoborneol ; the latter is also the main product when reducing agents like aluminium isoproxide and lithium aluminium hydride are employed.

Stereo Chemistry. Borneol has been found to possess the *endo*-configuration in which the *gem*-dimethyl bridge lies above the plane of the cyclohexane ring whereas hydroxyl group lies below the plane (Fig. 1.1).

Isoborneol has been found to possess the *exo*-configuration in which the bridge and the hydroxyl group lie above the plane of the cyclohexane ring (Fig. 1.1).

CamScanner

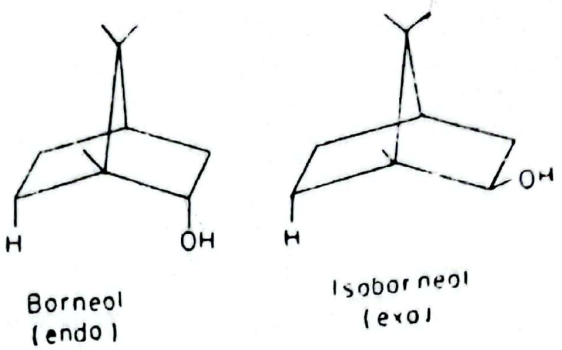
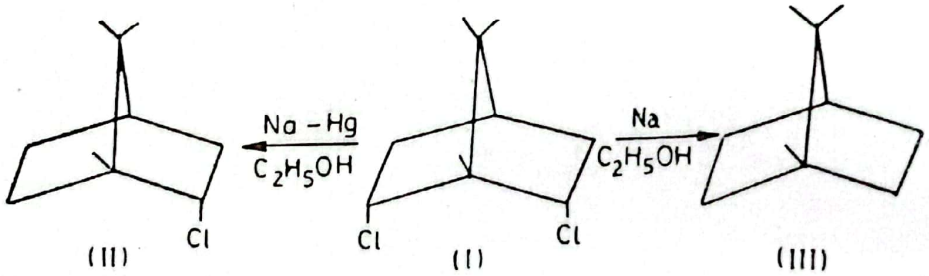


Fig. 1.1

The configurations of borneol and isoborneol shown in Fig. 1.1 have been assigned mainly on the basis of relative rates of reaction exhibited by the hydroxyl group. Borneol undergoes esterification more readily than isoborneol and the esters of borneol undergo hydrolysis more readily than those of isoborneol. These different rates reveal that the *hydroxyl group in borneol is less sterically hindered then in isoborneol.*

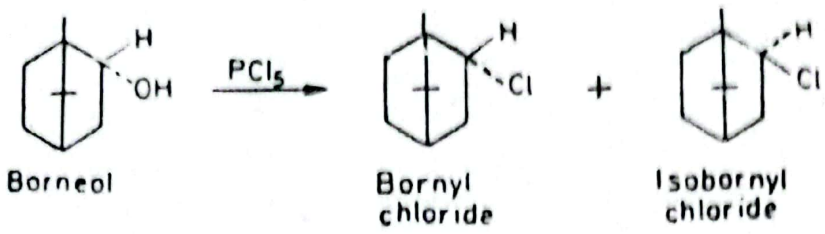
Further support to the structures given in Fig. 1.1 comes from the work of Kwart *et al.* (1956) who established the structure (I) for bornyl dichloride. This group of workers also showed that bornyl dichloride (I), when reduced with sodium amalgam and ethanol, yields bornyl chloride (II) and (I) when reduced with sodium and ethanol yields bornane (III),



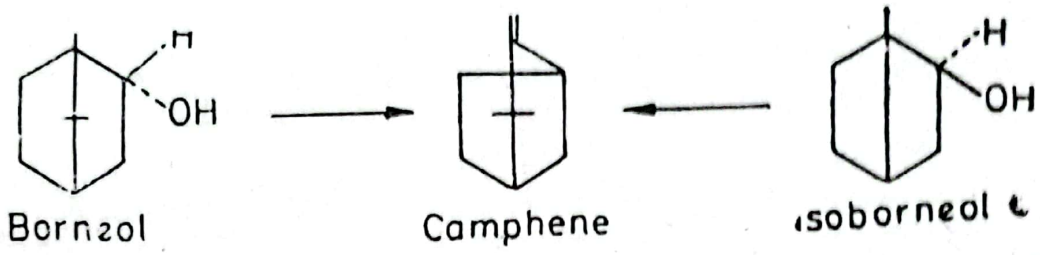
Mass spectrometry has also been used to confirm the stereochemistry of borneols shown in Fig. 1.1. The mass spectra of borneol and isoborneol are expected to be similar because both have similar structure but have different stereochemistries. In actual practice, this has been found to be same, *i.e.*, both possess the same peaks but the relative intensities of the these peaks are different. The maximum difference in intensities is shown by molecular ions, *i.e.*, the intensities are much stronger for borneol than for isoborneol. This may be explained on the basis of the fact that the molecular ion of the borneol is the more stable one due to smaller 1,3-steric interaction between OH and Me in isoborneol.

Important Reactions of Borneols

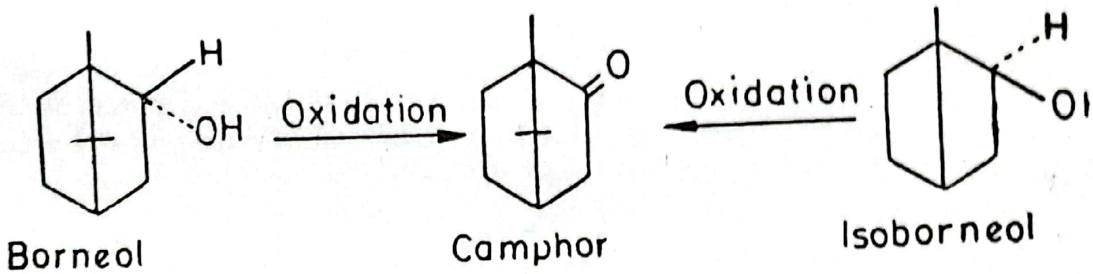
1. *With PCl_5.* When borneol is treated with phosphorus pentachloride, it yields a mixture of bornyl and isobornyl chlorides.



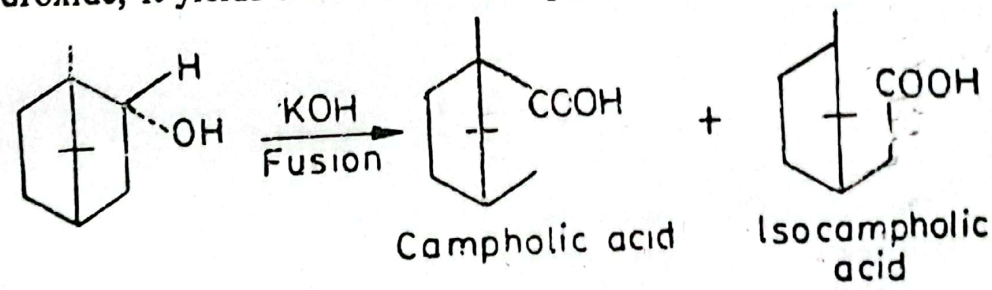
2. *Action of dehydrating agents.* When dehydrated, both borneol and isoborneol are dehydrated to camphene. However, the dehydration of isoborneol takes place more readily than borneol.



3. *Action of oxidising agents.* When oxidised, both borneol and isoborneol, yield camphor. However, copper catalyst can dehydrogenate only borneol but cannot dehydrogenate isoborneol.



4. *Action of KOH.* When borneol is fused with potassium hydroxide, it yields a mixture of campholic and isocampholic acids.



1.45. CAMPHENE

Introduction. It is a bicyclic monoterpenoid which occurs naturally in the (+)-, (−)- and (±)- forms. The (+)- form occurs in ginger, rosemary and spike oils. The (−)-form occurs in citronella and valerian oils.

It is a solid which melts at 51-52°C and has a smell of turpentine and camphor.

Constitution. This has been deduced on the basis of following analytical and synthetical evidences :

1. *Molecular formula.* From analytical data and molecular weight determination, molecular formula has been found to be $C_{10}H_{16}$.

2. *Presence of one double bond.* As camphene adds on one mole of bromine and also one mole of hydrogen chloride, this shows that camphene contains **one double bond**.

$$C_{10}H_{16}.Br_2 \xleftarrow{Br_2} C_{10}H_{16} \xrightarrow{HCl} C_{10}H_{16}.HCl$$

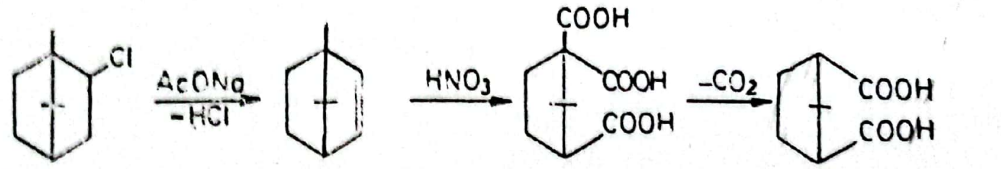
Camphene dibromide — Camphene — Camphene hydrochloride

The presence of a double bond in camphene is further supported by its methods of preparation such as (*i*) by the removal of a molecule of hydrogen chloride from bornyl and isobornyl chlorides with sodium acetate or (*ii*) by the dehydration of the borneols with potassium hydrogen phthalate.

(3) *Camphene as a bicyclic compound.* As camphene contains one double bond, the molecular formula of its parent saturated hydrocarbon would be $C_{10}H_{18}$ which belongs to the general formula C_nH_{2n-2} of bicyclic compounds. This reveals that camphene is bicyclic.

(4) *Oxidative degradation.* When camphene is oxidised with dilute nitric acid, it produces apocamphoric acid $C_9H_{14}O_4$ and carboxyapocamphoric acid $C_{10}H_{14}O_6$. The formation of the latter acid having the same number of carbon atoms as camphene reveals that the double bound is present in the ring of camphene. Further, the fact that carboxyapocamphoric acid, when heated above its melting point, is converted into apocamphoric acid implies that the former contains two carboxyl groups which are attached to the same carbon atom.

The above mentioned facts could be explained if structure (I) is assigned to camphene.



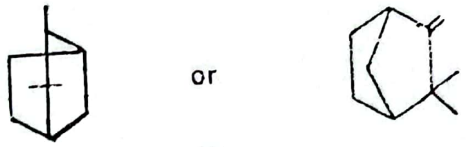
Bornyl chloride — Camphene (I) — Carboxyapocamphoric acid — Apocamphoric acid

The structure of apocamphoric acid was later proved by synthesis (Komppa, 1901).

(5) *Objection against structure (I).* Wagner (1890) did not agree to the structure (I) of camphene. He carried out the oxidation of camphene with dilute permanganate and got camphene glycol, $C_{10}H_{16}(OH)_2$. As the latter compound is a saturated glycol, Wagner argued that camphene is a bicylic compound [same as is structure (I)]. Then, wagner carried out further oxidation of camphene glycol. He got camphenic acid $C_{10}H_{16}O_4$ (a dibasic acid) and camphenylic acid, $C_{10}H_{16}O_3$.

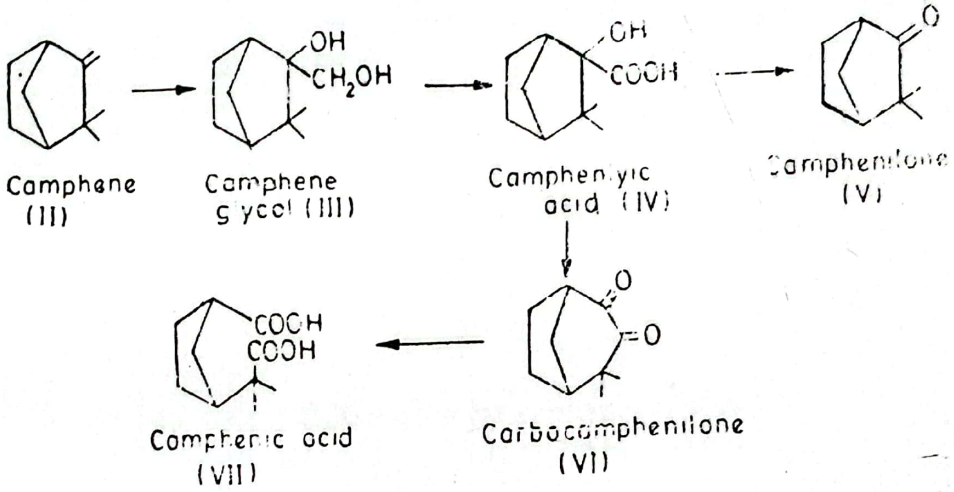
The latter acid is a hydroxy-monobasic acid which on oxidation with lead dioxide yields camphenilone.

Wagner argued that the structure (I) of camphene could not explain the above mentioned reactions. Therefore he suggested another structure (II) for camphene.



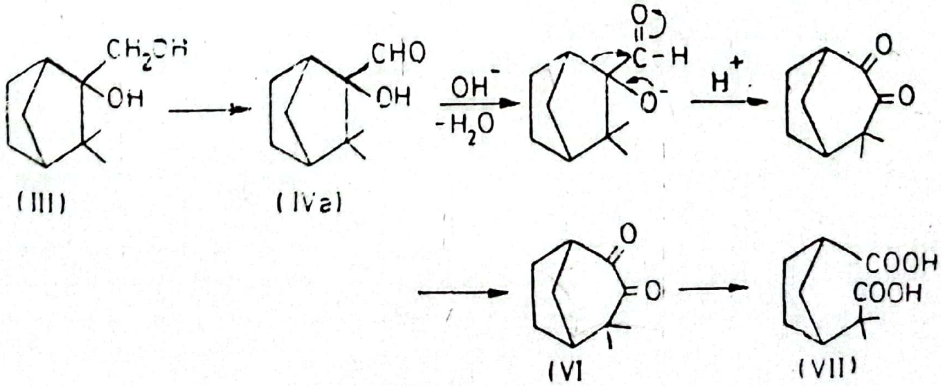
(II)

The structure (II) could explain the formation of camphene glycol, camphenylic acid and camphenilone as follows :



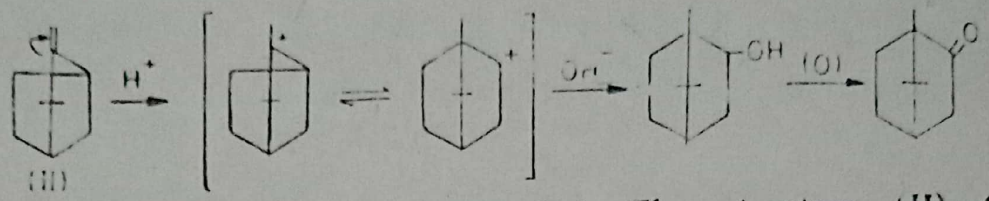
(6) *Limitation of structure (II).* By the structure (II), the formation of (III), (IV) and (V) could be explained. But this could not explain the formation of (VII).

In order to explain the formation of (VII), various mechanisms were proposed. However, the accepted mechanism is given by by Mayo (1959) who suggested that camphene glycol (III) is first of all oxidised to the aldehyde (IV A) and that this is then converted into (VII) via (VI) by involving the acyloin rearrangement.



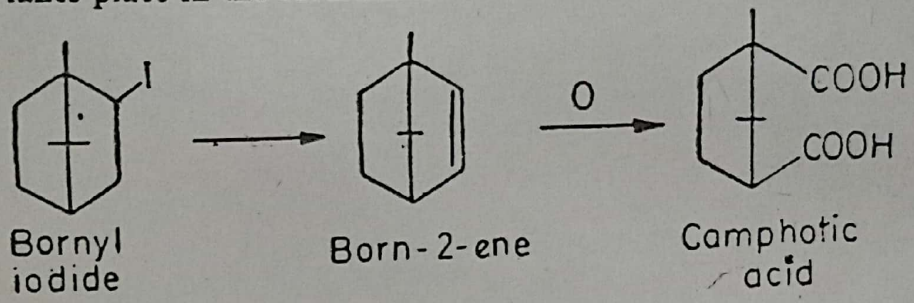
On oxidation with chromic acid (chromium trioxide in water or acetic acid), camphene yields camphor. This conversion could

be explained with structure (II) by involving the Wagner-Meerwein rearrangement.

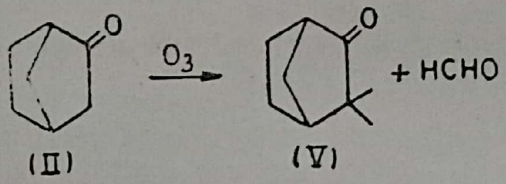


(7) *Confirmation of structure (II).* The structure (II) for camphene has been supported by the following facts :

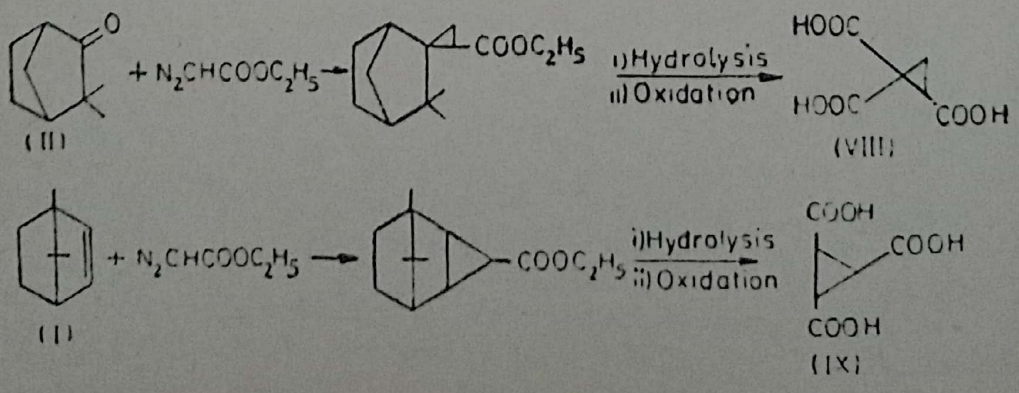
(*a*) When bornyl iodide is treated with ethanolic potassium hydroxide at 170°C, it yields born-2-ene (bornylene) $C_{10}H_{16}$ (m.p. 98°C) as well as camphene (51—52°C). Born-2-ene, when treated with permanganate, undergoes oxidation readily to form camphoric acid. This could be explained if born-2-ene has structure (I) which was originally assigned to camphene. Also, no rearrangement takes place in the formation of born-2-ene.



(*b*) When camphene is ozonolysed, it yields camphenilone (V) and formaldehyde. The formation of these products could only be explained by the Wagner formula (II) for camphene.

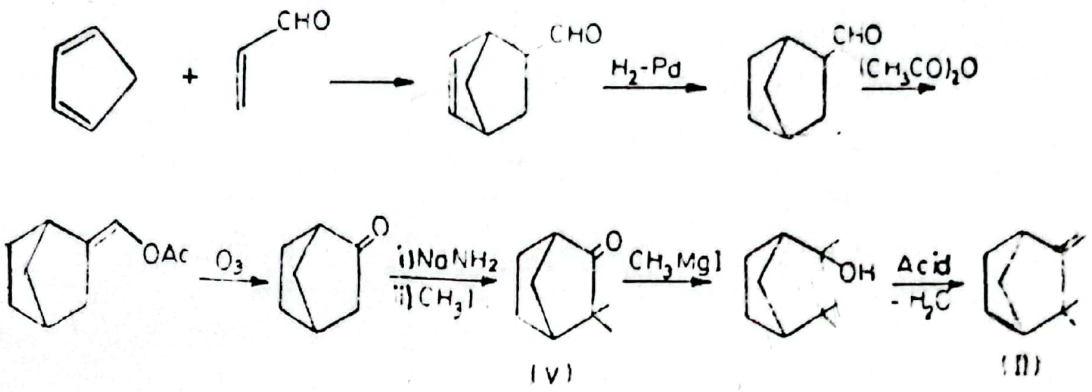


(*c*) Camphene when reacted with diazoacetic ester yields the product which upon hydrolysis, followed by oxidation yields cyclopropane—1, 1, 2—tricarboxylic acid (VIII). The formation of the (VIII) could be explained by structure (II) but not (I) because (I) (born-2-ene) would yield cyclopropane-1,2,3-tricarboxylic acid (IX).



(*d*) Camphenic acid was synthesised by Lipp (1914). This synthesis showed that it has the structure (VII) assigned to it by Wagner.

8. *Synthesis.* Finally, the structure (II) of camphene has been confirmed by its synthesis given by Diels and Alder (1928—1931) from cyclopentadiene.



1.46. NORBORNANE (FENCHANE) GROUP

The most important member of this group of terpenoids is fenchone which is a ketone and will be discussed in the following article.

1.47. FENCHONE

Introduction. It belongs to norbornane group of terpenoids. It is a ketone which is isomeric with camphor. It occurs in the oil of fennel and oil of thuja. It is optically active liquid having b.p. 192—193°C and both its enantiomers occur naturally.

It resembles camphor in many properties but is distinguished from it by its greater resistance to oxidation.

It is interesting to note that fenchone as well as its related compounds do not obey the isoprene rule.

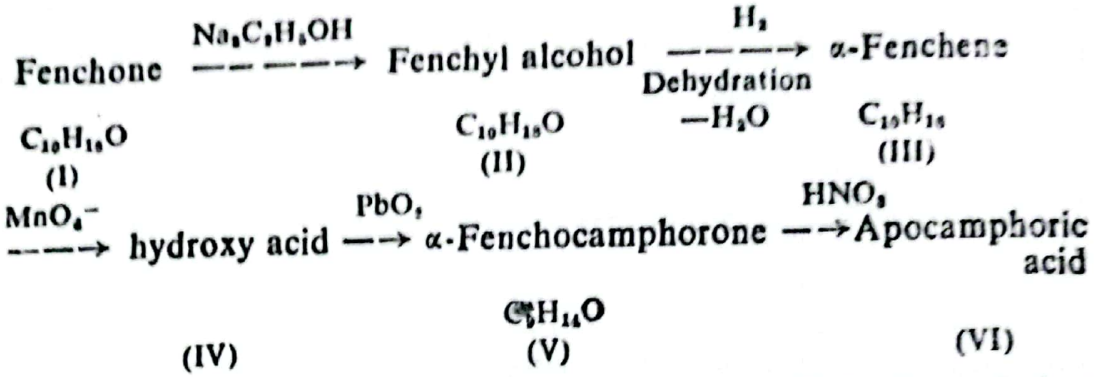
Constitution. The constitution of fenchone has been established on the basis of following analytical and synthetic evidences :

1. From analytical data, the molecular formula of fenchone has been found to be $C_{10}H_{16}O$.

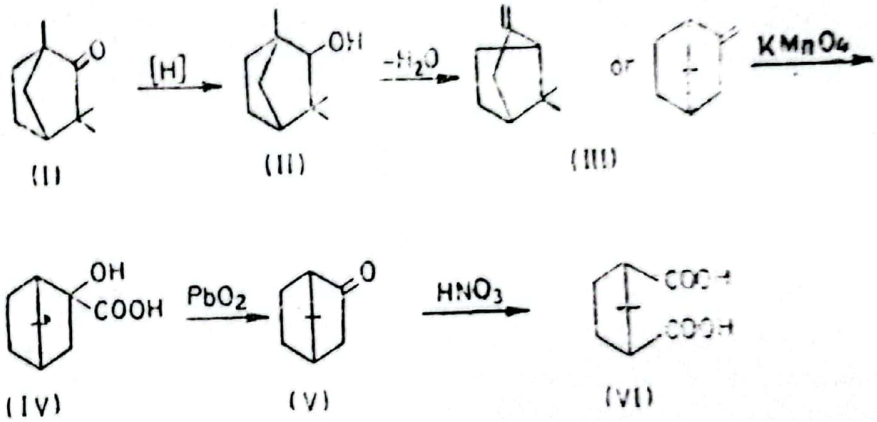
2. By the usual tests, it is shown that fenchone is a saturated ketone.

3. The fenchone (I), when reduced with sodium and alcohol, yields fenchyl alcohol, $C_{10}H_{18}O$ (II) which on dehydration under the influence of acids yields α-fenchene, $C_{10}H_{16}$ (III). The latter compound (III) when oxidised with permanganate yields the hydroxy-acid (IV) which on treatment with lead dioxide is converted into α-fenchocamphorone, $C_9H_{14}O$ (V). This on further oxidation with

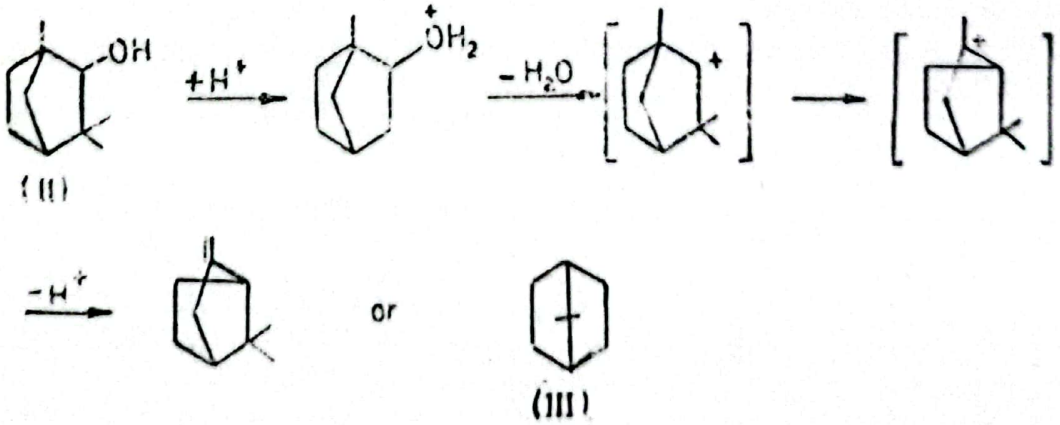
nitric acid yields apocamphoric acid (VI), a compound of known structure.



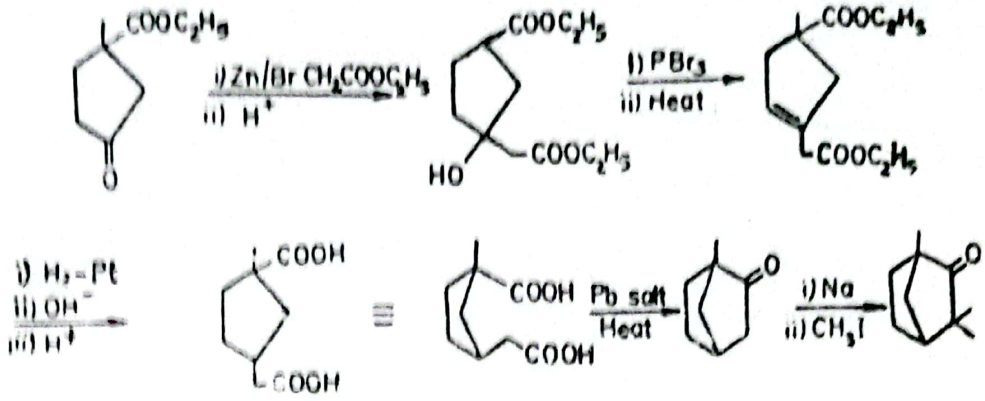
All the above reactions were carried out by Wallach *et al.* but it was Semmler (1909) who assigned the correct structure to fenchone. On the basis of this structure, all the foregoing reactions could be explained as follows :



It is interesting to note that the dehydration of fenchyl alcohol (II) to α-fenchene (III) takes place *via* a Wagner-Meerwein rearrangement. The mechanism for this rearrangement may be depicted as follows :



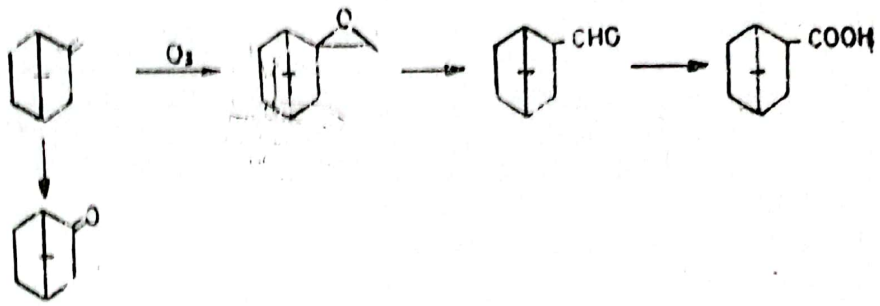
4. *Synthesis.* Finally, the structure (I) of fenchone has been confirmed by its synthesis given by Ruzicka (1917).



Reactions. These are as follows :

(*i*) The most interesting reaction is the dehydration of fenchyl alcohol (II) to α-fenchene (III). The mechanism of this reaction is given in step (3) [see above].

(*ii*) Another interesting reaction is the ozonolysis of fenchene because it yields some unexpected products. This reaction was studied by Komppa *et al* (1929).



1.48. SESQUITERPENOIDS

This group of terpenoids incorporates the hydrocarbons of the general formula $C_{15}H_{24}$ and their oxygenated derivatives. In general, sesquiterpenoids form the higher boiling fraction of the essential oils and this fraction constitutes their chief source.

On the basis of the number of rings present, the sesquiterpenoids are divided into four groups :

(*a*) Acyclic sesquiterpenoids,

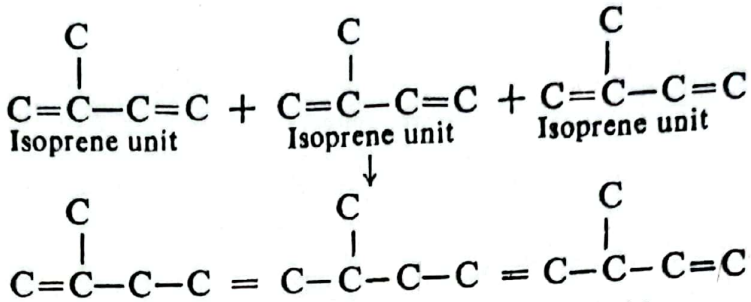
(*b*) Monocyclic sesquiterpenoids,

(*c*) Bicyclic sesquiterpenoids, and

(*d*) Tricyclic sesquiterpenoids.

Wallach (1887) suggested that three isoprene units are present in the structure of all the sesquiterpenoids, This has been found in most of the sesquiterpenoids because some exceptions to this are known.

When isoprene rule is applied to acyclic sesquiterpenoids, it is found that three isoprene units are linked head to tail to form these terpenoids and each of these will contain **four** double bonds. Each isoprene unit contains two double bonds but one disappears when each pair is connected. This is shown as follows :



Carbon skeleton of Acyclic Sesquiterpenoids.

When acyclic sesquiterpenoid is converted into a monocyclic sesquiterpenoid, one more double bond is consumed and the latter, therefore, contains **three double bonds**. Similarly, it can be proved that a bicyclic structure contains **two double bonds** and a tricyclic **one**.

From the number of double bonds present in the sesquiterpenoid molecule, it becomes possible to elucidate the nature of the terpenoid skeleton. Similarly, from the calculation of the molecular refractions for various types of structures (Table 1.5) it becomes possible to distinguish the sesquiterpenoid hydrocarbon structures. The refraction values are quite helpful to elucidate the structures of new sesquiterpenoids, e.g., zingiberene.

TABLE 1.5

Relation between Type of Sesquiterpenoid, Number of Double Bonds and Molecular Refraction

Type of sesquiterpenoid	Molecular refraction	Number of double bonds
1. Acyclic	69.5	4
2. Monocyclic	67.8	3
3. Bicyclic	66.1	2
4. Tricyclic	64.4	1

Although data given in Table 1.5 is quite helpful in elucidating the tentative structures of many sesquiterpenoids, yet their exact structures are established only by ozonolysis, oxidative degradations and sulphur or selenium dehydrogenation reactions.

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ACYCLIC SESQUITERPENOIDS

1.49. FARNESOL

Introduction. It is an acyclic sesquiterpenoid which occurs in the oil of ambrette seeds, rose, citronella, lime flowers and seville orange oil.

It is a liquid having b p. 120°C at 0.3 mm pressure.

Constitution. It was elucidated by Kerschbaum (1913) on the basis of following facts :

1. From analytical data, the molecular formula of fernesol has been found to be $C_{15}H_{26}O$.

2. When hydrogenated, farnesol adds on three moles of hydrogen to form hexahydro farnesol

$$\underset{\text{Farnesol}}{C_{15}H_{26}O} + 3H_2 \longrightarrow \underset{\text{Hexahydro farnesol}}{C_{15}H_{32}O}$$

The above reaction reveals that farnesol must possess three double bonds. This is further confirmed by the fact that farnesol adds on three moles of bromine to form hexabromo farnesol.

3. When oxidised with chromic acid, farnesol is converted into fernesal, $C_{15}H_{24}O$. As the latter compound is an aldehyde, it means that farnesol is a **primary alcohol.**

$$\underset{\text{Farnesol}}{C_{14}H_{23}CH_2OH} \xrightarrow[CrO_3]{(O)} \underset{\text{Farnesal}}{C_{14}H_{23}-CHO}$$

The complete structure of farnesol is elucidated as follows :

When oxidised with chromic acid, farnesol (I) is oxidised to farnesol (II), $C_{15}H_{24}O$. The latter compound is an aldehyde which upon treatment with hydroxylamine yields oxime that upon treatment with acetic anhydride undergoes dehydration to form cyanide (III). This when hydrolysed with alkali yields a mixture of farnesenic acid (IV) $C_{15}H_{24}O_2$ and ketone (V), $C_{13}H_{22}O$.

$$\underset{\text{Farnesol (I)}}{C_{14}H_{23}CH_2OH} \xrightarrow[(O)]{CrO_3} \underset{\text{Farnesal (II)}}{C_{14}H_{23}CHO} \xrightarrow{NH_2OH}$$

$$C_{14}H_{23}CH=NOH \xrightarrow[\text{Dehydration}]{Ac_2O}$$

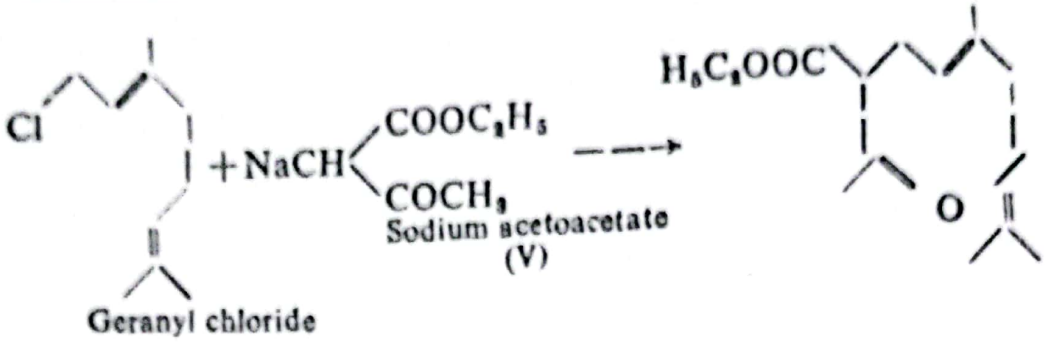
Farnesal oxime

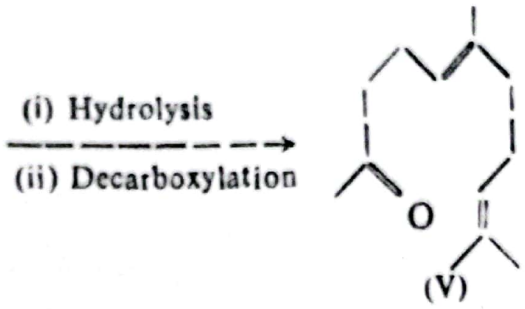
$$C_{14}H_{23}C\equiv N \xrightarrow[\text{hydrolysis}]{\text{Alkaline}} \underset{\text{Farnesenic acid (IV)}}{C_{14}H_{23}COOH} + \underset{\text{Ketone (V)}}{C_{13}H_{22}O}$$

Farnesal nitrile (I)

The ketone (V) was found to be dihydro-pseudoionone (geranylacetone) which is identical with dihydro-pseudo-ionone

(geranyl acetone) obtained from geranyl chloride and sodium acetoacetate.



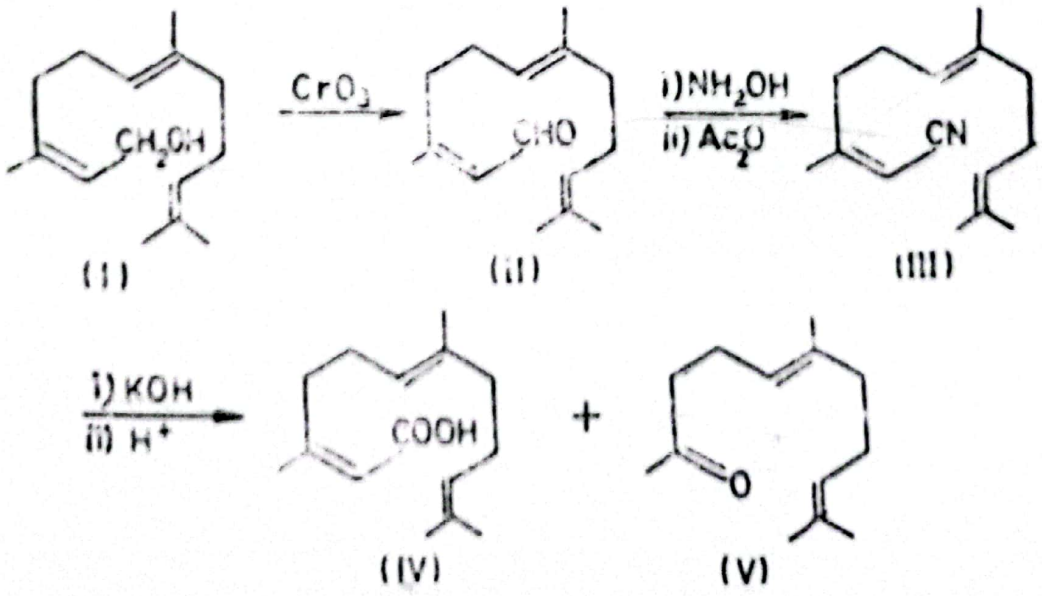


In the formation of this ketone (V), two carbon atoms are removed from its precursor. This reaction is considered to be a characteristic of α, β—unsaturated carbonyl compounds and, thus, it is concluded that the precursor is farnesenic acid and hence its nitrile is an α, β—unsaturated compound.

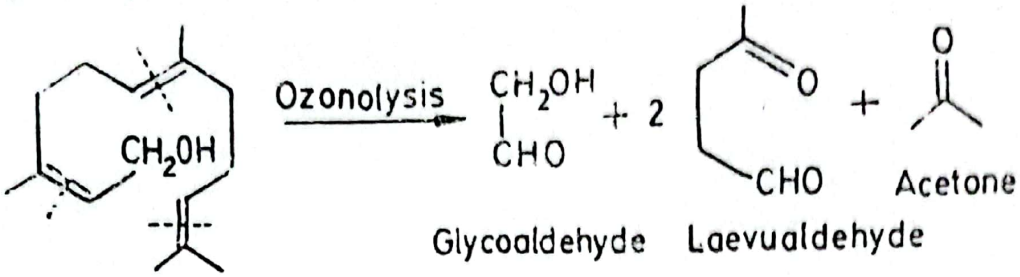
$$R-\overset{|}{C}=CH-COR' \xrightarrow[(+H_2O)]{OH^-} R-\overset{|}{C}=O+CH_3COR'$$

where R=H, or alkyl radical.

All the above reactions could be formulated as follows on the basis of the known structure of geranyl acetone.



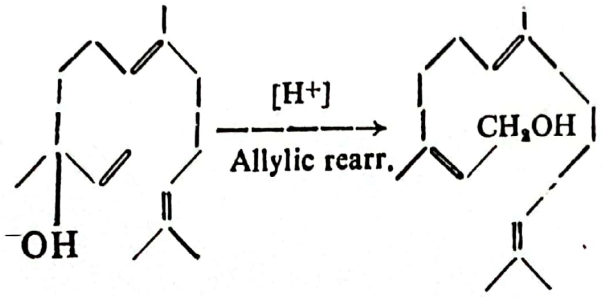
(5) The structure (I) for farnesol has been further confirmed by its ozonolysis to acetone, laevualdehyde and glycoaldehyde (Harries, 1913).



(I)

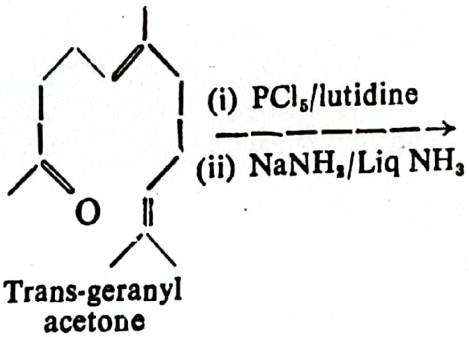
6. *Synthesis.* Finally, the structure (I) of farnesol has been confirmed by its synthesis.

(*a*) *Ruzicka's synthesis* (1923). According to this synthesis, farnesol is prepared by treating synthetic nerolidol with an acid. This preparation is analogous to the conversion of linalool to geraniol. Nerolidol is already synthesised from geranyl chloride.

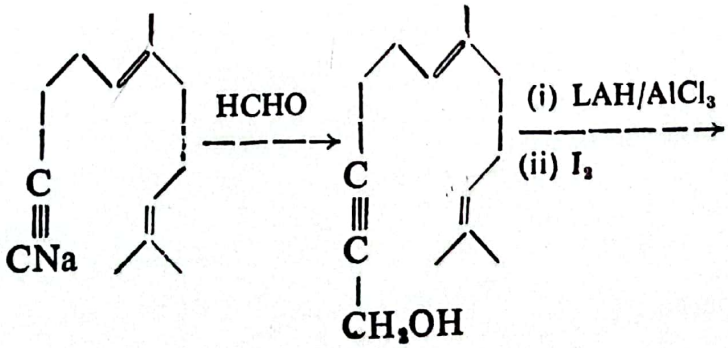


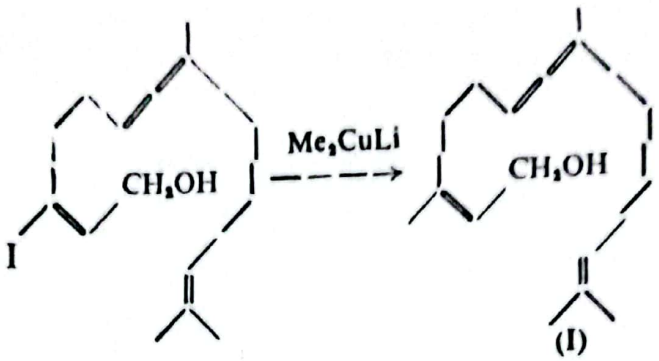
Nerolidol Farnesol (I)

(*b*) *Corey's synthesis* (1967). This is a recent synthesis of farnesol.



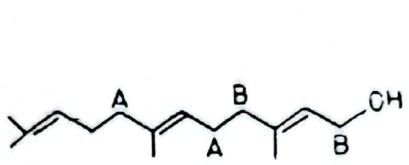
Trans-geranyl acetone



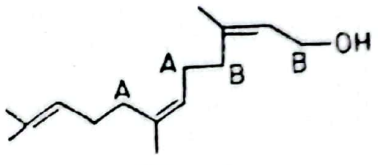


(I)

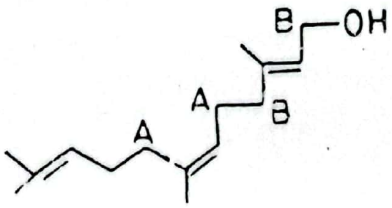
7. *Isomerism in farnesol.* Various workers have prepared a number of geometrical isomers of farnesol.



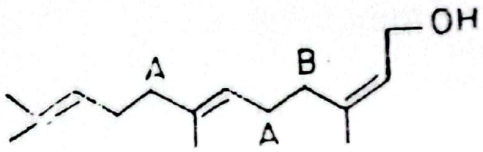
Trans-trans



Cis-cis



Trans-cis



Cis - trans

In the above geometrical isomers, A and B denote each pair. Also, the prefixes *cis* and *trans* have been used to represent the positions of methylene groups (in the main chain) with respect to each other for each double bond (in the chain) and at the end of the chain the position of the functional group with respect to the methylene group of the chain.

1.50. NEROLIDOL

Introduction. It is an acyclic sesquiterpenoid which is isomeric with farnesol. This occurs naturally in the (+)- form in the oil of neroli which is obtained from the flowers of bitter orange.

It is a liquid which boils at 125—127°C at 4.5 mm pressure. It is optically active.

Constitution. Ruzicka (1923) showed that the relationship between nerolidol and farnesol is the same as that between linalool and geraniol. However, the constitution of nerolidol is elucidated on the basis of following analytical and synthetic evidences :

1. From analytical data, the molecular formula of nerolidol has been found to be $C_{15}H_{26}O$.

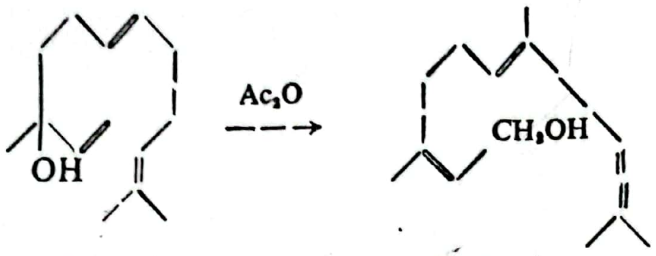
2. As nerolidol adds on three moles of hydrogen and bromine, this means that it contains three double bonds.

3. When treated with acetic anhydride, nerolidol is isomerised to farnesol. This reaction is similar to the isomerisation of linalool to geraniol. Moreover, as linalool yields citral on oxidation with chromic acid, nerolidol yields farnesal on oxidation under similar conditions. These facts clearly reveal that relationship between nerolidol and farnesol is the same as that between linalool and geraniol.

$$\text{Geraniol} \xleftarrow{Ac_2O} \text{Linalool} \xrightarrow{CrO_3} \text{Citral}$$

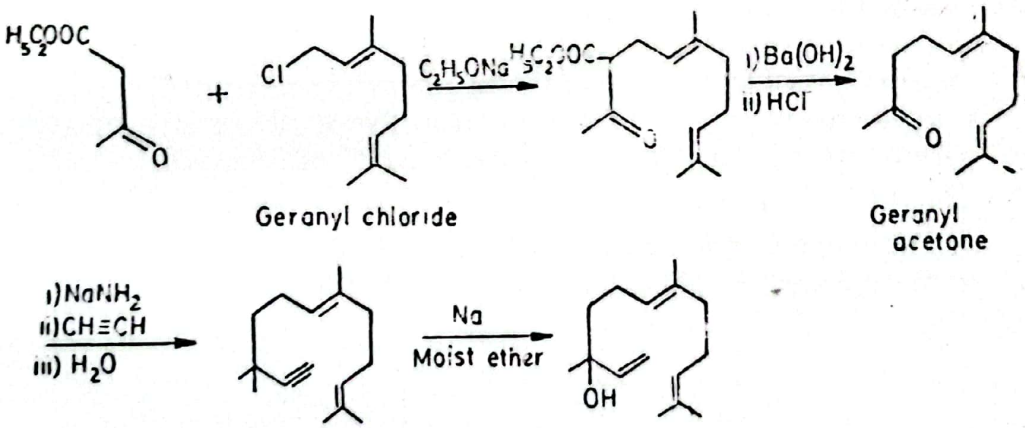
$$\text{Farnesol} \xleftarrow{Ac_2O} \text{Nerolidol} \xrightarrow{CrO_3} \text{Farnesal}$$

On the basis of analogous relation and the known structure of farnesol, nerolidol was assigned the structure (I).

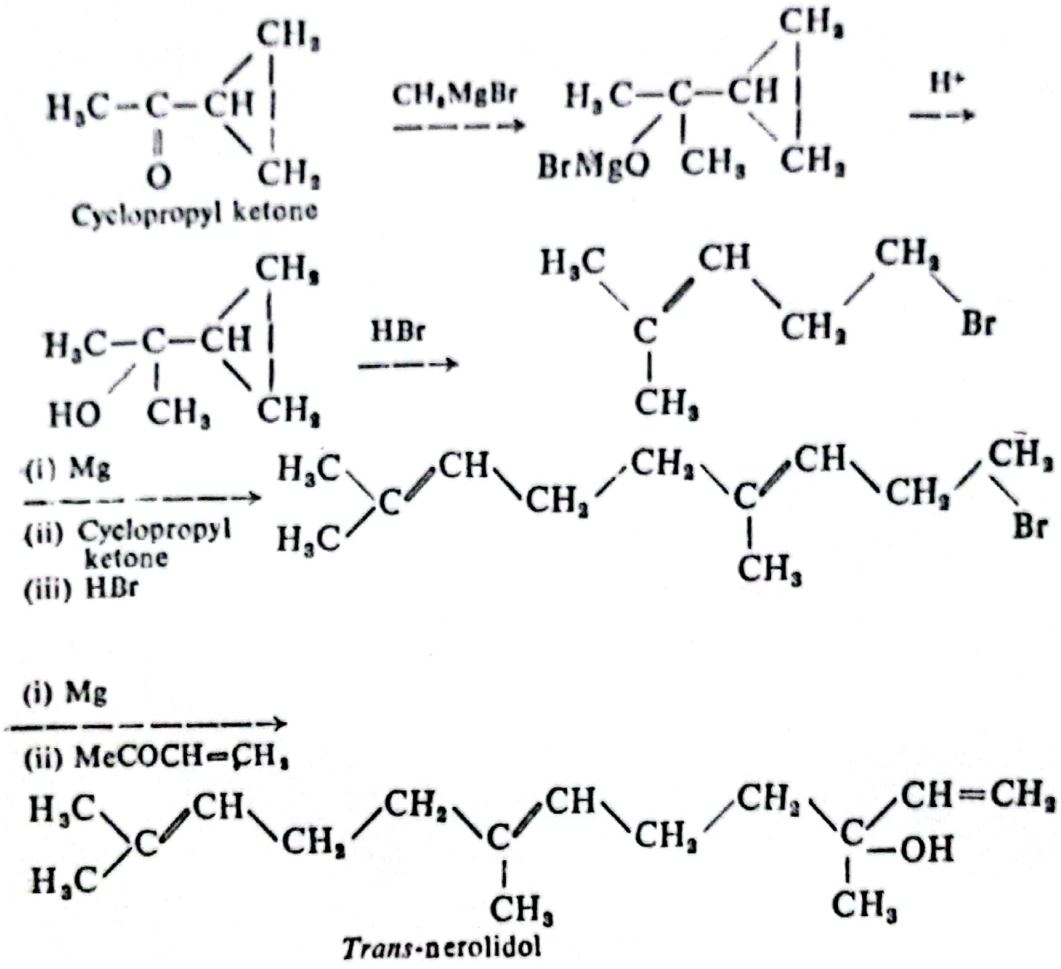


(+) Nerolidol (I) — Farnesol

4. *Synthesis.* The structure (I) for nerolidol has been confirmed by its synthesis given by Ruzicka (1923) starting from geranyl chloride as follows :



Trans-nerolidol was synthesised by Julia *et al.* (1959, 1960) in which isoprene units are repeatitively introduced by reaction between a Grignard's reagent and cyclopropyl methyl ketone. The carbinol so produced, when treated with hydrobromic acid, undergoes rearrangement.

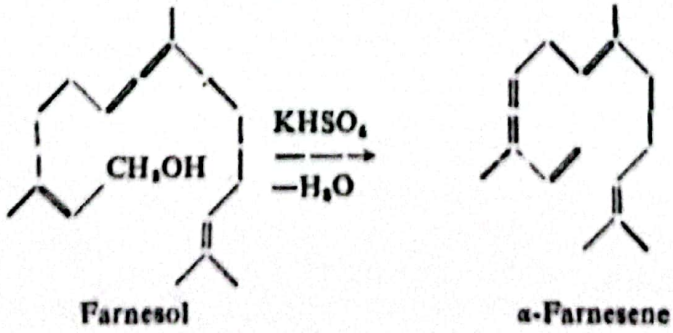


When *trans-norolidol* is *treated* with acid, it produces a mixture of **trans-trans** and **cis-trans**-farnesol. by the allylic rearrangement.

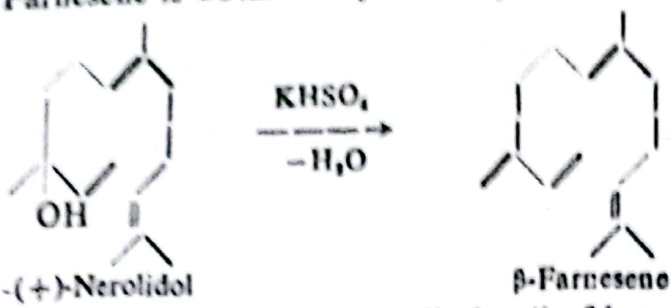
1.51. FARNESENES

The parent hydrocarbons of the two isomeric alcohols farnesol and nerolidol are α- and β- farnesenes respectively. Their structures have been elucidated by their formation from farnesol and nerolidol.

α-Farnesene is obtained by the dehydration of farnesol with potassium hydrogen sulphate (Harries *et al.* 1913).



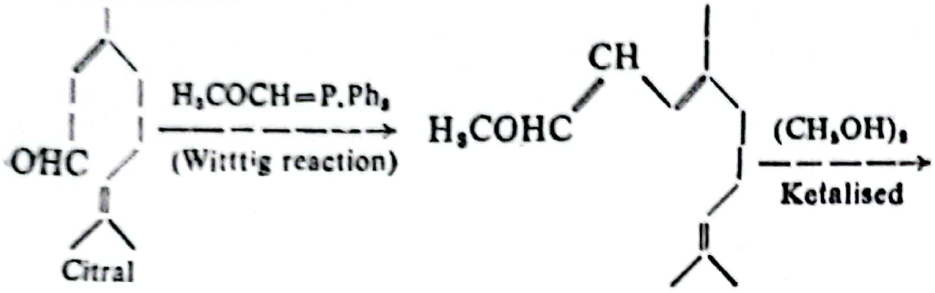
β- Farnesene is obtained by the dehydration of nerolidol.

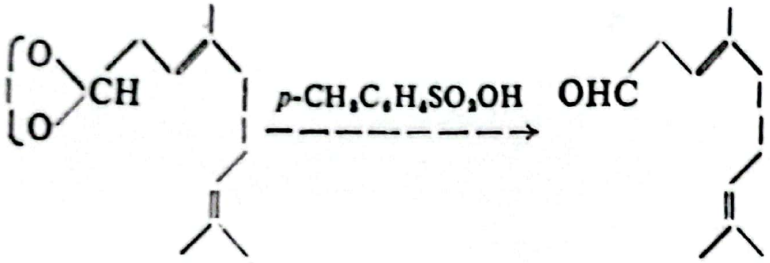


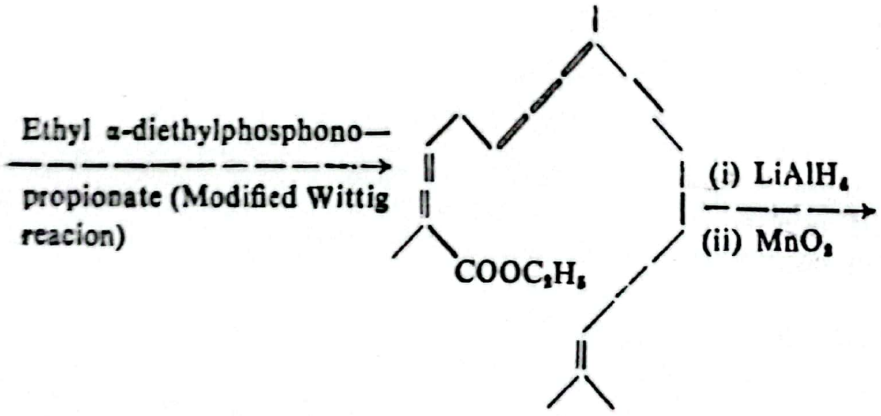
β-Farnesene also occurs naturally in oil of hops.

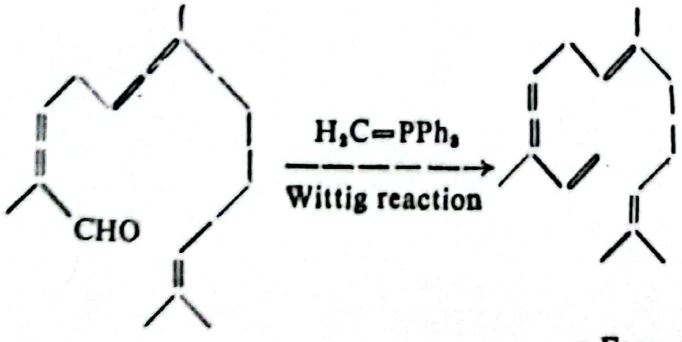
Synthesis. α- and β-Farnesenes have been synthesised by Vig. O.P. *et al.* in 1970.

α-*Farnesene.*



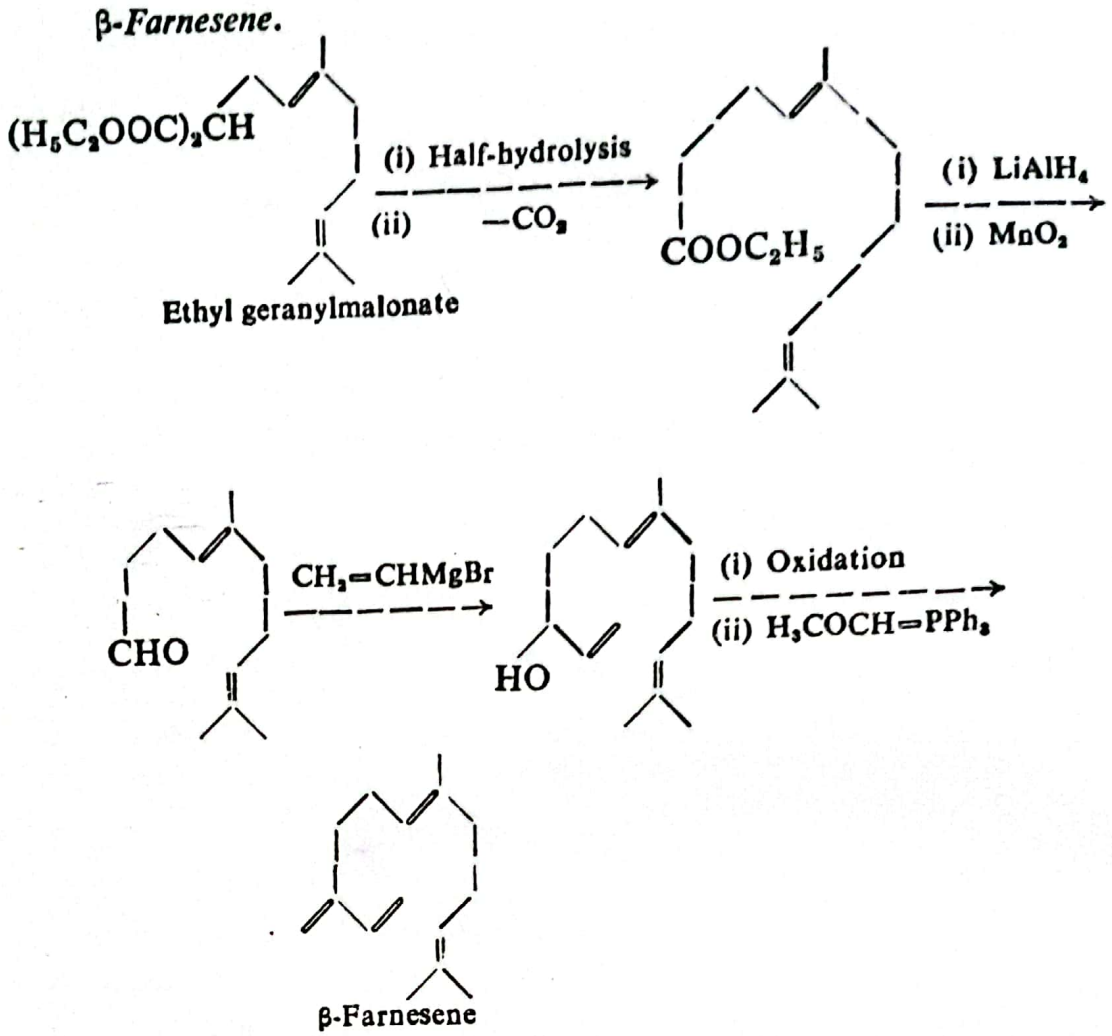






α-Farnesene

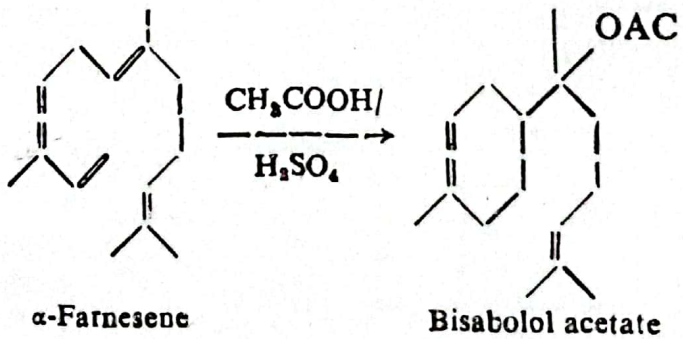
β-*Farnesene.*



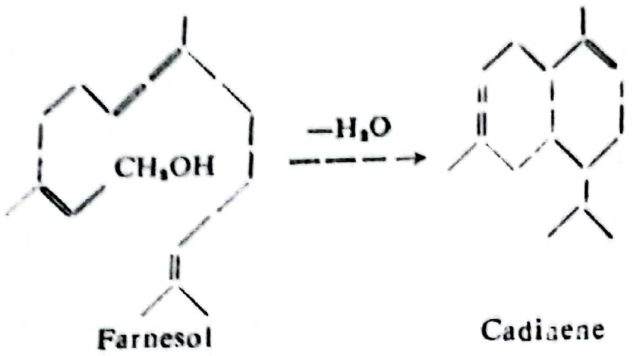
1.52. CYCLISATION OF ACYCLIC SESQUITERPENOIDS

The following conversions illustrate the relationship between acyclic and monocyclic sesquiterpenoids.

(*a*) *α-Farnesene to bisabolol.* When α-farnesene is treated with cold acetic acid in the presence of sulphuric acid for some hours, it is converted into α-bisabolol acetate.



(*b*) *Farnesol into cadinene.* When dehydrated, farnesol is converted into bicyclic sesquiterpenoid called cadinene.



MONOCYCLIC SESQUITERPENOIDS

1.53. BISABOLENE

Introduction It is a monocyclic sesquiterpenoid which occurs in the essential oils of myrrh, carrot and bargamot. It is isomeric with zingiberene. It is a liquid which boils at 133—134°C at 12 mm pressure.

Constitution. This was determined by Ruzicka *et al.* (1925) on the basis of following analytical and synthetic evidences.

1. From analytical data, the molecular formula of bisabolene has been found to be $C_{15}H_{24}$.

2. When treated with hydrogen chloride, bisabolene adds on its three moles to form trihydrochloride.

The latter compound, when heated with sodium acetate in acetic acid, gives the parent hydrocarbon, bisabolene.

$$\underset{\text{Bisabolene}}{C_{15}H_{24}} \xrightleftharpoons[CH_3COONa/\,CH_3COOH]{3\,HCl} \underset{\text{Bisabolene trihydrochloride}}{C_{15}H_{24}.3HCl}$$

The above reacion indicates that bisabolene contains three double bonds. Thus, the molecular formula of saturated parent hydrocarbon of bisabolene is $C_{15}H_{30}$ which corresponds to the general formula C_nH_{2n} for monocyclic hydrocarbons and, therefore, bisabolene is monocyclic.

3. Its exact structure was established on the basis of fact that nerolidol on dehydration yields a mixture of α- and β-farnesenes. This mixture when treated with formic acid yields a monocyclic sesquiterpenoid (or possibly a mixture) which reacts with hydrogen chloride to yield bisabolene trihydrochloride. The latter compound when treated with sodium acetate in acetic acid loses its three molecules of hydrogen chloride to produce bisabolene ; thus bisabolene might be (I), (II) or (III) because all three would yield the same bisabolene trihydrochloride.

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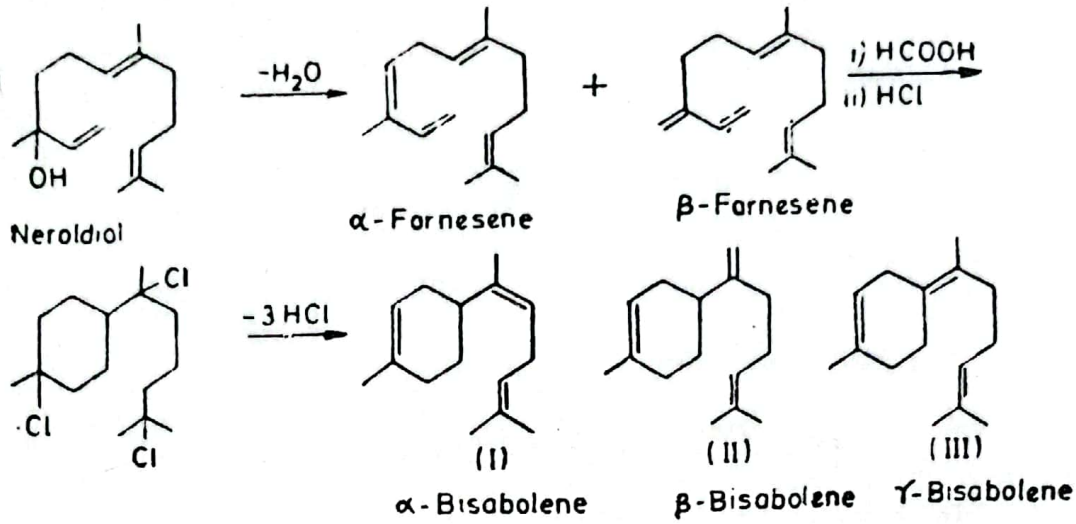
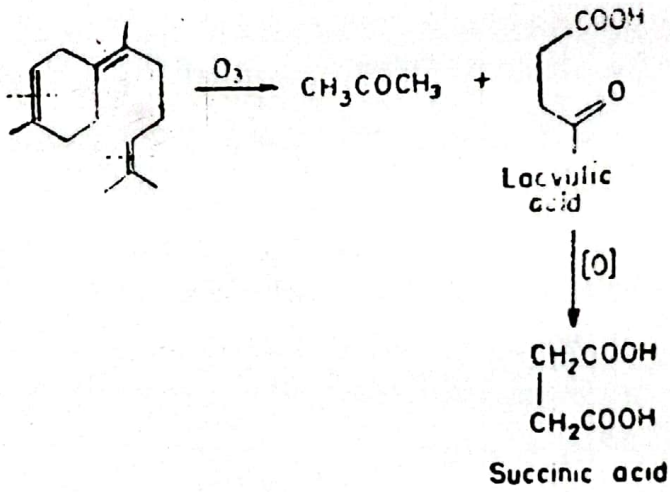
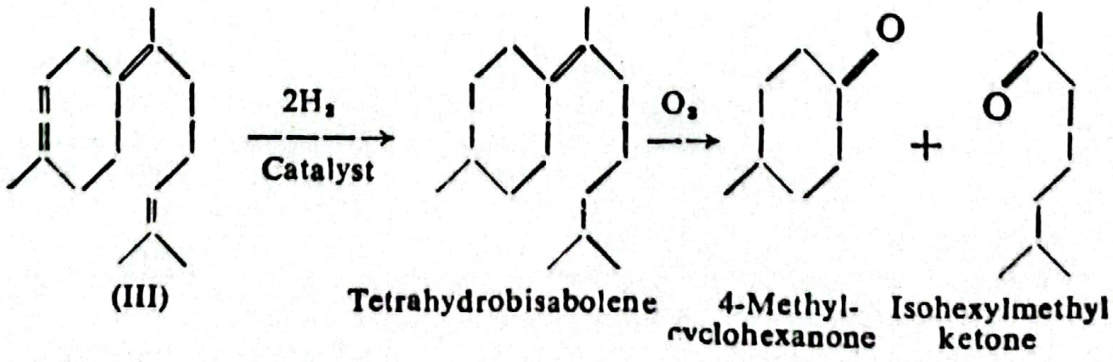
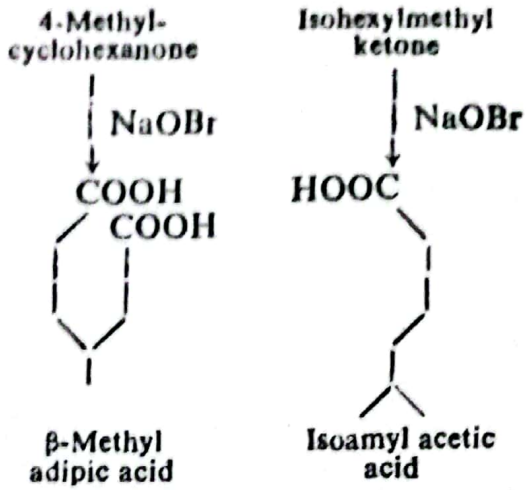


4. When bisabolene is ozonolysed, it yields acetone, laevulic acid and a small amount of succinic acid. The formation of these products could only be explained by structure (III).



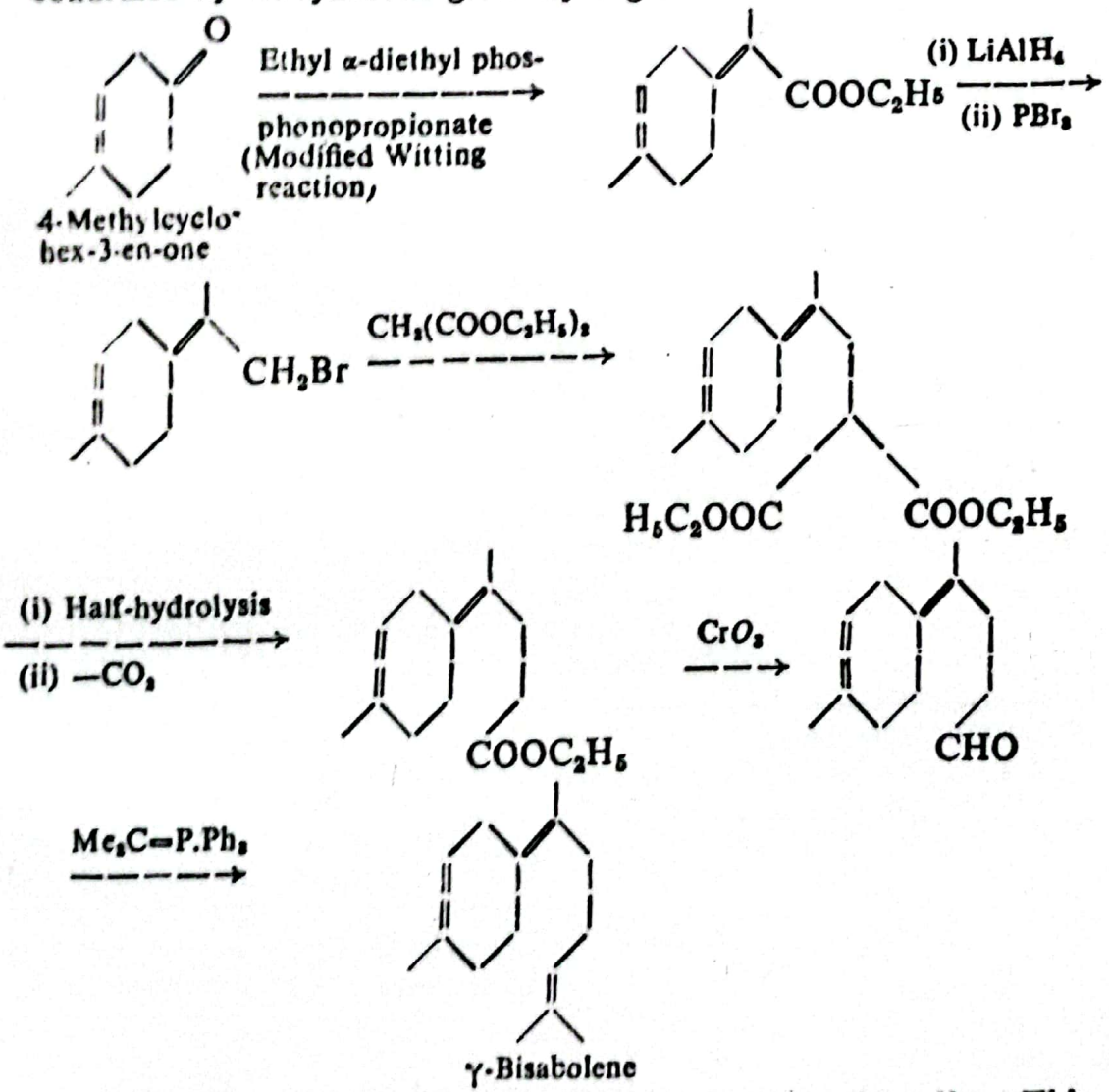
5. Catalytic reduction of bisabolene in neutral solution yields tetrahydrobisabolene. The latter compound on ozonolysis yields isohexylmethyl ketone and 4-methylcyclohexanone. These two compounds when further oxidised with sodium hypobromite yield isoamyl acetic acid and β-methyl adipic acid respectively. The formation of these compounds could only explained on the basis of structure (III) for bisabolene.



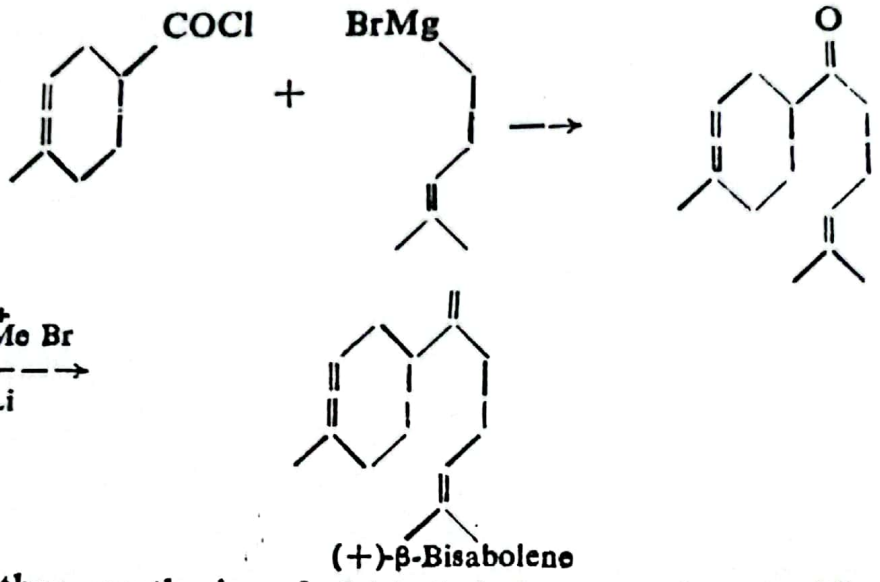


6. Ruzicka *et al.* (1929) showed that synthetic and natural bisabolene consisted mainly of the γ-isomer. However, α and β-isomers also occur naturally but to a small extent only.

7. *Synthesis.* Finally, the structure of γ-bisabolene has been confirmed by its synthesis given by Vig *et al.*

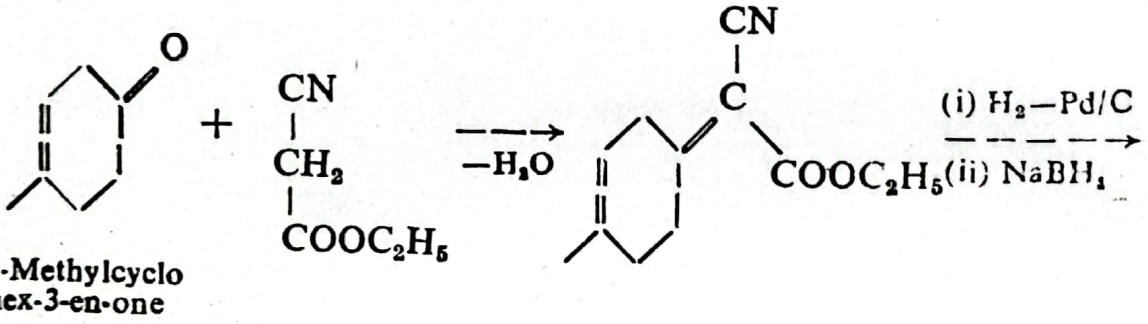


β-*Bisabolene.* This has been found to occur naturally. This has been synthesised by Manjarrez *et al.* (1966) *via* the Wittig reaction.

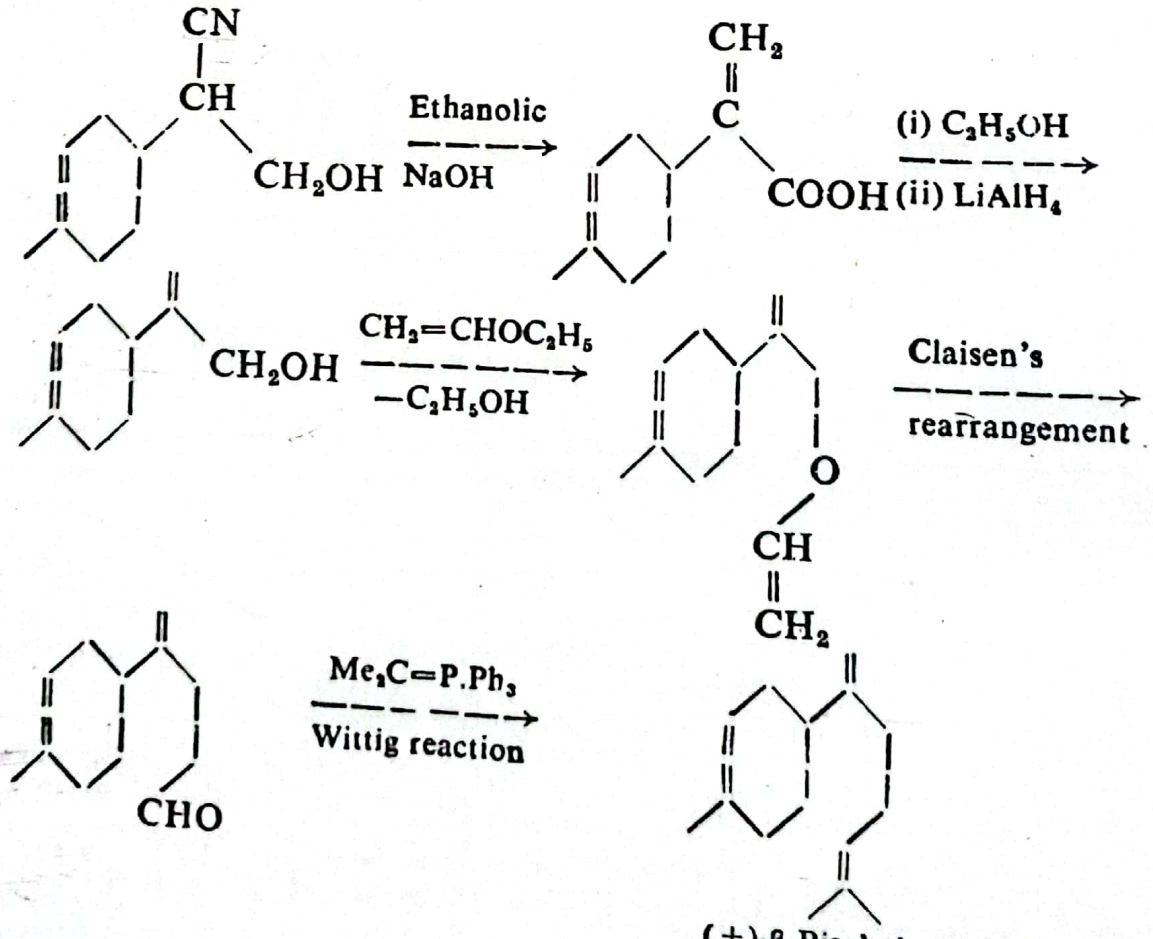


(+)-β-Bisabolene

Another synthesis of β-bisabolene was given by Vig *et al.* (1967).

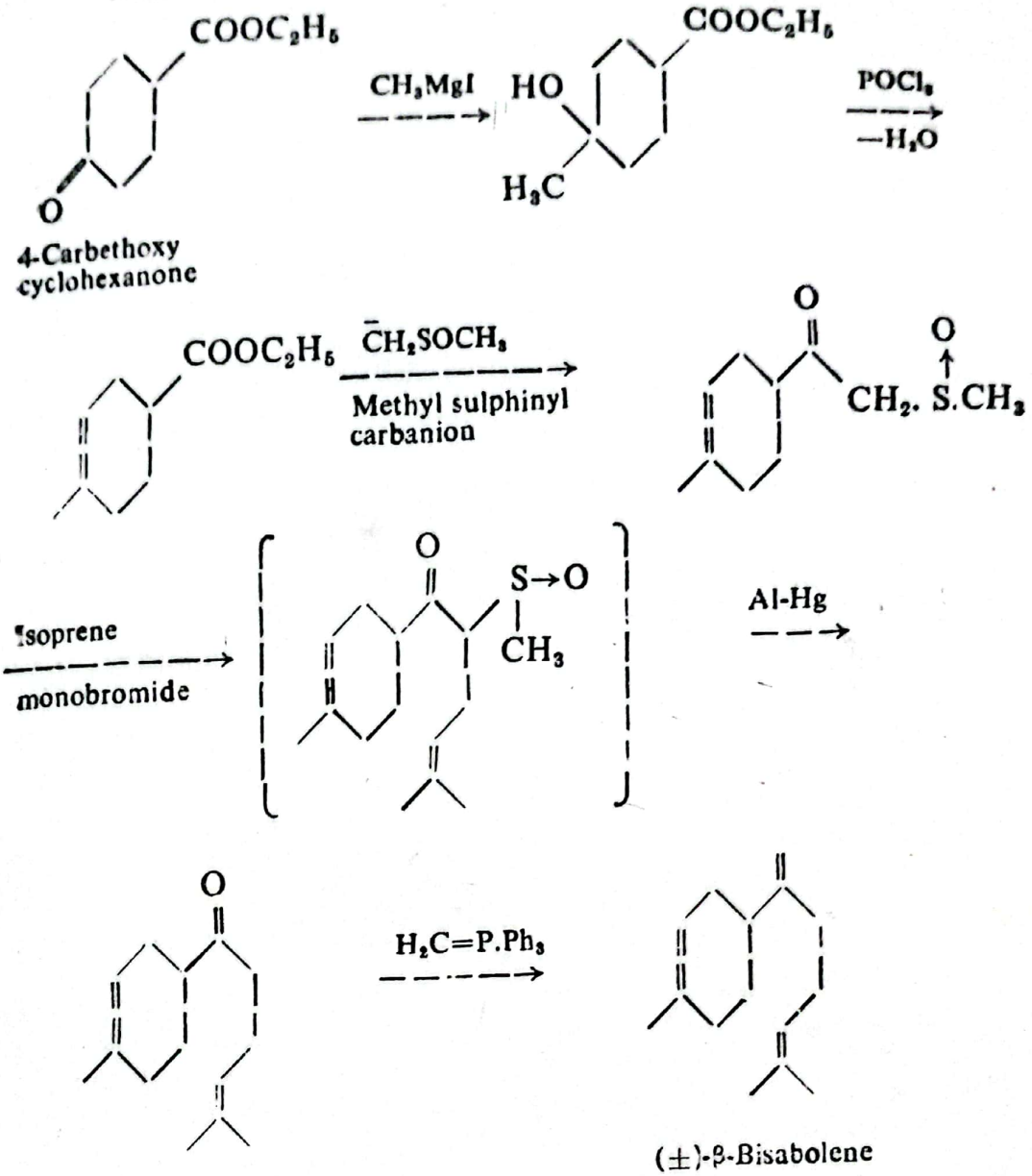


4-Methylcyclo hex-3-en-one



(±)-β-Bisabolene

Another synthesis was also given by Vig *et al.* in 1971.



1.54. ZINGIBERENE

Introduction. It is a monocyclic sesquiterpenoid which occurs in the (−) form in ginger oil. It is an optically active liquid having b.p. 134°C at 14 mm pressure.

Constitution. It is based on the following analytical and synthetic evidences :

1. From analytical data and molecular weight determination, the molecular formula of zingiberene has been found to be $C_{15}H_{24}$.

2. As zingiberene forms dihydrochloride with hydrogen chloride, this shows that it contains **two double bonds.** The molecular refraction, however, reveals the presence of **three double bonds.** This has been confirmed by the fact that zingiberene when

reduced with hydrogen in the presence of platinum catalyst] yields hexahydrozingiberene, $C_{15}H_{30}$.

$$\underset{\text{Zingiberene}}{C_{15}H_{24}} + 3H_2 \xrightarrow[\text{Black}]{Pt} \underset{\text{Hexahydro zingiberene}}{C_{15}H_{30}}$$

As the saturated parent hydrocarbon of zingiberene possesses the formula $C_{15}H_{30}$ which corresponds to C_nH_{2n} for monocyclic compounds and, therefore, zingiberene is a monocyclic compound.

3. It is interesting to note that zingiberene when reduced by means of sodium and ethanol yields dihydrozingiberene $C_{15}H_{26}$. This reaction shows that two of the double bonds are probably conjugated. This has been confirmed by the following facts :

(*i*) It forms an adduct with maleic anhydride.

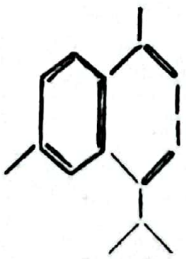
(*ii*) Zingiberene shows optical exaltation whereas dihydrozingberene does not, i.e, dihydrozingberene has molecular refraction of 68.37 which is similar to calculated value (68.25) of dihydrozingiberene on the assumption that dihydrozingiberene contains isolated double bonds which are produced by the reduction of one of the conjagated double bonds of zingiberene.

$$\underset{\substack{\text{Zingiberene}\\\text{(Shows optical}\\\text{exaltation)}}}{C_{15}H_{24}} + 4[H] \longrightarrow \underset{\substack{\text{Dihydrozingiberene}\\\text{(does not show optical}\\\text{exaltation because it}\\\text{contains two isolated}\\\text{double bonds)}}}{C_{15}H_{26}}$$

(*iii*) The λ_{max} of zingiberene is 260 (ε 2700) nm. But the calculated value of λ_{max} on the basis of a homoannular conjugated diene system is 253+...nm whereas the value for a heteroannular system is 214+...nm. Therefore, λ_{max} also favours the conjugated system of double bonds in zingiberene.

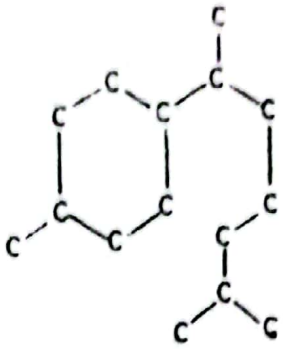
4. When zingiberene is heated with sulphur, it undergoes dehydrogenation to form cadalene. The latter compound is of known structure and has been proved to be 1, 6-dimethyl-4-isopropyl naphthalene.

$$\underset{\text{Zingiberene}}{C_{15}H_{24}} \xrightarrow[\text{Heated}]{S}$$

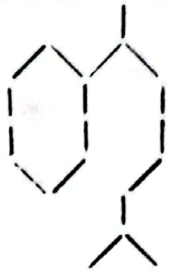


Cadalene

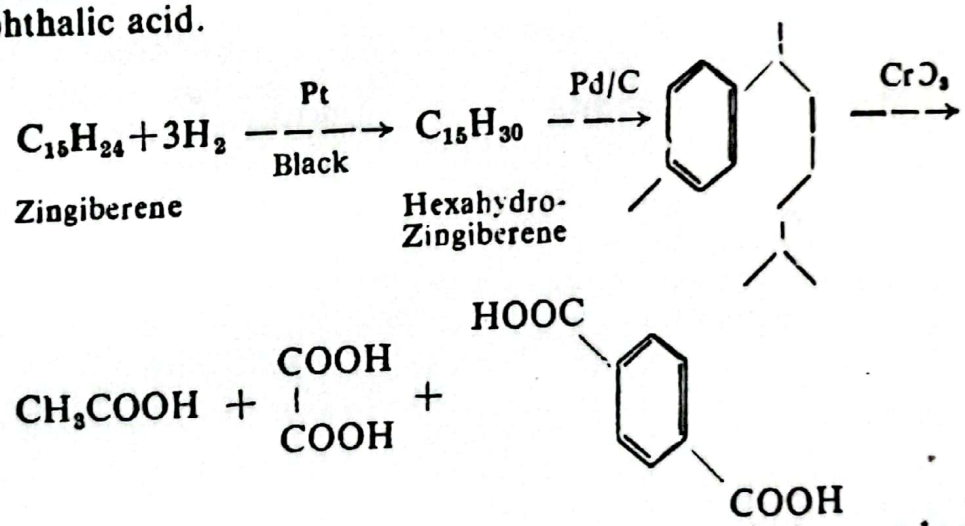
Hence in zingiberene, the following cadalene carbon skeleton must be present :



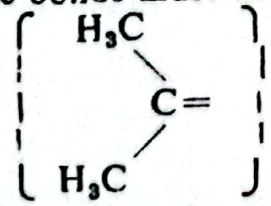
5. When zingiberene is ozonolysed, it yields acetone, laevulic acid and succinic acid. But these products are also obtained from bisabolene. Therefore, it appears that both zingiberene and bisabolene must have the same carbon skeleton.



The above carbon skeleton of zingiberene has been confirmed by the fact that hexahydrozingiberene when dehydrogenated over palladised charcoal yields 6-*p*-tolyl-2-methyl heptane (I) which upon oxidation with chromic acid yields acetic acid, oxalic acid and terephthalic acid.



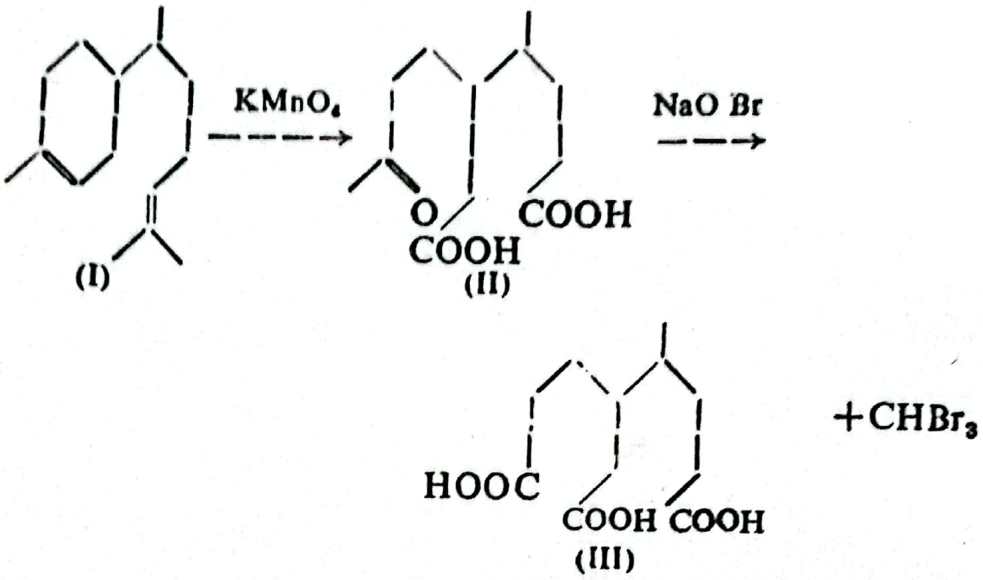
6. During ozonolysis of zingiberene, one of the products is acetone. The formation of this product reveals that one of the double bonds must be present as isopropylidene group, *i.e.*,



in zingiberene.

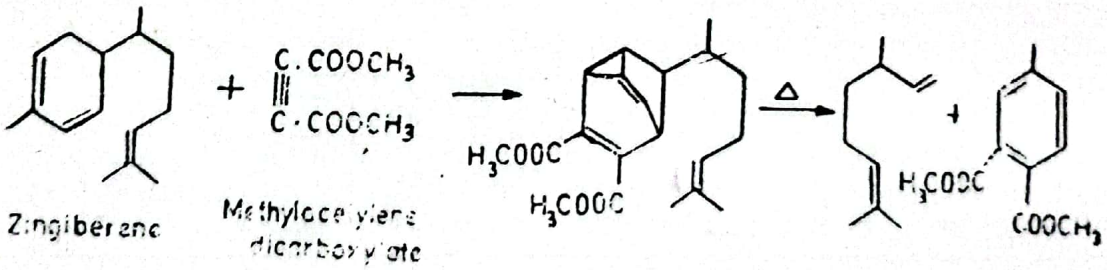
7. When the oxidation of dihydrozingiberene (I) is carried out with permanganate, it yields a ketodicarboxylic acid, $C_{12}H_{20}O_5$ (II). The latter compound when oxidised with sodium hypobromite yields a tricarboxylic acid, $C_{11}H_{18}O_6$ (III), along with bromoform.

The formation of bromoform reveals that the compound (II) must have a methyl ketone group (CH_3CO) Now if (I) is considered to be the structure of dihydrozingiberene, the foregoing reactions may be formulated as follows :



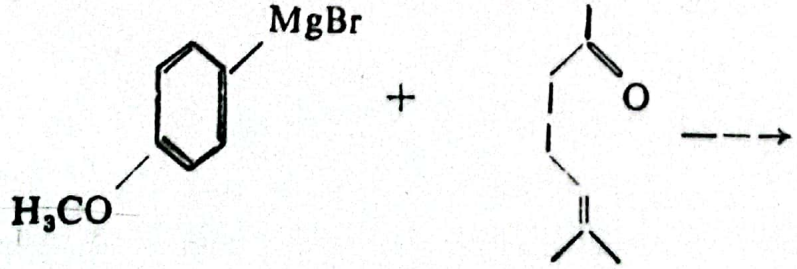
8. From the above sequence of reactions, the position of two of the three double bonds have been ascertained. Now the question arises : what is the position of the third double bond ? This can be ascertained as follows :

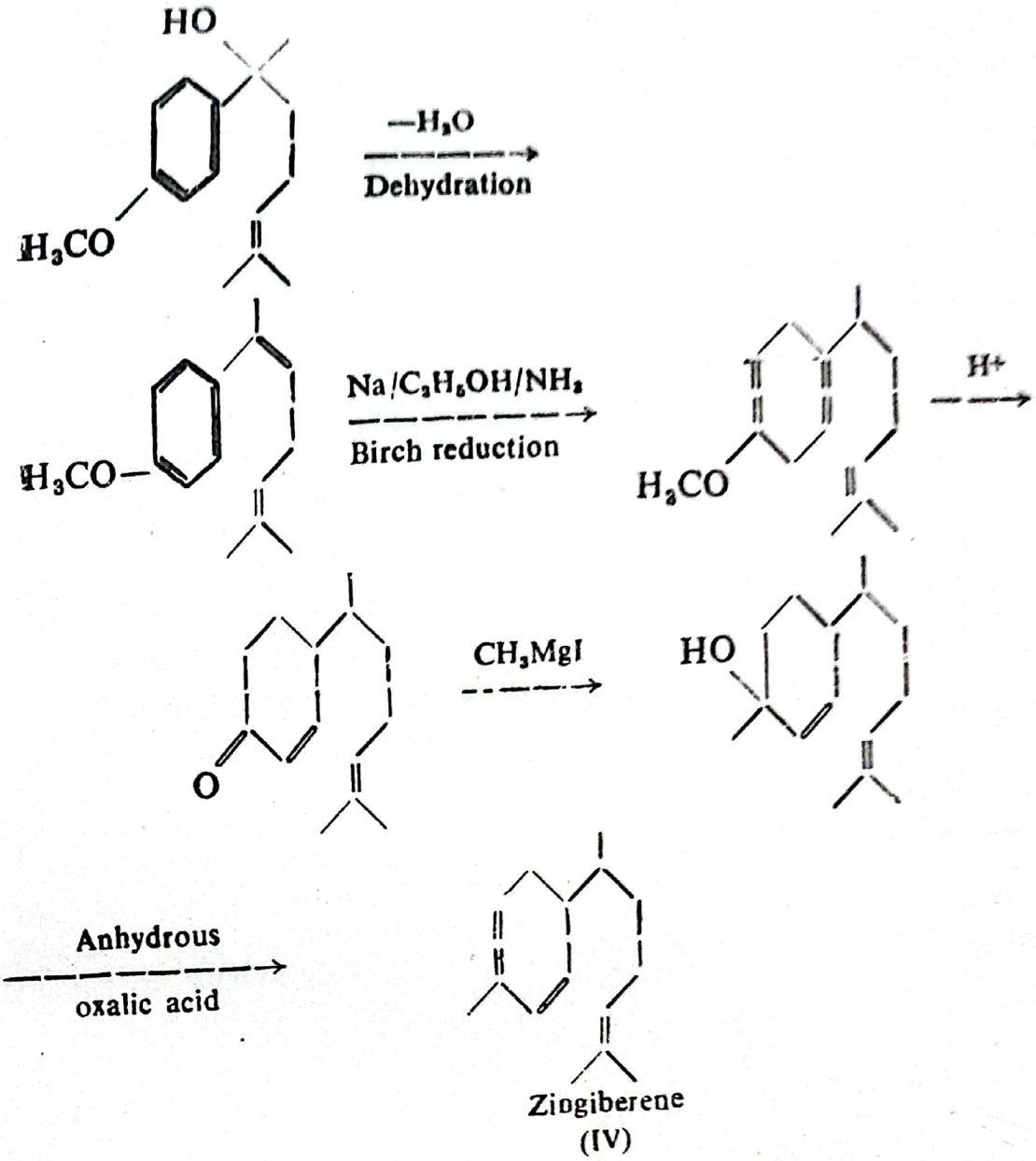
With methyl acetylenedicarboxylate, zingiberene forms an adduct (which was not isolated) but when hydrolysed it yields 1, 6-dimethylocta-3, 6-diene and methyl 4-methylphthalate. All the reactions could be explained if (IV) is the correct structure of zingiberene.



(IV)

9. *Synthesis.* Finally, the structure (IV) of zingiberene has been confirmed by its synthesis given by Bhattacharya and Mukerjee from methylheptenone and *p*-methoxymethyl magnesium bromide.





Zingiberene
(IV)

10. *Stereochemistry.* If the structure (IV) of zingiberene is examined, it is found to contain two chiral centres. As the acyclic chiral centre is stereochemically related to that in (+)-citronellal whereas the cyclic chiral centre to that in (−)-α-phellandrene, it means that (−)-zingiberene possesses the absolute configuration (IV).

BICYCLIC SESQUITERPENOIDS

1.55. α-CADINENE

Introduction. It is a bicyclic sesquiterpenoid which occurs abundantly in oil of cubebs, oleum cadinum and various other essential oils. It is an optically active liquid having b.p. 134−136°C at 11 mm pressure

Although cadinene is not employed as such, yet essential oils containing α-cadinene are used for the scenting of soaps.

Constitution. This is established on the basis of following facts :

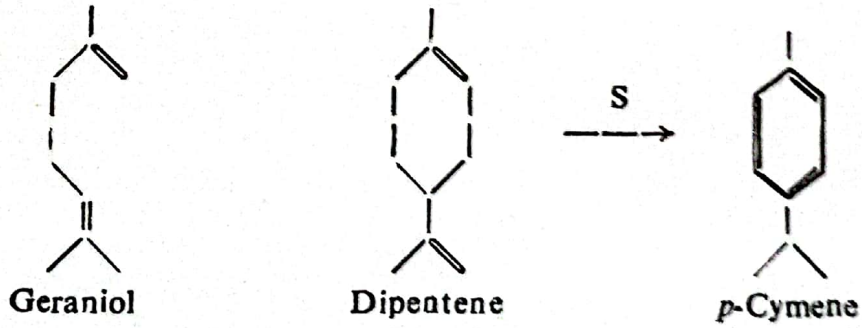
1. From analytical data, the molecular formula of α-cadinene has been found to be $C_{15}H_{24}$.

2. When catalytically reduced, it takes up two moles of hydrogen and is converted into tetrahydrocadinene. With hydrogen chloride, it also takes up its two moles to form the fully saturated cadinene dihydrochloride. Both these reactions reveal that α-cadinene contains **two double bonds** and, therefore, it is bicyclic, *i.e.*, it contains two rings in its molecule.

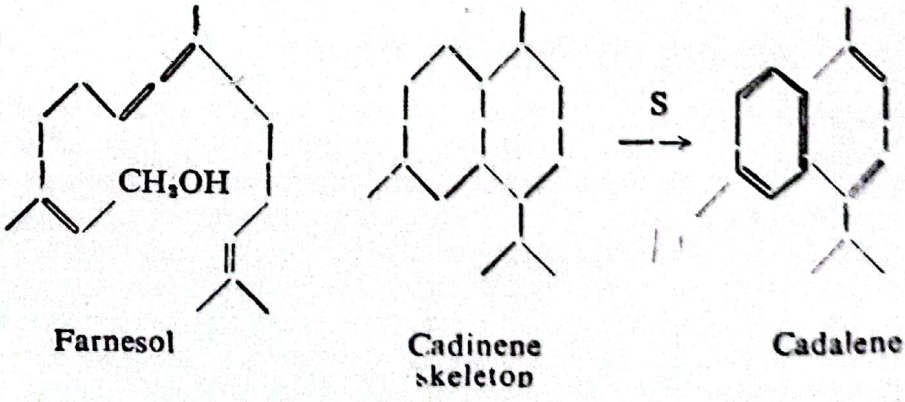
$$\underset{\text{Cadinene dihydrochloride}}{C_{15}H_{24}.2\,HCl} \longleftarrow \underset{\alpha\text{-Cadinene}}{C_{15}H_{24}} \xrightarrow{2H_2} \underset{\text{Tetrahydrocadinene}}{C_{15}H_{28}}$$

Further, molecular refractivity measurement and spectral analysis reveal that the two double bonds in α-cadinene are not present in a conjugated system.

3. When dehydrogenation of α-cadinene is carried out with sulphur, it yields cadalene, $C_{15}H_{18}$. However, cadalene does not add on bromine but forms a picrate. This reveals that cadalene is an aromatic compound. Ruzicka postulated that the relationship of farnesol to cadinene was similar to that of geraniol to dipentene. Further, dipentene, when dehydrogenated, with sulphur yields *p*-cymene. This reveals that if the above analogy is correct, then cadalene should be 1, 6-dimethyl-4-isopropylnaphthalene.

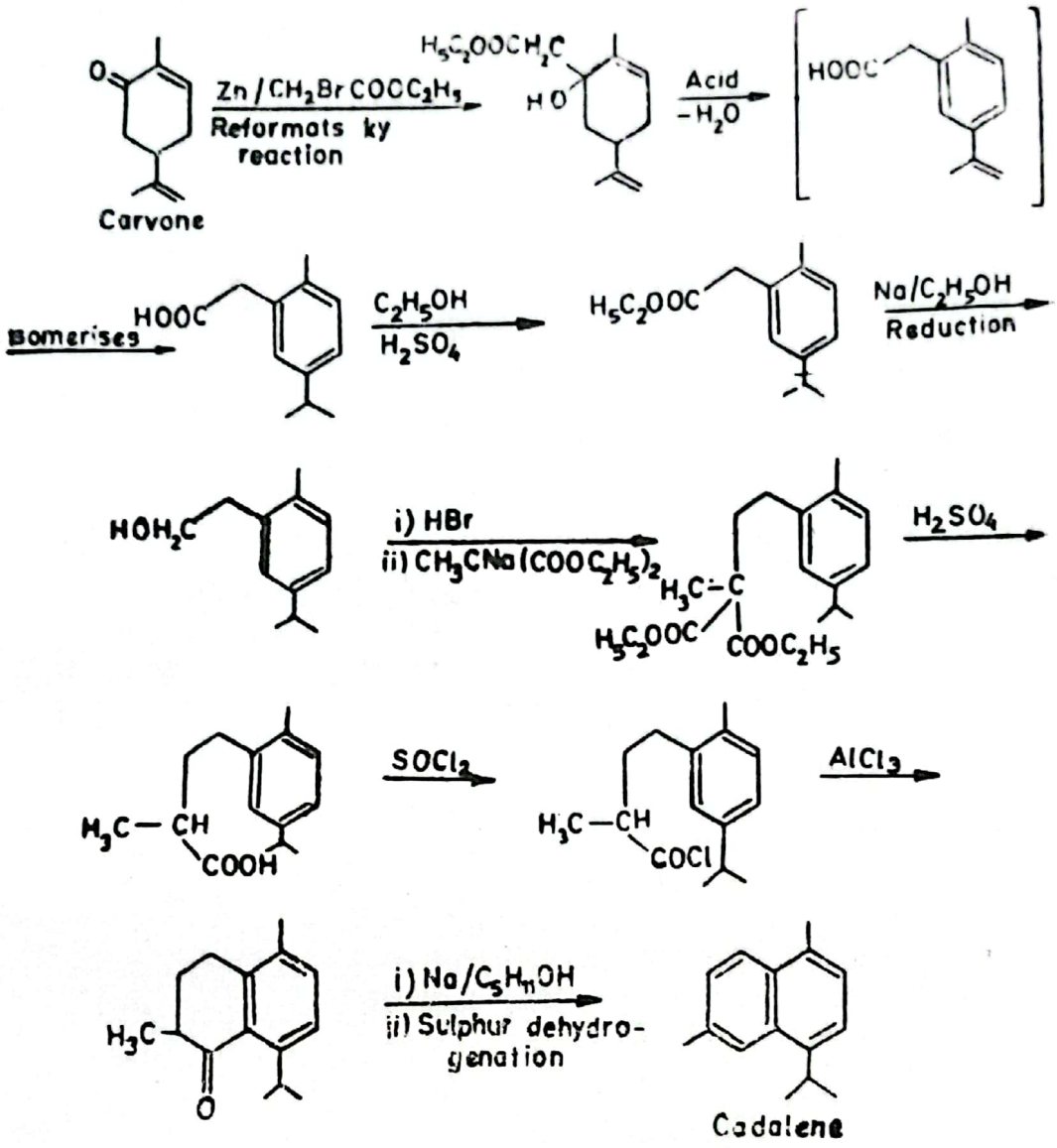


Geraniol — Dipentene — *p*-Cymene



Farnesol — Cadinene skeleton — Cadalene

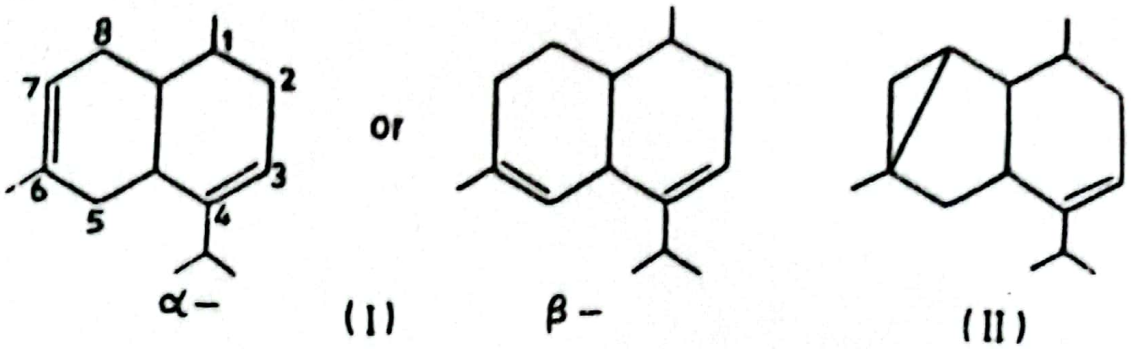
Ruzicka (1922) synthesised 1, 6-dimethyl-4-isopropylnaphthalene and showed that it was identical with cadalene.



4. Now the next problem is to ascertain the positions of the two double bonds in cadinene. When the molecular refraction of cadinene is determined, the value so obtained reveals that there is no optical exaltation and hence the two double bonds are not conjugated. .This is further confirmed by the fact that cadinene does not get reduced with sodium and amyl alcohol, and also does not exhibit strong absorption in the ultraviolet region.

When cadinene is ozonolysed, it yields a compound which has the same number of carbon atoms as cadinene. This shows that the two double bonds are not present in the side chain but are present in the ring systems. Further, they should be present in different rings because if they are present in the same ring, carbon would have been lost on ozonolysis.

On the basis of above facts, Ruzicka *et al.* (1924) proposed the structure (I) (α or β) for cadinene which is based on its relationship with copaene (II) because the latter compound (II) adds on two moles of hydrogen chloride to form copaene dihydrochloride which



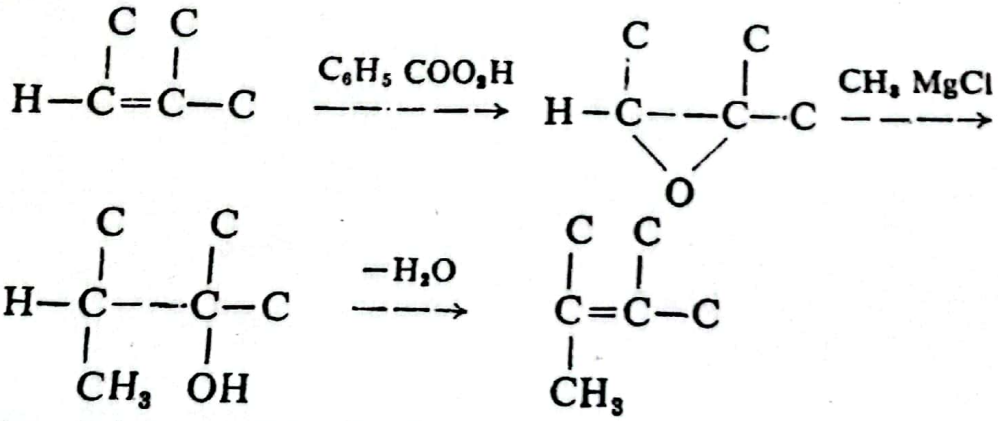
(I) (II)

Structure of cadinene Structure of copaene

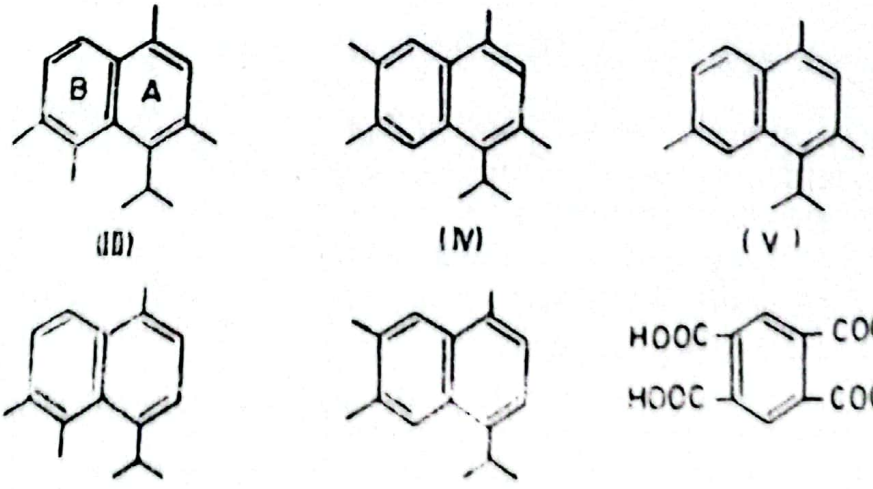
has been found to be identical with cadinene dihydrochloride. It is interesting to note that both α- and β- structures yield the same dihydrochloride as (II).

5. Upto 1942, structure (I) (α or β) was considered to be the correct structure of cadinene. However, this structure was rejected by Campbell and Soffer who reinvestigated the problem. These workers prepared cadinene monoxide and dioxide by treating cadinene with perbenzoic acid. They treated these oxides separately with excess of methylmagnesium chloride and then dehydrogenated the product with selenium. They got monomethylcadalene from cadinene monoxide and a dimethylcadalene from cadinene dioxide.

The conversion of cadinene into its methyl derivative as well as its dimethyl derivative takes place according to the following scheme.



The positions of the double bonds in cadinene remain the same as the positions of methyl groups in monomethyl cadinene and



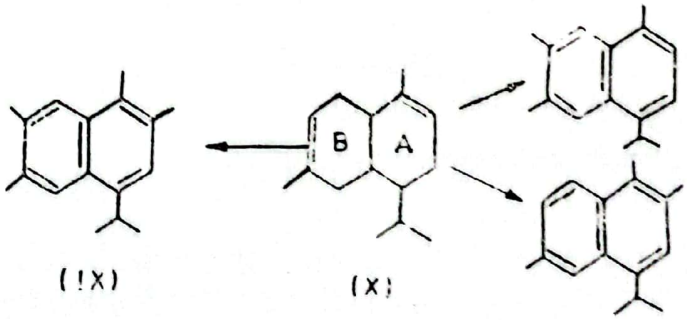
(III) (IV) (V)

(VI) (VII) (VIII)

dimethyl cadinene. If Ruzicka structure (I) for cadinene is correct, then the expected formulae for dimethylcadalene would be (III) and (IV) for its α- and β-isomers. Similarly, the expected formulae for monomethylcadalenes would be (V) for (α or β), (VI) (from α) and (VII) from (β).

In order to arrive at the correct structure of cadinene, Campbell and Soffer carried out the oxidation of dimethylcadalene, first with chromic acid and then with nitric acid and they got pyromellitic acid (benzene−1, 2, 4, 5−tetracarboxylic acid) (VIII). The formation of the latter product (VIII) ruled out (III) as the structure of dimethyl-cadalene but favoured the structure (IV) in which the two methyl groups are located at positions 6 and 7 in ring B, Therefore, the double bond in cadinene in ring B is 6, 7. This also rejected structure (VI).

If the position of double bond in ring A remained same as in structure (I), then it would be expected that dimethylcadalene was (IV) whereas monomethyl cadalene was either (V) or (VII). Campbell and Soffer were successful in synthesising (IV) and (VII) and found that each was different from the methylcadalenes they had got from cadinene. This proved that the structures (IV) and (VII) were not correct and the position of double bond in ring A could not be 3, 4. The only formula for dimethylene cadalene which could yield (VIII) on oxidation would be (IX). Campbell and Soffer synthesised compound (IX) and this was found to be identical with the dimethyl-cadalene obtained from cadinene. Hence, cadinene must be (IX). Therefore, the correct structure of cadinene must be (X). On the basis of this structure, the formula of mono-methylcadalene may be written as follows :



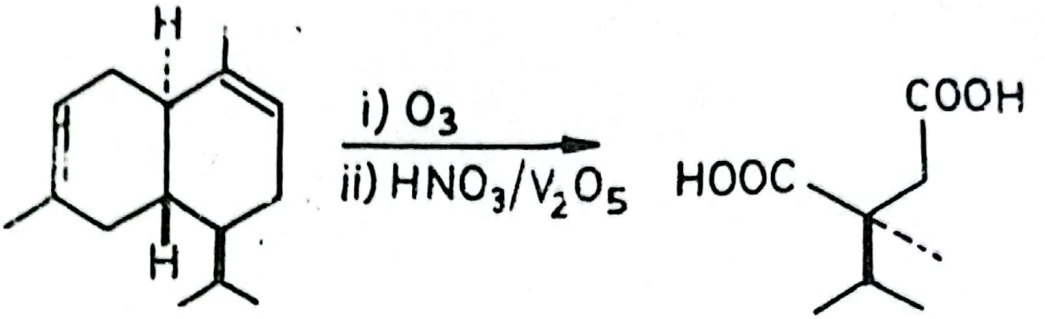
(IX) (X)

(XI)

When cadiene (X) is oxidised, it is expected to give two monoxides (one due to oxidation of ring A or B) and one of these would yield (VII). The latter compound was different from the monomethylcadalene obtained from cadinene. Hence, if the correct structure of cadinene is (X), then the formula of monomethyl-cadalene obtained from cadinene must be (XI). Campbell and Soffer synthesised (XI) and they found this compound to be identical with the compound obtained from cadinene. Hence, (X) is the correct structure of α-cadinene.

6. From X-ray analysis of (−) cadinene dihydrobromide, it was found that the two rings possess the trans-decalin structure and

the isopropyl group is having the *cis*-position with respect to the hydrogen atom at the nearer ring junction. Further, (−)—cadinene when ozonolysed followed by oxidation with nitric acid in the presence of vanadium pentoxide yielded D (+) isopropylsuccinic acid (XII). This established the absolute configuration of the ring carbon atom attached to the isopropyl group.



(X) (XII)

DITERPENOIDS

1.56. PHYTOL

Introduction. It is an acyclic diterpenoid which makes up about one third of the chlorophyll molecule. This also forms part of the molecules of vitamins E and K.

Phytol is produced by the alkaline hydrolysis of chlorophyll.

Phytol is an optically active compound which boils at 145°C at 0.03 mm pressure.

Constitution. The constitution of phytol has been elucidated on the basis of analytical and synthetic evidences as follows :

1. The **molecular formula** of phytol has been found to be $C_{20}H_{40}O$.

2. When it is catalytically hydrogenated, it adds on one mole of hydrogen to form dihydrophytol indicating that phytol contains **one double bond.**

$$\underset{\text{Phytol}}{C_{20}H_{40}O} + H_2 \xrightarrow{Pt} \underset{\text{Dihydrophytol}}{C_{20}H_{42}O}$$

3. When phytol is oxidised with chromic acid or PCl_5, it yields a monocarboxylic acid called phytenic acid which has the same number of carbon atoms, thereby indicating that the oxygen atom in it is present as a **primary alcoholic group.**

4. The molecular formula of the fully saturated parent hydrocarbon of phytol will be $C_{20}H_{42}$ which corresponds to the general formula C_nH_{2n+2} of acyclic compounds. Thus, phytol is an **acyclic compound.**

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5. When phytol is ozonolysed, it yields glycol aldehyde and a saturated ketone, $C_{18}H_{36}O$ (I). The formation of these products on the ozonolysis of phytol could be formulated if following is the part structure for phytol.

$$\underset{\text{Phytol}}{C_{17}H_{35}C{=}CH.\,CH_2OH} \xrightarrow{O_3} \underset{\text{Ketone (I)}}{C_{17}H_{35}CO} + \underset{\text{Glycolaldehyde}}{OHC.\,CH_2OH}$$

As the ketone (I) $C_{17}H_{35}CO$ undergoes haloform reaction with sodium hypobromite, this means that the ketone contains a methyl group attached to the ketone group. Thus, the part structure of ketone may be represented as follows :

$$\underset{(I)}{C_{16}H_{33}\overset{\overset{CH_3}{|}}{C}{=}O} \xrightarrow{NaOBr} C_{16}H_{33}COOH + CHBr_3$$

On the basis of the structure of this ketone, the part structure of phytol may be represented as follows :

$$C_{16}H_{33}\overset{\overset{CH_3}{|}}{C}{=}CH.\,CH_2OH$$

6. From molecular formula of phytol, it is indicated that it may be composed of **four reduced isoprene units**. If it is so, and assuming that there is a head to tail linking of the isoprene units, the structure of the saturated ketone (I) may be written as follows :

$$\begin{array}{l} \overset{\overset{CH_3}{|}}{CH}.CH_2.\,CH_2.\,CH_2.\,\overset{\overset{CH_3}{|}}{CH}.\,CH_2.\,CH_2\,CH_2.\,\overset{\overset{CH_3}{|}}{CH}.\,CH_2.\,CH_2.\,CH_2CO \\ \quad\overset{|}{CH_3} \end{array}$$

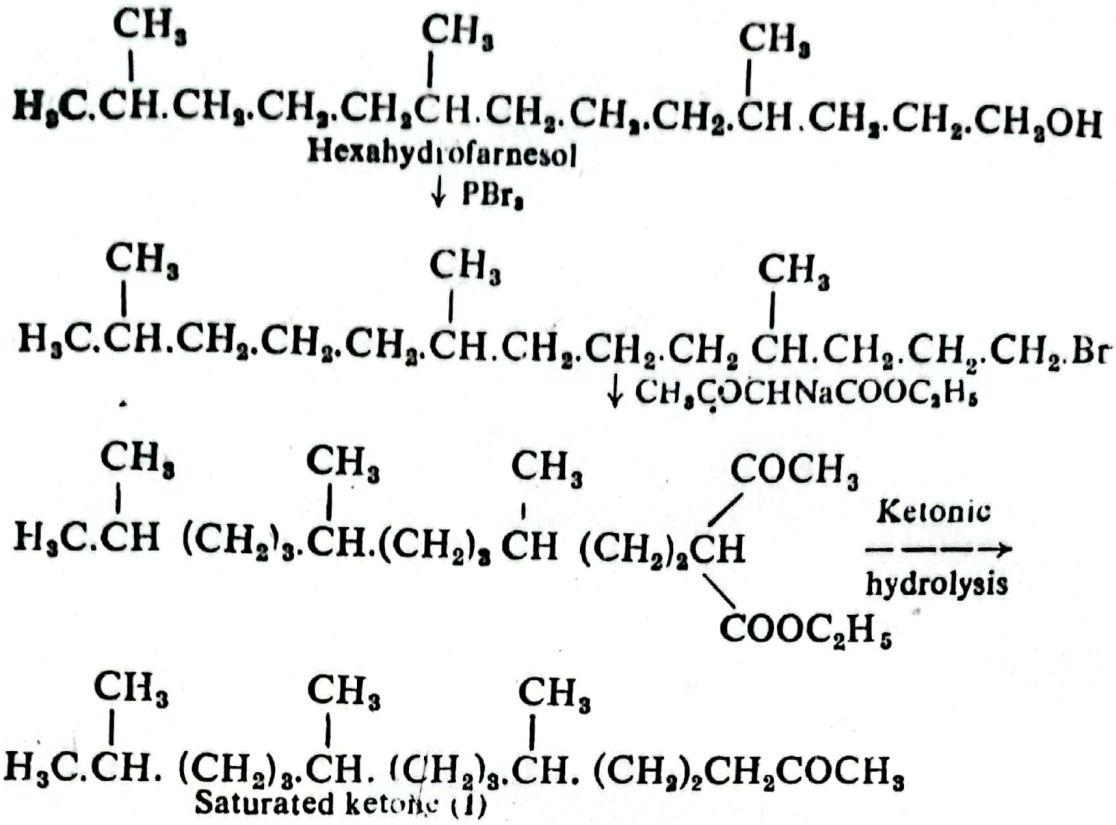
(I)

7. The above structure for ketone (I) has been confirmed by its various syntheses. Two of these are as follows :

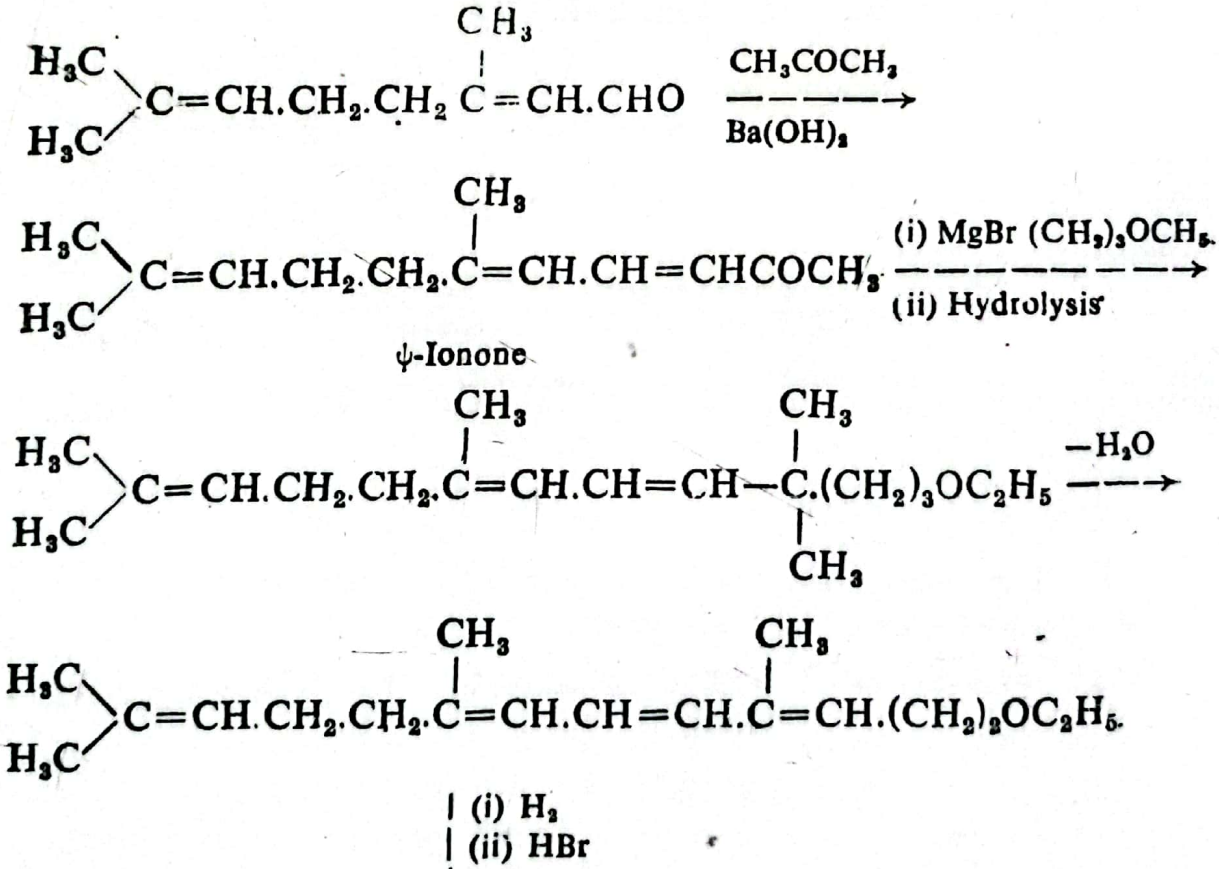
(*a*) **Fischer et al. (1928) synthesis.** This synthesis starts from farnesol.

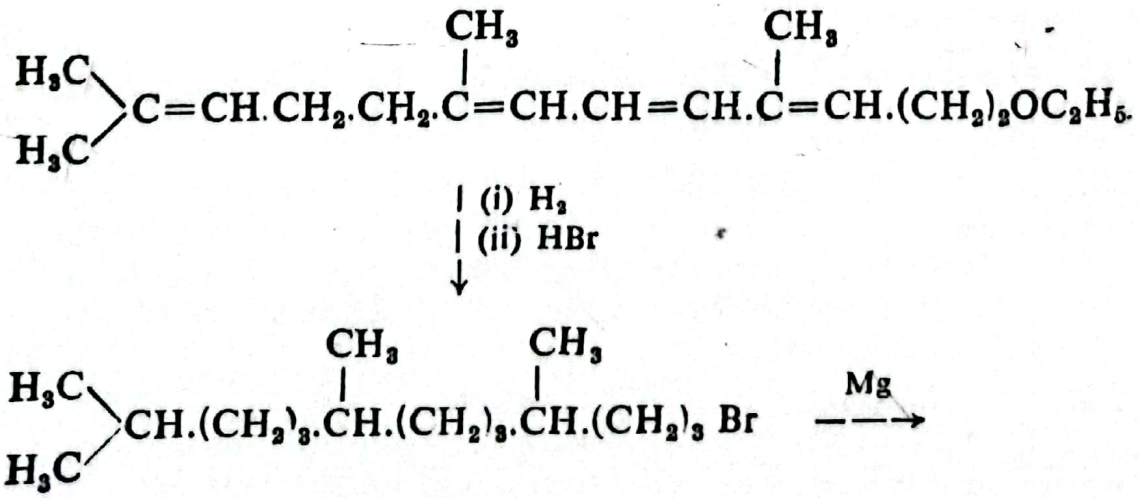
The latter compound on catalytic dehydrogenation yields hexahydrofarnesol which when treated with phosphorus tribromide yields hexahydrofarnesyl bromide. This latter compound, on treatment with sodioacetoacetic ester followed by ketonic hydrolysis, yields the saturated ketone.

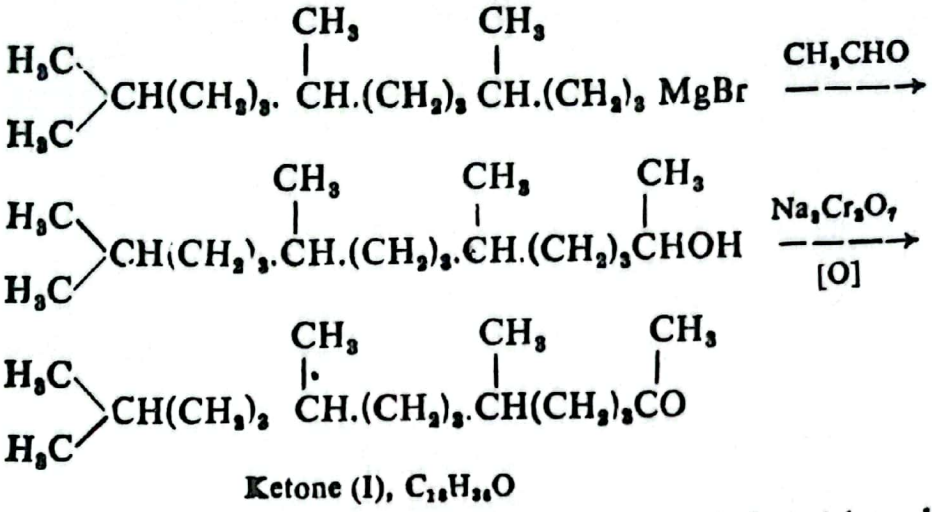
$$\underset{\text{Farnesol}}{H_3C.\overset{\overset{CH_3}{|}}{C}{=}CH.CH_2.CH_2.\overset{\overset{CH_3}{|}}{C}{=}CH.CH_2.CH_2.\overset{\overset{CH_3}{|}}{C}{=}CH.CH_2OH} \xrightarrow{H_2/Pd}$$



(*b*) **Smith and Sprung synthesis (1943).** This starts from citral.

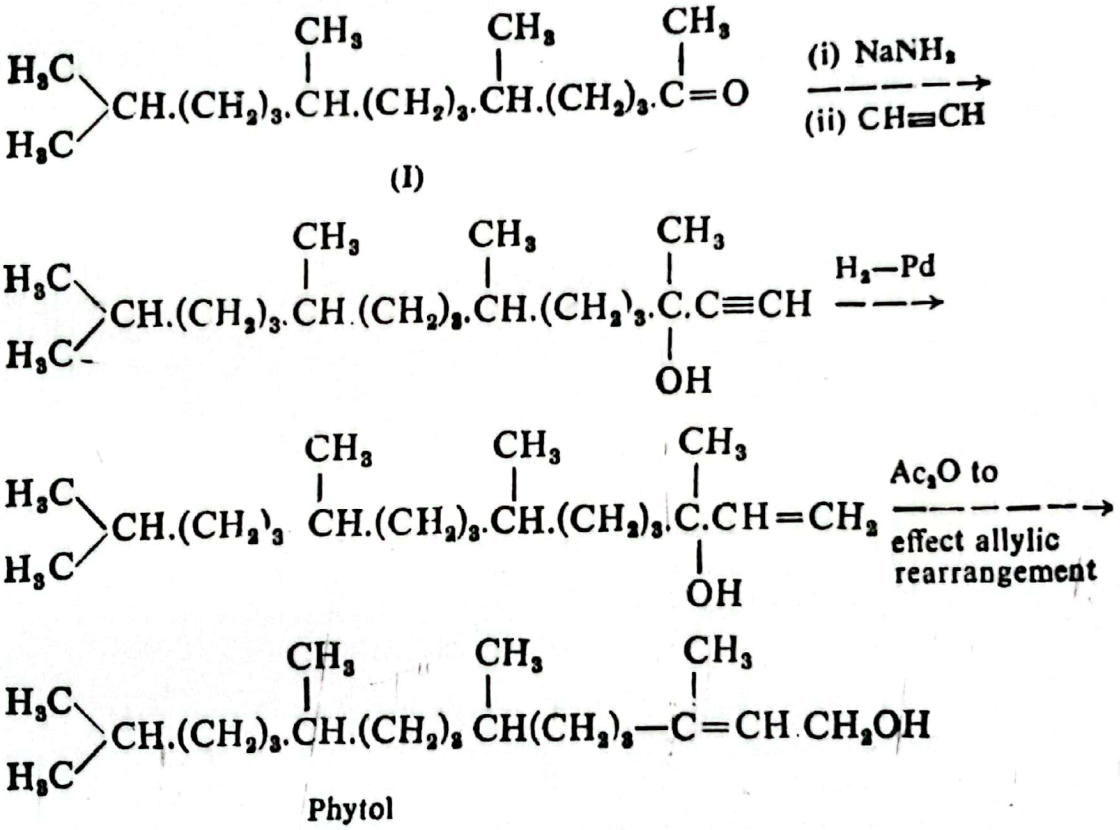






Ketone (I), $C_{18}H_{36}O$

7. *Synthesis.* Finally, the structure of phytol has been confirmed by its synthesis from ketone (I) given by Fischer (1929).



Phytol

8. In the phytol molecule, there are two **chiral centres** C_7 and C_{11}. Karrer *et al.* (1943) isolated natural phytol from nettles. The natural phytol is very **weakly dextrorotatory** and is synthesised by Weedon *et al.* (1959) who have assigned (R)-configurations to the two chiral centres. From ORD studies, Djerassi *et al.* have also assigned the (R)-configuration to C_7. They did it on the basis of the fact that (S)-2-methylbutanal (related to L-glyceraldehyde) showed a positive ORD curve whereas the degraded phytol (—C,Me—CHO) exhibited a negative ORD curve. Therefore, the (R)-configuration was assigned to C_7 of phytol. Weedon *et al.* (1959) assigned the trans configuration to the double bond by means of NMR spectroscopy.

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1.57. ABIETIC ACID

Introduction. It is a tricyclic diterpenoid which is the main constituent of rosin or colophony. It does not occur naturally in the original rosin which has been obtained from oleo resins but is formed from labile precursors during the collection, storage and steam distillation of the oleoresin.

Its melting point is 172–175°C.

Constitution of Abietic Acid

1. From analytical data, the **molecular formula** of abietic acid has been found to be $C_{20}H_{30}O_2$.

2. As abietic acid gives effervescence with sodium bicarbonate, this shows that it contains a **carboxylic group**. Moreover, as it is very difficult to esterify acid, this shows that the carboxylic group is attached to a **tertiary carbon atom**. This is further confirmed by the fact that abietic acid when warmed with concentrated sulphuric acid evolves carbon monoxide.

3. As abietic acid adds on two moles of hydrogen in the presence of catalyst to form tetrahydro abietic acid, this shows that it contains two double bonds.

$$\underset{\text{Abietic acid}}{C_{19}H_{29}COOH} \xrightarrow[Ni]{2\,H_2} \underset{\text{Tetrahydroabietic acid}}{C_{19}H_{33}CO\,OH}$$

4. If —COOH group is considered as a substituent group, the parent hydrocarbon of the abietic acid would be $C_{19}H_{34}$ corresponding to the general formula C_nH_{2n-4} for tricyclic compounds and, therefore, abietic acid is a tricyclic compound.

5. When abietic acid is dehydrogenated with sulphur, selenium or palladised charcoal, it yields **retene** having molecular formula $C_{18}H_{18}$ having m p. 99°C. By oxidative degradation, retene (I) has been shown to be 1-methyl-7-isopropylphenanthrene.

The retene (I) when oxidised yields retenequinone (II). The latter compound when oxidised with alkaline permanganate yields the compound (III) which on further oxidation with dichromate yields (IV). The latter compound when heated with concentrated potassium hydroxide yields (V).

In the conversion of (II) to (III), benzilic acid arrangement is involved. But (V) forms a cyclic anhydride whereas (IV) does not. Therefore, one carboxylic group in (IV) must be **ortho** to the centre ring. As (III) has been formed by the loss of one carbon atom only from (II), this reveals that the carboxylic group has been derived from a methyl group present at position 1 which in turn has been inherited from structure (I) of retene. This shows that one alkyl group must be present at position 1 in the phenanthrene nucleus of (I). When (IV) is heated, it yields fluorenone ; (V) when

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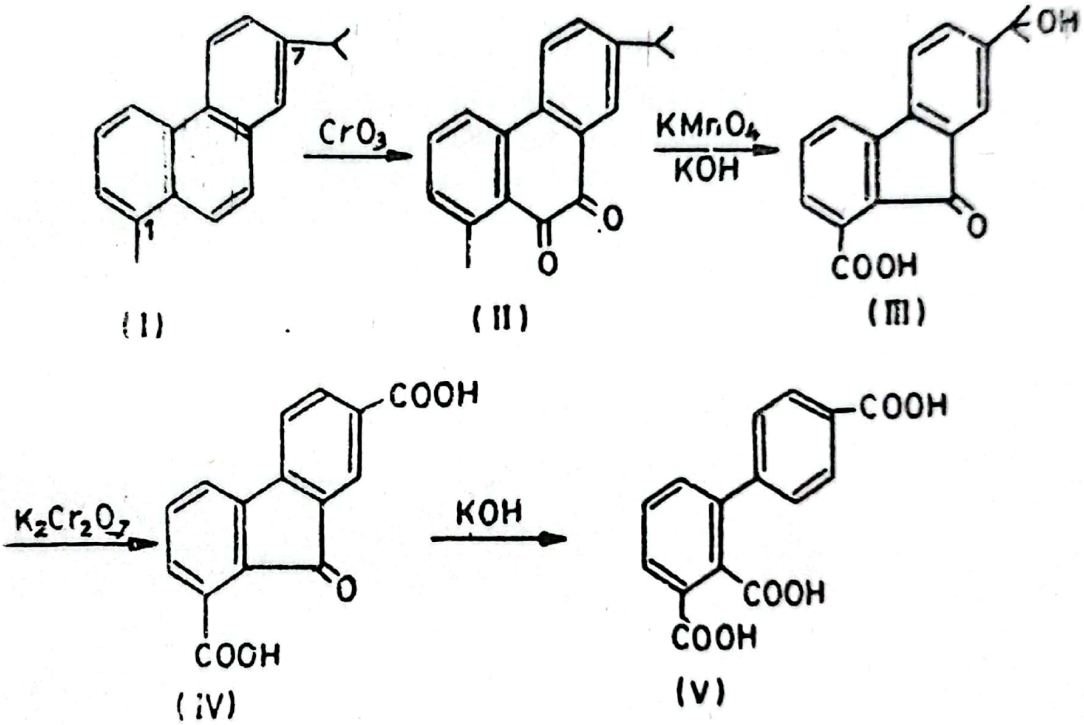
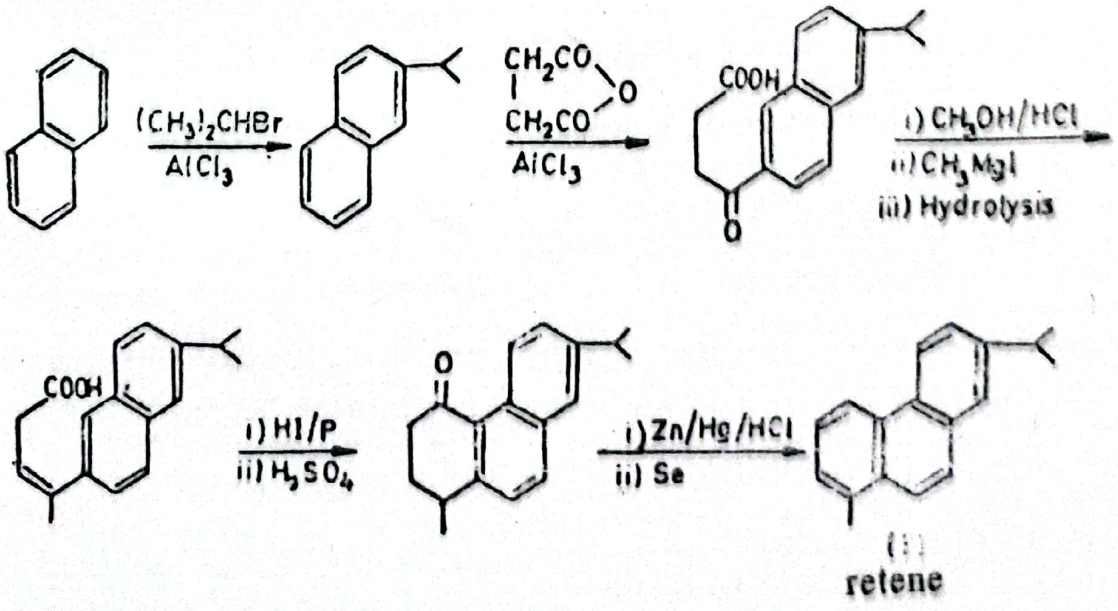
heated yields biphenyl. Hence the carbon skeletons of (IV) and (V) are established.

Now the next problem is to locate the position of the isopropyl group in retene (I). This has been ascertained by fusing (III) with potassium hydroxide. The product has been shown to be 4-isopropylbiphenyl by oxidation to biphenyl-4-carboxylic acid. Therefore, an isopropyl group is present at position 7 in retene.

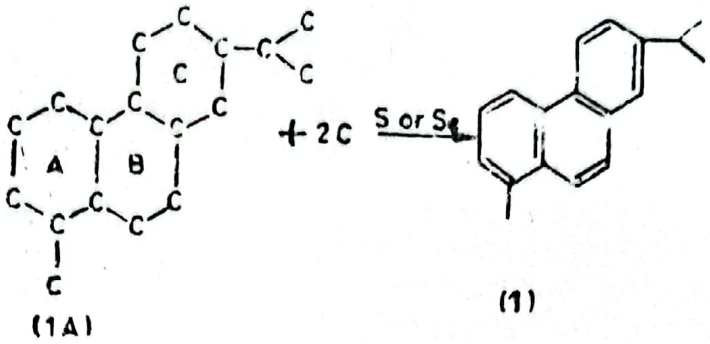
All the above reactions may be written as follows by considering the (I) as the structure of retene.



The structure (I) for retene has been confirmed by its synthesis given by Haworth *et al.* (1932).



From the formation of retene (I) from the dehydrogenation of abietic acid, it follows that the latter may possess the following carbon skeleton (IA).



(1A) (1)

In abietic acid, there are 20 carbon atoms. Out of these, the positions of 18 carbon atoms have been established by retene formation whereas the positions of remaining two carbon atoms are satisfied by one tartiary carboxyl group and one angular methyl group. In order to establish the exact structure of abietic acid, one must know the following facts :

(*a*) The position of the carboxyl group

(*b*) The position of the angular methyl group

(*c*) The position of two double bonds.

We will ascertain the above facts one by one as follows :

(*a*) **Position of the carboxylic group.** Ruzicka *et al.* (1922) ascertained the position of carboxyl group to be 4 *i.e.*, C_4 in abietic acid by the following work :

Abictic acid when methylated yields methyl abietate ($C_{19}H_{29}COOCH_3$) which on reduction with sodium and ethanol yields abietinol ($C_{19}H_{29}CH_2OH$). The latter compound when treated with phosphorus pentachloride loses a molecule of water to yield methylabietin ($C_{20}H_{30}$) which on distillation with sulphur is converted into homoretene.

$$\underset{\text{Abietic acid}}{C_{19}H_{29}COOH} \xrightarrow[HCl]{CH_3OH} \underset{\text{Methyl abietate}}{C_{19}H_{29}COOCH_3} \xrightarrow{Na/C_2H_5OH} \underset{\text{Abietinol}}{C_{19}H_{29}CH_2OH}$$

$$\xrightarrow[-H_2O]{PCl_5} \underset{\text{Methyl abietin}}{C_{20}H_{30}} \xrightarrow[\text{Distillation}]{S} \underset{\text{Homoretene}}{C_{19}H_{20}}$$

As homoretene ($C_{19}H_{20}$) contains one CH_2 group more than retene ($C_{18}H_{18}$) and on oxidation with alkaline potassium ferricyanide yields phenanthrene -1, 7- dicarboxylic acid which is also obtained from the oxidation of retene under similar conditions, this means that the homoretene must be 1-ethyl-7-isopropyl phenanthrene in which ethyl group must be present at position 1 instead of methyl group in retene.

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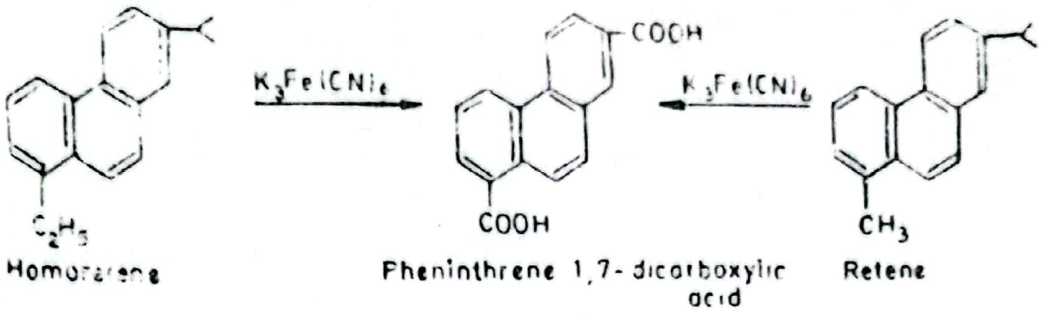
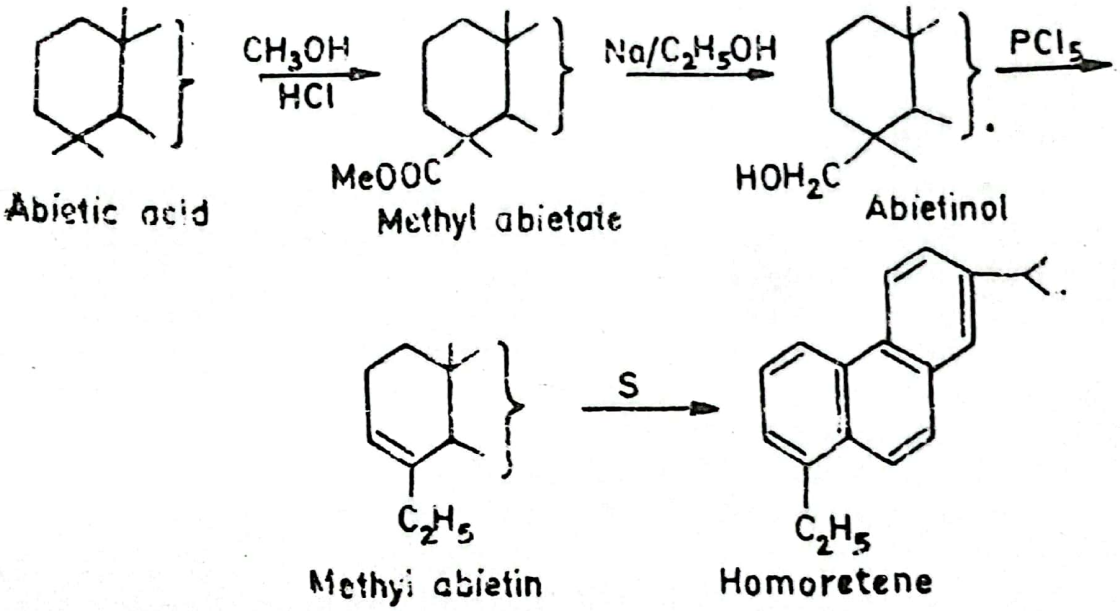


The structure of homoretene has been confirmed by its synthesis given by Haworth *et al.* (1932). This synthesis is same as that for retene with the only difference that ethylmagnesium iodide is used instead of methylmagnesium iodide in the synthesis of homoretene.

The formation of an ethyl group in homoretene has been explained by considering that abietinol undergoes a Wagner-Meerwein rearrangement on dehydration.



(*b*) *Position of the angular methyl group.* When abietic acid (VI) is oxidised with potassium permanganate, it yields a mixture of products among which are two tricarboxylic acids, $C_{11}H_{16}O_6$ (VII) and $C_{12}H_{18}O_6$ (VIII). The compound (VII) on dehydrogenation with selenium yields *m*-xylene whereas the compound (VIII) forms hemimellitene. In both these dehydrogenations, there occurs a loss of three carbon atoms and if it is assumed that the lost carbon atoms were the three carboxyl groups, then two methyl groups in (VII) and (VIII) must be present in the **meta-position.** Further, as each of the two compounds (VII) and (VIII) contains the methyl group originally present in abietic acid (position 4), it means that the acids (VII) and (VIII) must possess ring A of abetic acid. All these results can be explained if there is an angular methyl group at position 10 which will be lost during sulphur dehydrogenation of abietic acid.

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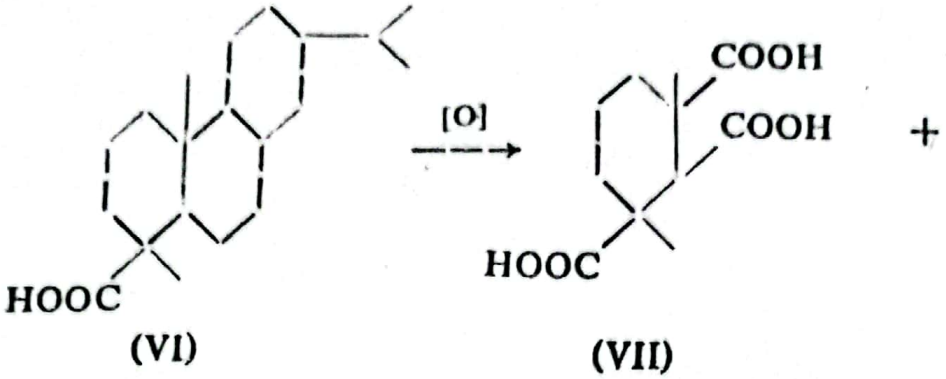
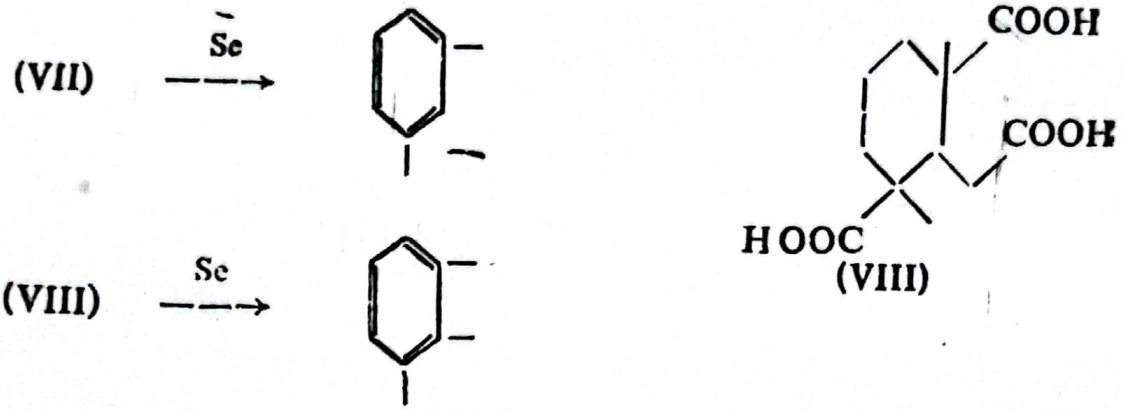


(VI) (VII)



In the above reactions, the structure (VI) is assumed to be the carbon skeleton of abietic acid.

(c) *Position of the two double bonds.* We have already proved that abietic acid contains two double bonds. Ruzicka *et al.* (1932) showed that abietic acid forms an adduct with maleic acid at above 100°C, indicating that the two double bonds are conjugated. Later on, it was also reported that levopimaric acid also forms the same adduct at room temperature. In order to explain this result, it was argued that abietic acid first undergoes isomerisation to yield levopimaric acid at above 100°C which then forms the adduct. However, this reaction cannot be accepted as evidence for conjugation in abietic acid.

The UV spectrum of abietic acid exhibits a maximum at 238 (ϵ 16000) nm. This value is indicative of the fact that the double bonds may be conjugated. But the basic value for a homoannular diene system is 253 nm. Therefore, it means that the two double bonds are not present in the same ring. This is supported by the fact that the calculated value for the structure assigned to abietic acid on this basis is $214+4\times5+5=239$ nm.

When oxidation of abietic acid is carried out with potassium permanganate, it yields various products, including isobutyric acid. The formation of the latter product reveals that one double bond is present in ring C whose position is either 12, 13- or 13, 14-. If the double bond is present in the 12, 13 position, then the double bond which is conjugated with it must be present in the same ring (9, 11 or 8, 14). On the other hand, if one double bond is present in 13, 14- position, then the other double bond may be either in ring B or ring C. But we have already proved that the two double bonds

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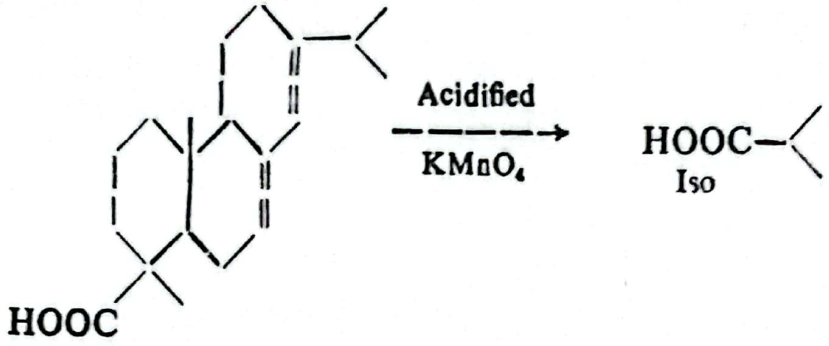
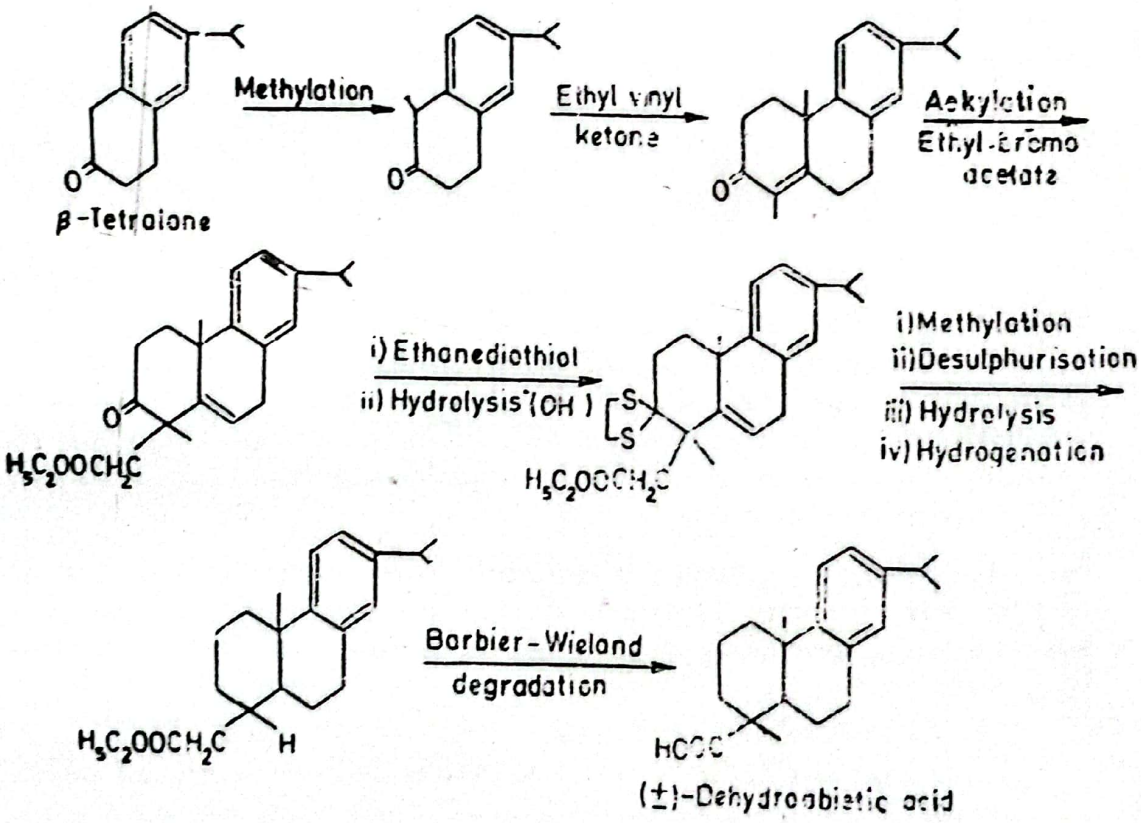
are present in different rings. Therefore, their positions may be 7, 8 and 13, 14. However, actual positions are these, *i.e.*, 7, 8 and 13, 14 because these have been confirmed by oxidative studies on abietic acid by Ruzicka *et al.* (1938—1941).

6. On the basis of the above facts, the structure of abietic acid may be represented as follows :



Abietic acid

7. **Synthesis.** The stereochemistry of abetic acid has been confirmen by stork *et.al.* (1956) who synthesised (±) dehydroabietic acid (The latter compound is also obtained by heating abietic acid in the presence of kalladium on charcoal at 250—275° C).



TRITERPENOIDS

1.58. SQUALENE

Introduction. It is an acyclic triterpinoid which has been isolated from the liver oils of sharks. It is also found in olive oil and several other vegetable oils. In small quantities, squalene has been detected in leaves.

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It is a liquid which boils at 240—242° C at 4 mm pressure. Its structure has been elucidated mainly by Heilborn.

Constitution. This has been elucidated on the basis of following analytical and synthetic evidences :

1. From analytical data, the **molecular formula** of squalene has been found to be $C_{30}H_{50}$.

2. When squalene is hydrogenated catalytically, it adds on six moles of hydrogen to form per hydrosqualene, $C_{30}H_{62}$. This reaction reveals that squalene has *six double bonds.*

$$\underset{\text{Squalene}}{C_{30}H_{50}} + 6H_2 \xrightarrow[Ni]{\text{Catalyst}} \underset{\text{Perhydro squalene}}{C_{30}H_{62}}$$

The presence of six double bonds in squalene has been further confirmed by the fact that squalene when treated with hydrogen chloride adds on six moles of the reagent, *i.e.*, hydrogen chloride. Further, squalene hexachloride has been separated into three crystalline forms, revealing that squalene itself is also a *mixture of 3 isomers.*

3. The molecular formula of fully hydrogenated squalene, *i.e.*, perhydrosqualene, is $C_{30}H_{62}$ which corresponds to C_nH_{2n+2} for acyclic compounds. Hence, squalene is an open chain compound.

4. Since squalene cannot be reduced by sodium and amyl alcohol, it means that there are no *conjugated double bonds* present in the molecule.

5. When squalene is treated with boiling acetic anhydride containing 1 per cent sulphuric acid or with formic acid, it gets cyclised to bicyclic, tricyclic and finally tetracyclic compound having four, three and two double bonds respectively. This reaction suggests that squalene is a terpenoid. This is further suggested by the fact that squalene on hydrolysis yields amylene (C_5H_{10}) as one of the hydrolysis products.

6. When squalene is oxidised with chromyl chloride in carbon tetrachloride, it yields formaldehyde, acetaldehyde and succinic acid, revealing that squalene contains the following three fragments :

$$-\overset{|}{C}=CH_2 \xrightarrow{[O]} HCHO \xrightarrow{[O]} HCOOH$$

$$-CH=\underset{|}{\overset{|}{C}}- \xrightarrow{[O]} CH_3CHO$$

$$=\underset{|}{\overset{|}{C}}-CH_2-CH_2-\overset{|}{C}= \xrightarrow{[O]} COOH.CH_2.CH_2.COOH$$

7. Squalene when ozonolysed yields various products such as carbon dioxide, formaldehyde, formic acid, succinic acid,

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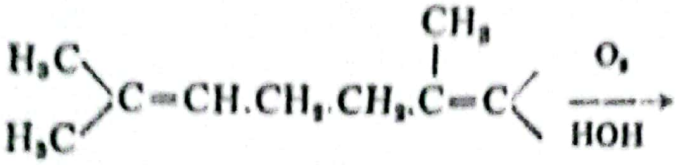
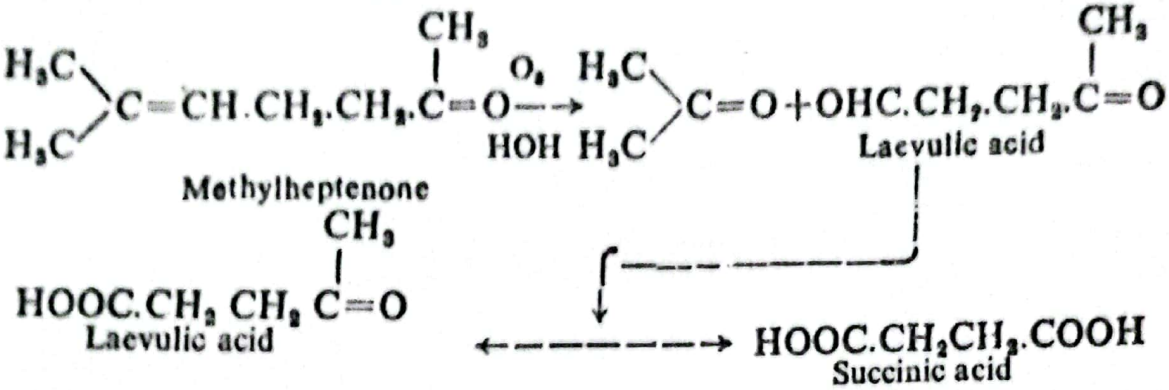
acetone, laevulic acid and methylheptenone. The formation of laevulic acid reveals that the following skeleton (I) is present in squalene.

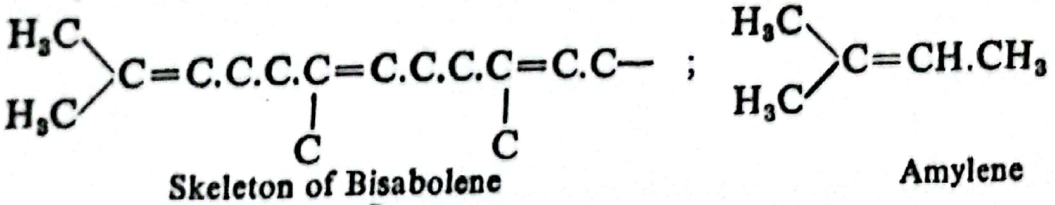


(I)

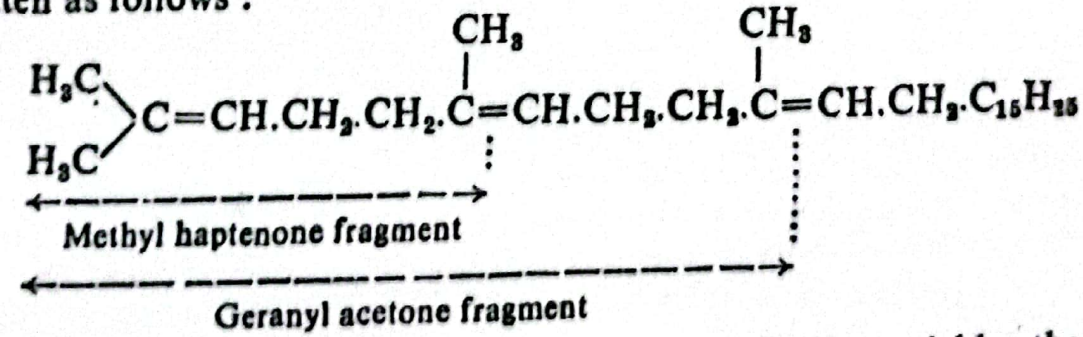


It is important to mention here that the structure (I) could not explain the ozonolysis products such as CO_2, HCHO and HCOOH.

8. When squalene is subjected to dry distillation, it yields a mixture of hydrocarbons, among which one with lowest boiling point has been shown to be *amylene*, C_5H_{10}. while the other with high boiling point has been shown to be bisabolene. The formation of bisabolene suggests that carbon skeleton of bisabolene is present is squalene.

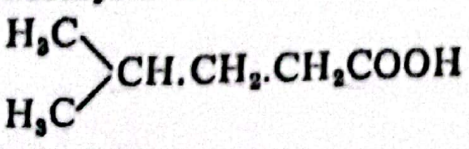


On this basis, the part structure (II) for squalene may be written as follows :

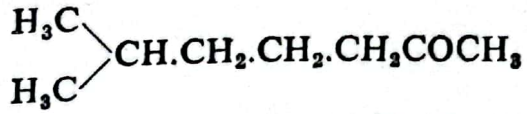


9. Ozonolysis of decahydrosqualene, $C_{30}H_{60}$, yields the following compounds (the formation of these could lead to the structure of squalene).

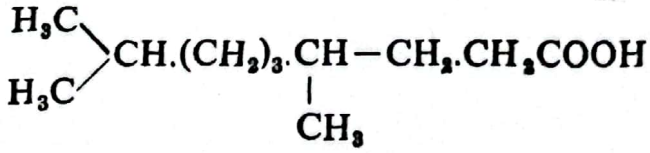
(*i*) γ-Methyl-n-valeric acid



(*ii*) Methylisohexyl ketone

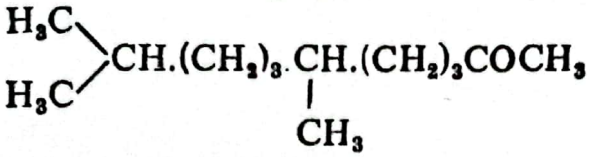


(*iii*) 4,8-Dimethyl nonic acid ($C_{10}H_{21}COOH$)

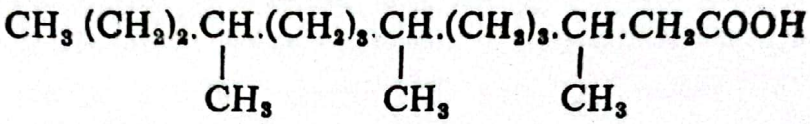


(*iv*) Haxahydro-ψ-ion one (Geranylacetone)

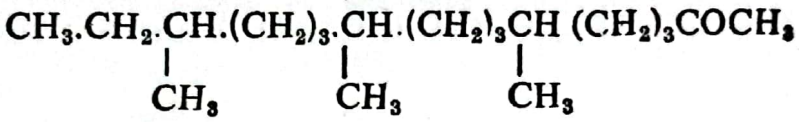
($C_{13}H_{26}O$)



(*v*) Acid of the formula, $C_{16}H_{33}COOH$

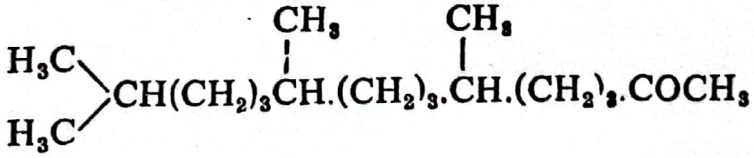


(*vi*) Ketone of the formula, $C_{19}H_{38}O$

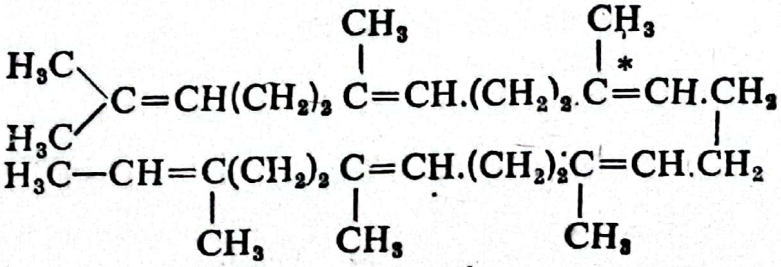


or

Ketone of the formula, $C_{18}H_{36}O$

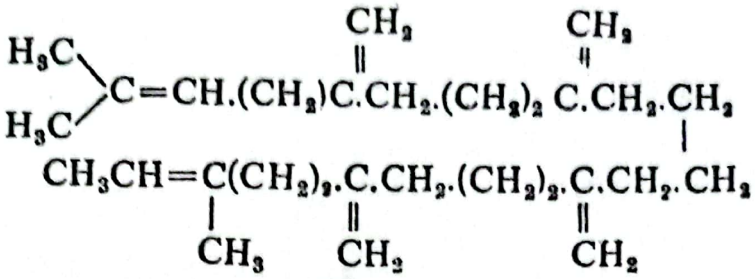


As the compounds (*iv*) $C_{13}H_{26}O$ and (*v*) $C_{16}H_{33}COOH$ could account for all the carbon atoms of the squalene, it follows that the double bond of decahydrosqualene must be present between C_{13} and C_{14}. Thus, by joining the two fragments (*iv*) and (*v*) with a double bond between C_{13} and C_{14}, the structure of decahydrosqualene is obtained. The positions of the remaining **five double bonds** in squalene have been ascertained from the fact that squalene is a isoprenoid and, therefore, its structure may be represented as follows :



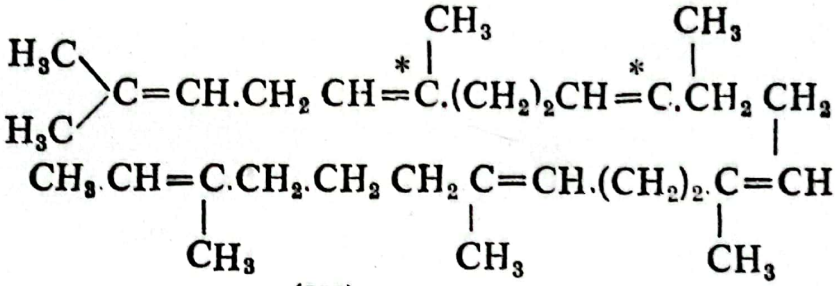
(II) Structure of squalene

10. In the above structure (II), double bond represented by asterisk (*) is the double bond of decahydrosqualene which breaks up at this double bond to give rise compounds (*iv*) and (*v*). However, this structure could not explain the formation of the compounds (*i*), (*iii*) and (*vi*) of point 9 Also, the structure (II) could not explain the formation of CO_2, HCHO and HCOOH by the ozonolysis of methylene group. In order to explain the formation of these products, Douben and Bradlow postulated the structure (III) for squalene which contains exo-methylene groups.



(III)

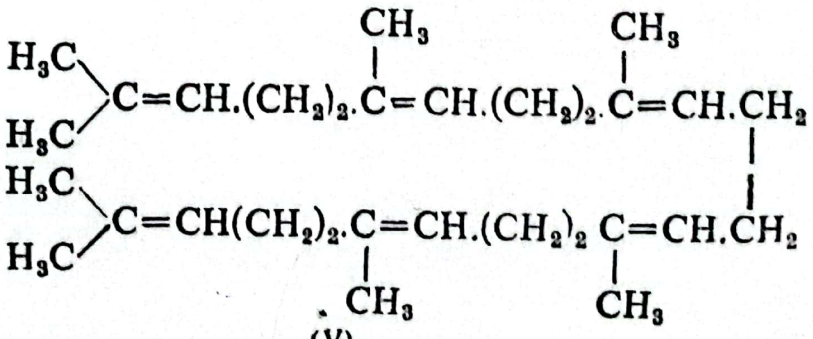
11. Similar to structure (II), the structure (III) could not explain the formation of the compounds of step 9. But the structures (iii) $C_{10}H_{21}COOH$ and (vi) $C_{19}H_{38}O$ account for all the thirty carbon atoms of squalene. Therefore, another structure (IV) was postulated by squalene.



(IV)

In the structure (IV), the double bonds marked by asterisks are the site of attack in decahydrosqualene during the oxidation to yield (I) γ-methylvaleric acid, (iii) the acid $C_{10}H_{21}COOH$ and (iv) the ketone $C_{19}H_{38}O$.

12. All the three structures (II), (III) and (IV) could not explain the formation of the ketone $C_{18}H_{36}O$, However, its formation could be explained by symmetrical structure (V) which is as follows :



(V)

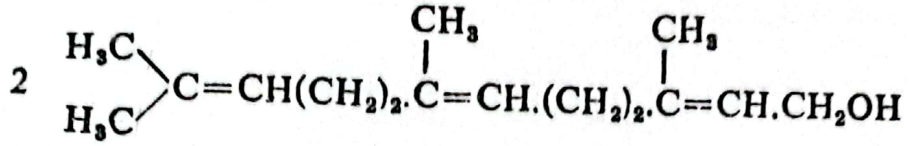
The structure (V) has been confirmed by its synthesis as given by Karrer *et al.* (1931).

13. From the above mentioned facts, it may be concluded that all the reactions of squalene could not be explained by a single structure but by a mixture of four isomeric forms, *viz.*, (II), (III), (IV) and (V).

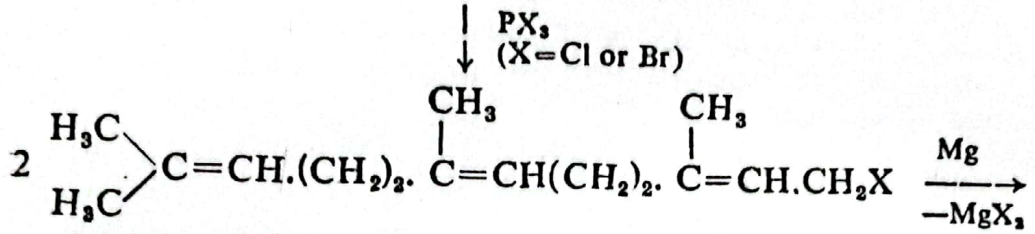
From step (2), it is evident that squalence is a mixture of only three isomeric forms. But structure (II) is discarded because all the reactions given by this structure may be easily explained by structure (IV). Therefore, squalene is only a mixture of (III), (IV) and (V) isomeric forms.

14. **Synthesis.** The symmetrical structure (V) of squalene has been confirmed by its various syntheses :

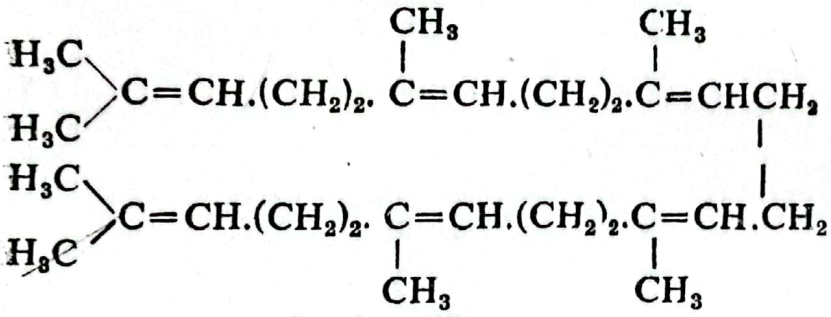
(*a*) *Karrer and Halfenstein (1931)*



2 Moles of farnesyl alcohol

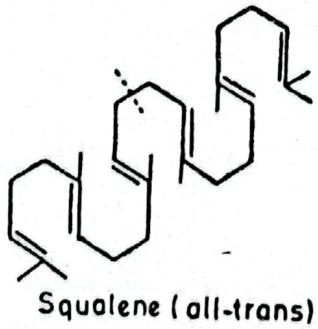


2 Moles of farnesyl bromide or chloride



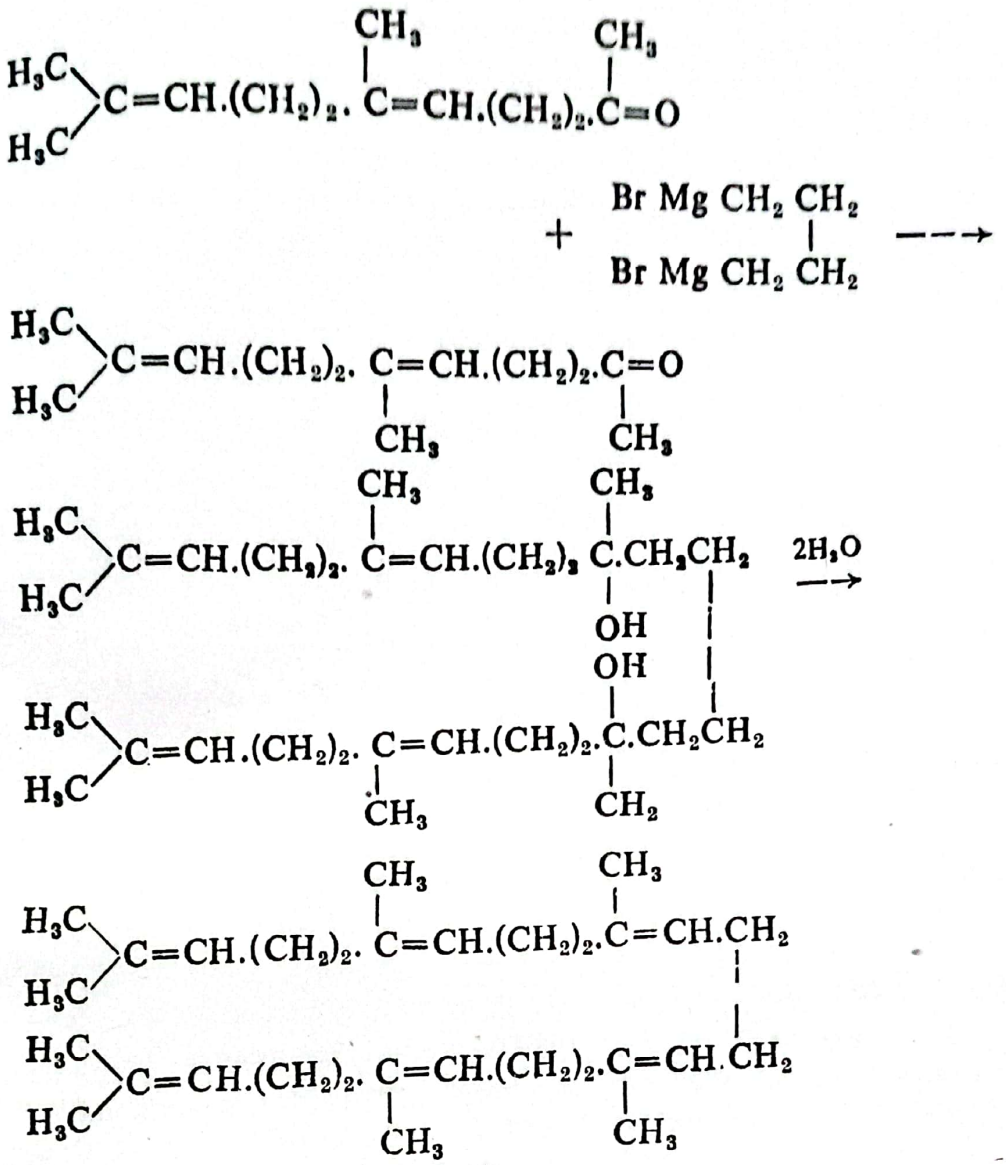
Squalene (V)

or



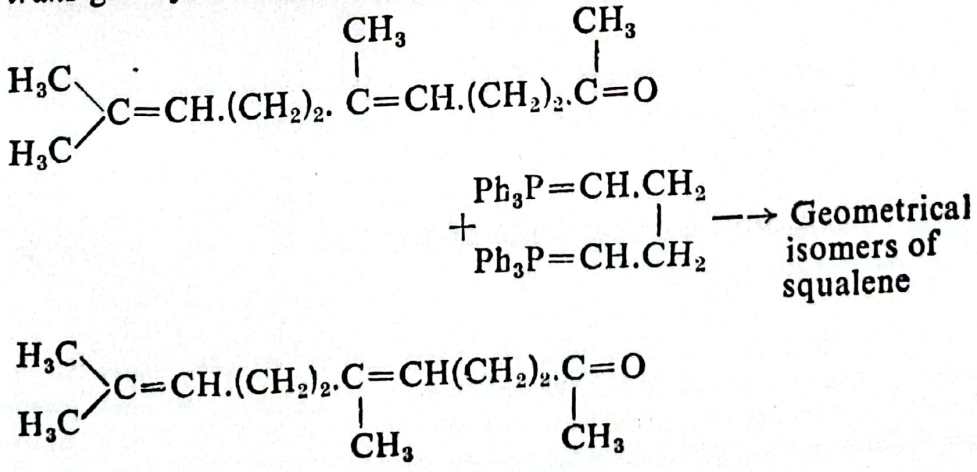
Squalene (all-trans)

(*b*) Squalene can be synthesised by the condensation of two moles of geranylacetone with bis-Grignard derivative of 1, 4- dibromobutane.



Squalene (V)

(c) *Whiting's synthesis (1958).* According to this synthesis, squalene is synthesised by means of the Wittig's reaction from pure *trans* geranyl acetone and triphenylphosphoronylide.



2 Moles of *trans*-geranylacetone

From the geometrical isomers of squalene, Whiting *et al.* isolated 12.5 per cent pure all-*trans*-squalene *via* the thiourea complex. This isomer was identical with natural squalene.

The triphenylphosphoronylide required for this synthesis has been prepared by the reaction of 1, 4-dibromobutane and triphenyl phosphine in the following way :

$$BrCH_2CH_2CH_2CH_2Br + 2Ph_3P \longrightarrow Ph_3\overset{+}{P}CH_2.CH_2.CH_2.CH_2\overset{+}{P}Ph_3 \; 2Br^-$$

$$\xrightarrow[-2HBr]{Li\ C_4H_9} \begin{matrix} Ph_3P{=}CH.CH_2 \\ | \\ | \\ Ph_3P{=}CH.CH_2 \end{matrix}$$

PENTATERPENOIDS

1.59. β-AMYRIN

Introduction. Historically, the series of which β-amyrin is an important member has the basic carbon framework called oleanane, which itself derives its name from oleanolic acid.

It occurs most abundantly in the plant kingdom either in the free state or as an ester. Sometimes, it is also associated with its isomer, *i.e.*, α-amyrin. The main sources of β-amyrin are Manila elemi, resin, shea nut fat, seeds and resins of grapes, etc.

Constitution of β-Amyrin. This has been elucidated on the basis of following analytical and synthetic evidences :

1. The **molecular formula** of β-amyrin has been found to be $C_{30}H_{50}O$.

2. A double bond at 12,13 position is almost universally present in the members of this series. But this bond is extremely inert. Therefore, it does not undergo the usual reactions of a double bond. Due to this reason, early work did not reveal its presence.

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Thus, bromination of β-amyrin acetate gave a substitution rather than addition product. Furthermore β-amyrin was recovered unchanged from attempted hydrogenation experiments (even at 280°C under a pressure of 80 atmosphere of hydrogen in the presence of platinum). However, the presence of the double bond (unreactive) was finally confirmed by molecular refraction measurements, perbenzoic acid and a positive test with tetranitromethane. The tests are as follows :

When β-amyrin is treated with perbenzoic acid, it yields monoepoxide. The formation of this product reveals that β-amyrin contains one **double bond**. Its pesence is further confirmed by the fact the β-amyrin produces a yellow colour with tetranitromethane.

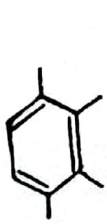
3. The fact, that the double bond is flanked by a methylene group, was proved as follows :

As β-amyrin acetate underwent allylic oxidation to the enone (λ_{max} 249 nm) which on reduction followed by dehydration yielded the diene which on the basis of U.V. spectrum (λ_{max} 280 nm) must be homoannular.

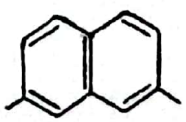
4. As β-amyrin, when oxidised, yields a ketone, this indicates that the only oxygen atom in β-amyrin should be present as a secondary alcoholic group.

5. The parent hydrocarbon corresponding to β-amyrin should be $C_{30}H_{52}$ which corresponds to C_nH_{2n-8} for pentacyclic compounds and hence β-amyrin is a **pentacylic compound.**

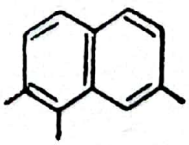
6. The selenium dehydrogenation of β-amynin yields the following products along with the usual products of dehydrogenation :



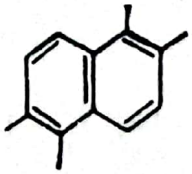
1, 2, 3, 4-Tetramethyl benzene (I)



3, 6-Dimethyl napthhalene (II)



1, 2, 7-Trimethyl naphthalene (III)



1, 2, 5, 6-Tetramethyl naphthalene (IV)

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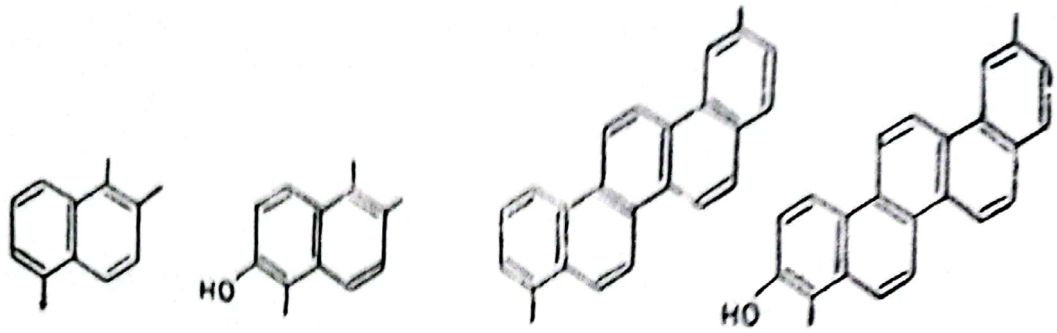
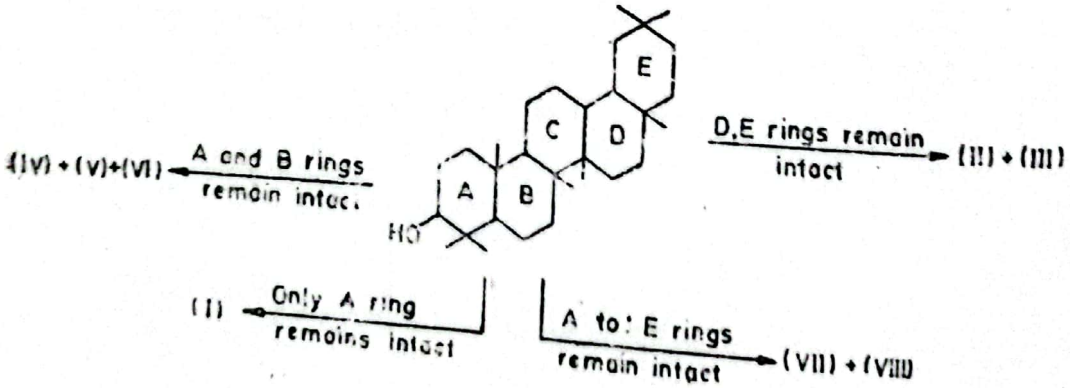


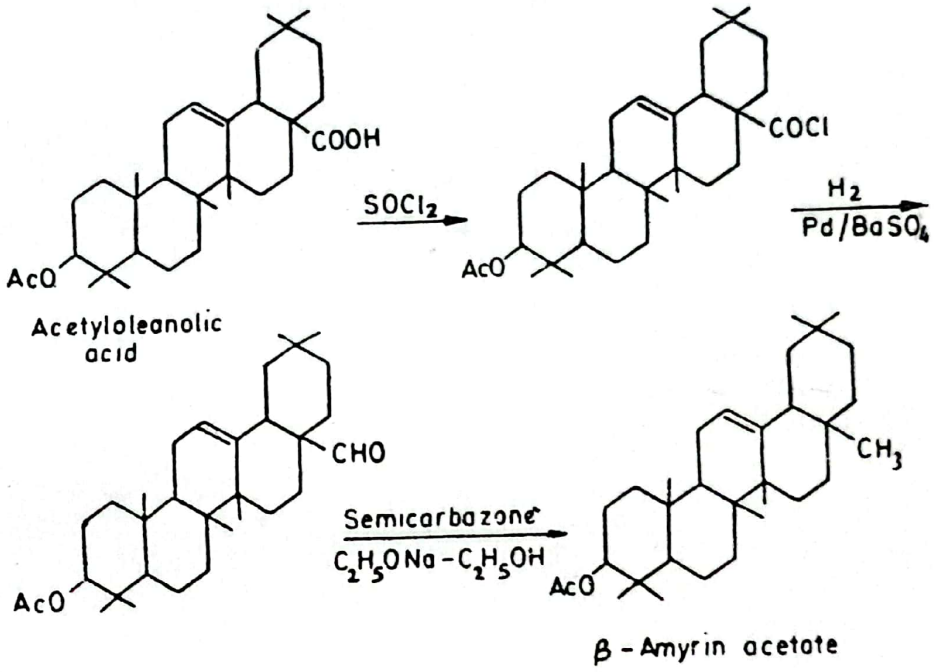
1, 2, 5-Trimethyl naphthalene (V) | 6-Hydroxy-1, 2, 5-trimethyl naphthalene (VI) | 1, 8-Dimethyl-picene (VII) | 2-Hydroxy-1, 8-dimethyl picene (VIII)

The structures of all the above products have been ascertained by doing comparison with the synthetic compounds of known structures.

From the products obtained from selenium dehydrogenation of β-amyrin, it follows that a picene carbon skeleton is present in β-amyrin. Further from the products (VI) and (VIII), it may be concluded that β-amyrin carries secondary alcoholic group at C-2.



7. Now the next problem is to ascertain the **position of double bond** in β-amyrin. This has been achieved by the degradation of oleanolic acid, a related member of β-amyrin group. These two compounds, *i.e.*, β-amyrin and oleanolic acid are closely related to each other ; this relationship has been confirmed by the fact that the latter can be converted into the former by converting the —COOH group of oleanolic acid into the—CH_3 group of β-amyrin by the standard method as given below :



When oleanolic acid is oxidatively degraded, it yields a lactone dicarboxylic acid which on pyrolysis gives the naphthalene derivatives containing rings A+B and D+E. The formation of these products reveals that the double bond must be present in the ring C. Further, it was also found that the double bond is present in between C_{12} and C_{18} in the ring C.

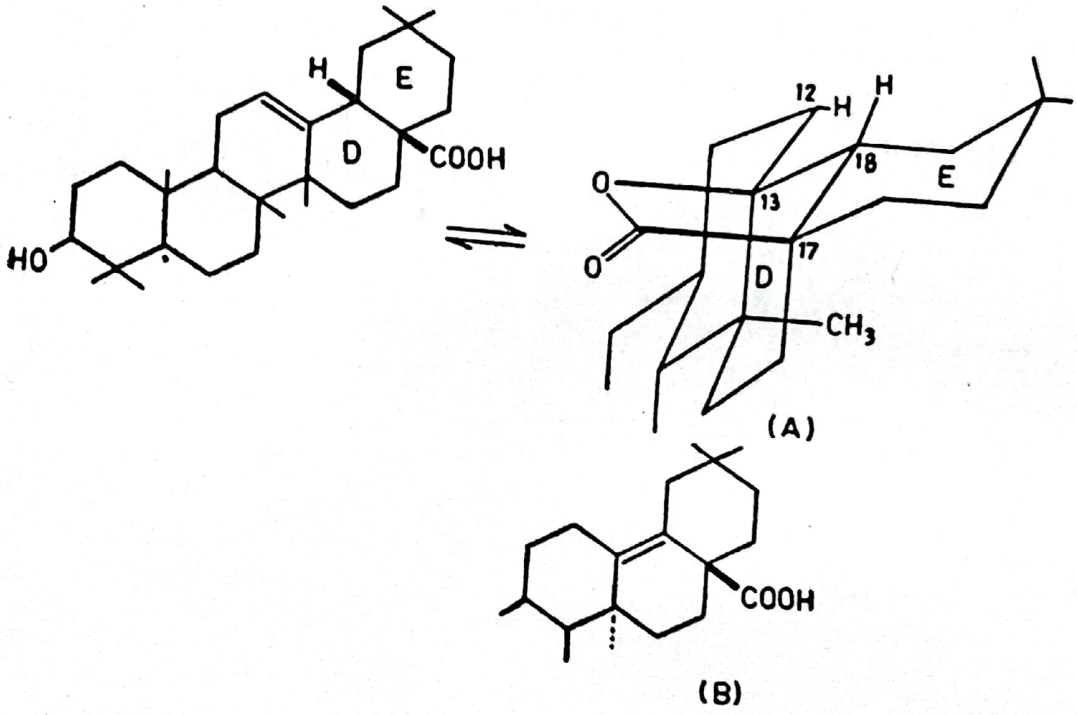
8. A few points regarding the stereochemistry of β-amyrin may be made. β-Amyrin has eight asymmetric centres.

(*i*) A/B ring junction is trans.

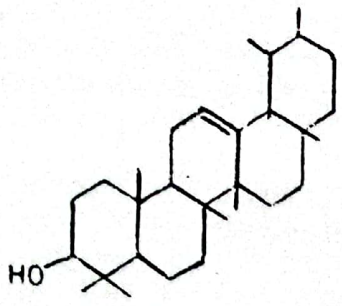
(*ii*) The stereo chemical features of ring A get support by the usual dehydration rearrangement with phosphorus pentachloride

(*iii*) The stereo chemistry of D/E ring junoture has been determined from the lactone formation studies in oleanolic acid.

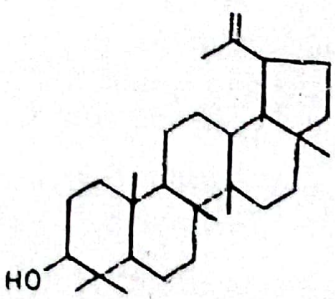
This acid when treated with hydrogen chloride in chloroform solution yielded an equilibrium mixture which contained 24% of lactone (A). The formation of this lactone is only possible when the carboxyl group at C-17 is axial, thus β-oriented. Failure of the lactone to undergo elimination to (B) (more stable position of the double bond) rather than to oleanolic acid revealed, therefore, that the hydrogen at C 18 is *cis* to the oxygen at C-13. This is in keeping with the generalisation that a preferred pathway is *trans* diaxial elimination, here involving C-12 axial hydrogen leading to the starting compound. This leads to the conclusion that the rings D and E in oleanolic acid must be *cis* fused.



The structures of other two pentacyclic terpenoids are as follows :



α-Amyrin



Lupeol